

Goodness-of-fit & residual plots

We use the DHARMA package to simulate, calculate and plot residuals.

Bru-Big combination

```
# For each behavior, we test the best model for residuals, uniformity, overdispersion, and inflation  
# calculate the residuals for the model
```

```
m1 <- simulateResiduals(fittedModel = additive_indirect_brubig, plot = F)
```

```
residuals(m1)
```

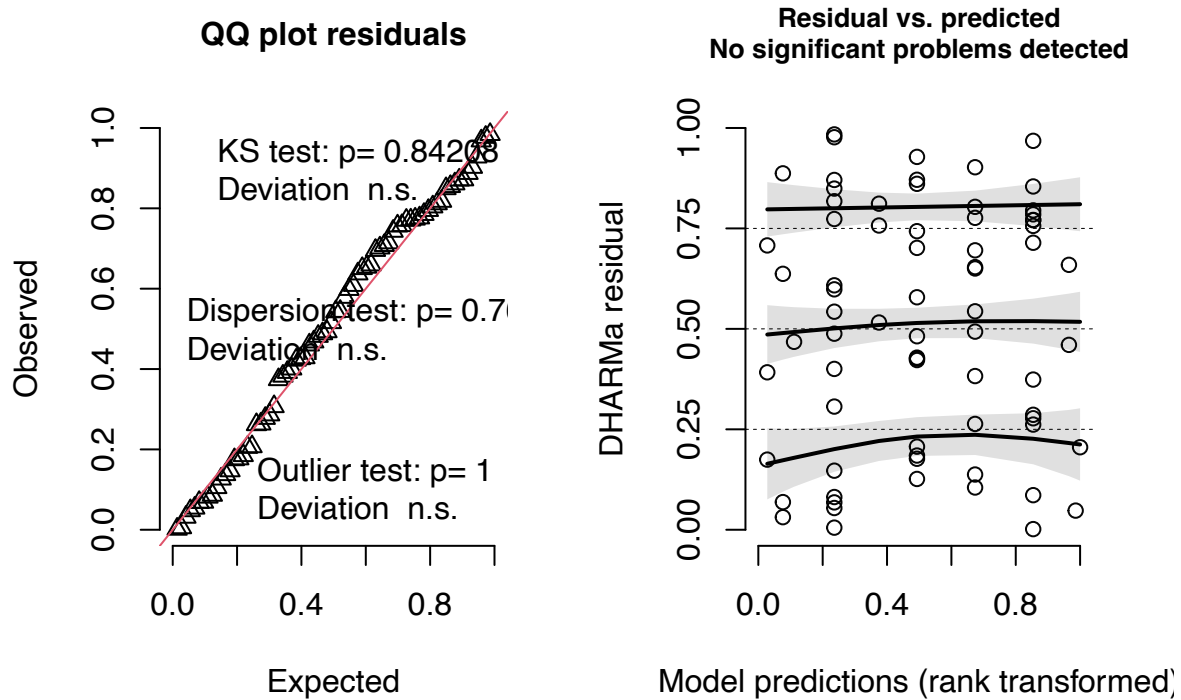
```
## [1] 0.286147522 0.261789168 0.068873789 0.068128401 0.005251602 0.054128933  
## [7] 0.081427452 0.086126174 0.391870214 0.105158853 0.184767229 0.047647234  
## [13] 0.373799717 0.147032733 0.206869186 0.467721159 0.636968599 0.277444826  
## [19] 0.137484944 0.493074722 0.176680720 0.031381607 0.428407370 0.422333957  
## [25] 0.515765693 0.001630627 0.481630005 0.382530000 0.707654047 0.887067454  
## [31] 0.543764812 0.578681029 0.598344749 0.542748977 0.126442296 0.773405869  
## [37] 0.487968671 0.174550430 0.701598925 0.608357104 0.743100942 0.426053526  
## [43] 0.870591285 0.205405048 0.653747325 0.695472579 0.306625191 0.776365952  
## [49] 0.803738018 0.650259821 0.400341400 0.817729590 0.784987533 0.757263061  
## [55] 0.984000000 0.976926097 0.871072385 0.928000000 0.861018270 0.770543445  
## [61] 0.659374677 0.460196760 0.811577342 0.902273537 0.263388235 0.794919396  
## [67] 0.756249027 0.714282588 0.849119871 0.771070282 0.968000000 0.854415379
```

```
#plot simulation output
```

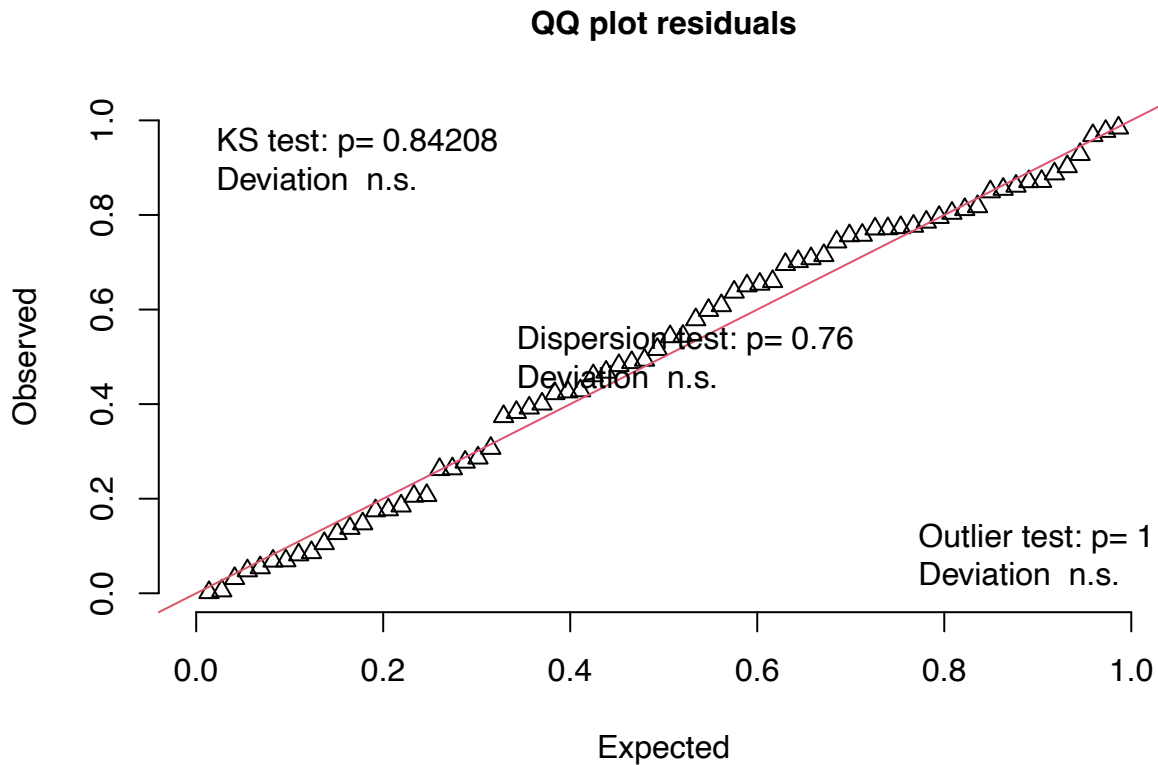
```
plot(m1)
```

```
## Warning in newton(lsp = lsp, X = G$X, y = G$y, Eb = G$Eb, UrS = G$UrS, L = G$L,  
## : Fitting terminated with step failure - check results carefully
```

DHARMA residual



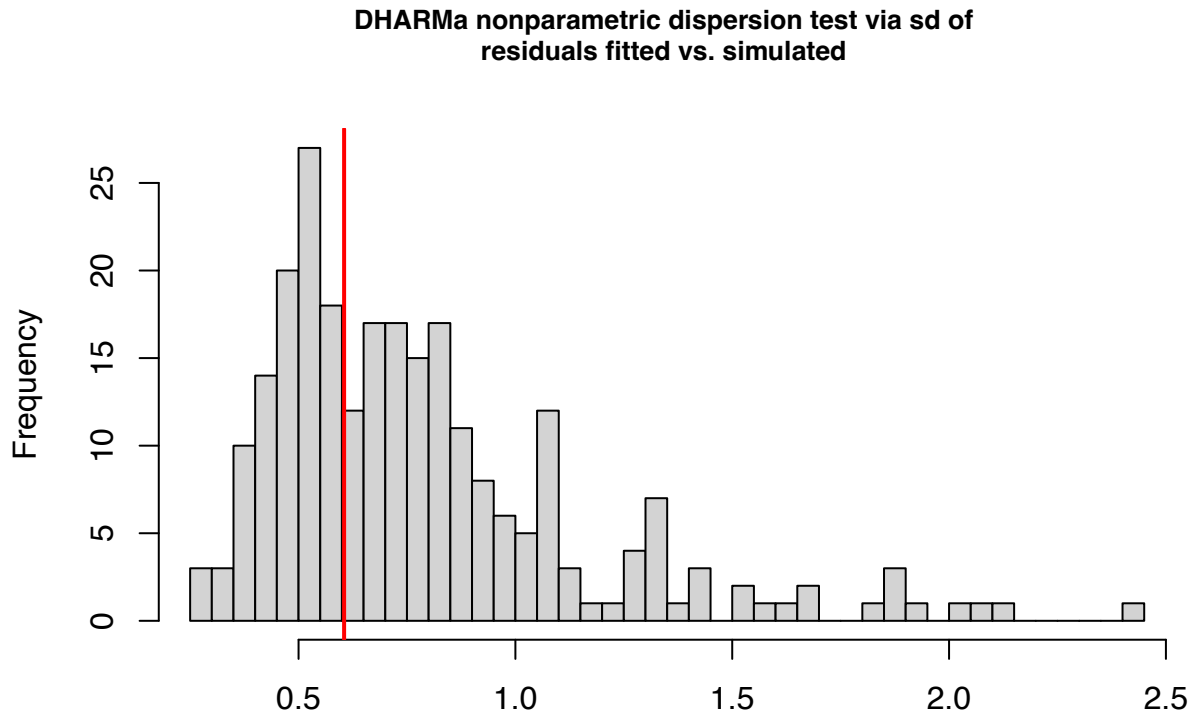
#tests if the overall distribution conforms to expectations.
`testUniformity(m1)`



 ## Exact one-sample Kolmogorov-Smirnov test

```
##
## data: simulationOutput$scaledResiduals
## D = 0.070473, p-value = 0.8421
## alternative hypothesis: two-sided
```

```
testDispersion(m1)
```

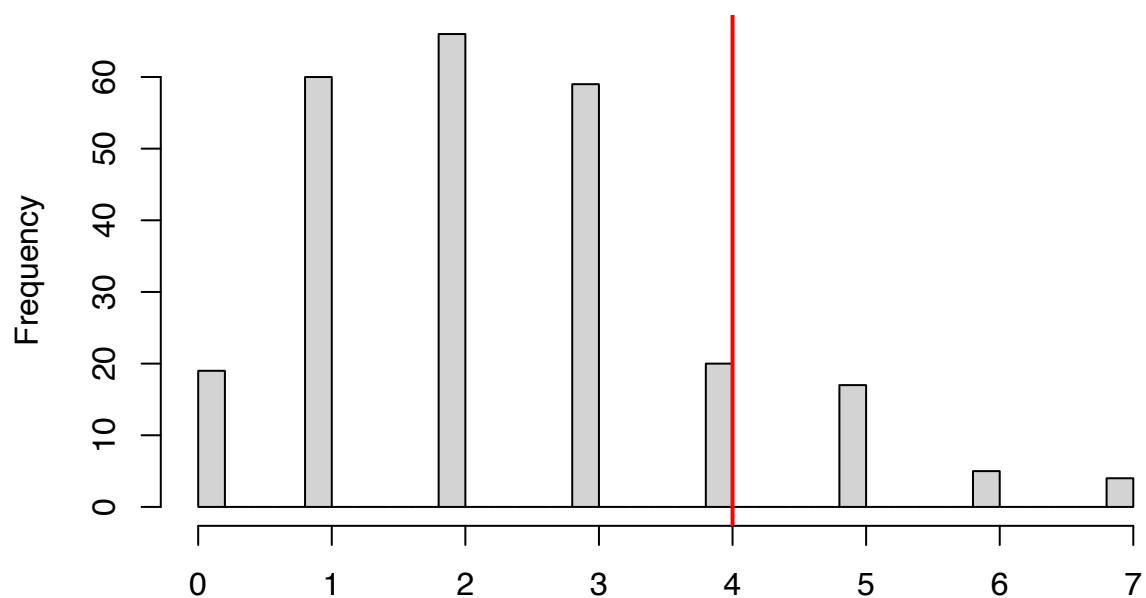


Simulated values, red line = fitted model. p-value (two.sided) = 0.76

```
##
## DHARMa nonparametric dispersion test via sd of residuals fitted vs.
## simulated
##
## data: simulationOutput
## dispersion = 0.76831, p-value = 0.76
## alternative hypothesis: two.sided
```

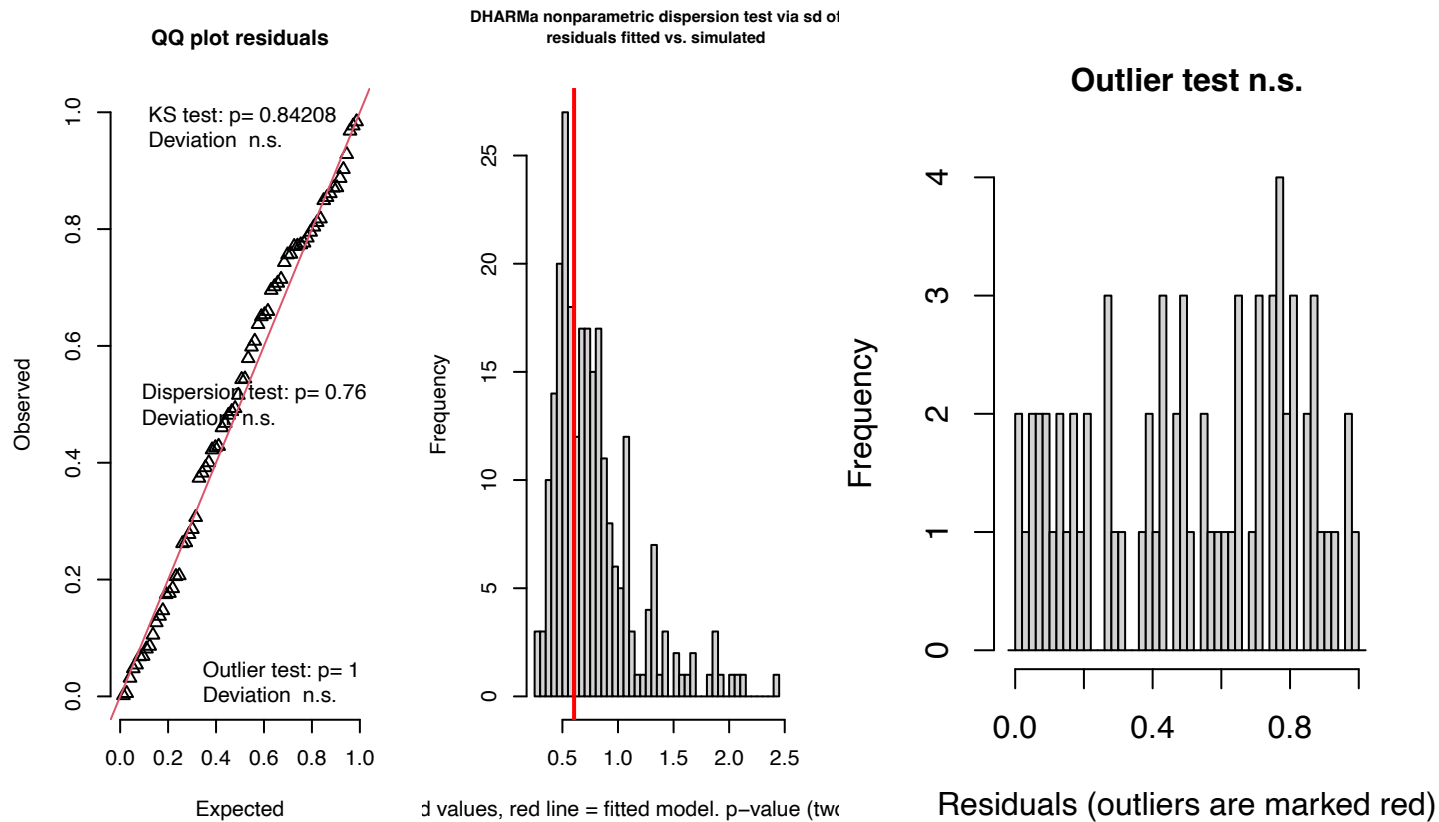
```
testZeroInflation(m1)
```

DHARMA zero-inflation test via comparison to expected zeros with simulation under H0 = fitted model



Simulated values, red line = fitted model. p-value (two.sided) = 0.368

```
##
## DHARMA zero-inflation test via comparison to expected zeros with
## simulation under H0 = fitted model
##
## data: simulationOutput
## ratioObsSim = 1.6892, p-value = 0.368
## alternative hypothesis: two.sided
testResiduals(m1)
```



```
## $uniformity
##
## Exact one-sample Kolmogorov-Smirnov test
##
## data: simulationOutput$scaledResiduals
## D = 0.070473, p-value = 0.8421
## alternative hypothesis: two-sided
##
##
## $dispersion
##
## DHARMA nonparametric dispersion test via sd of residuals fitted vs.
## simulated
##
## data: simulationOutput
## dispersion = 0.76831, p-value = 0.76
## alternative hypothesis: two.sided
##
##
## $outliers
##
## DHARMA bootstrapped outlier test
##
## data: simulationOutput
## outliers at both margin(s) = 0, observations = 72, p-value = 1
## alternative hypothesis: two.sided
## percent confidence interval:
## 0.00000000 0.02777778
```

```

## sample estimates:
## outlier frequency (expected: 0.0059722222222222 )
##
## $uniformity
##
## Exact one-sample Kolmogorov-Smirnov test
##
## data: simulationOutput$scaledResiduals
## D = 0.070473, p-value = 0.8421
## alternative hypothesis: two-sided
##
##
## $dispersion
##
## DHARMA nonparametric dispersion test via sd of residuals fitted vs.
## simulated
##
## data: simulationOutput
## dispersion = 0.76831, p-value = 0.76
## alternative hypothesis: two.sided
##
##
## $outliers
##
## DHARMA bootstrapped outlier test
##
## data: simulationOutput
## outliers at both margin(s) = 0, observations = 72, p-value = 1
## alternative hypothesis: two.sided
## percent confidence interval:
## 0.00000000 0.02777778
## sample estimates:
## outlier frequency (expected: 0.0059722222222222 )
##
##
# calculate the residuals for the model
m2 <- simulateResiduals(fittedModel = interaction_indirect_brubig, plot = F)

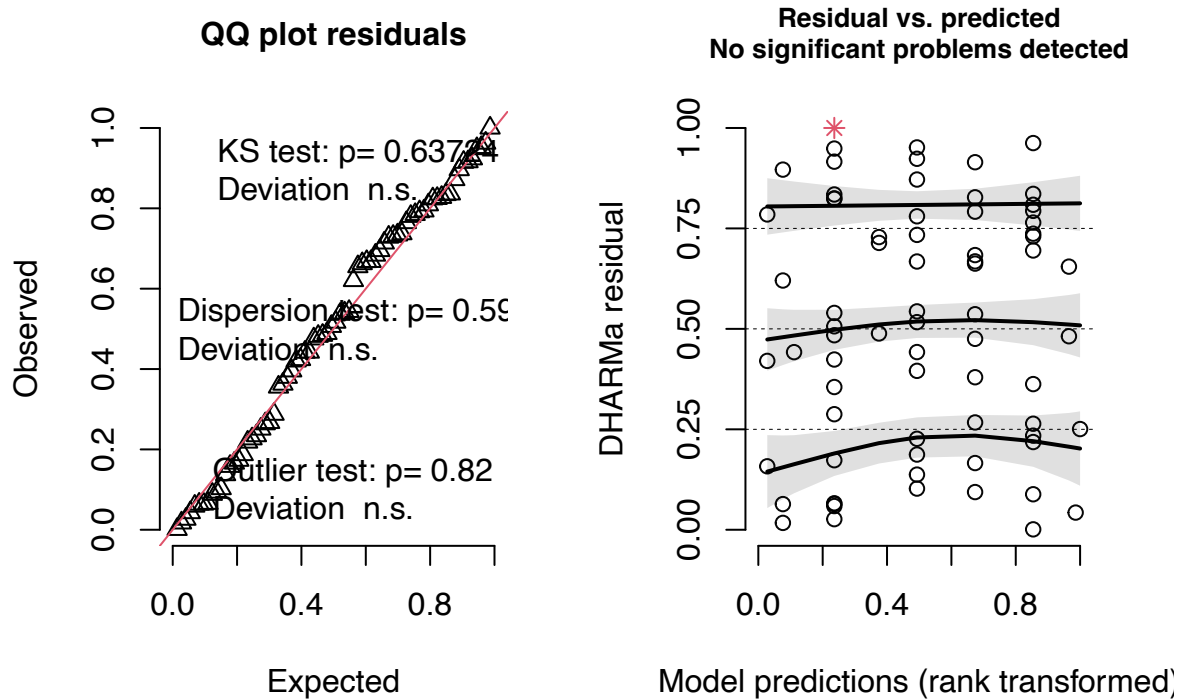
residuals(m2)

## [1] 0.23424969 0.21833841 0.01707752 0.06590199 0.02637831 0.05893169
## [7] 0.06404155 0.08835028 0.42026992 0.09381576 0.13730231 0.04275564
## [13] 0.36251774 0.17304178 0.18716519 0.44206390 0.62036429 0.26389542
## [19] 0.16600390 0.53643316 0.22641505 0.06386368 0.44224109 0.39549833
## [25] 0.48861544 0.00101238 0.54357031 0.37961827 0.78477062 0.89640040
## [31] 0.47516574 0.66764657 0.50632347 0.48416464 0.10248779 0.82481141
## [37] 0.35523688 0.15789924 0.73391221 0.53971090 0.78012966 0.51674164
## [43] 0.83444916 0.25067417 0.66243784 0.68364351 0.28760079 0.79172768
## [49] 0.82741992 0.66796761 0.42376335 0.82371977 0.76455549 0.72845577
## [55] 1.00000000 0.94871402 0.92331638 0.95156181 0.87166092 0.80888879
## [61] 0.65493332 0.48163625 0.71392730 0.91477141 0.26711094 0.73692547
## [67] 0.73045782 0.69507033 0.91612649 0.79370572 0.96242268 0.83600000

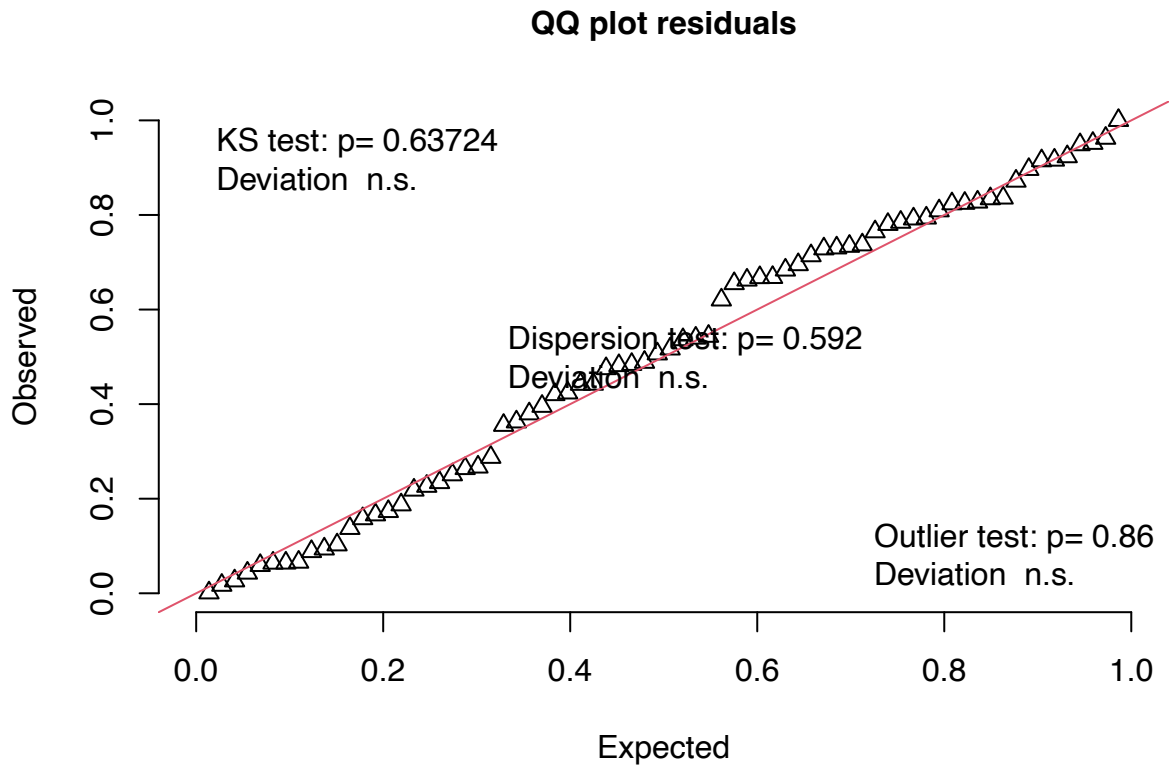
#plot simulation output
plot(m2)

```

DHARMa residual



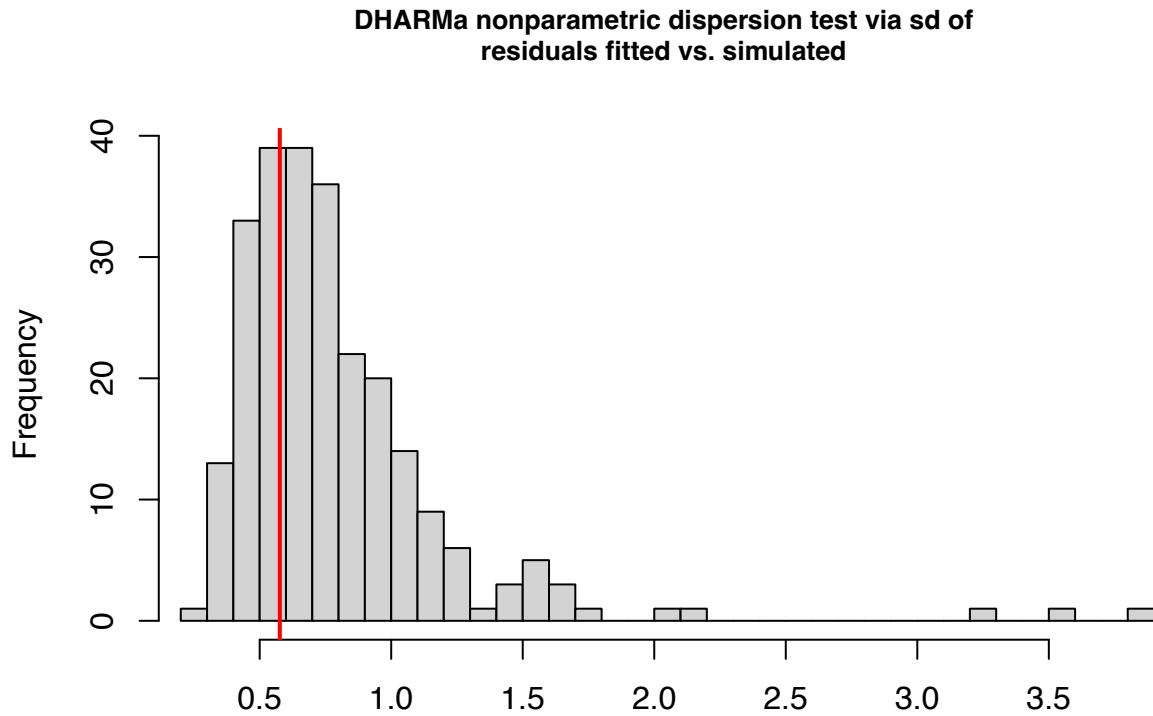
#tests if the overall distribution conforms to expectations.
`testUniformity(m2)`



 ## Exact one-sample Kolmogorov-Smirnov test

```
##
## data:  simulationOutput$scaledResiduals
## D = 0.085489, p-value = 0.6372
## alternative hypothesis: two-sided
```

```
testDispersion(m2)
```

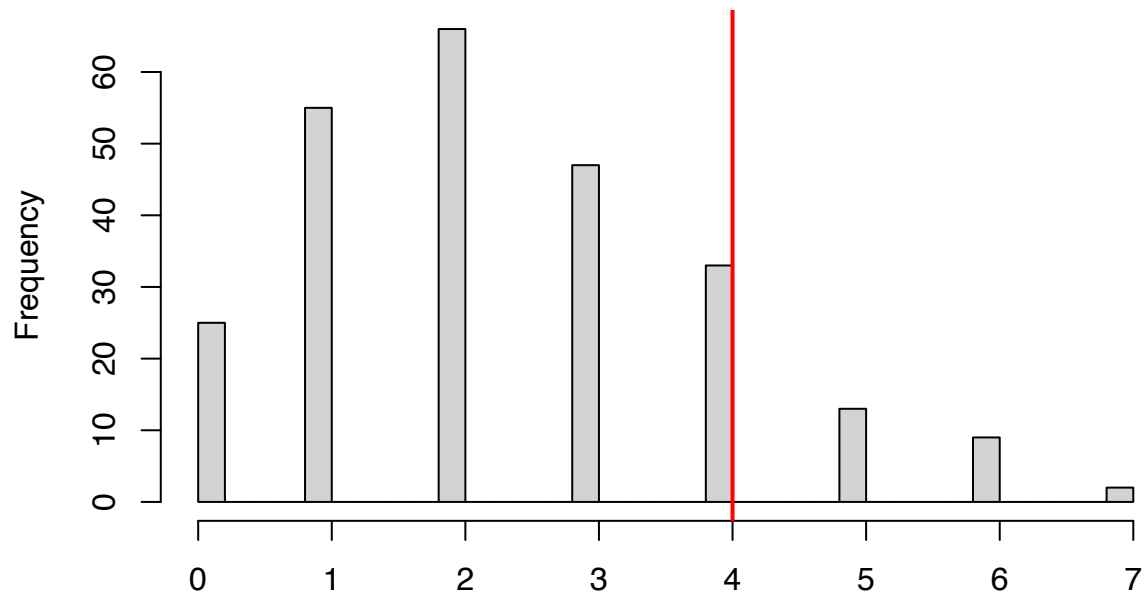


Simulated values, red line = fitted model. p-value (two.sided) = 0.592

```
##
## DHARMa nonparametric dispersion test via sd of residuals fitted vs.
## simulated
##
## data:  simulationOutput
## dispersion = 0.72539, p-value = 0.592
## alternative hypothesis: two.sided
```

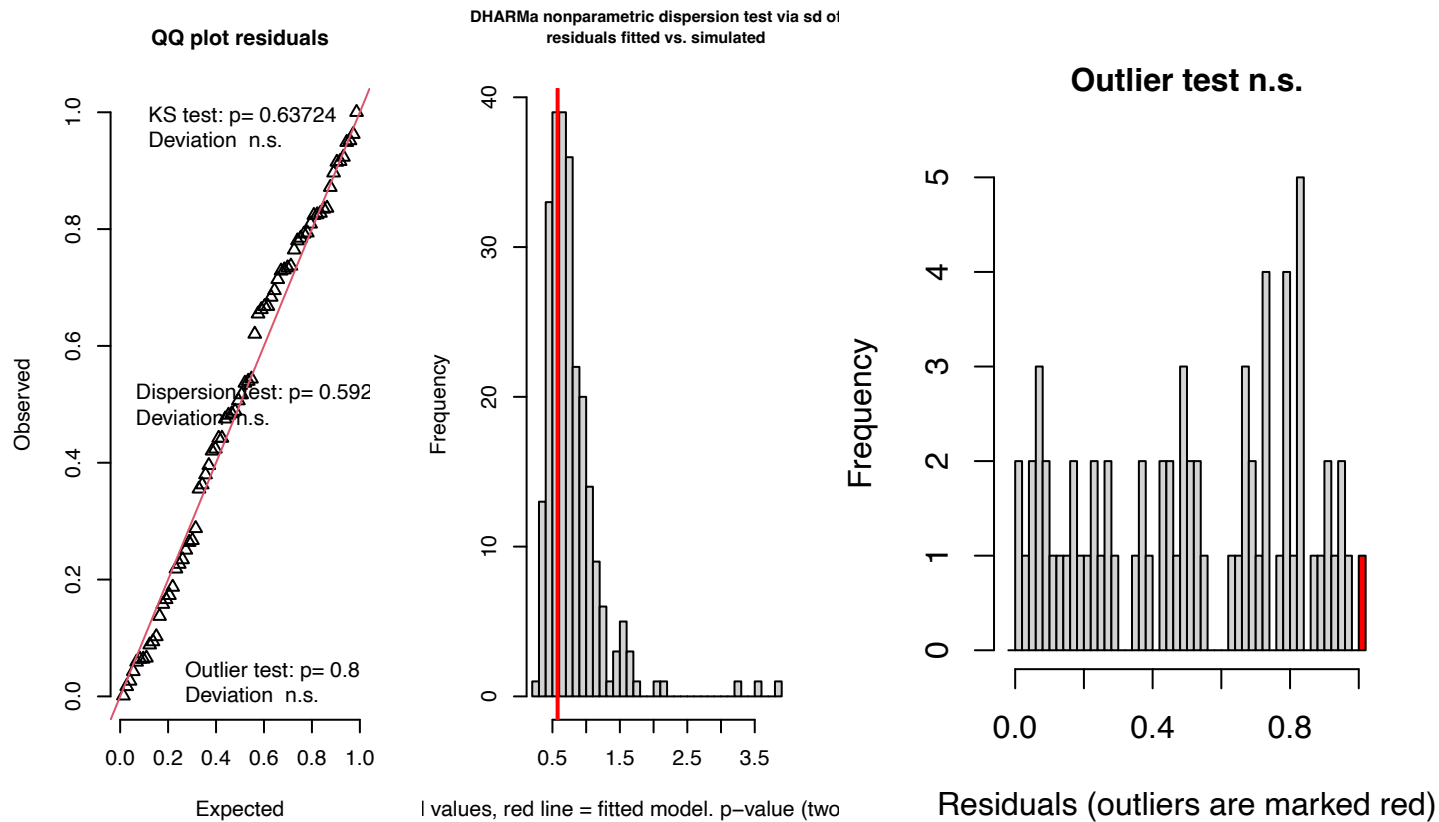
```
testZeroInflation(m2)
```

**DHARMA zero-inflation test via comparison to
expected zeros with simulation under H0 = fitted
model**



Simulated values, red line = fitted model. p-value (two.sided) = 0.456

```
##
## DHARMA zero-inflation test via comparison to expected zeros with
## simulation under H0 = fitted model
##
## data: simulationOutput
## ratioObsSim = 1.6863, p-value = 0.456
## alternative hypothesis: two.sided
testResiduals(m2)
```



```
## $uniformity
##
## Exact one-sample Kolmogorov-Smirnov test
##
## data: simulationOutput$scaledResiduals
## D = 0.085489, p-value = 0.6372
## alternative hypothesis: two-sided
##
##
## $dispersion
##
## DHARMA nonparametric dispersion test via sd of residuals fitted vs.
## simulated
##
## data: simulationOutput
## dispersion = 0.72539, p-value = 0.592
## alternative hypothesis: two.sided
##
##
## $outliers
##
## DHARMA bootstrapped outlier test
##
## data: simulationOutput
## outliers at both margin(s) = 1, observations = 72, p-value = 0.92
## alternative hypothesis: two.sided
## percent confidence interval:
## 0.00000000 0.03506944
```

```

## sample estimates:
## outlier frequency (expected: 0.008472222222222 )
##                                0.01388889

## $uniformity
##
## Exact one-sample Kolmogorov-Smirnov test
##
## data: simulationOutput$scaledResiduals
## D = 0.085489, p-value = 0.6372
## alternative hypothesis: two-sided
##
##
## $dispersion
##
## DHARMA nonparametric dispersion test via sd of residuals fitted vs.
## simulated
##
## data: simulationOutput
## dispersion = 0.72539, p-value = 0.592
## alternative hypothesis: two.sided
##
##
## $outliers
##
## DHARMA bootstrapped outlier test
##
## data: simulationOutput
## outliers at both margin(s) = 1, observations = 72, p-value = 0.92
## alternative hypothesis: two.sided
## percent confidence interval:
## 0.00000000 0.03506944
## sample estimates:
## outlier frequency (expected: 0.008472222222222 )
##                                0.01388889

# calculate the residuals for the model
m7 <- simulateResiduals(fittedModel = additive_direct_brubig, plot = F)

residuals(m7)

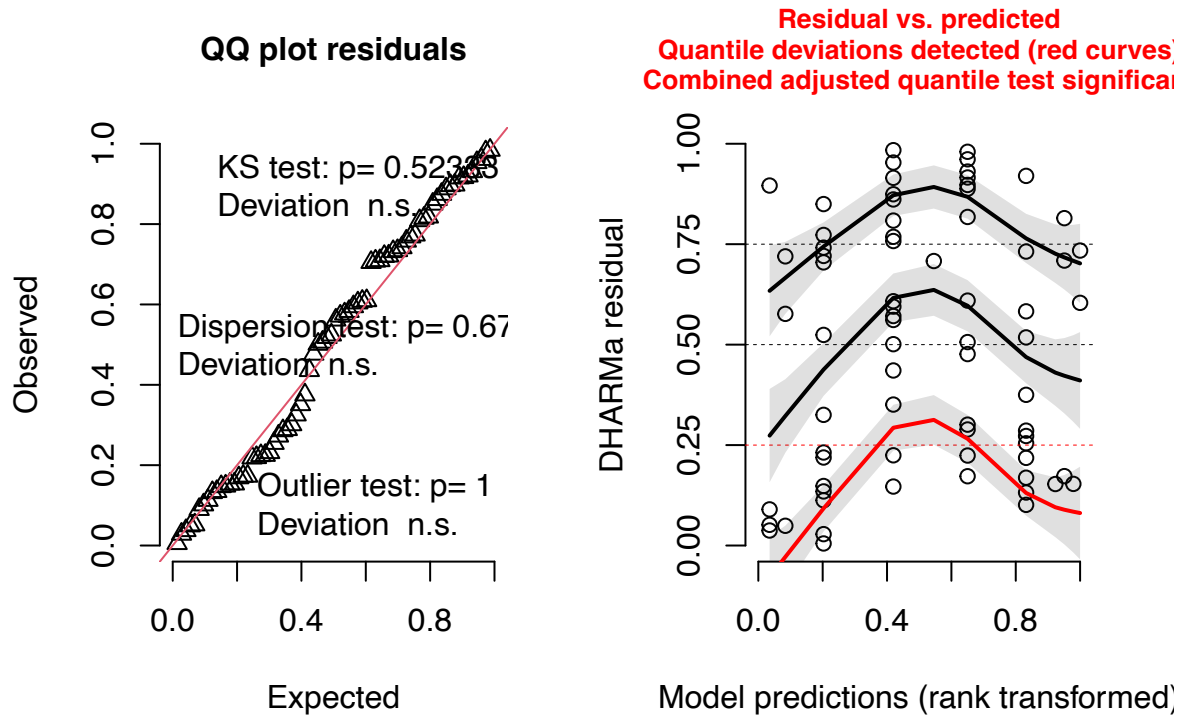
## [1] 0.609574644 0.288820493 0.037368159 0.435665194 0.146743520 0.350873702
## [7] 0.561372269 0.172468618 0.049160256 0.101271398 0.112595921 0.153406028
## [13] 0.223934010 0.571153725 0.219023538 0.708097191 0.089937740 0.300882866
## [19] 0.218198923 0.517995240 0.524196212 0.896000000 0.028356833 0.325212851
## [25] 0.172739403 0.506339782 0.134650854 0.272582921 0.576591569 0.051558216
## [31] 0.131872849 0.231517514 0.808789922 0.501105703 0.704692799 0.594620491
## [37] 0.757853446 0.719695059 0.148186531 0.608049584 0.005150675 0.741366525
## [43] 0.224586906 0.153089311 0.375075868 0.168779655 0.768328433 0.254412490
## [49] 0.285949328 0.582850778 0.874297307 0.915123647 0.476482455 0.709217636
## [55] 0.953270294 0.861355478 0.772987070 0.720719868 0.849812367 0.817783103
## [61] 0.604000000 0.734157296 0.814827848 0.731699662 0.920000000 0.897413795
## [67] 0.888000000 0.930813591 0.984000000 0.961239513 0.916000000 0.980000000

#plot simulation output
plot(m7)

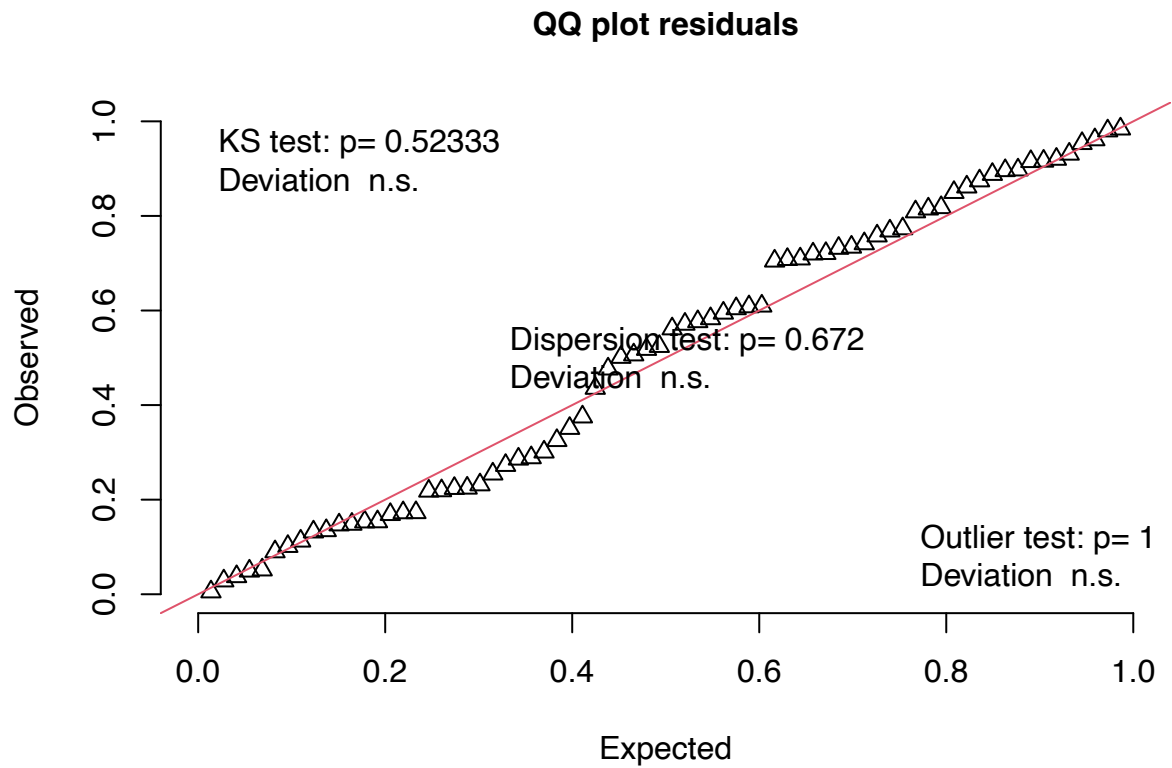
```

```
## qu = 0.25, log(sigma) = -2.435802 : outer Newton did not converge fully.
## Warning in newton(lsp = lsp, X = G$X, y = G$y, Eb = G$Eb, UrS = G$UrS, L = G$L,
## : Fitting terminated with step failure - check results carefully
```

DHARMA residual

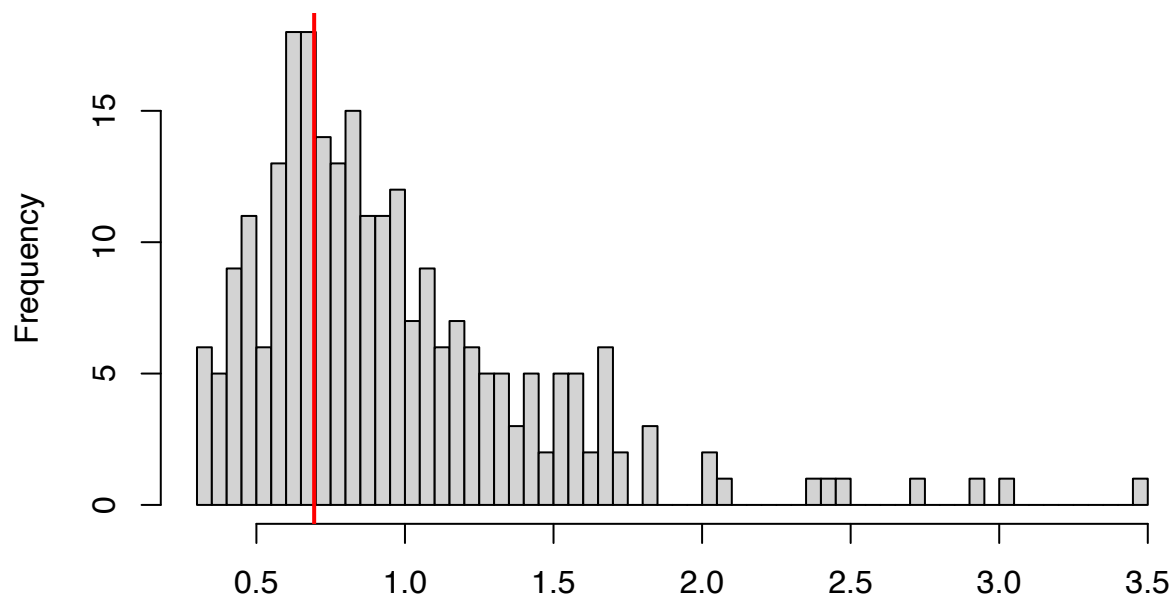


```
#tests if the overall distribution conforms to expectations.
testUniformity(m7)
```



```
##  
## Exact one-sample Kolmogorov-Smirnov test  
##  
## data: simulationOutput$scaledResiduals  
## D = 0.093582, p-value = 0.5233  
## alternative hypothesis: two-sided  
testDispersion(m7)
```

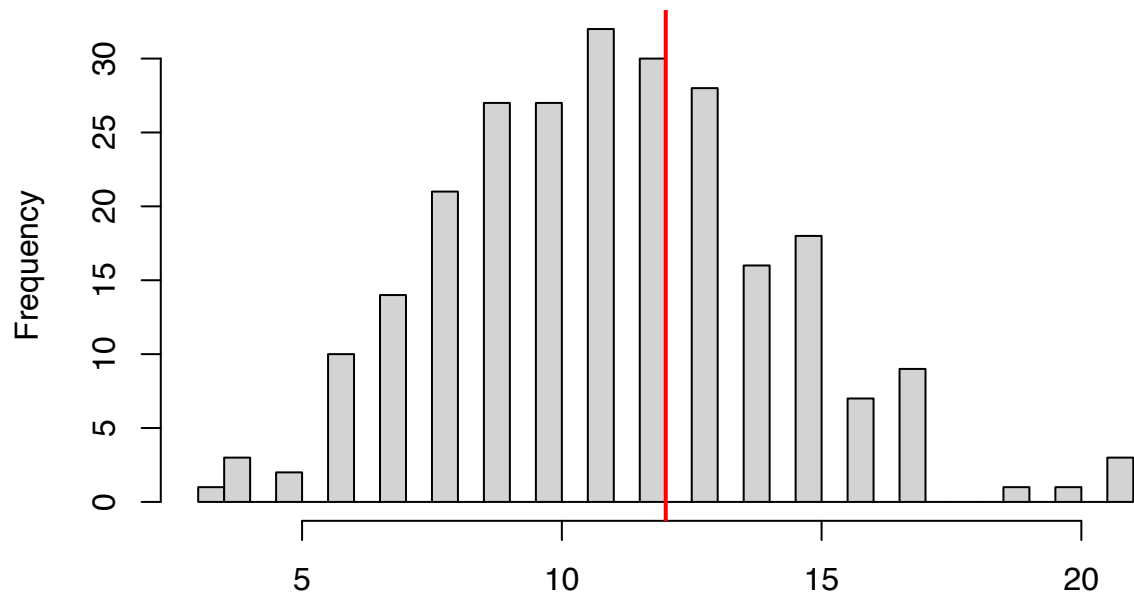
DHARMa nonparametric dispersion test via sd of residuals fitted vs. simulated



Simulated values, red line = fitted model. p-value (two.sided) = 0.672

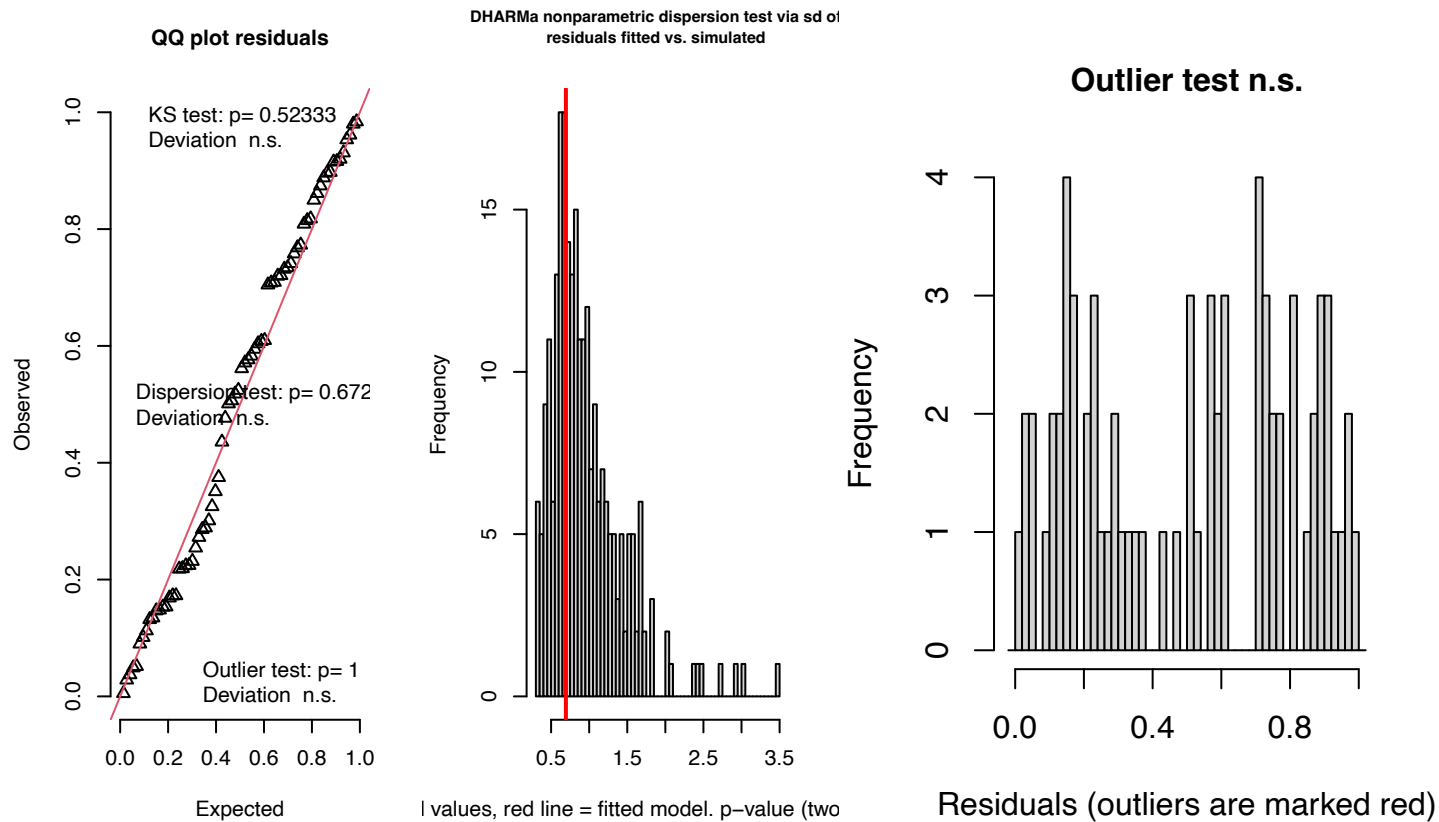
```
##
## DHARMa nonparametric dispersion test via sd of residuals fitted vs.
## simulated
##
## data:  simulationOutput
## dispersion = 0.7224, p-value = 0.672
## alternative hypothesis: two.sided
testZeroInflation(m7)
```

DHARMA zero-inflation test via comparison to expected zeros with simulation under H0 = fitted model



Simulated values, red line = fitted model. p-value (two.sided) = 0.904

```
##
## DHARMA zero-inflation test via comparison to expected zeros with
## simulation under H0 = fitted model
##
## data: simulationOutput
## ratioObsSim = 1.071, p-value = 0.904
## alternative hypothesis: two.sided
testResiduals(m7)
```



```
## $uniformity
##
## Exact one-sample Kolmogorov-Smirnov test
##
## data: simulationOutput$scaledResiduals
## D = 0.093582, p-value = 0.5233
## alternative hypothesis: two-sided
##
##
## $dispersion
##
## DHARMA nonparametric dispersion test via sd of residuals fitted vs.
## simulated
##
## data: simulationOutput
## dispersion = 0.7224, p-value = 0.672
## alternative hypothesis: two.sided
##
##
## $outliers
##
## DHARMA bootstrapped outlier test
##
## data: simulationOutput
## outliers at both margin(s) = 0, observations = 72, p-value = 1
## alternative hypothesis: two.sided
## percent confidence interval:
## 0.00000000 0.02777778
```

```

## sample estimates:
## outlier frequency (expected: 0.007361111111111111 )
##
## 0

## $uniformity
##
## Exact one-sample Kolmogorov-Smirnov test
##
## data: simulationOutput$scaledResiduals
## D = 0.093582, p-value = 0.5233
## alternative hypothesis: two-sided
##
##
## $dispersion
##
## DHARMA nonparametric dispersion test via sd of residuals fitted vs.
## simulated
##
## data: simulationOutput
## dispersion = 0.7224, p-value = 0.672
## alternative hypothesis: two.sided
##
##
## $outliers
##
## DHARMA bootstrapped outlier test
##
## data: simulationOutput
## outliers at both margin(s) = 0, observations = 72, p-value = 1
## alternative hypothesis: two.sided
## percent confidence interval:
## 0.00000000 0.02777778
## sample estimates:
## outlier frequency (expected: 0.007361111111111111 )
##
## 0

# calculate the residuals for the model
m8 <- simulateResiduals(fittedModel = interaction_direct_brubig, plot = F)

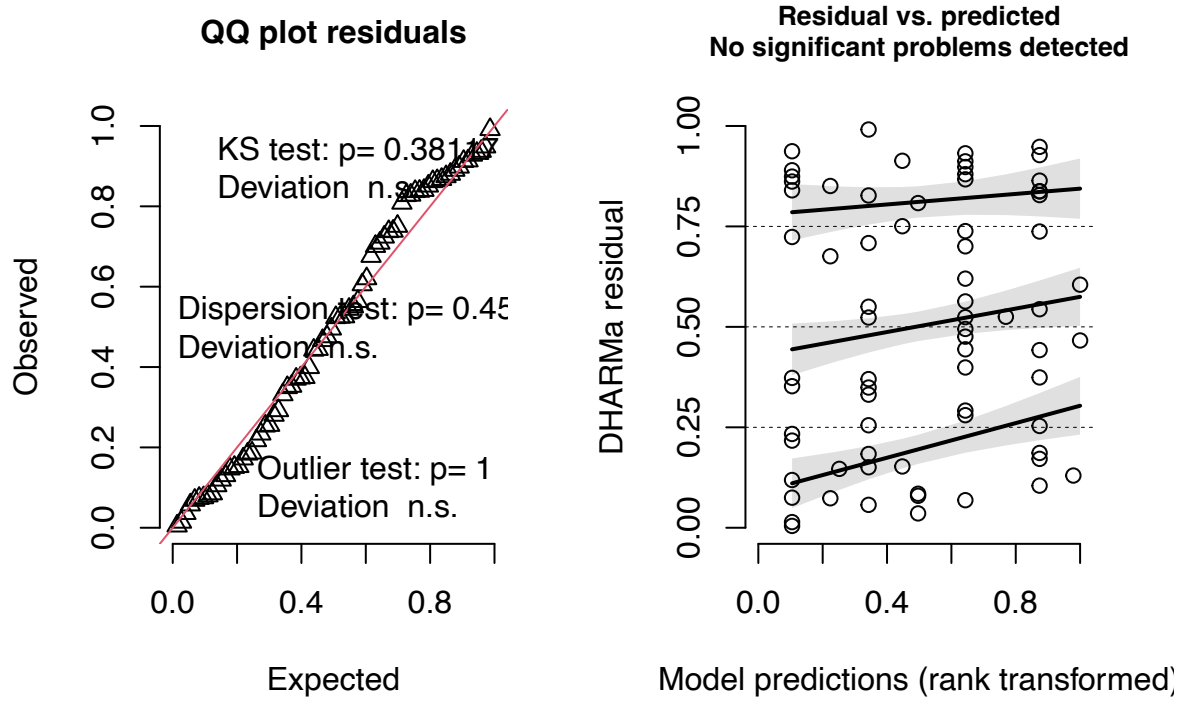
residuals(m8)

## [1] 0.54393588 0.25336962 0.03573855 0.39892272 0.06856238 0.28015776
## [7] 0.47562638 0.10479603 0.07335440 0.05700541 0.07488430 0.13000298
## [13] 0.17147801 0.49480888 0.37297366 0.52514363 0.07956073 0.18616410
## [19] 0.25505965 0.55008014 0.72367937 0.80811676 0.00461898 0.35232496
## [25] 0.15252043 0.44220502 0.23345492 0.34899695 0.67568429 0.08466123
## [31] 0.15052757 0.21693996 0.73833727 0.44409481 0.83980078 0.52419170
## [37] 0.62000000 0.85061528 0.11884995 0.56333489 0.01409962 0.86169661
## [43] 0.29167896 0.14656279 0.52321793 0.18363412 0.70045815 0.33125290
## [49] 0.36996224 0.70854891 0.89783948 0.88000000 0.37414503 0.75046734
## [55] 0.91200000 0.86718697 0.87397418 0.88983093 0.93714527 0.73730706
## [61] 0.46614579 0.60499377 0.91333169 0.82703315 0.99124941 0.83704905
## [67] 0.83773613 0.82800000 0.93200000 0.94800000 0.86400000 0.92751520

#plot simulation output
plot(m8)

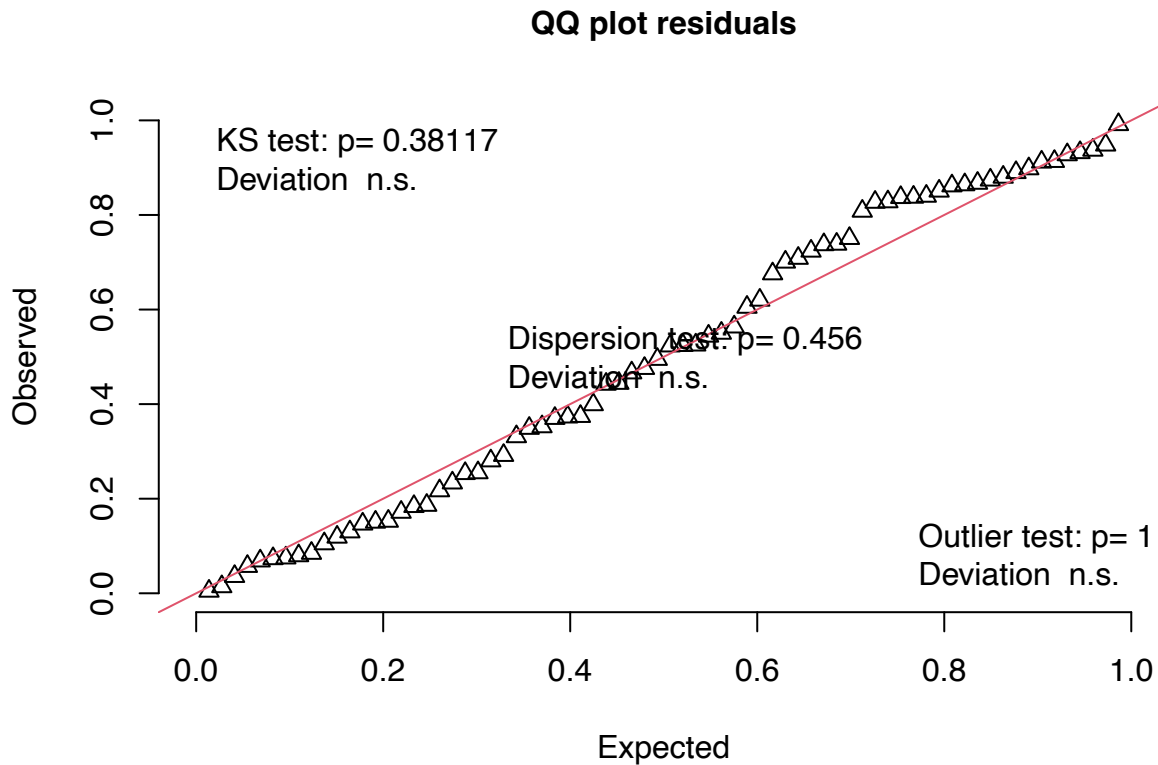
```

DHARMA residual



#tests if the overall distribution conforms to expectations.

`testUniformity(m8)`

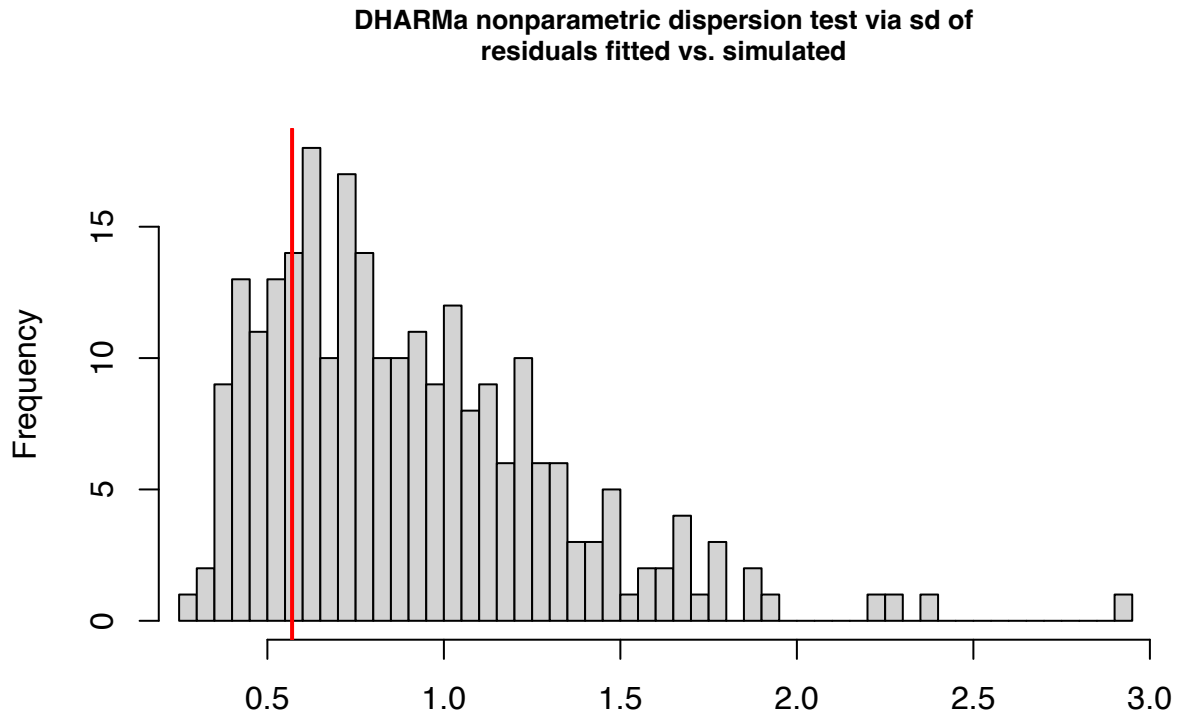


##

Exact one-sample Kolmogorov-Smirnov test

```
##
## data: simulationOutput$scaledResiduals
## D = 0.10481, p-value = 0.3812
## alternative hypothesis: two-sided
```

```
testDispersion(m8)
```

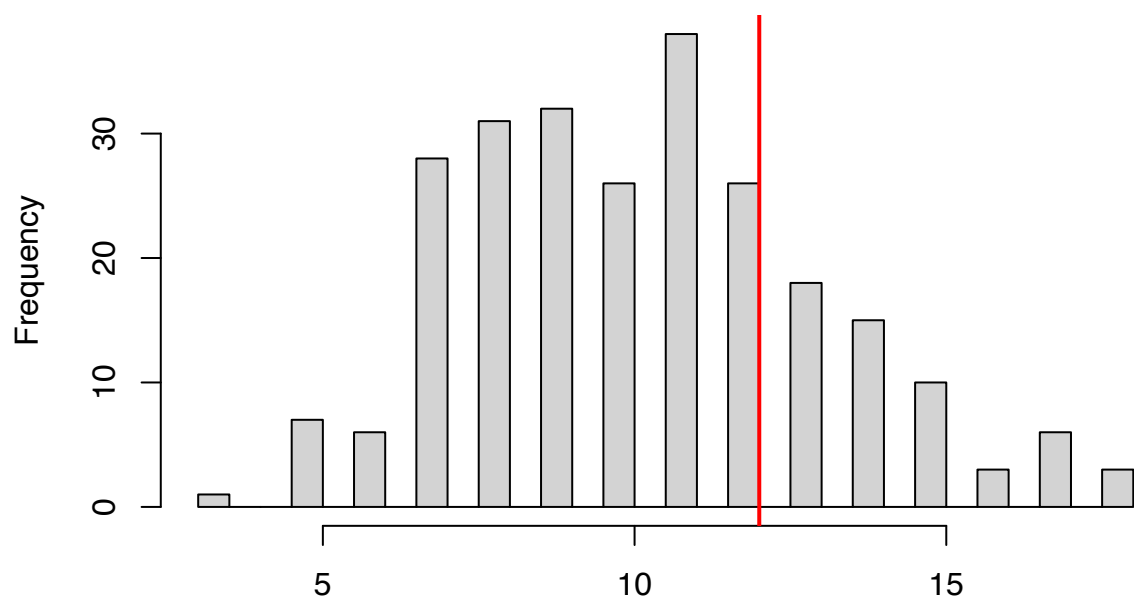


Simulated values, red line = fitted model. p-value (two.sided) = 0.456

```
##
## DHARMa nonparametric dispersion test via sd of residuals fitted vs.
## simulated
##
## data: simulationOutput
## dispersion = 0.63152, p-value = 0.456
## alternative hypothesis: two.sided
```

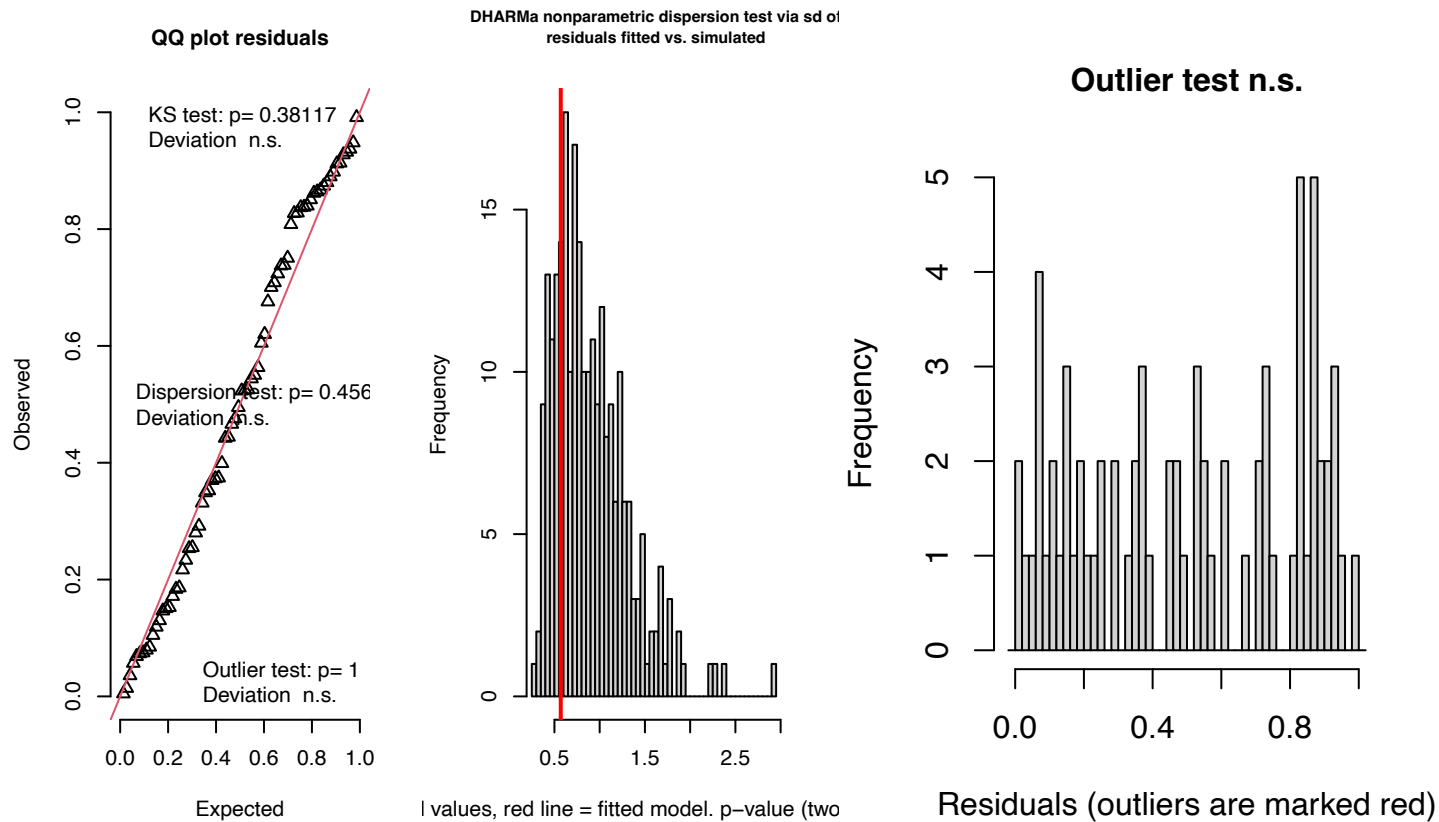
```
testZeroInflation(m8)
```

DHARMA zero-inflation test via comparison to expected zeros with simulation under H0 = fitted model



Simulated values, red line = fitted model. p-value (two.sided) = 0.648

```
##
## DHARMA zero-inflation test via comparison to expected zeros with
## simulation under H0 = fitted model
##
## data: simulationOutput
## ratioObsSim = 1.1565, p-value = 0.648
## alternative hypothesis: two.sided
testResiduals(m8)
```



```
## $uniformity
##
## Exact one-sample Kolmogorov-Smirnov test
##
## data: simulationOutput$scaledResiduals
## D = 0.10481, p-value = 0.3812
## alternative hypothesis: two-sided
##
##
## $dispersion
##
## DHARMA nonparametric dispersion test via sd of residuals fitted vs.
## simulated
##
## data: simulationOutput
## dispersion = 0.63152, p-value = 0.456
## alternative hypothesis: two.sided
##
##
## $outliers
##
## DHARMA bootstrapped outlier test
##
## data: simulationOutput
## outliers at both margin(s) = 0, observations = 72, p-value = 1
## alternative hypothesis: two.sided
## percent confidence interval:
## 0.00000000 0.03506944
```

```

## sample estimates:
## outlier frequency (expected: 0.00527777777777778 )
##
## 0

## $uniformity
##
## Exact one-sample Kolmogorov-Smirnov test
##
## data: simulationOutput$scaledResiduals
## D = 0.10481, p-value = 0.3812
## alternative hypothesis: two-sided
##
##
## $dispersion
##
## DHARMA nonparametric dispersion test via sd of residuals fitted vs.
## simulated
##
## data: simulationOutput
## dispersion = 0.63152, p-value = 0.456
## alternative hypothesis: two.sided
##
##
## $outliers
##
## DHARMA bootstrapped outlier test
##
## data: simulationOutput
## outliers at both margin(s) = 0, observations = 72, p-value = 1
## alternative hypothesis: two.sided
## percent confidence interval:
## 0.00000000 0.03506944
## sample estimates:
## outlier frequency (expected: 0.00527777777777778 )
##
## 0

# calculate the residuals for the model
m19 <- simulateResiduals(fittedModel = additive_harass_brubig, plot = F)

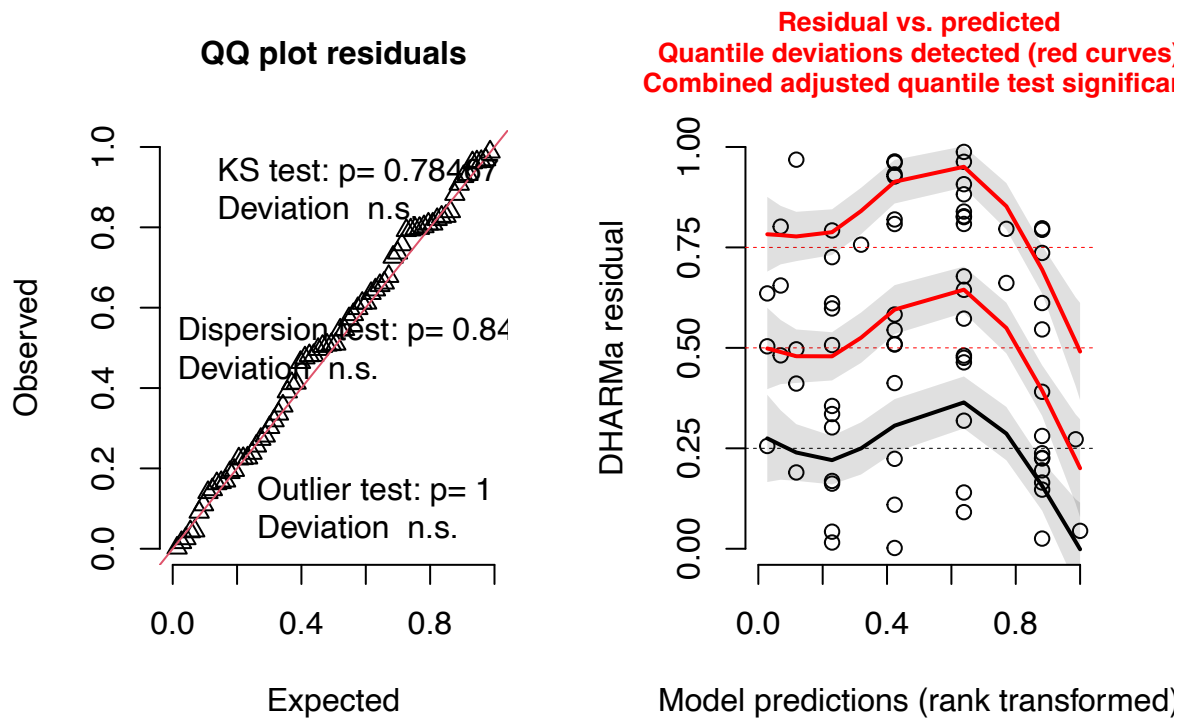
residuals(m19)

## [1] 0.001967273 0.223814563 0.411179843 0.480769738 0.140064069 0.463883007
## [7] 0.678006612 0.509511831 0.255806973 0.302018743 0.147245937 0.272445934
## [13] 0.544479449 0.644413371 0.390690915 0.756831135 0.496368335 0.508272350
## [19] 0.162110508 0.335328316 0.164863079 0.968367294 0.195158724 0.025300752
## [25] 0.655227526 0.808239347 0.224662630 0.015573120 0.635561357 0.189922592
## [31] 0.506995373 0.280736830 0.882216260 0.091180892 0.736300406 0.475226696
## [37] 0.825345543 0.503864418 0.237734697 0.572751181 0.226153929 0.793700590
## [43] 0.318728551 0.044561466 0.355448462 0.168761003 0.839218213 0.042519944
## [49] 0.611077596 0.725771511 0.907375407 0.827594354 0.109587854 0.801720824
## [55] 0.963101629 0.808349939 0.545932160 0.611622035 0.797633094 0.931730636
## [61] 0.661541875 0.796184349 0.481375585 0.792226419 0.598009235 0.583029510
## [67] 0.412371943 0.964000000 0.987698083 0.926558637 0.819558109 0.960798723

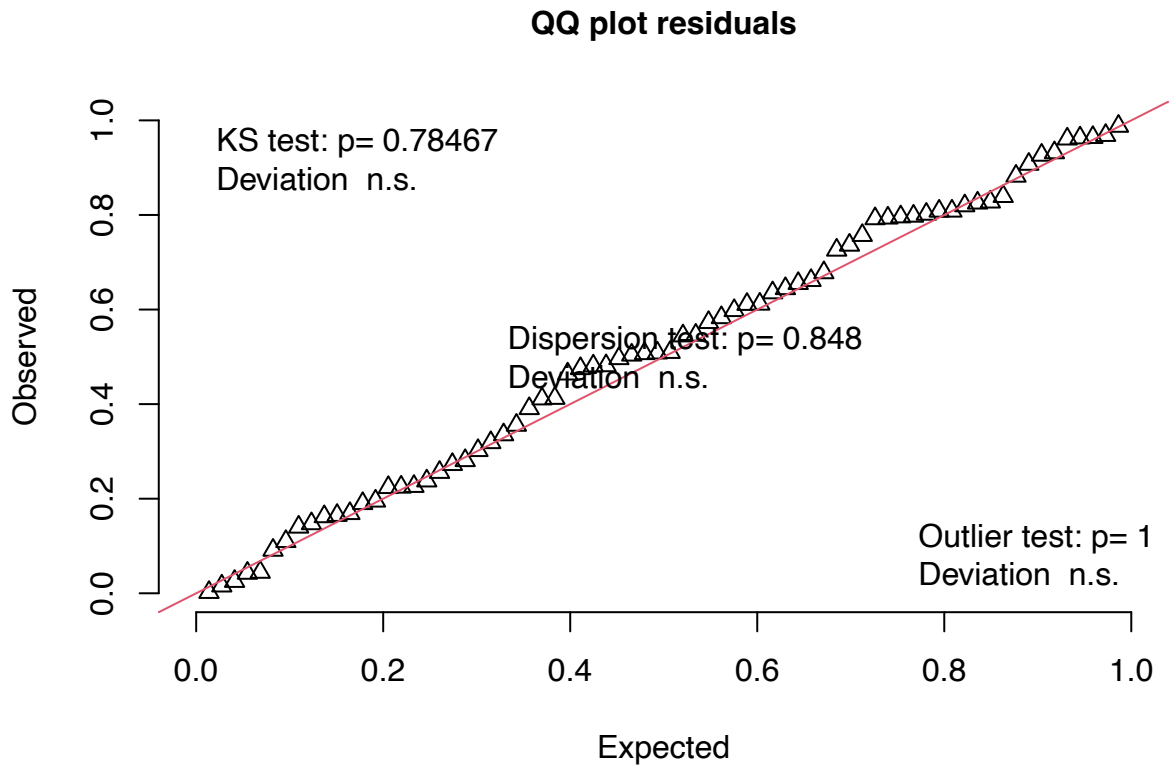
#plot simulation output
plot(m19)

```

DHARMA residual



#tests if the overall distribution conforms to expectations.
`testUniformity(m19)`

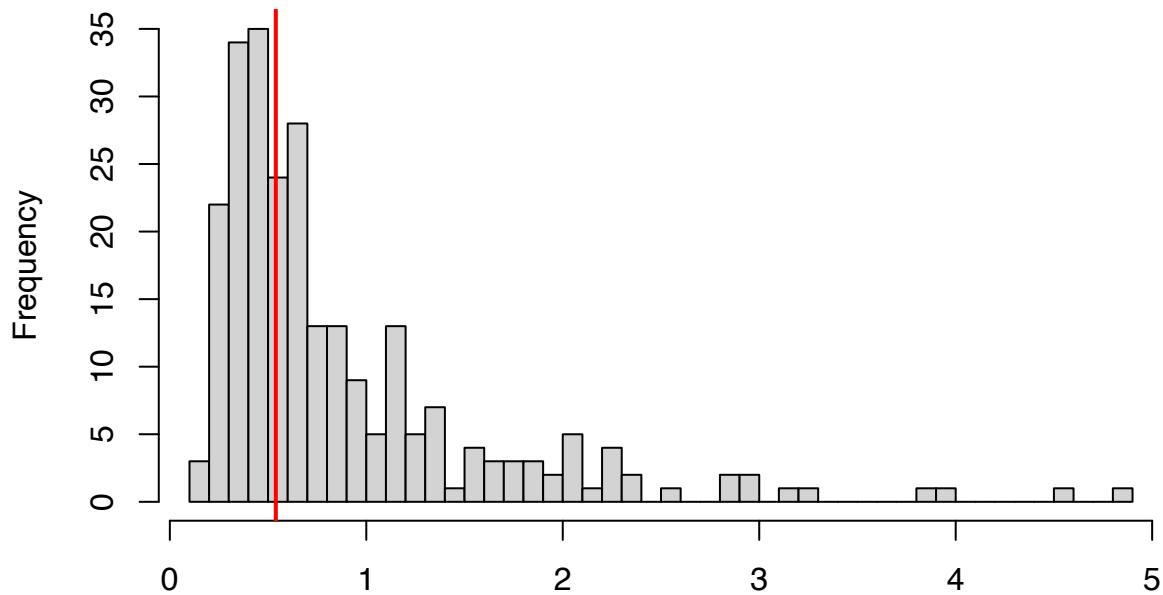


 ## Exact one-sample Kolmogorov-Smirnov test

```
##
## data: simulationOutput$scaledResiduals
## D = 0.074994, p-value = 0.7847
## alternative hypothesis: two-sided
```

```
testDispersion(m19)
```

DHARMa nonparametric dispersion test via sd of residuals fitted vs. simulated

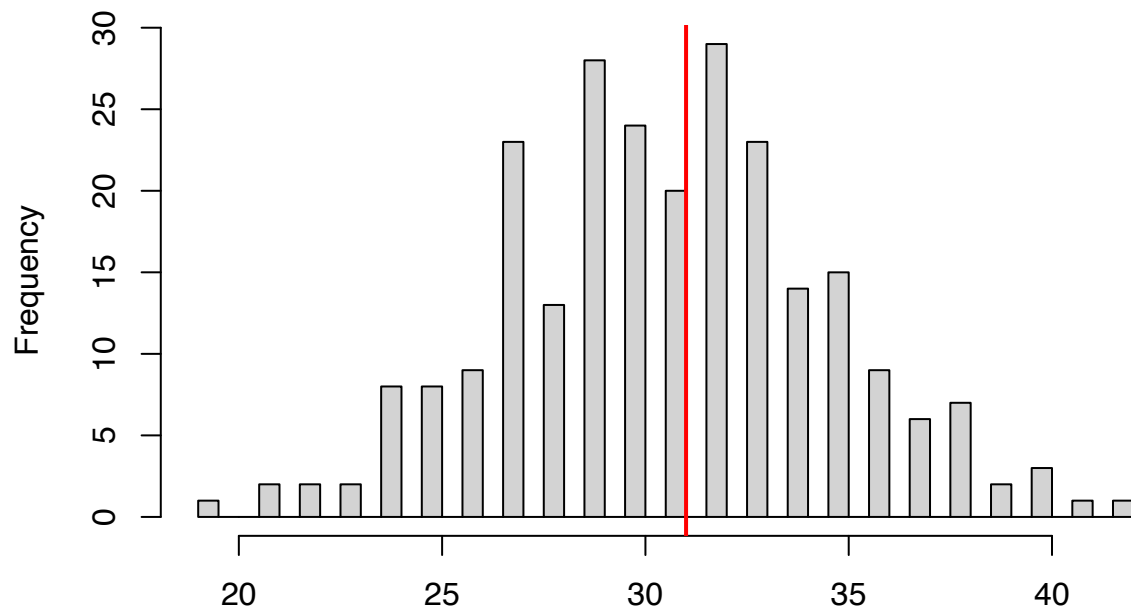


Simulated values, red line = fitted model. p-value (two.sided) = 0.848

```
##
## DHARMa nonparametric dispersion test via sd of residuals fitted vs.
## simulated
##
## data: simulationOutput
## dispersion = 0.60948, p-value = 0.848
## alternative hypothesis: two.sided
```

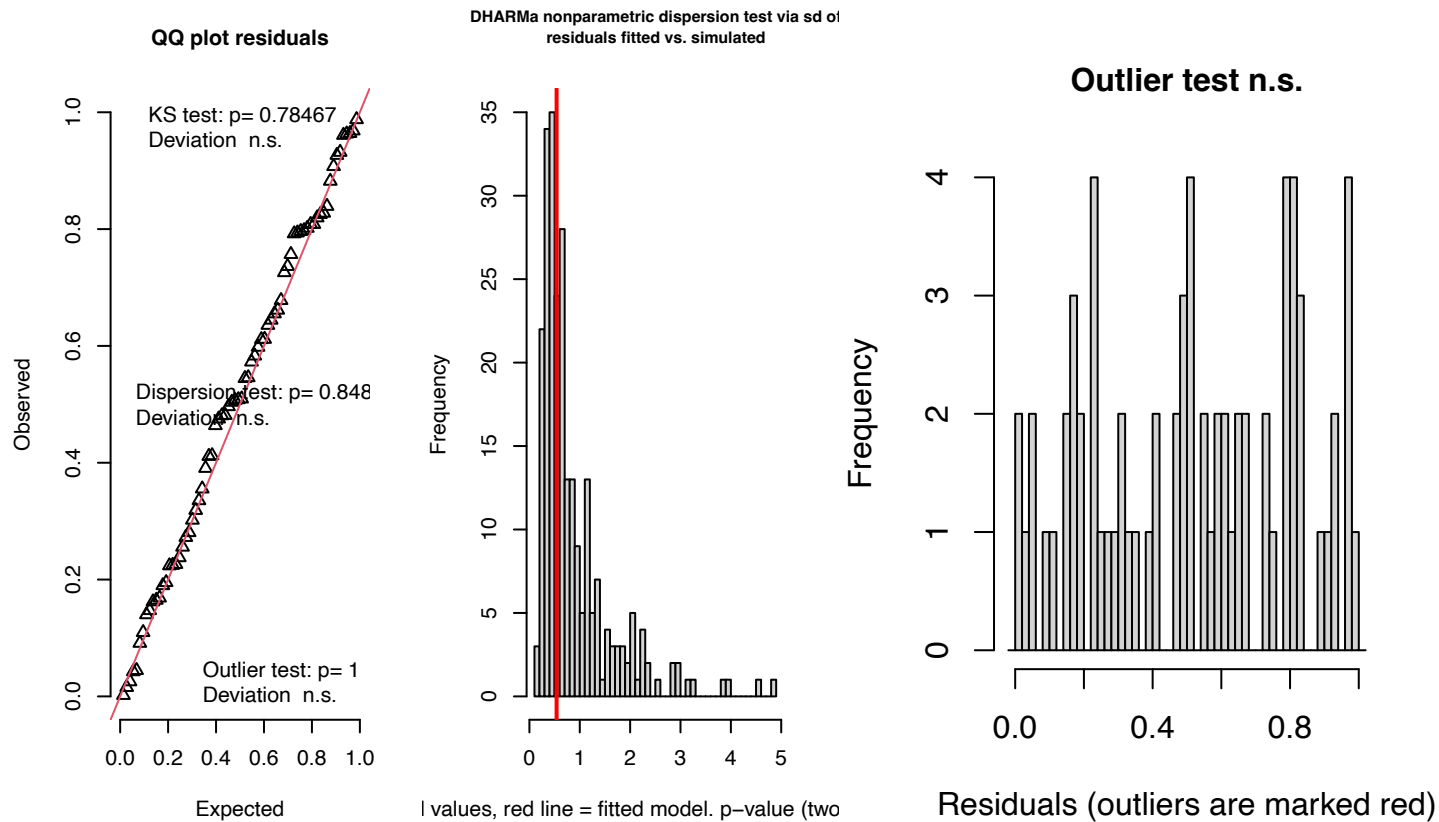
```
testZeroInflation(m19)
```

DHARMA zero-inflation test via comparison to expected zeros with simulation under H0 = fitted model



Simulated values, red line = fitted model. p-value (two.sided) = 1

```
##
## DHARMA zero-inflation test via comparison to expected zeros with
## simulation under H0 = fitted model
##
## data: simulationOutput
## ratioObsSim = 1.0071, p-value = 1
## alternative hypothesis: two.sided
testResiduals(m19)
```



```
## $uniformity
##
## Exact one-sample Kolmogorov-Smirnov test
##
## data: simulationOutput$scaledResiduals
## D = 0.074994, p-value = 0.7847
## alternative hypothesis: two-sided
##
##
## $dispersion
##
## DHARMA nonparametric dispersion test via sd of residuals fitted vs.
## simulated
##
## data: simulationOutput
## dispersion = 0.60948, p-value = 0.848
## alternative hypothesis: two.sided
##
##
## $outliers
##
## DHARMA bootstrapped outlier test
##
## data: simulationOutput
## outliers at both margin(s) = 0, observations = 72, p-value = 1
## alternative hypothesis: two.sided
## percent confidence interval:
## 0.00000000 0.02777778
```

```

## sample estimates:
## outlier frequency (expected: 0.0056944444444444 )
##
## 0

## $uniformity
##
## Exact one-sample Kolmogorov-Smirnov test
##
## data: simulationOutput$scaledResiduals
## D = 0.074994, p-value = 0.7847
## alternative hypothesis: two-sided
##
##
## $dispersion
##
## DHARMA nonparametric dispersion test via sd of residuals fitted vs.
## simulated
##
## data: simulationOutput
## dispersion = 0.60948, p-value = 0.848
## alternative hypothesis: two.sided
##
##
## $outliers
##
## DHARMA bootstrapped outlier test
##
## data: simulationOutput
## outliers at both margin(s) = 0, observations = 72, p-value = 1
## alternative hypothesis: two.sided
## percent confidence interval:
## 0.00000000 0.02777778
## sample estimates:
## outlier frequency (expected: 0.0056944444444444 )
##
## 0

# calculate the residuals for the model
m20 <- simulateResiduals(fittedModel = interaction_harass_brubig, plot = F)

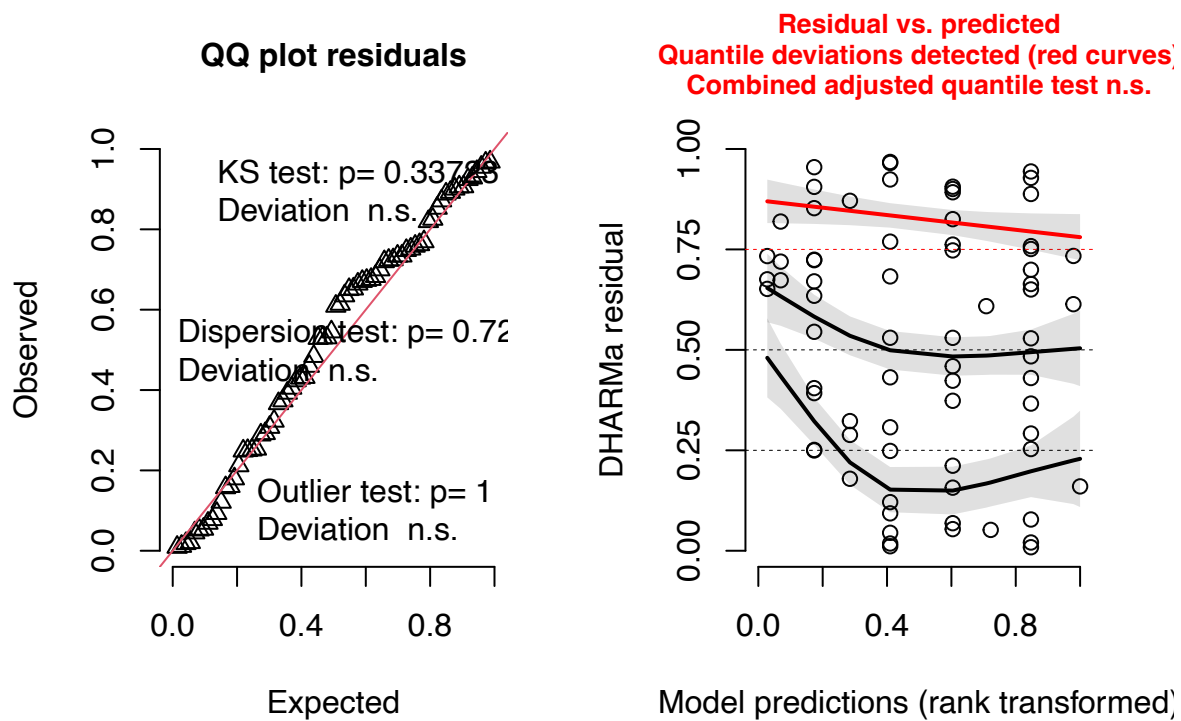
residuals(m20)

## [1] 0.211711588 0.053844831 0.322697851 0.366319237 0.020581975 0.291792162
## [7] 0.528740984 0.373296633 0.675592266 0.544939278 0.018086965 0.160207058
## [13] 0.529954196 0.483869439 0.431649466 0.608769214 0.288333880 0.423245376
## [19] 0.249645902 0.724417821 0.120990424 0.871026802 0.093093766 0.044679220
## [25] 0.719638405 0.747524399 0.248445638 0.404771731 0.733417054 0.179143034
## [31] 0.670499134 0.530135026 0.699364786 0.077804878 0.924102135 0.253182210
## [37] 0.751049283 0.651744136 0.307365649 0.429906561 0.011252798 0.967746588
## [43] 0.008764249 0.051927908 0.392360507 0.251034813 0.758168839 0.634635206
## [49] 0.723175828 0.905646479 0.888000000 0.663968787 0.157068950 0.819412944
## [55] 0.928000000 0.650349162 0.682850075 0.769357415 0.965100197 0.900351711
## [61] 0.613688428 0.733790849 0.673231164 0.954687141 0.852689309 0.459281341
## [67] 0.068648246 0.906145787 0.944000000 0.825308595 0.762055856 0.892263575

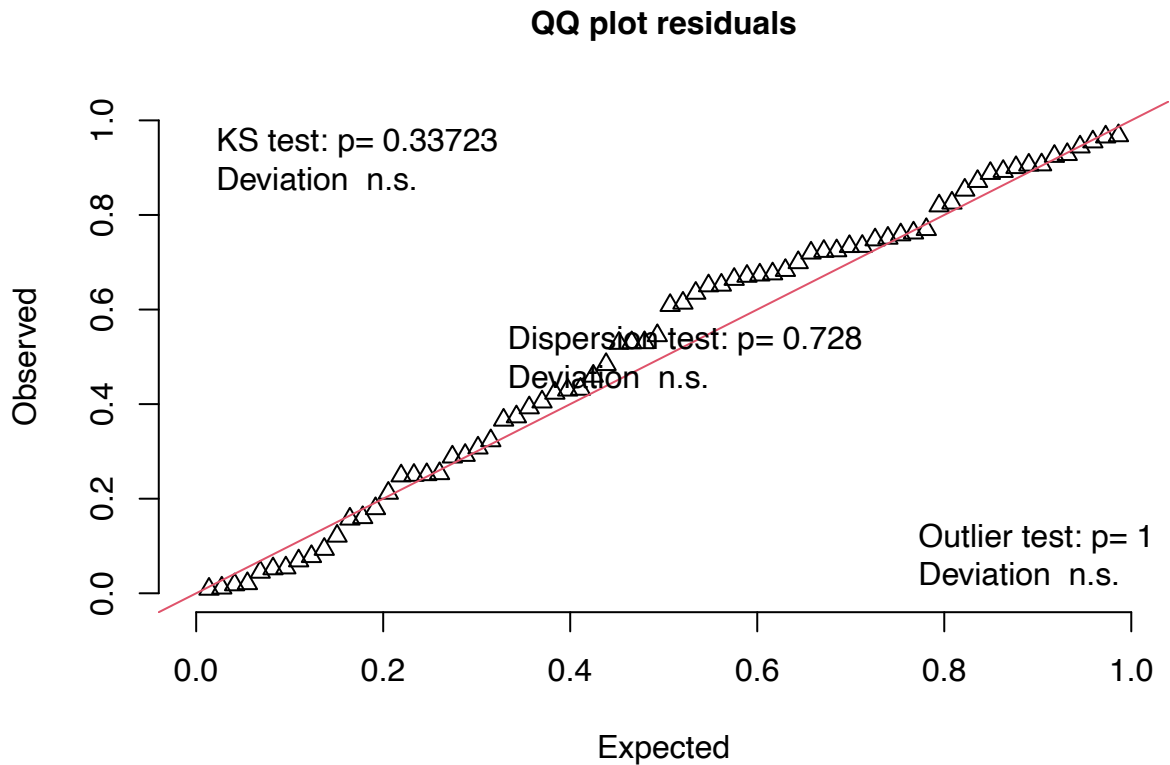
#plot simulation output
plot(m20)

```

DHARMA residual



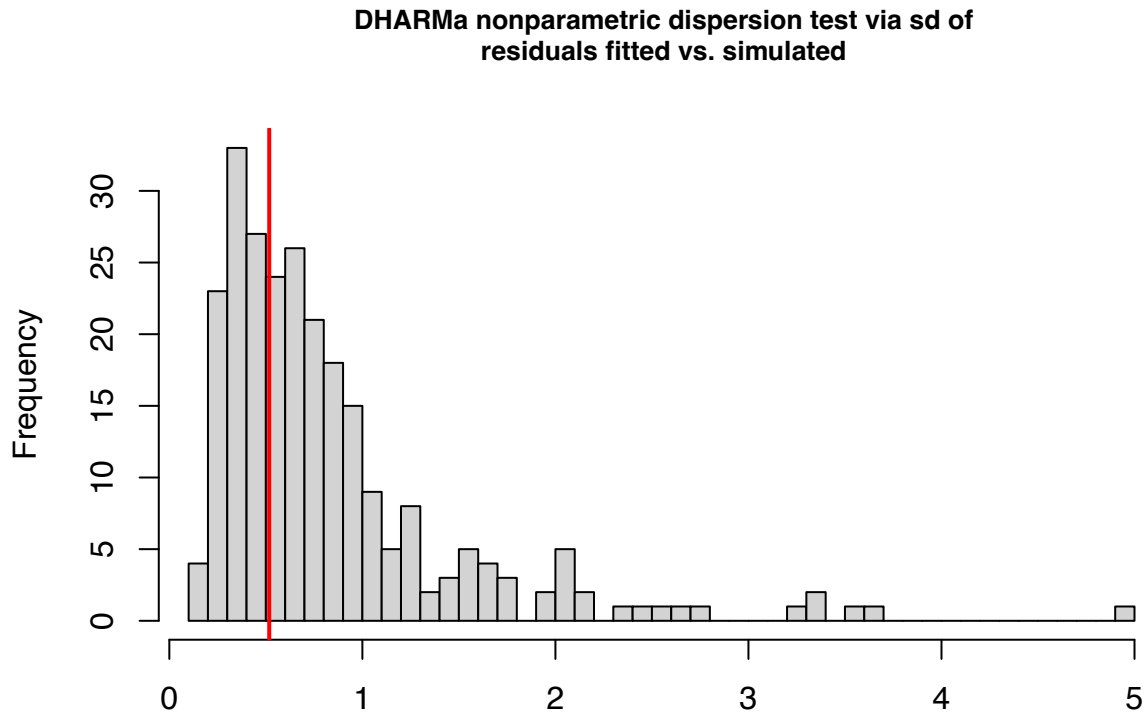
#tests if the overall distribution conforms to expectations.
`testUniformity(m20)`



 ## Exact one-sample Kolmogorov-Smirnov test

```
##
## data:  simulationOutput$scaledResiduals
## D = 0.10877, p-value = 0.3372
## alternative hypothesis: two-sided
```

```
testDispersion(m20)
```

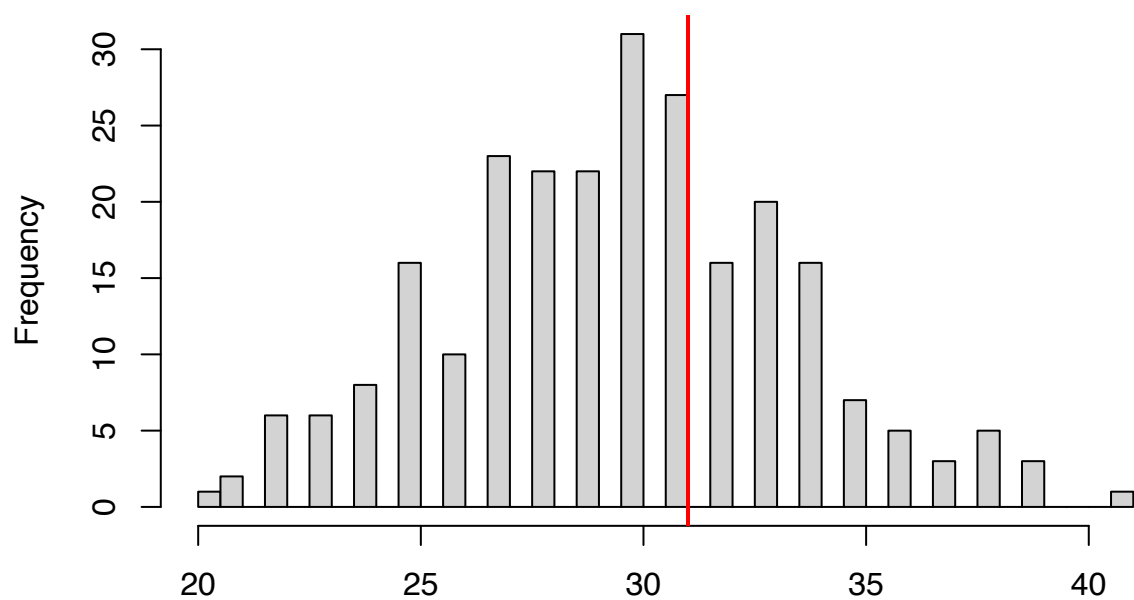


Simulated values, red line = fitted model. p-value (two.sided) = 0.728

```
##
## DHARMa nonparametric dispersion test via sd of residuals fitted vs.
## simulated
##
## data:  simulationOutput
## dispersion = 0.61185, p-value = 0.728
## alternative hypothesis: two.sided
```

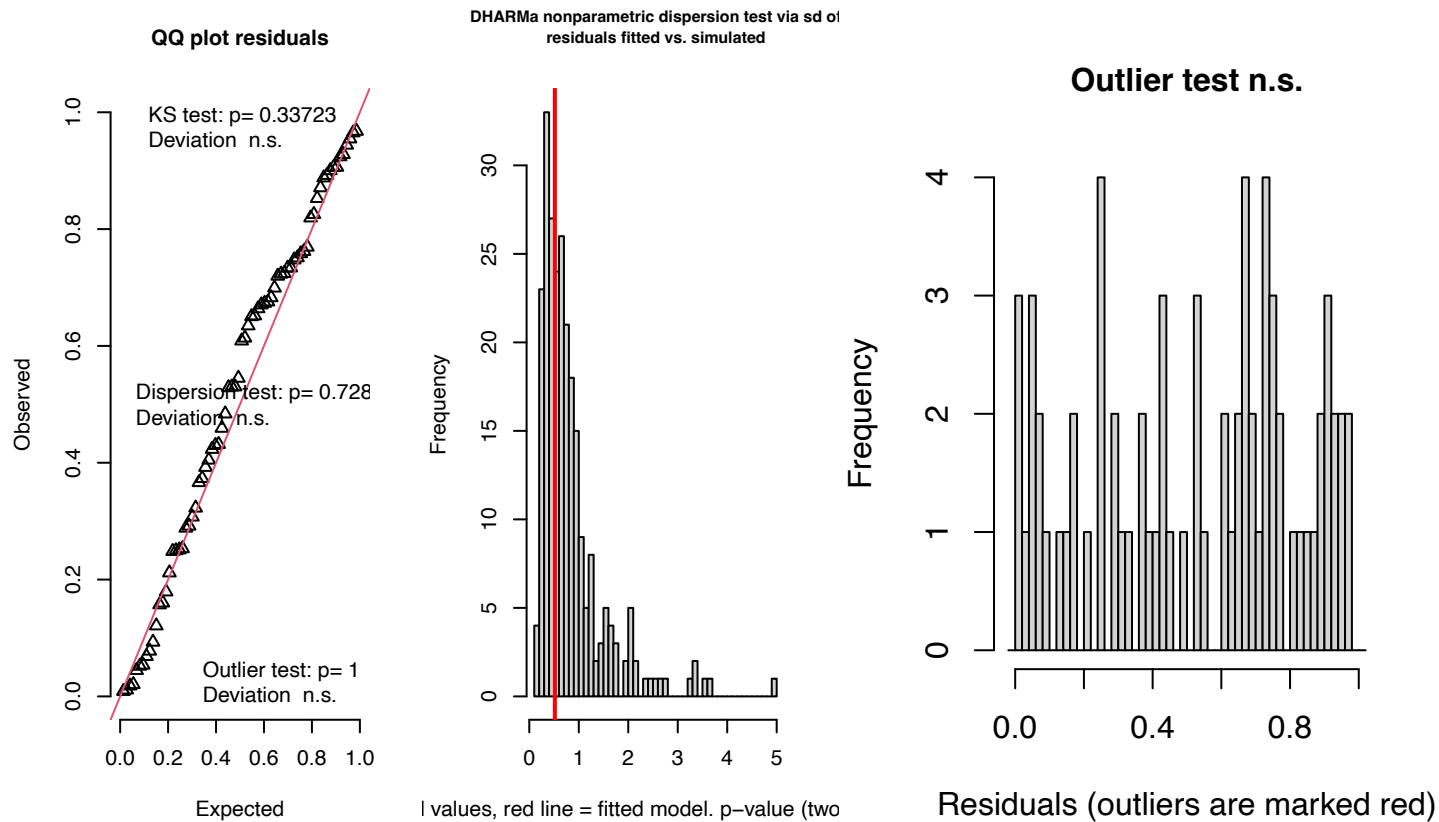
```
testZeroInflation(m20)
```

**DHARMA zero-inflation test via comparison to
expected zeros with simulation under H0 = fitted
model**



Simulated values, red line = fitted model. p-value (two.sided) = 0.824

```
##
## DHARMA zero-inflation test via comparison to expected zeros with
## simulation under H0 = fitted model
##
## data: simulationOutput
## ratioObsSim = 1.0436, p-value = 0.824
## alternative hypothesis: two.sided
testResiduals(m20)
```



```
## $uniformity
##
## Exact one-sample Kolmogorov-Smirnov test
##
## data: simulationOutput$scaledResiduals
## D = 0.10877, p-value = 0.3372
## alternative hypothesis: two-sided
##
##
## $dispersion
##
## DHARMA nonparametric dispersion test via sd of residuals fitted vs.
## simulated
##
## data: simulationOutput
## dispersion = 0.61185, p-value = 0.728
## alternative hypothesis: two.sided
##
##
## $outliers
##
## DHARMA bootstrapped outlier test
##
## data: simulationOutput
## outliers at both margin(s) = 0, observations = 72, p-value = 1
## alternative hypothesis: two.sided
## percent confidence interval:
## 0.00000000 0.02777778
```

```

## sample estimates:
## outlier frequency (expected: 0.0044444444444444 )
##
## $uniformity
##
## Exact one-sample Kolmogorov-Smirnov test
##
## data: simulationOutput$scaledResiduals
## D = 0.10877, p-value = 0.3372
## alternative hypothesis: two-sided
##
##
## $dispersion
##
## DHARMA nonparametric dispersion test via sd of residuals fitted vs.
## simulated
##
## data: simulationOutput
## dispersion = 0.61185, p-value = 0.728
## alternative hypothesis: two.sided
##
##
## $outliers
##
## DHARMA bootstrapped outlier test
##
## data: simulationOutput
## outliers at both margin(s) = 0, observations = 72, p-value = 1
## alternative hypothesis: two.sided
## percent confidence interval:
## 0.00000000 0.02777778
## sample estimates:
## outlier frequency (expected: 0.0044444444444444 )
##
##
# calculate the residuals for the model
m13 <- simulateResiduals(fittedModel = additive_resp_brubig, plot = F)

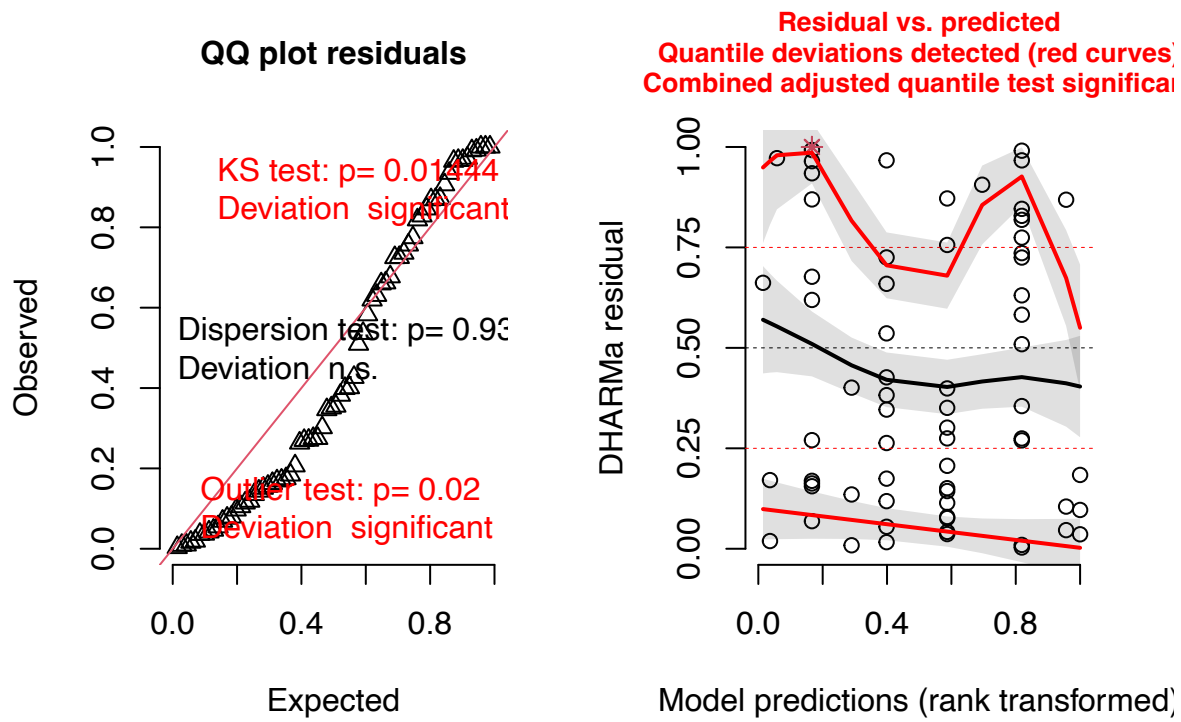
residuals(m13)

## [1] 1.000000000 1.000000000 0.990743259 0.818685354 0.966690408 0.992435862
## [7] 0.183760363 0.659425164 0.399052716 0.972173854 1.000000000 0.828592729
## [13] 0.206451983 0.906053259 0.046340382 0.964602003 0.346320879 0.966741269
## [19] 0.871890376 0.868708792 0.274722573 0.150745223 0.401013054 0.869465480
## [25] 0.350717476 0.536095019 0.096596190 0.104943612 0.174515593 0.301453062
## [31] 0.845964694 0.269705113 0.041992026 0.724769479 0.582563954 0.035983423
## [37] 0.143586596 0.274522213 0.114044783 0.756132560 0.003123244 0.661881640
## [43] 0.725071794 0.118483308 0.355122744 0.382345539 0.054741968 0.263199377
## [49] 0.509151140 0.735815318 0.677036603 0.008618607 0.774416735 0.630732814
## [55] 0.079173449 0.036545212 0.076305342 0.934531658 0.019228490 0.170890907
## [61] 0.135133823 0.426644085 0.016433010 0.068711122 0.169093615 0.162574633
## [67] 0.009608666 0.155574534 0.619261397 0.270288459

#plot simulation output
plot(m13)

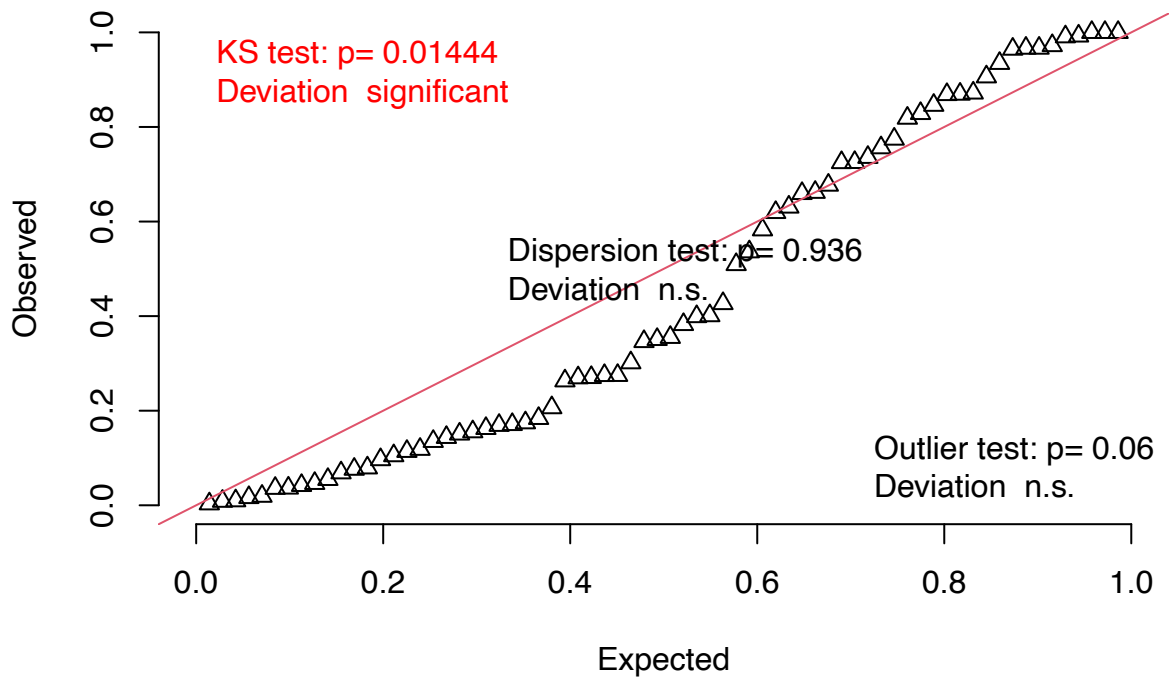
```

DHARMA residual



#tests if the overall distribution conforms to expectations.
`testUniformity(m13)`

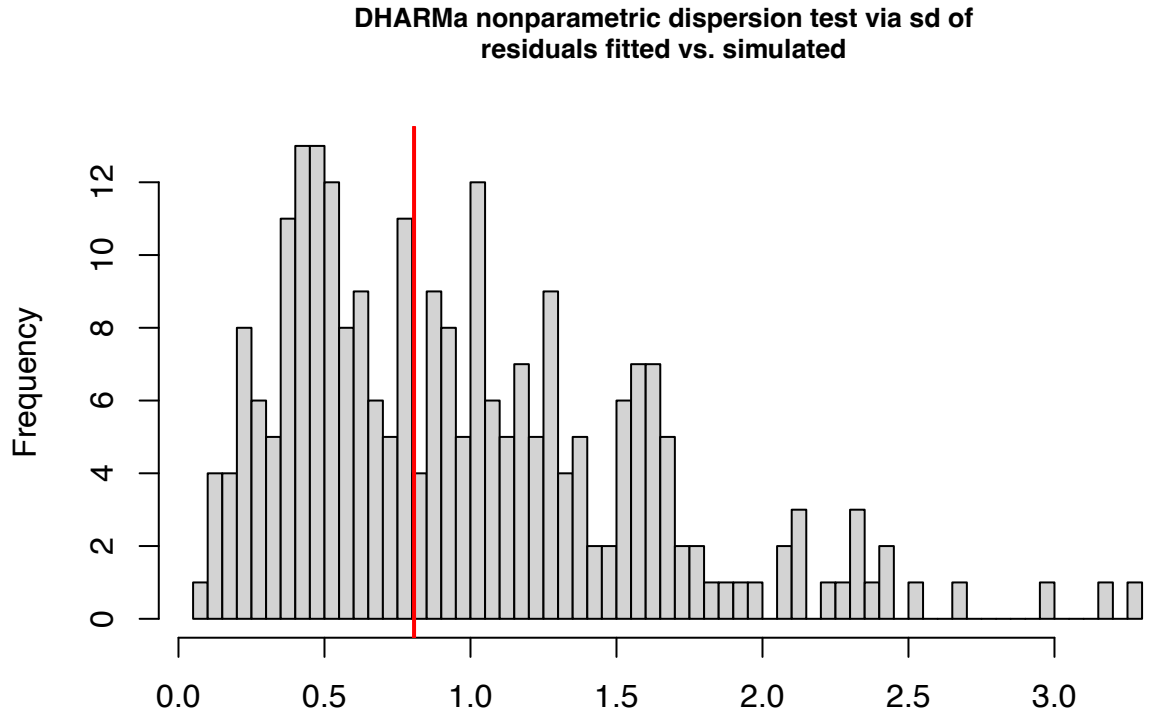
QQ plot residuals



 ## Asymptotic one-sample Kolmogorov-Smirnov test

```
##
## data: simulationOutput$scaledResiduals
## D = 0.18767, p-value = 0.01444
## alternative hypothesis: two-sided
```

```
testDispersion(m13)
```

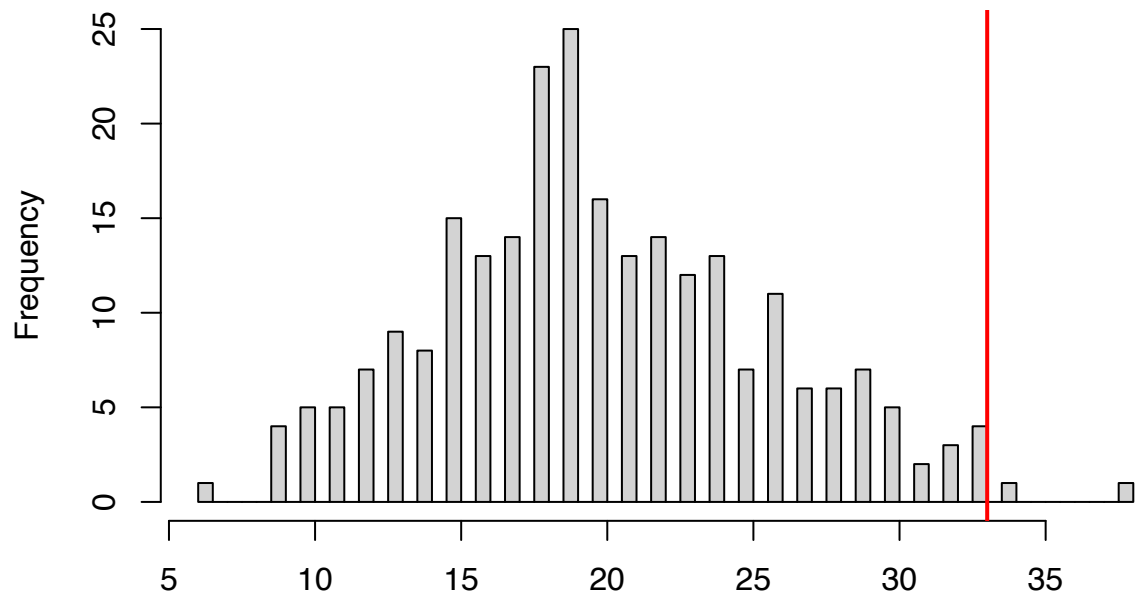


Simulated values, red line = fitted model. p-value (two.sided) = 0.936

```
##
## DHARMa nonparametric dispersion test via sd of residuals fitted vs.
## simulated
##
## data: simulationOutput
## dispersion = 0.82406, p-value = 0.936
## alternative hypothesis: two.sided
```

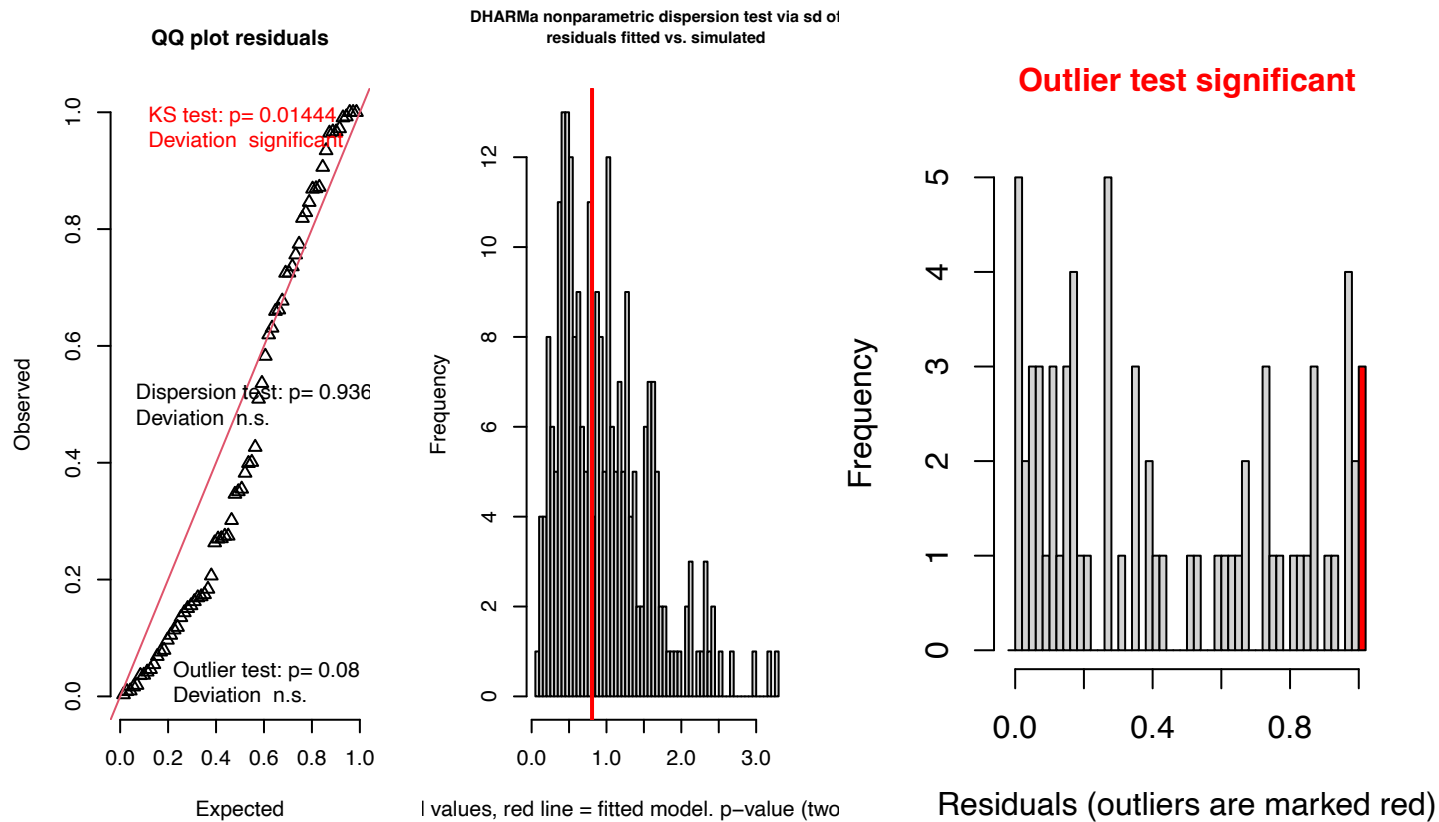
```
testZeroInflation(m13)
```

DHARMA zero-inflation test via comparison to expected zeros with simulation under H0 = fitted model



Simulated values, red line = fitted model. p-value (two.sided) = 0.048

```
##
## DHARMA zero-inflation test via comparison to expected zeros with
## simulation under H0 = fitted model
##
## data: simulationOutput
## ratioObsSim = 1.6451, p-value = 0.048
## alternative hypothesis: two.sided
testResiduals(m13)
```



```
## $uniformity
##
## Asymptotic one-sample Kolmogorov-Smirnov test
##
## data: simulationOutput$scaledResiduals
## D = 0.18767, p-value = 0.01444
## alternative hypothesis: two-sided
##
##
## $dispersion
##
## DHARMA nonparametric dispersion test via sd of residuals fitted vs.
## simulated
##
## data: simulationOutput
## dispersion = 0.82406, p-value = 0.936
## alternative hypothesis: two.sided
##
##
## $outliers
##
## DHARMA bootstrapped outlier test
##
## data: simulationOutput
## outliers at both margin(s) = 3, observations = 70, p-value = 0.02
## alternative hypothesis: two.sided
## percent confidence interval:
## 0.00000000 0.02857143
```

```

## sample estimates:
## outlier frequency (expected: 0.00442857142857143 )
##                                0.04285714

## $uniformity
##
## Asymptotic one-sample Kolmogorov-Smirnov test
##
## data: simulationOutput$scaledResiduals
## D = 0.18767, p-value = 0.01444
## alternative hypothesis: two-sided
##
##
## $dispersion
##
## DHARMA nonparametric dispersion test via sd of residuals fitted vs.
## simulated
##
## data: simulationOutput
## dispersion = 0.82406, p-value = 0.936
## alternative hypothesis: two.sided
##
##
## $outliers
##
## DHARMA bootstrapped outlier test
##
## data: simulationOutput
## outliers at both margin(s) = 3, observations = 70, p-value = 0.02
## alternative hypothesis: two.sided
## percent confidence interval:
## 0.00000000 0.02857143
## sample estimates:
## outlier frequency (expected: 0.00442857142857143 )
##                                0.04285714

# calculate the residuals for the model
m14 <- simulateResiduals(fittedModel = interaction_resp_brubig, plot = F)

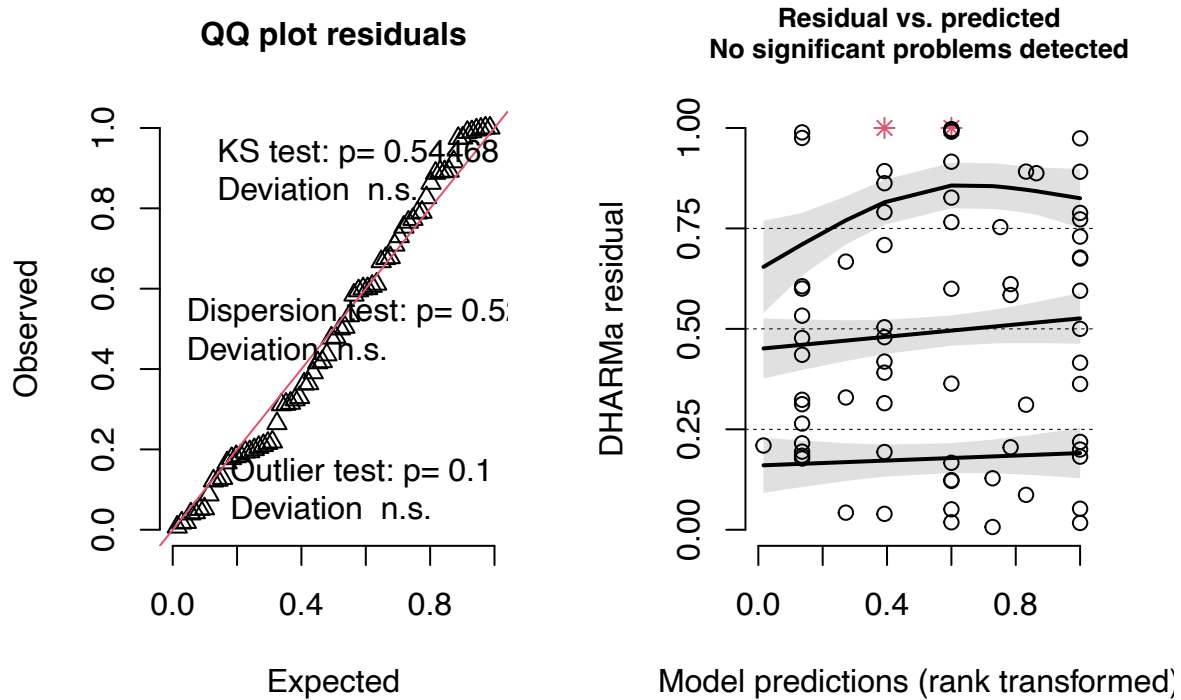
residuals(m14)

## [1] 1.000000000 0.997180582 0.974248271 0.788568369 0.891091251 0.995069807
## [7] 0.584223862 0.862539377 0.477283429 0.753368871 0.990714114 0.772567663
## [13] 0.177329389 0.887422606 0.311312077 0.916041001 0.418536531 1.000000000
## [19] 0.975160609 0.891420467 0.600240497 0.606108596 0.667420987 0.826687105
## [25] 0.194451915 0.790437611 0.611455067 0.086918489 0.315014266 0.184229767
## [31] 0.677475798 0.182036700 0.435463841 0.675105450 0.415823920 0.205284527
## [37] 0.323533093 0.199573090 0.215200444 0.989072841 0.052320361 0.209664194
## [43] 0.892694396 0.039578962 0.218706770 0.504036615 0.390985271 0.479264899
## [49] 0.362821659 0.594906092 0.599799533 0.329204005 0.729727624 0.500084281
## [55] 0.532825103 0.264353062 0.312507205 0.765715213 0.128112339 0.006833009
## [61] 0.042580551 0.708696196 0.193481820 0.166682264 0.123546058 0.018560663
## [67] 0.017003365 0.050846756 0.363755047 0.121637280

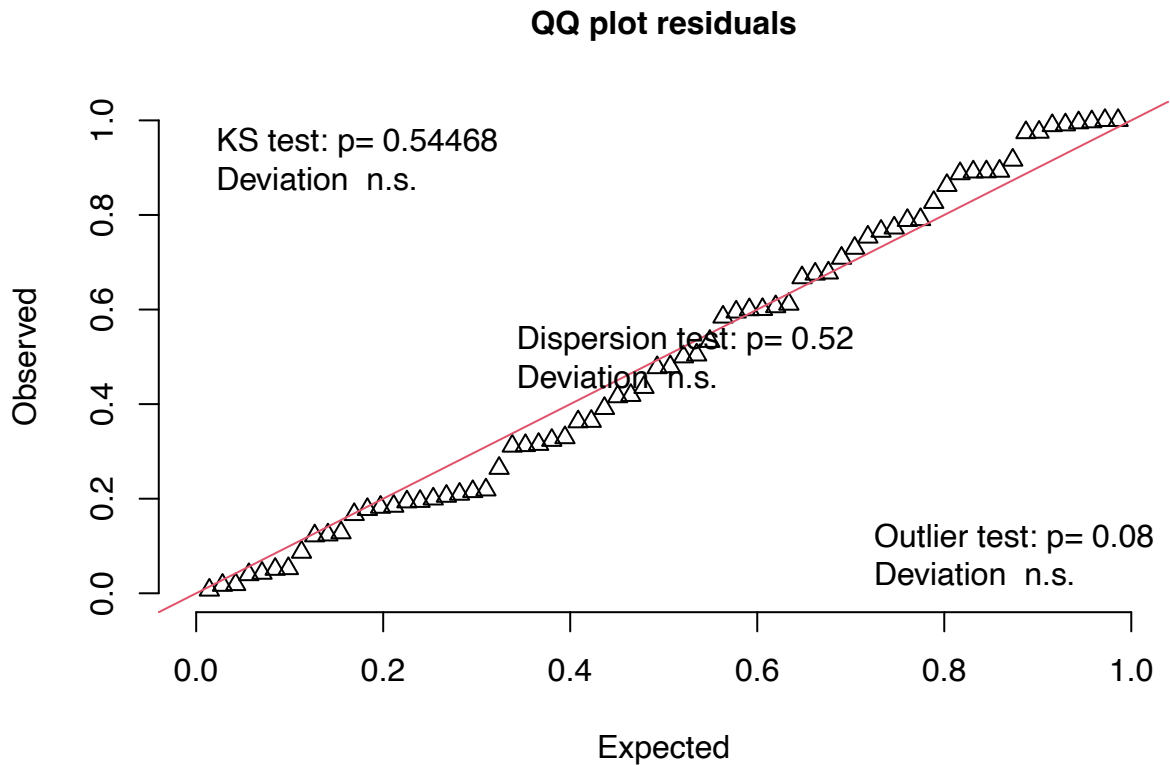
#plot simulation output
plot(m14)

```

DHARMa residual



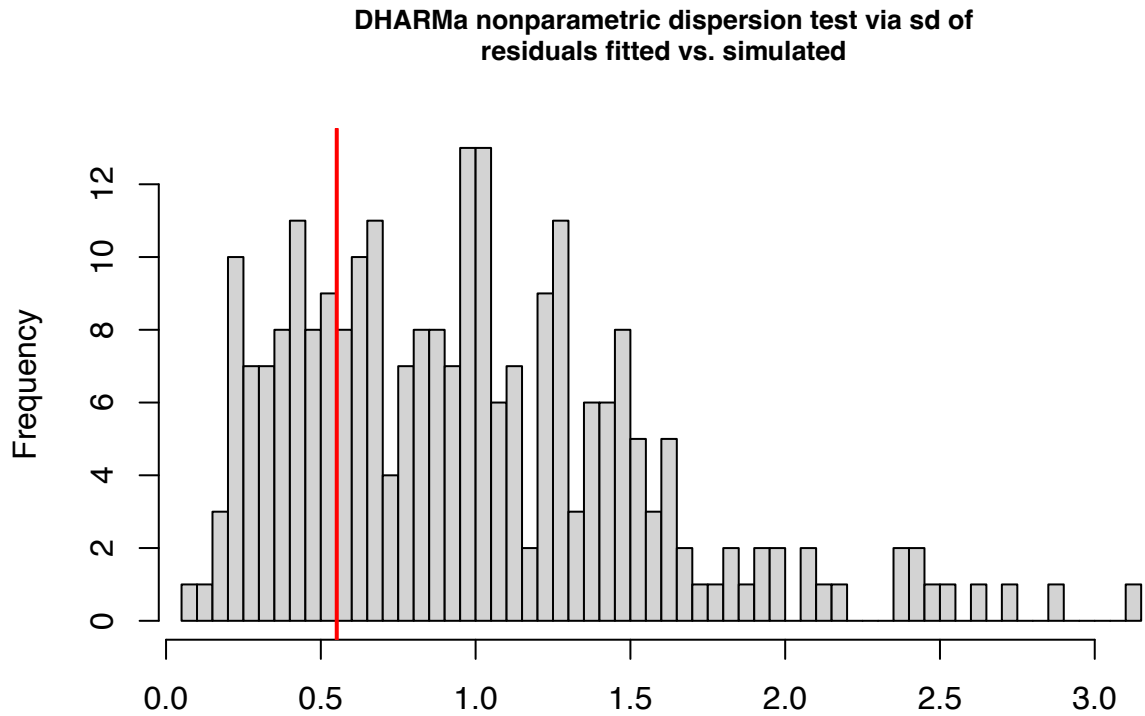
#tests if the overall distribution conforms to expectations.
`testUniformity(m14)`



 ## Asymptotic one-sample Kolmogorov-Smirnov test

```
##
## data:  simulationOutput$scaledResiduals
## D = 0.095579, p-value = 0.5447
## alternative hypothesis: two-sided
```

```
testDispersion(m14)
```

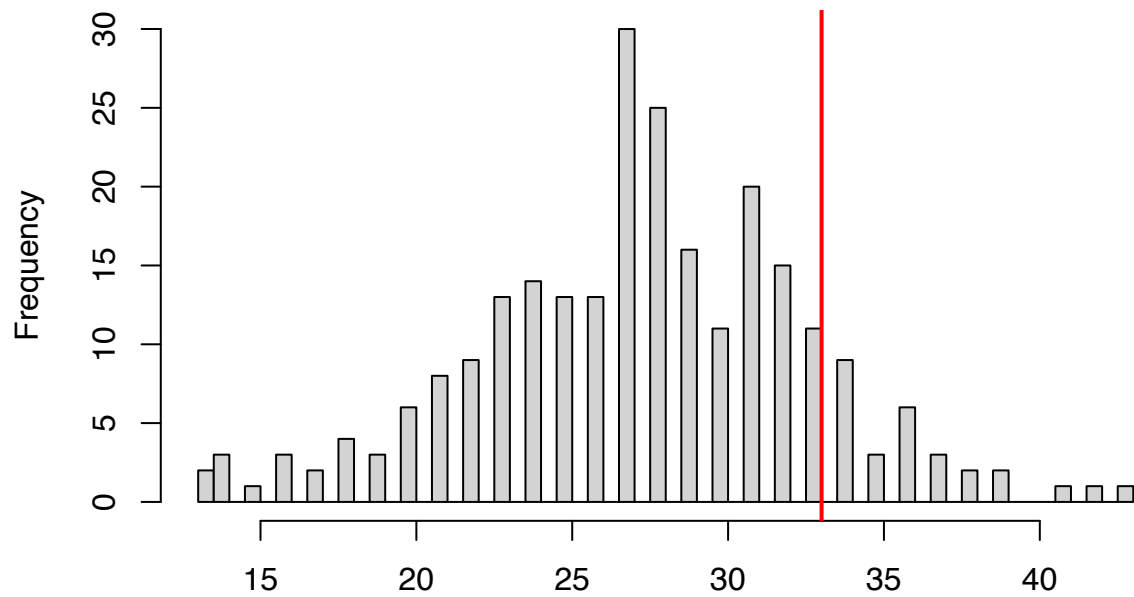


Simulated values, red line = fitted model. p-value (two.sided) = 0.52

```
##
## DHARMa nonparametric dispersion test via sd of residuals fitted vs.
## simulated
##
## data:  simulationOutput
## dispersion = 0.56136, p-value = 0.52
## alternative hypothesis: two.sided
```

```
testZeroInflation(m14)
```

DHARMA zero-inflation test via comparison to expected zeros with simulation under H0 = fitted model



Simulated values, red line = fitted model. p-value (two.sided) = 0.312

```
##
## DHARMA zero-inflation test via comparison to expected zeros with
## simulation under H0 = fitted model
##
## data: simulationOutput
## ratioObsSim = 1.2021, p-value = 0.312
## alternative hypothesis: two.sided
testResiduals(m14)
```



```

## sample estimates:
## outlier frequency (expected: 0.00228571428571429 )
##                                0.02857143

## $uniformity
##
## Asymptotic one-sample Kolmogorov-Smirnov test
##
## data: simulationOutput$scaledResiduals
## D = 0.095579, p-value = 0.5447
## alternative hypothesis: two-sided
##
##
## $dispersion
##
## DHARMA nonparametric dispersion test via sd of residuals fitted vs.
## simulated
##
## data: simulationOutput
## dispersion = 0.56136, p-value = 0.52
## alternative hypothesis: two.sided
##
##
## $outliers
##
## DHARMA bootstrapped outlier test
##
## data: simulationOutput
## outliers at both margin(s) = 2, observations = 70, p-value = 0.1
## alternative hypothesis: two.sided
## percent confidence interval:
## 0.00000000 0.03607143
## sample estimates:
## outlier frequency (expected: 0.00228571428571429 )
##                                0.02857143

# calculate the residuals for the model
m25 <- simulateResiduals(fittedModel = additive_rej_brubig, plot = F)

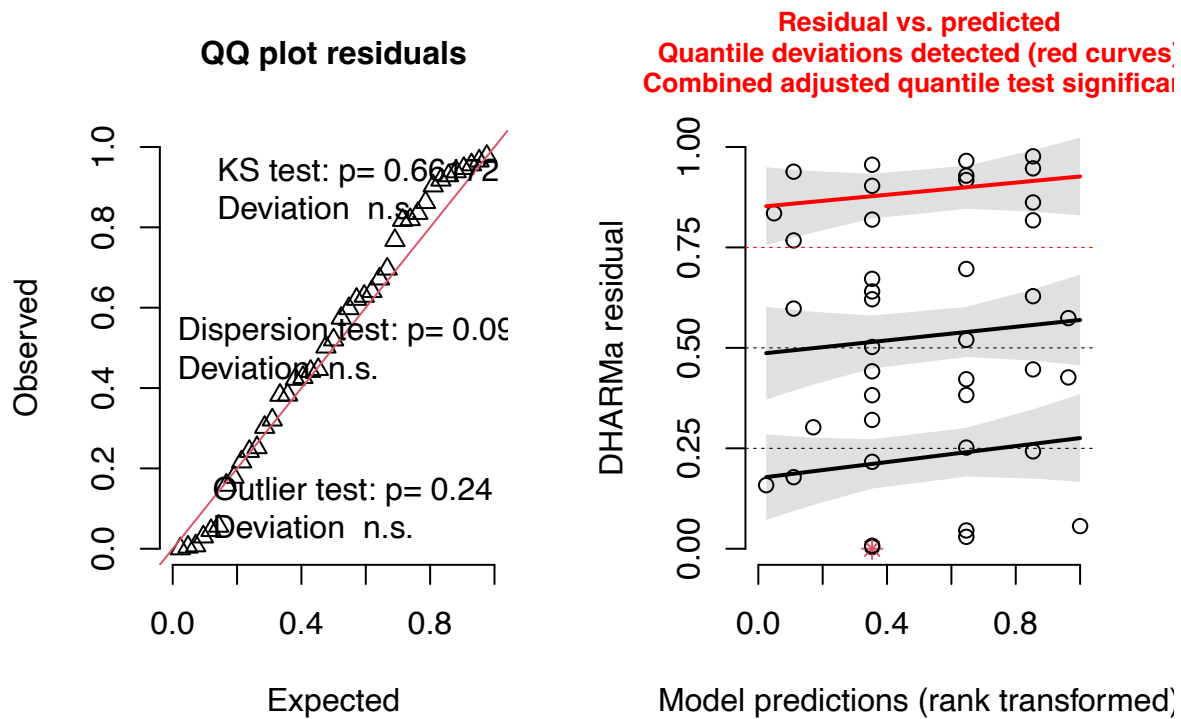
residuals(m25)

## [1] 0.216371365 0.502227056 0.441666512 0.382649280 0.056361658 0.421969434
## [7] 0.671705438 0.817470377 0.302374022 0.251776355 0.158260440 0.045569230
## [13] 0.628623941 0.004234258 0.946778314 0.320647874 0.955791939 0.640436841
## [19] 0.242441684 0.000000000 0.178173974 0.597811353 0.620971252 0.382158253
## [25] 0.834832170 0.819295213 0.007499891 0.861911636 0.446391107 0.976730243
## [31] 0.519943785 0.574071329 0.426464345 0.938448559 0.767419737 0.965278176
## [37] 0.928606149 0.903263075 0.917728112 0.029610888 0.696331650

#plot simulation output
plot(m25)

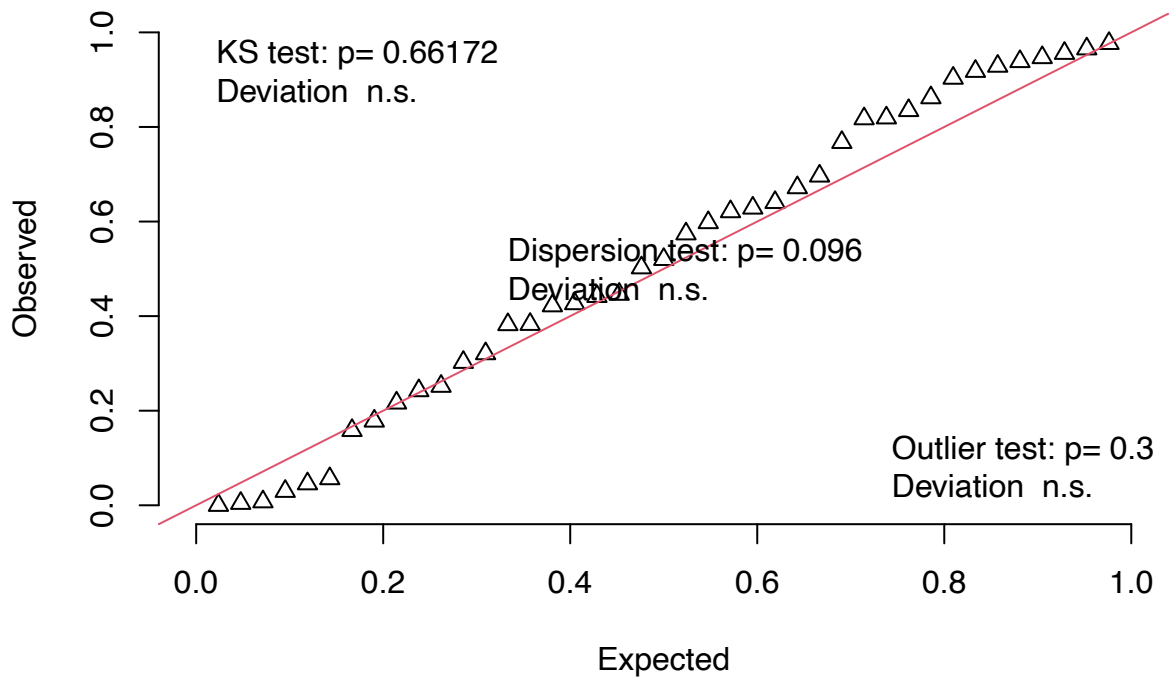
```

DHARMA residual



#tests if the overall distribution conforms to expectations.
`testUniformity(m25)`

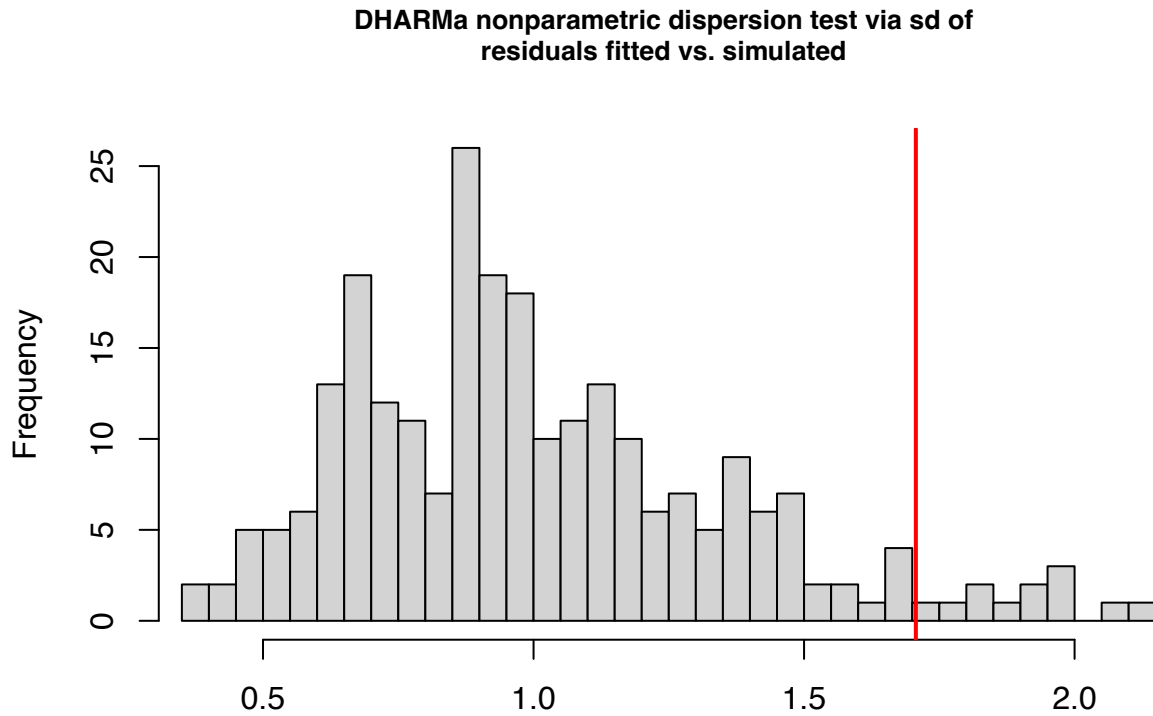
QQ plot residuals



 ## Exact one-sample Kolmogorov-Smirnov test

```
##
## data:  simulationOutput$scaledResiduals
## D = 0.11015, p-value = 0.6617
## alternative hypothesis: two-sided
```

```
testDispersion(m25)
```

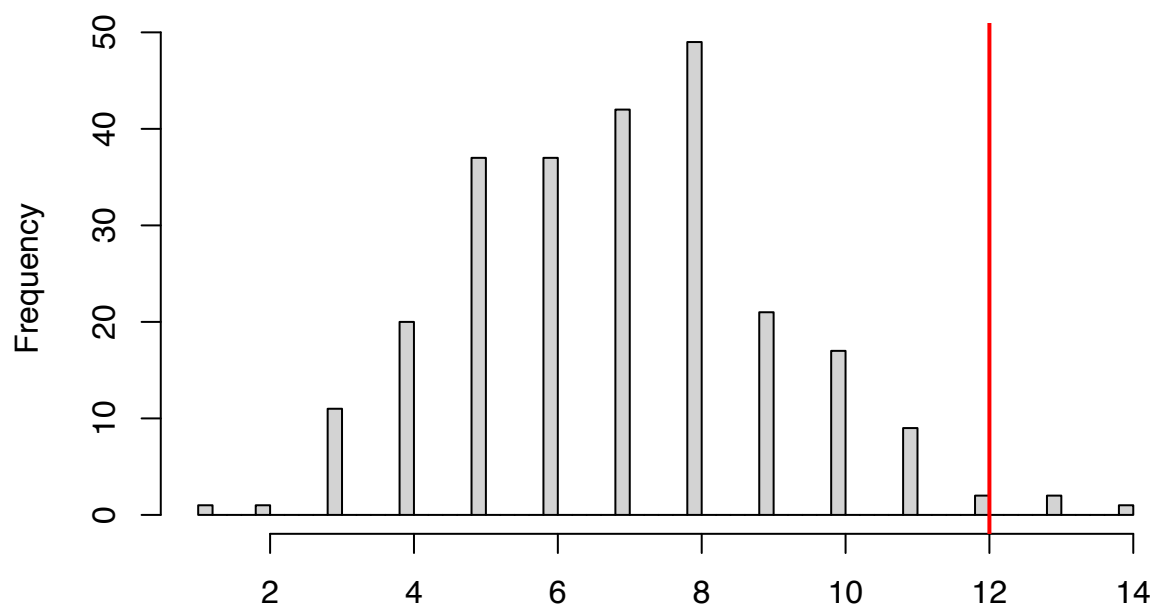


Simulated values, red line = fitted model. p-value (two.sided) = 0.096

```
##
## DHARMa nonparametric dispersion test via sd of residuals fitted vs.
## simulated
##
## data:  simulationOutput
## dispersion = 1.6929, p-value = 0.096
## alternative hypothesis: two.sided
```

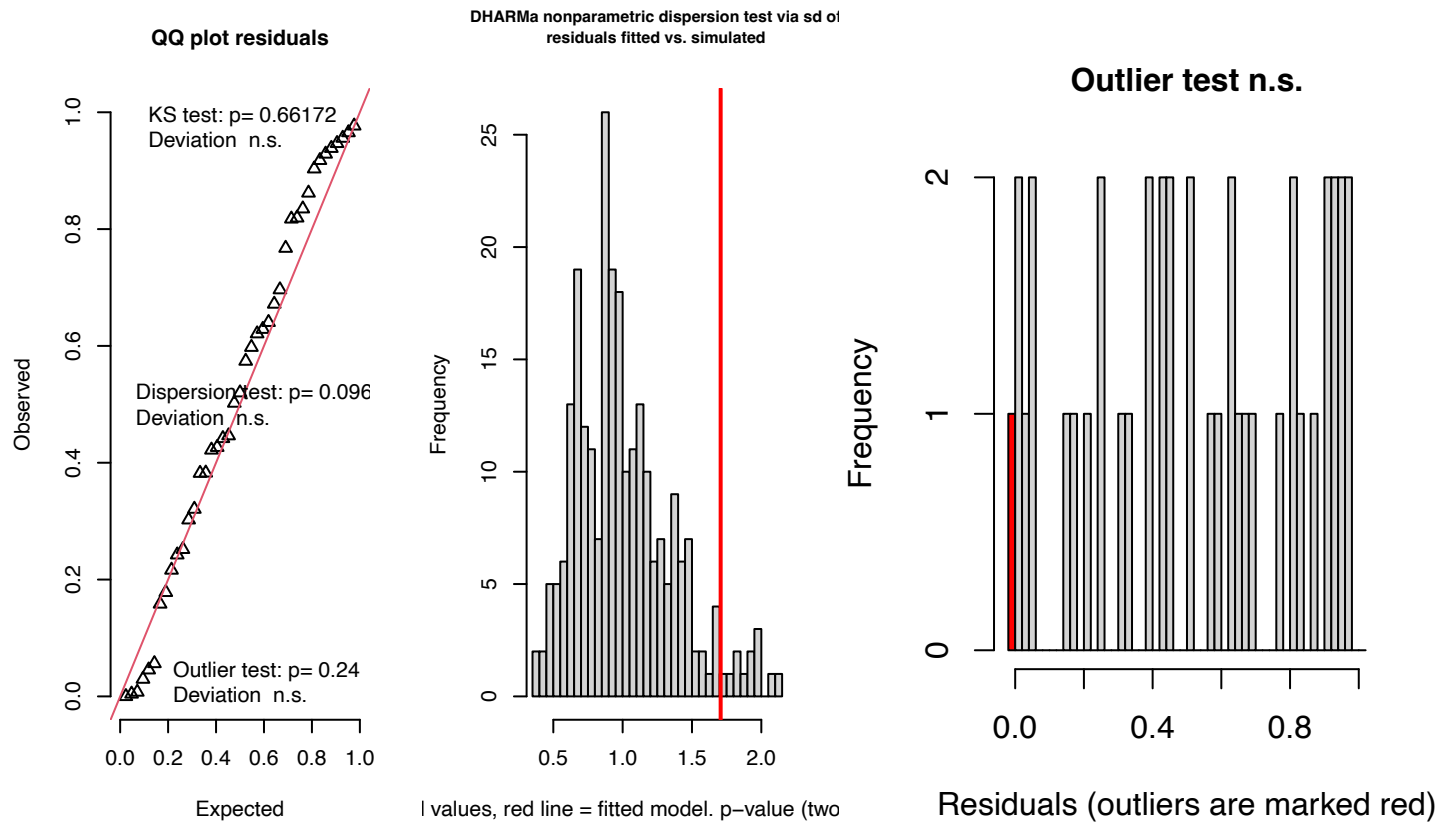
```
testZeroInflation(m25)
```

DHARMA zero-inflation test via comparison to expected zeros with simulation under H0 = fitted model



Simulated values, red line = fitted model. p-value (two.sided) = 0.04

```
##
## DHARMA zero-inflation test via comparison to expected zeros with
## simulation under H0 = fitted model
##
## data: simulationOutput
## ratioObsSim = 1.7331, p-value = 0.04
## alternative hypothesis: two.sided
testResiduals(m25)
```



```
## $uniformity
##
## Exact one-sample Kolmogorov-Smirnov test
##
## data: simulationOutput$scaledResiduals
## D = 0.11015, p-value = 0.6617
## alternative hypothesis: two-sided
##
##
## $dispersion
##
## DHARMA nonparametric dispersion test via sd of residuals fitted vs.
## simulated
##
## data: simulationOutput
## dispersion = 1.6929, p-value = 0.096
## alternative hypothesis: two.sided
##
##
## $outliers
##
## DHARMA bootstrapped outlier test
##
## data: simulationOutput
## outliers at both margin(s) = 1, observations = 41, p-value = 0.22
## alternative hypothesis: two.sided
## percent confidence interval:
## 0.00000000 0.02439024
```

```

## sample estimates:
## outlier frequency (expected: 0.00268292682926829 )
##                                0.02439024

## $uniformity
##
## Exact one-sample Kolmogorov-Smirnov test
##
## data: simulationOutput$scaledResiduals
## D = 0.11015, p-value = 0.6617
## alternative hypothesis: two-sided
##
##
## $dispersion
##
## DHARMA nonparametric dispersion test via sd of residuals fitted vs.
## simulated
##
## data: simulationOutput
## dispersion = 1.6929, p-value = 0.096
## alternative hypothesis: two.sided
##
##
## $outliers
##
## DHARMA bootstrapped outlier test
##
## data: simulationOutput
## outliers at both margin(s) = 1, observations = 41, p-value = 0.22
## alternative hypothesis: two.sided
## percent confidence interval:
## 0.00000000 0.02439024
## sample estimates:
## outlier frequency (expected: 0.00268292682926829 )
##                                0.02439024

# calculate the residuals for the model
m26 <- simulateResiduals(fittedModel = interaction_rej_brubig, plot = F)

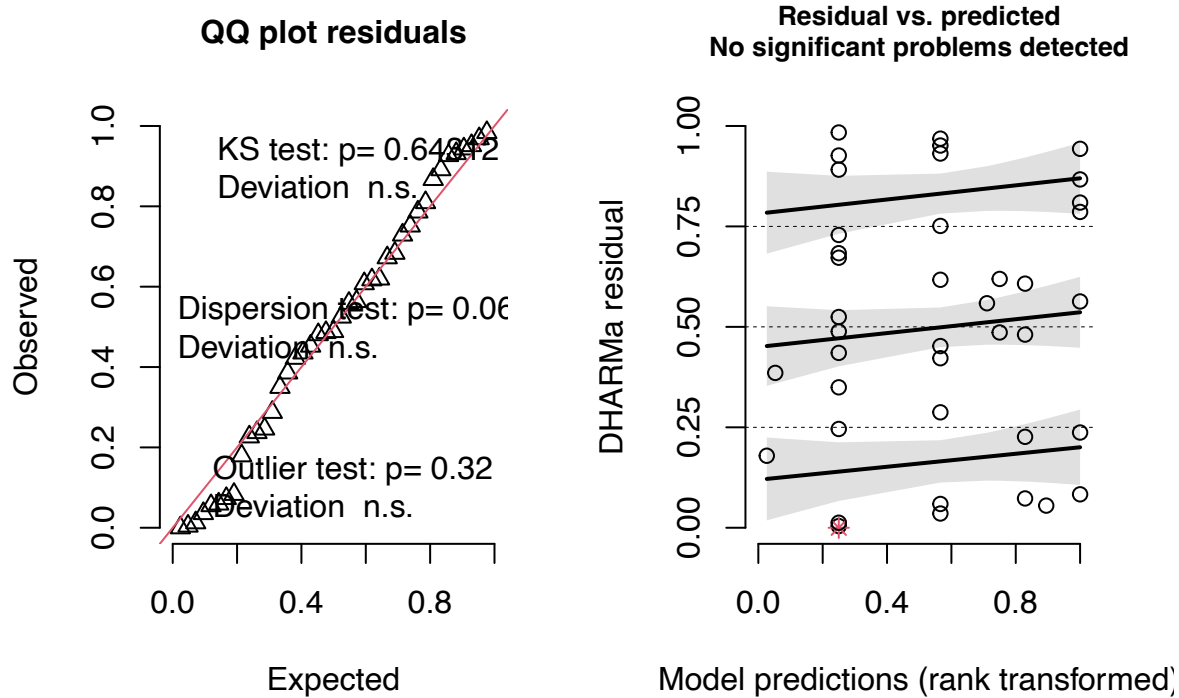
residuals(m26)

## [1] 0.245876552 0.524548259 0.488488745 0.452221876 0.055109177 0.421969434
## [7] 0.728700365 0.786034720 0.385462416 0.287321252 0.179206674 0.059216568
## [13] 0.563211113 0.004234258 0.866945784 0.349558748 0.983815057 0.672096021
## [19] 0.083254817 0.000000000 0.072986929 0.226320588 0.683332108 0.435200143
## [25] 0.558765940 0.891295213 0.012499819 0.809364737 0.237029278 0.943118372
## [31] 0.617224178 0.619265516 0.486083705 0.607609563 0.480751041 0.968434705
## [37] 0.951333403 0.926526150 0.931436867 0.035481184 0.751233196

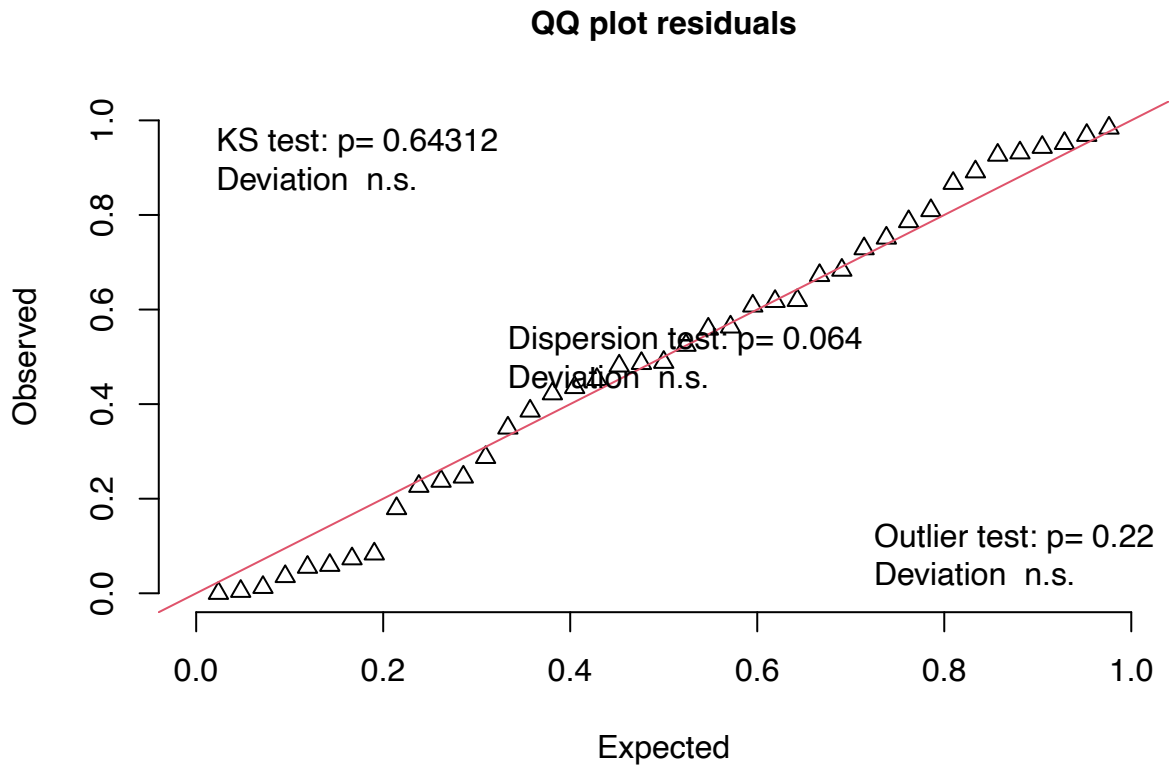
#plot simulation output
plot(m26)

```

DHARMA residual



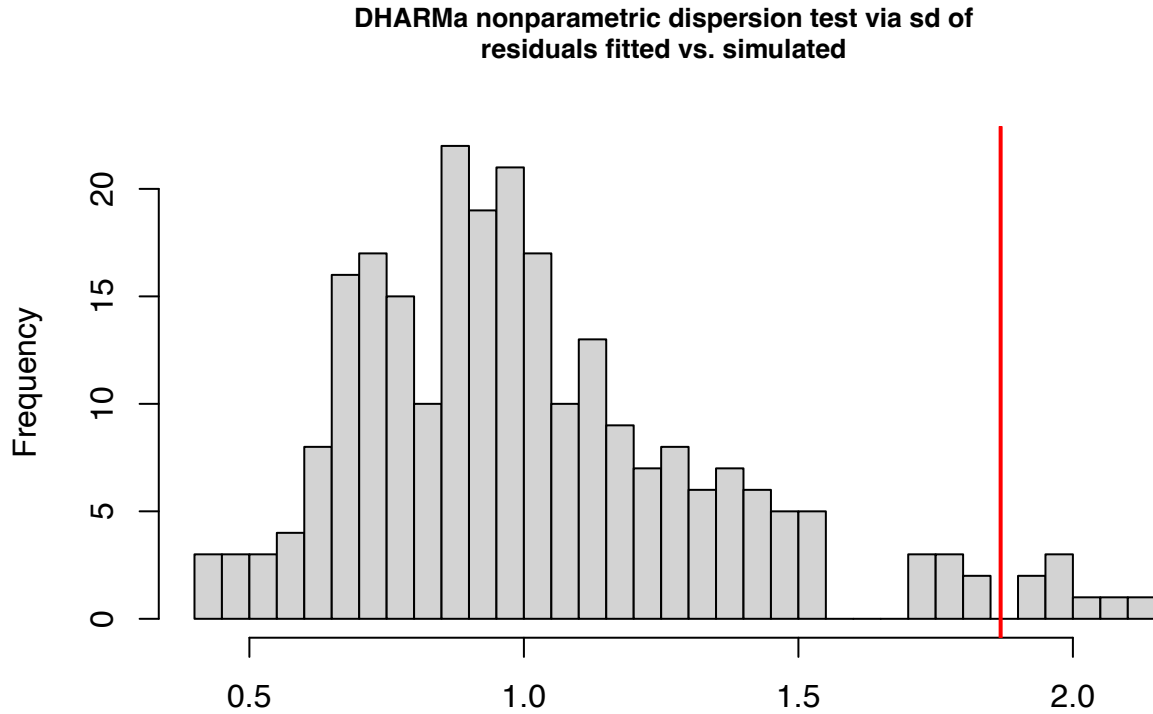
#tests if the overall distribution conforms to expectations.
`testUniformity(m26)`



 ## Exact one-sample Kolmogorov-Smirnov test

```
##
## data: simulationOutput$scaledResiduals
## D = 0.11187, p-value = 0.6431
## alternative hypothesis: two-sided
```

```
testDispersion(m26)
```

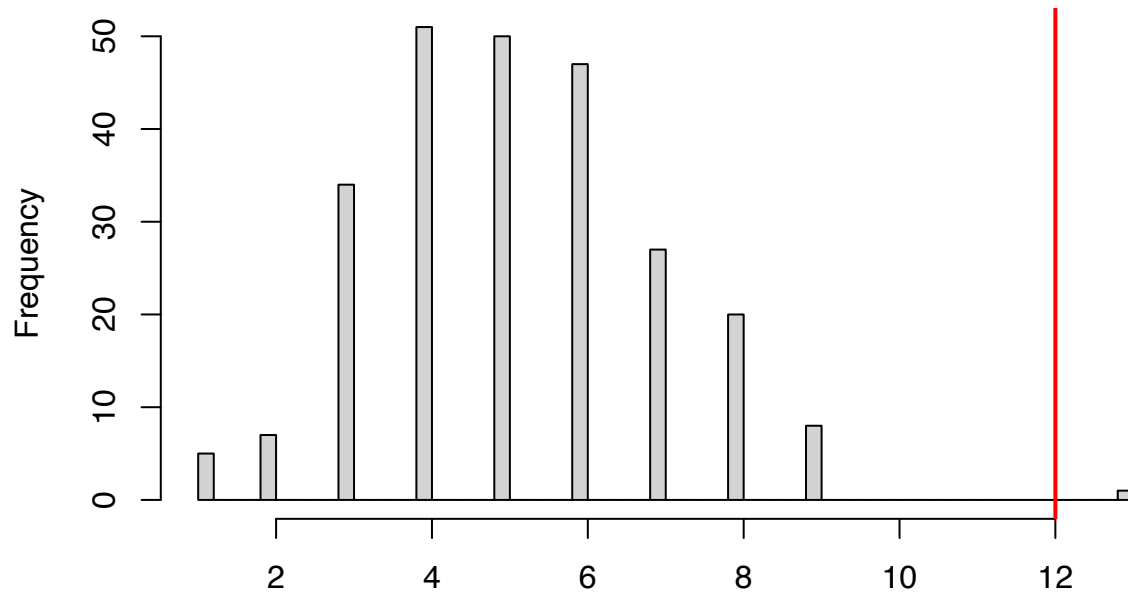


Simulated values, red line = fitted model. p-value (two.sided) = 0.064

```
##
## DHARMa nonparametric dispersion test via sd of residuals fitted vs.
## simulated
##
## data: simulationOutput
## dispersion = 1.8328, p-value = 0.064
## alternative hypothesis: two.sided
```

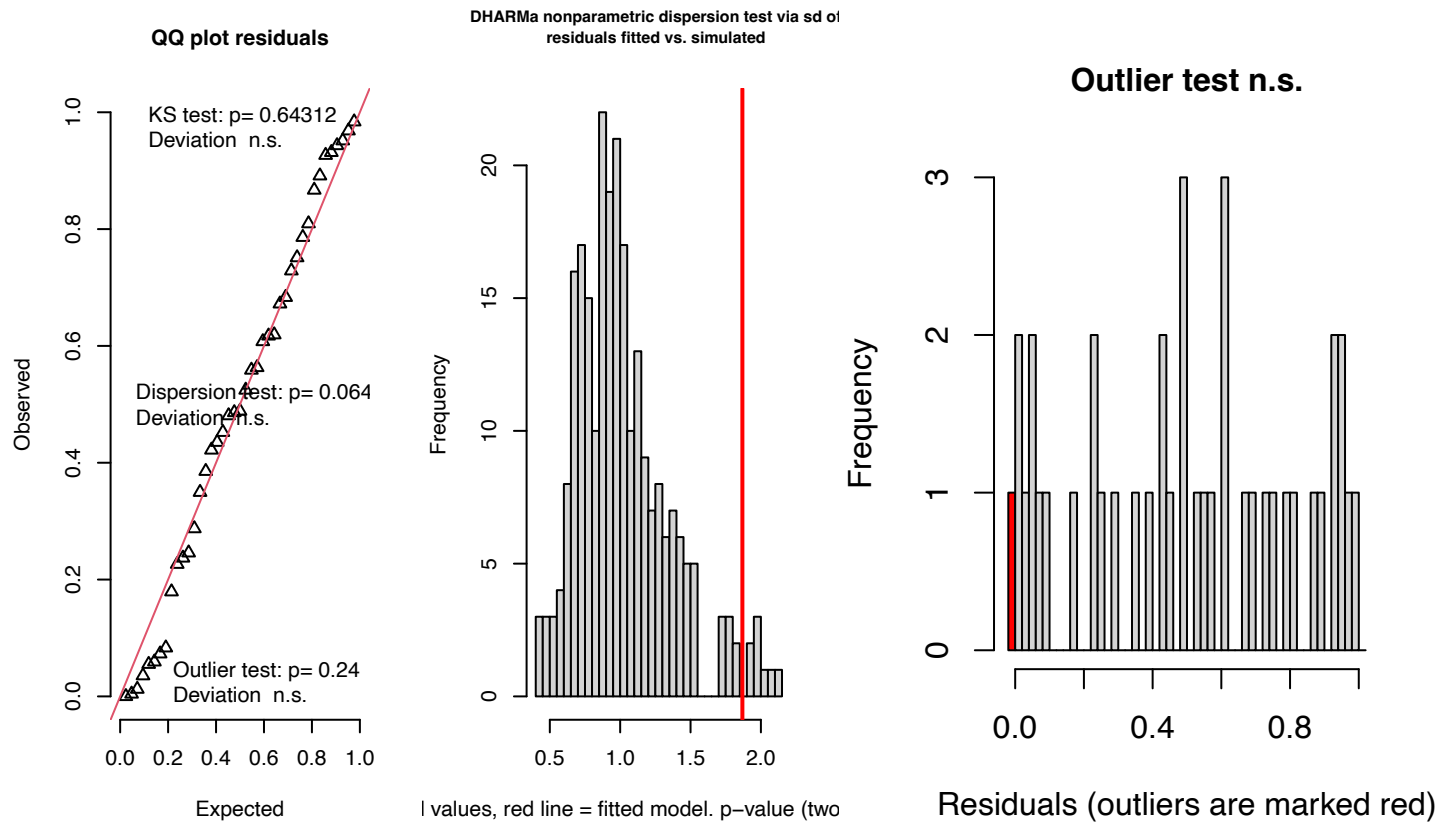
```
testZeroInflation(m26)
```

DHARMA zero-inflation test via comparison to expected zeros with simulation under H0 = fitted model



Simulated values, red line = fitted model. p-value (two.sided) = 0.008

```
##
## DHARMA zero-inflation test via comparison to expected zeros with
## simulation under H0 = fitted model
##
## data: simulationOutput
## ratioObsSim = 2.3238, p-value = 0.008
## alternative hypothesis: two.sided
testResiduals(m26)
```



```
## $uniformity
##
## Exact one-sample Kolmogorov-Smirnov test
##
## data: simulationOutput$scaledResiduals
## D = 0.11187, p-value = 0.6431
## alternative hypothesis: two-sided
##
##
## $dispersion
##
## DHARMA nonparametric dispersion test via sd of residuals fitted vs.
## simulated
##
## data: simulationOutput
## dispersion = 1.8328, p-value = 0.064
## alternative hypothesis: two.sided
##
##
## $outliers
##
## DHARMA bootstrapped outlier test
##
## data: simulationOutput
## outliers at both margin(s) = 1, observations = 41, p-value = 0.24
## alternative hypothesis: two.sided
## percent confidence interval:
## 0.00000000 0.02439024
```

```
## sample estimates:
## outlier frequency (expected: 0.00317073170731707 )
##                                0.02439024

## $uniformity
##
## Exact one-sample Kolmogorov-Smirnov test
##
## data: simulationOutput$scaledResiduals
## D = 0.11187, p-value = 0.6431
## alternative hypothesis: two-sided
##
##
## $dispersion
##
## DHARMA nonparametric dispersion test via sd of residuals fitted vs.
## simulated
##
## data: simulationOutput
## dispersion = 1.8328, p-value = 0.064
## alternative hypothesis: two.sided
##
##
## $outliers
##
## DHARMA bootstrapped outlier test
##
## data: simulationOutput
## outliers at both margin(s) = 1, observations = 41, p-value = 0.24
## alternative hypothesis: two.sided
## percent confidence interval:
## 0.00000000 0.02439024
## sample estimates:
## outlier frequency (expected: 0.00317073170731707 )
##                                0.02439024
```

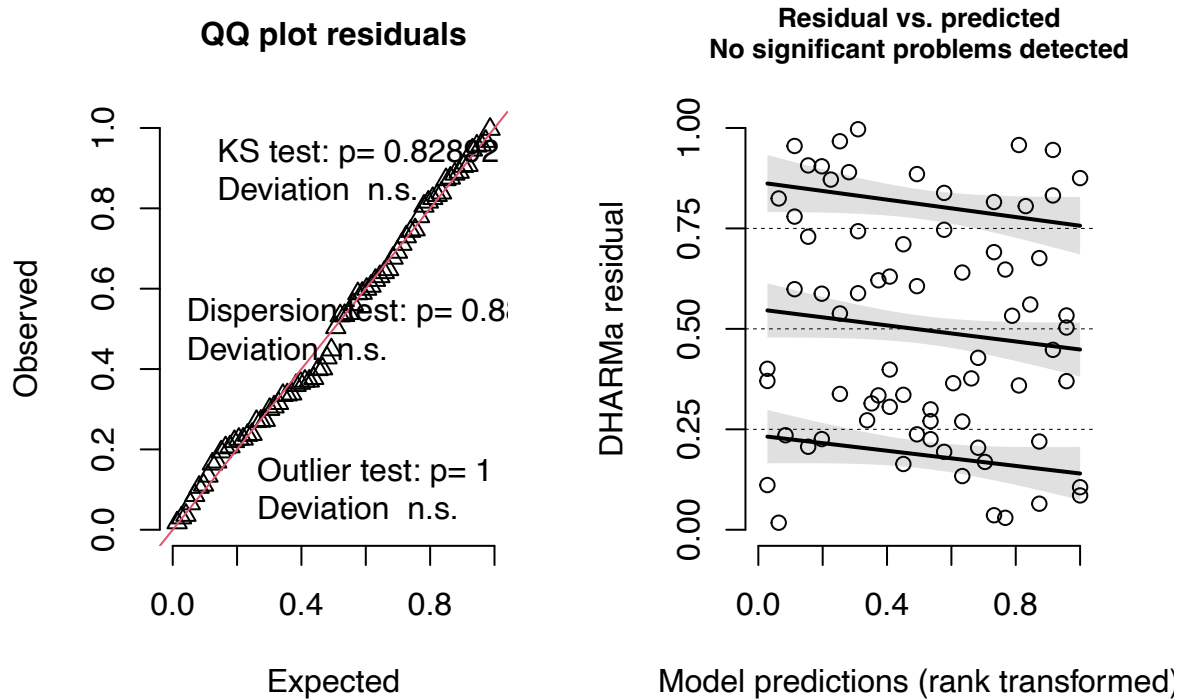
```
# calculate the residuals for the model
m31 <- simulateResiduals(fittedModel = glm_int_brubig, plot = F)
```

```
residuals(m31)
```

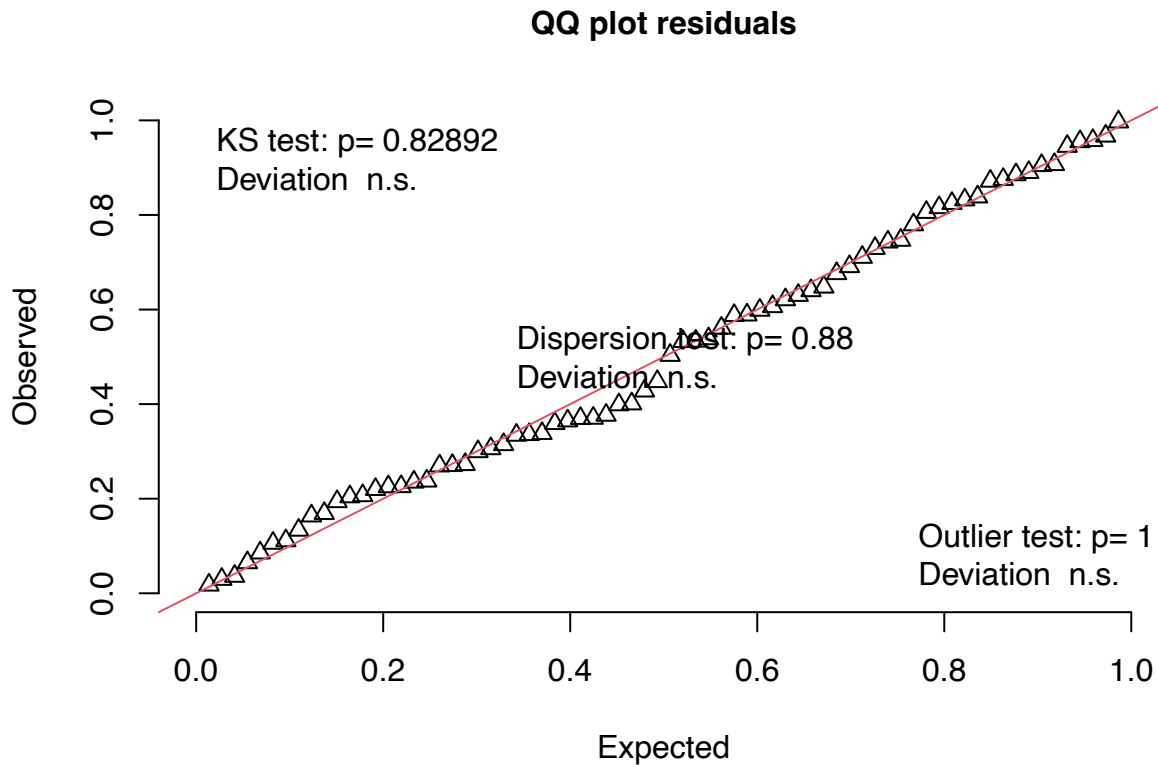
```
## [1] 0.36985175 0.53332340 0.16880053 0.69096046 0.23752399 0.88527797
## [7] 0.64766035 0.83207971 0.99684506 0.58755338 0.23524986 0.80555242
## [13] 0.94529015 0.03612392 0.11123091 0.56073186 0.74664128 0.95754140
## [19] 0.63013970 0.30587893 0.20654330 0.83836923 0.62088928 0.40066855
## [25] 0.87152944 0.44782623 0.82453293 0.90458567 0.58866077 0.19396409
## [31] 0.33804302 0.01751328 0.02966256 0.21965989 0.16369417 0.10591245
## [37] 0.06484081 0.74296596 0.72937058 0.81568410 0.33439873 0.71060982
## [43] 0.60634010 0.31451130 0.77941082 0.95519959 0.87528004 0.22571845
## [49] 0.96694680 0.53838444 0.26946850 0.67614721 0.29966093 0.89002640
## [55] 0.08543422 0.64015322 0.90676270 0.37033685 0.33584544 0.50333453
## [61] 0.36471419 0.37664946 0.27235062 0.59874078 0.39877640 0.27032819
## [67] 0.42792831 0.35915203 0.13360289 0.20415650 0.53265953 0.22539152
```

```
#plot simulation output
plot(m31)
```

DHARMA residual



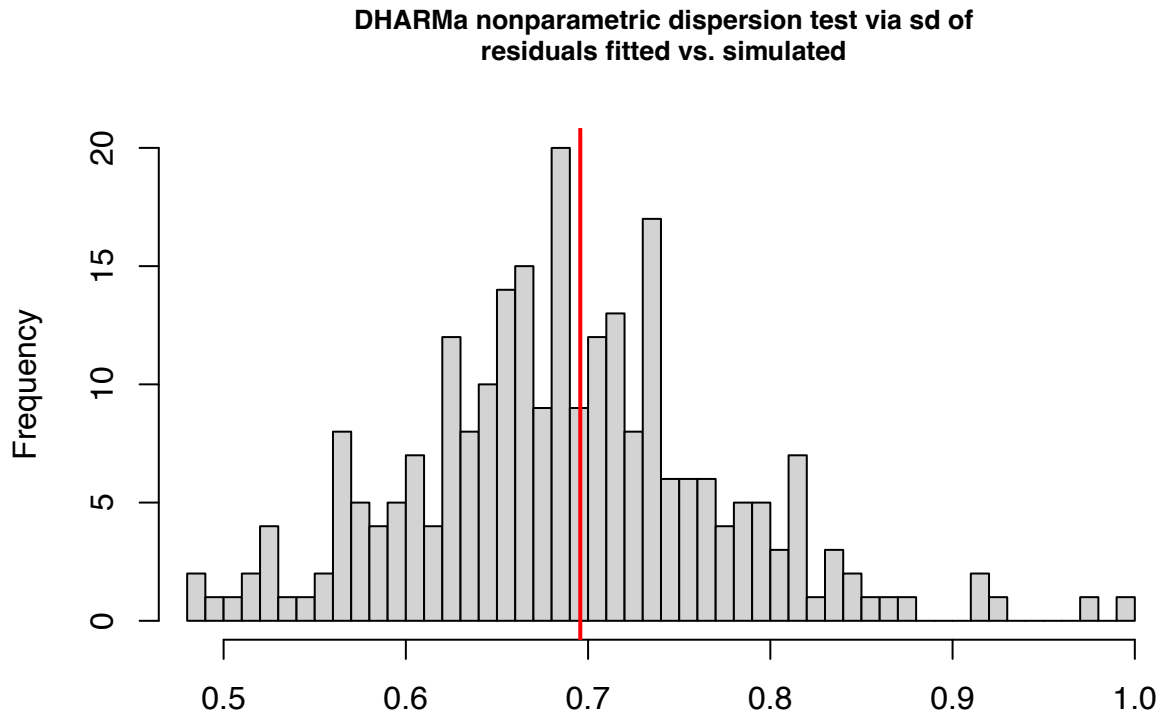
#tests if the overall distribution conforms to expectations.
`testUniformity(m31)`



 ## Exact one-sample Kolmogorov-Smirnov test

```
##
## data: simulationOutput$scaledResiduals
## D = 0.071554, p-value = 0.8289
## alternative hypothesis: two-sided
```

```
testDispersion(m31)
```

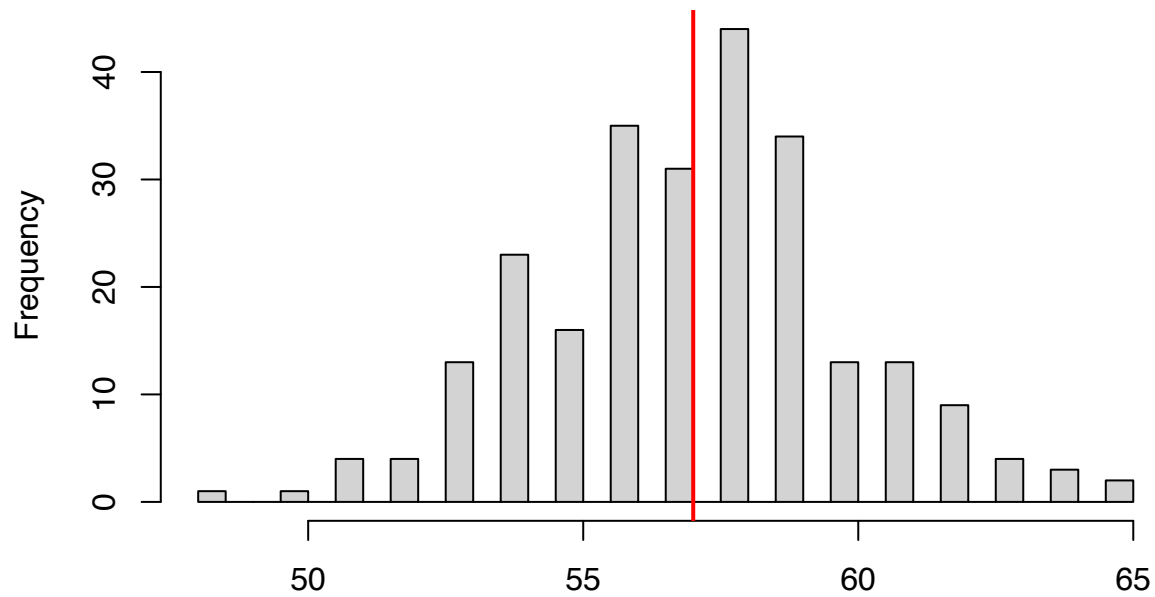


Simulated values, red line = fitted model. p-value (two.sided) = 0.88

```
##
## DHARMa nonparametric dispersion test via sd of residuals fitted vs.
## simulated
##
## data: simulationOutput
## dispersion = 1.0112, p-value = 0.88
## alternative hypothesis: two.sided
```

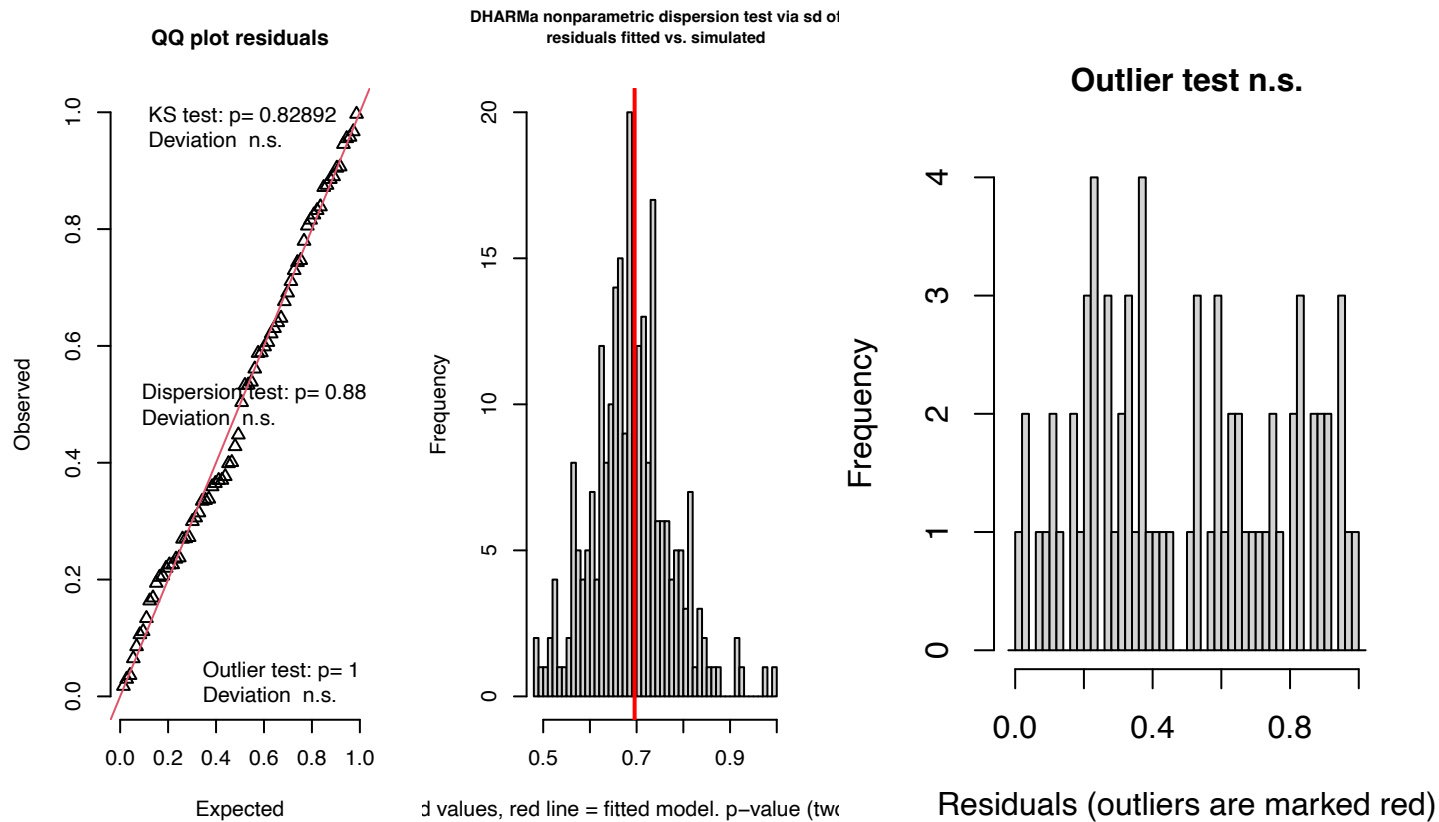
```
testZeroInflation(m31)
```

DHARMA zero-inflation test via comparison to expected zeros with simulation under H0 = fitted model



Simulated values, red line = fitted model. p-value (two.sided) = 1

```
##
## DHARMA zero-inflation test via comparison to expected zeros with
## simulation under H0 = fitted model
##
## data: simulationOutput
## ratioObsSim = 0.99574, p-value = 1
## alternative hypothesis: two.sided
testResiduals(m31)
```



```
## $uniformity
##
## Exact one-sample Kolmogorov-Smirnov test
##
## data: simulationOutput$scaledResiduals
## D = 0.071554, p-value = 0.8289
## alternative hypothesis: two-sided
##
##
## $dispersion
##
## DHARMA nonparametric dispersion test via sd of residuals fitted vs.
## simulated
##
## data: simulationOutput
## dispersion = 1.0112, p-value = 0.88
## alternative hypothesis: two.sided
##
##
## $outliers
##
## DHARMA bootstrapped outlier test
##
## data: simulationOutput
## outliers at both margin(s) = 0, observations = 72, p-value = 1
## alternative hypothesis: two.sided
## percent confidence interval:
## 0 0
```

```

## sample estimates:
## outlier frequency (expected: 0 )
##
## $uniformity
##
## Exact one-sample Kolmogorov-Smirnov test
##
## data: simulationOutput$scaledResiduals
## D = 0.071554, p-value = 0.8289
## alternative hypothesis: two-sided
##
##
## $dispersion
##
## DHARMA nonparametric dispersion test via sd of residuals fitted vs.
## simulated
##
## data: simulationOutput
## dispersion = 1.0112, p-value = 0.88
## alternative hypothesis: two.sided
##
##
## $outliers
##
## DHARMA bootstrapped outlier test
##
## data: simulationOutput
## outliers at both margin(s) = 0, observations = 72, p-value = 1
## alternative hypothesis: two.sided
## percent confidence interval:
## 0 0
## sample estimates:
## outlier frequency (expected: 0 )
##
##
# calculate the residuals for the model
m32 <- simulateResiduals(fittedModel = glm_add_brubig, plot = F)

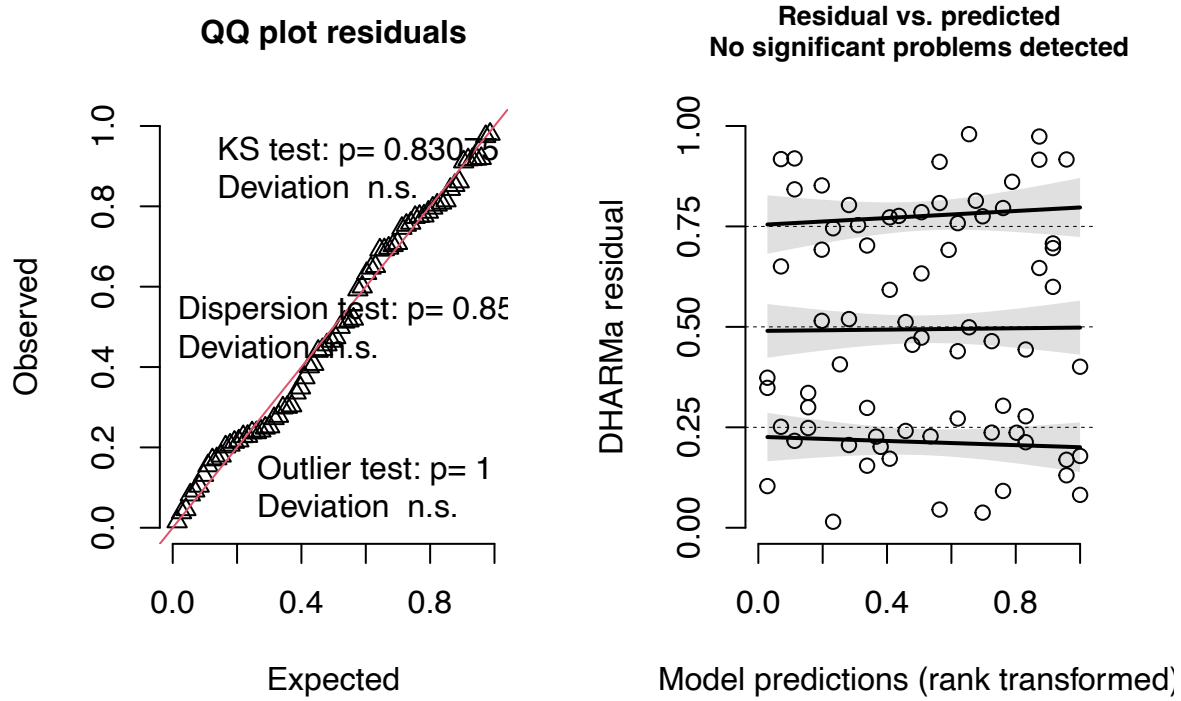
residuals(m32)

## [1] 0.59954915 0.69604616 0.22685931 0.80818235 0.25129350 0.91754354
## [7] 0.77517375 0.91603985 0.78551390 0.51939719 0.20137388 0.86110887
## [13] 0.97373927 0.04531521 0.10366721 0.81386943 0.84227022 0.97961987
## [19] 0.44361835 0.21289173 0.17184403 0.91918462 0.46442518 0.37342309
## [25] 0.75300144 0.64644989 0.74537777 0.80327208 0.47328326 0.21634457
## [31] 0.27178659 0.01499137 0.03757257 0.30359497 0.08184709 0.16921644
## [37] 0.09163454 0.63300699 0.59224891 0.91075230 0.23675430 0.40078394
## [43] 0.65043756 0.23651250 0.69211681 0.85203804 0.91660292 0.20585523
## [49] 0.75808629 0.43932171 0.29834013 0.79572363 0.33562024 0.77610302
## [55] 0.13060633 0.70252713 0.77256182 0.34811664 0.17866978 0.70765893
## [61] 0.40672315 0.45538409 0.22768511 0.51491707 0.27754837 0.29993556
## [67] 0.51202983 0.49895618 0.15447834 0.24095215 0.69146486 0.24825733

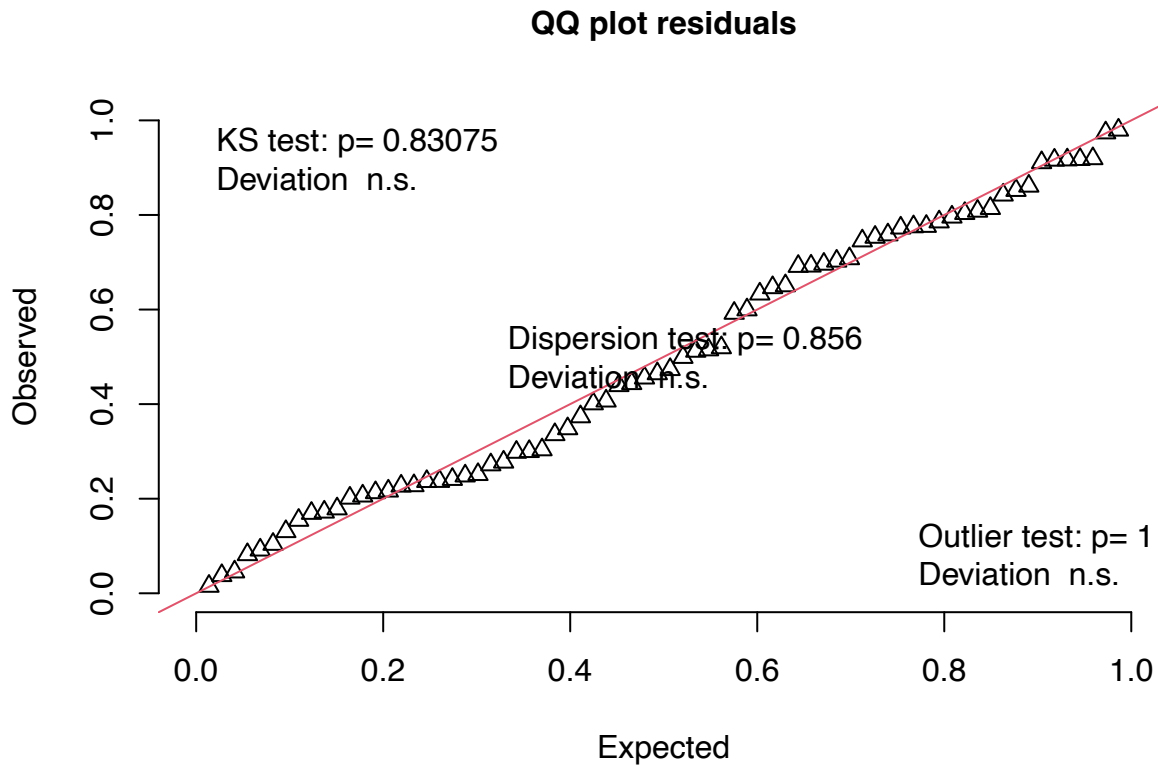
#plot simulation output
plot(m32)

```

DHARMa residual



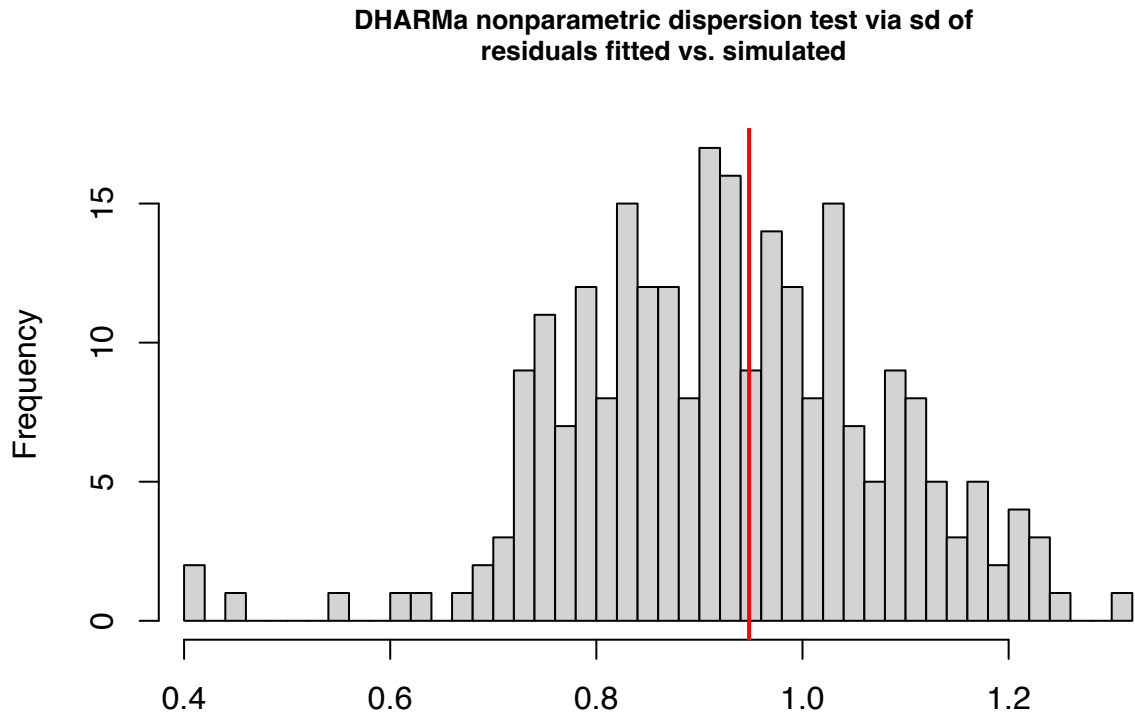
#tests if the overall distribution conforms to expectations.
`testUniformity(m32)`



 ## Exact one-sample Kolmogorov-Smirnov test

```
##
## data: simulationOutput$scaledResiduals
## D = 0.071405, p-value = 0.8307
## alternative hypothesis: two-sided
```

```
testDispersion(m32)
```

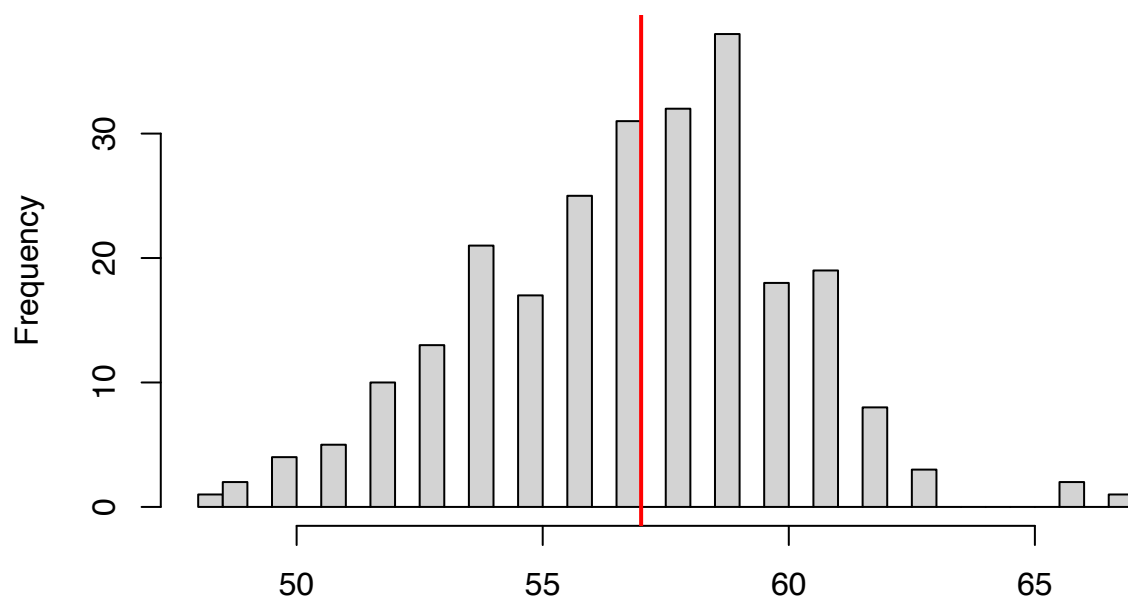


Simulated values, red line = fitted model. p-value (two.sided) = 0.856

```
##
## DHARMa nonparametric dispersion test via sd of residuals fitted vs.
## simulated
##
## data: simulationOutput
## dispersion = 1.0248, p-value = 0.856
## alternative hypothesis: two.sided
```

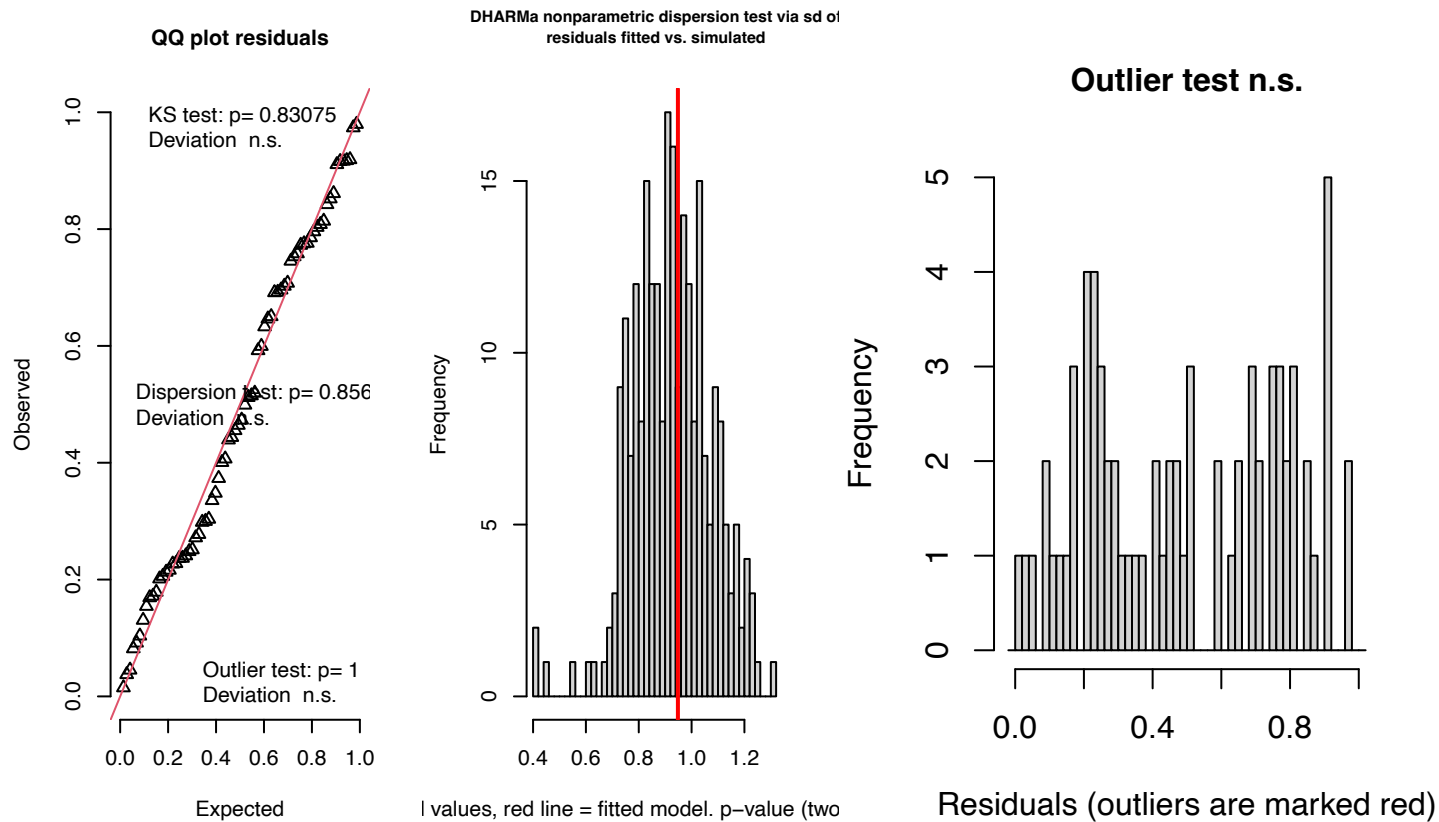
```
testZeroInflation(m32)
```

DHARMA zero-inflation test via comparison to expected zeros with simulation under H0 = fitted model



Simulated values, red line = fitted model. p-value (two.sided) = 1

```
##
## DHARMA zero-inflation test via comparison to expected zeros with
## simulation under H0 = fitted model
##
## data: simulationOutput
## ratioObsSim = 0.99881, p-value = 1
## alternative hypothesis: two.sided
testResiduals(m32)
```



```
## $uniformity
##
## Exact one-sample Kolmogorov-Smirnov test
##
## data: simulationOutput$scaledResiduals
## D = 0.071405, p-value = 0.8307
## alternative hypothesis: two-sided
##
##
## $dispersion
##
## DHARMA nonparametric dispersion test via sd of residuals fitted vs.
## simulated
##
## data: simulationOutput
## dispersion = 1.0248, p-value = 0.856
## alternative hypothesis: two.sided
##
##
## $outliers
##
## DHARMA bootstrapped outlier test
##
## data: simulationOutput
## outliers at both margin(s) = 0, observations = 72, p-value = 1
## alternative hypothesis: two.sided
## percent confidence interval:
## 0 0
```

```
## sample estimates:
## outlier frequency (expected: 0 )
##                                0

## $uniformity
##
##   Exact one-sample Kolmogorov-Smirnov test
##
## data:  simulationOutput$scaledResiduals
## D = 0.071405, p-value = 0.8307
## alternative hypothesis: two-sided
##
##
## $dispersion
##
##   DHARMA nonparametric dispersion test via sd of residuals fitted vs.
##   simulated
##
## data:  simulationOutput
## dispersion = 1.0248, p-value = 0.856
## alternative hypothesis: two.sided
##
##
## $outliers
##
##   DHARMA bootstrapped outlier test
##
## data:  simulationOutput
## outliers at both margin(s) = 0, observations = 72, p-value = 1
## alternative hypothesis: two.sided
## percent confidence interval:
##    0 0
## sample estimates:
## outlier frequency (expected: 0 )
##                                0
```

Bru-Mol combination

```
# For each behavior, we test the best model for residuals, uniformity, overdispersion, and inflation
```

```
# calculate the residuals for the model
```

```
m3 <- simulateResiduals(fittedModel = additive_indirect_brumol, plot = F)
```

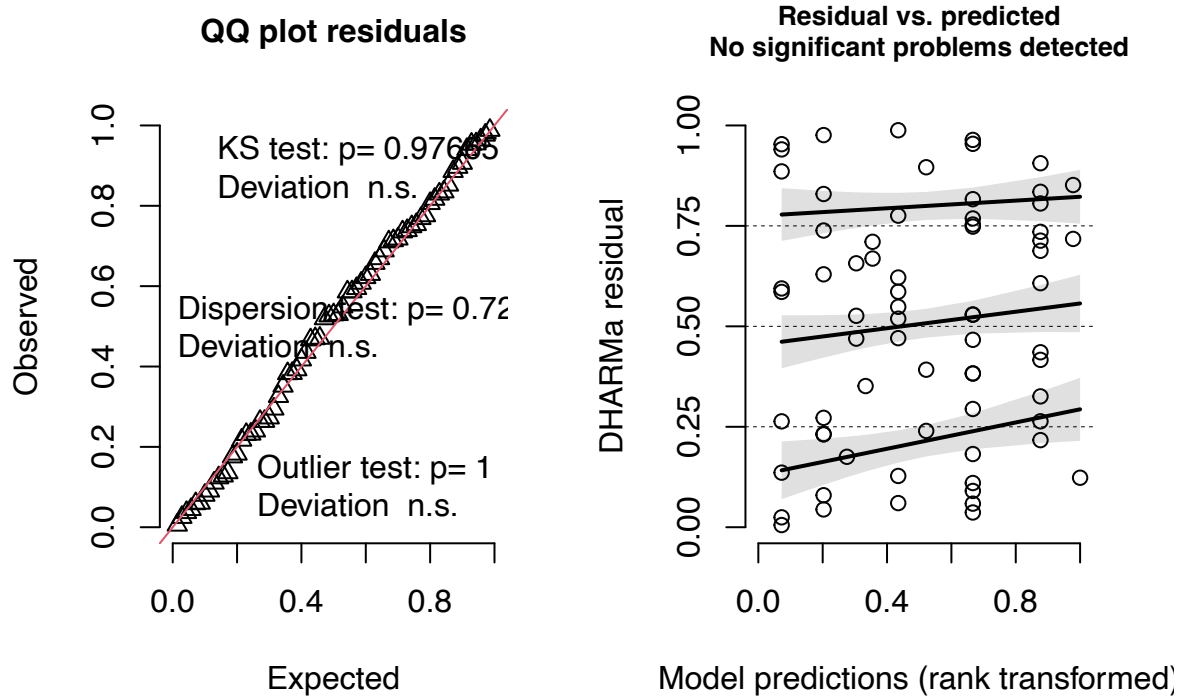
```
residuals(m3)
```

```
## [1] 0.263501559 0.024654163 0.079700225 0.044282107 0.110082767 0.036726226
## [7] 0.005366117 0.135902234 0.090300776 0.272251096 0.058724032 0.216810826
## [13] 0.175319739 0.127686031 0.123193618 0.263661857 0.231938302 0.526232295
## [19] 0.182201318 0.351414125 0.239777392 0.594282580 0.392105153 0.629506309
## [25] 0.416558930 0.435454264 0.470622231 0.060075625 0.519046350 0.585720402
## [31] 0.586161675 0.621546354 0.607431954 0.529516373 0.466506146 0.753437914
## [37] 0.735244520 0.885618543 0.382361155 0.710367703 0.528860954 0.668655882
## [43] 0.976000000 0.748192653 0.548585316 0.688007966 0.294322530 0.829411177
## [49] 0.325655710 0.940353299 0.713330413 0.805866417 0.775432818 0.383000723
## [55] 0.834819622 0.768445988 0.230991957 0.953374304 0.656889842 0.905965196
```

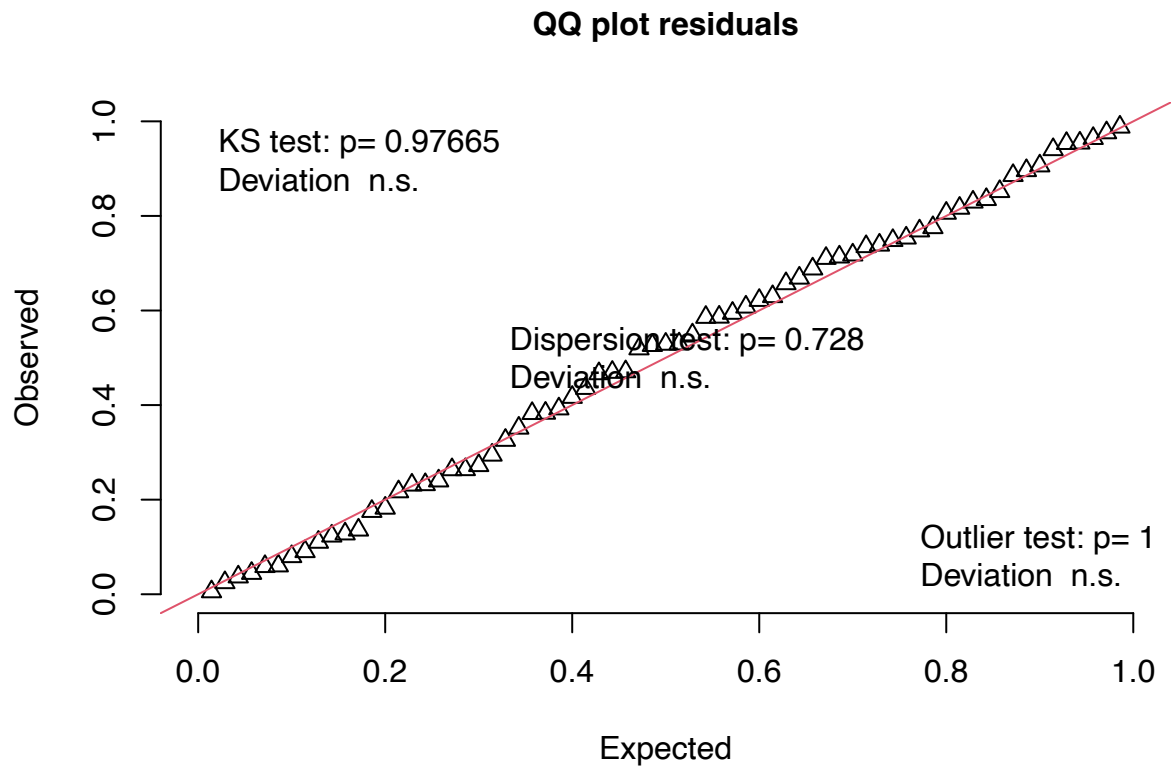
```
## [61] 0.738482320 0.964000000 0.954261629 0.988000000 0.851706028 0.717590422
## [67] 0.469430142 0.896000000 0.816437689
```

```
#plot simulation output
plot(m3)
```

DHARMA residual

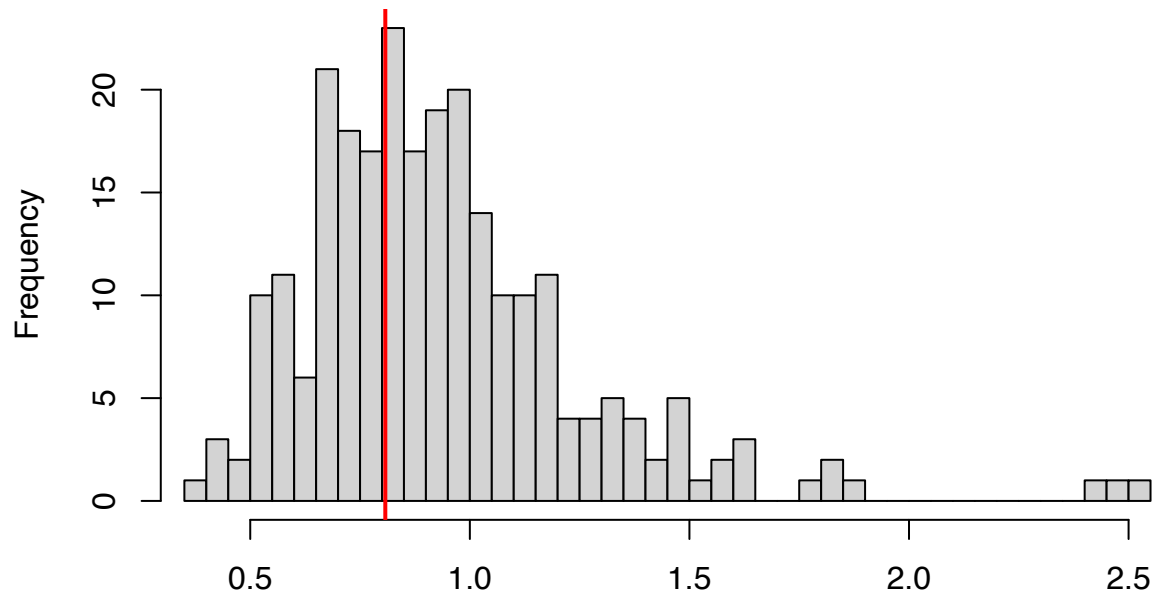


```
#tests if the overall distribution conforms to expectations.
testUniformity(m3)
```



```
##
## Exact one-sample Kolmogorov-Smirnov test
##
## data: simulationOutput$scaledResiduals
## D = 0.055278, p-value = 0.9766
## alternative hypothesis: two-sided
testDispersion(m3)
```

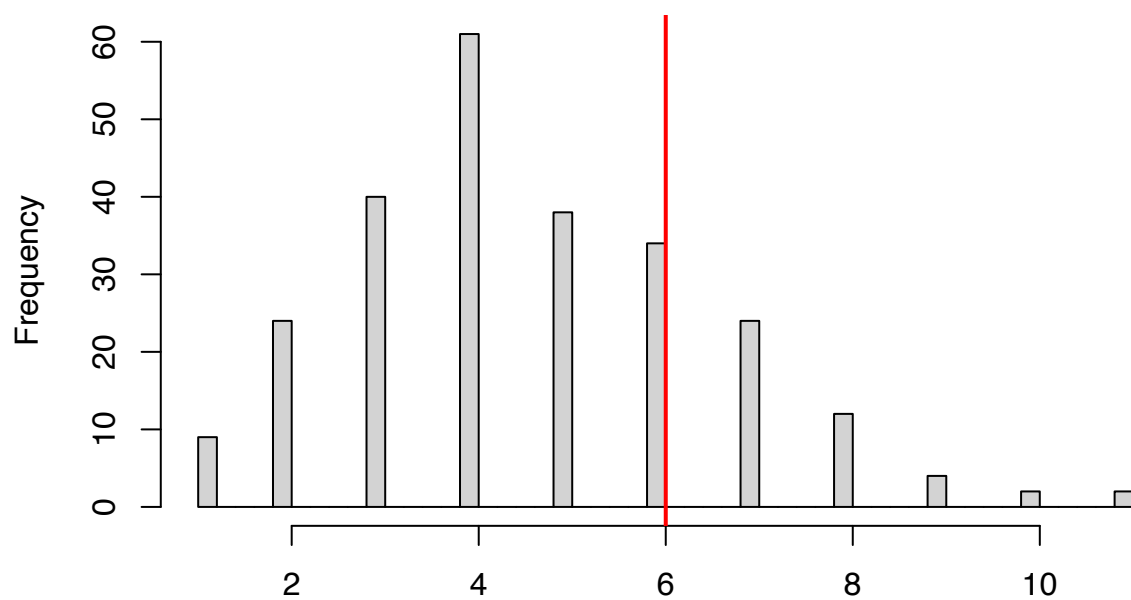
**DHARMA nonparametric dispersion test via sd of
residuals fitted vs. simulated**



Simulated values, red line = fitted model. p-value (two.sided) = 0.728

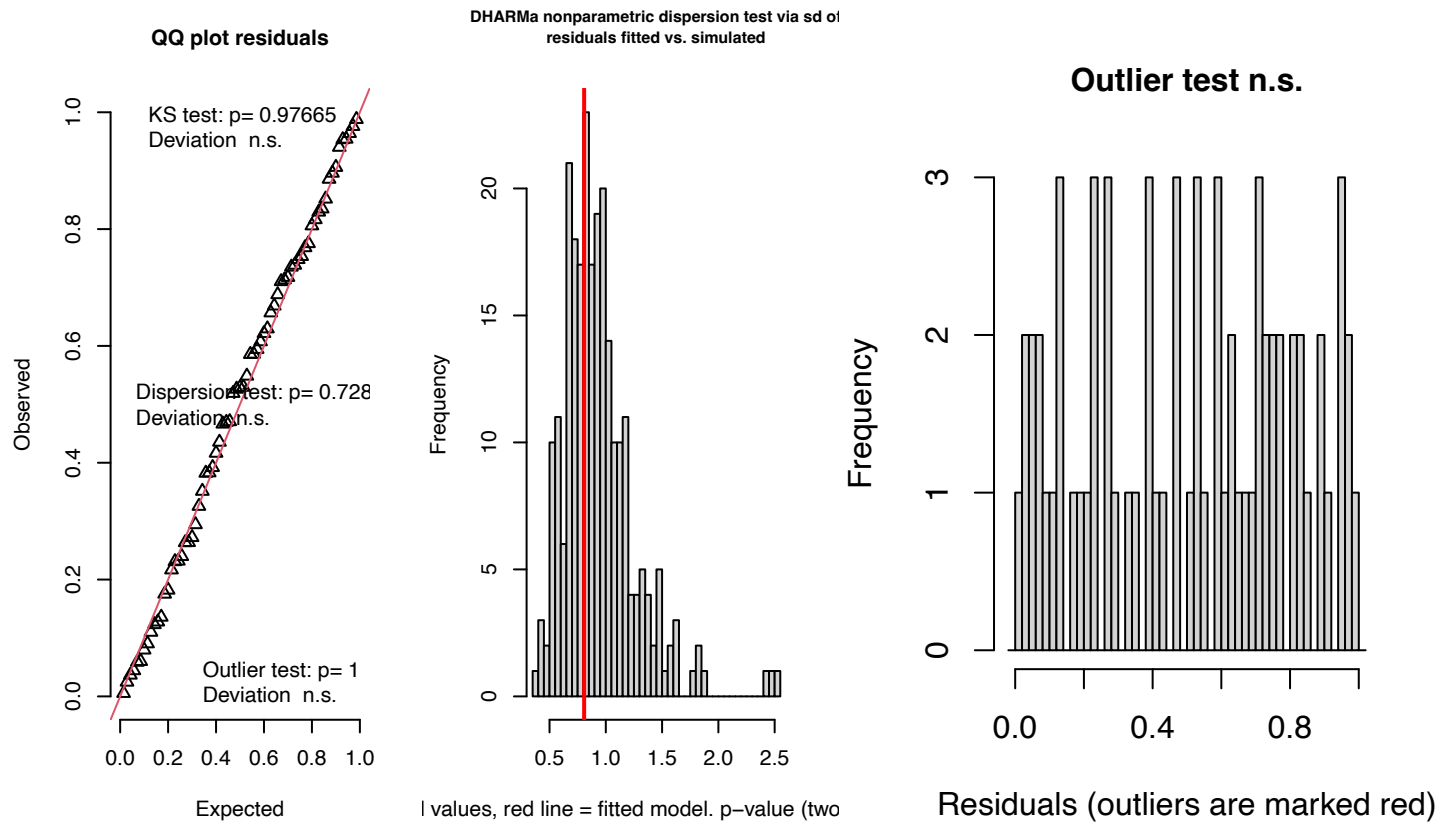
```
##
## DHARMA nonparametric dispersion test via sd of residuals fitted vs.
## simulated
##
## data:  simulationOutput
## dispersion = 0.85904, p-value = 0.728
## alternative hypothesis: two.sided
testZeroInflation(m3)
```

**DHARMA zero-inflation test via comparison to
expected zeros with simulation under H0 = fitted
model**



Simulated values, red line = fitted model. p-value (two.sided) = 0.624

```
##  
## DHARMA zero-inflation test via comparison to expected zeros with  
## simulation under H0 = fitted model  
##  
## data: simulationOutput  
## ratioObsSim = 1.2965, p-value = 0.624  
## alternative hypothesis: two.sided  
testResiduals(m3)
```



```
## $uniformity
##
## Exact one-sample Kolmogorov-Smirnov test
##
## data: simulationOutput$scaledResiduals
## D = 0.055278, p-value = 0.9766
## alternative hypothesis: two-sided
##
##
## $dispersion
##
## DHARMA nonparametric dispersion test via sd of residuals fitted vs.
## simulated
##
## data: simulationOutput
## dispersion = 0.85904, p-value = 0.728
## alternative hypothesis: two.sided
##
##
## $outliers
##
## DHARMA bootstrapped outlier test
##
## data: simulationOutput
## outliers at both margin(s) = 0, observations = 69, p-value = 1
## alternative hypothesis: two.sided
## percent confidence interval:
## 0.00000000 0.02898551
```

```

## sample estimates:
## outlier frequency (expected: 0.00608695652173913 )
##
## 0

## $uniformity
##
## Exact one-sample Kolmogorov-Smirnov test
##
## data: simulationOutput$scaledResiduals
## D = 0.055278, p-value = 0.9766
## alternative hypothesis: two-sided
##
##
## $dispersion
##
## DHARMA nonparametric dispersion test via sd of residuals fitted vs.
## simulated
##
## data: simulationOutput
## dispersion = 0.85904, p-value = 0.728
## alternative hypothesis: two.sided
##
##
## $outliers
##
## DHARMA bootstrapped outlier test
##
## data: simulationOutput
## outliers at both margin(s) = 0, observations = 69, p-value = 1
## alternative hypothesis: two.sided
## percent confidence interval:
## 0.00000000 0.02898551
## sample estimates:
## outlier frequency (expected: 0.00608695652173913 )
##
## 0

# For each behavior, we test the best model for residuals, uniformity, overdispersion, and inflation

# calculate the residuals for the model
m4 <- simulateResiduals(fittedModel = interaction_indirect_brumol, plot = F)

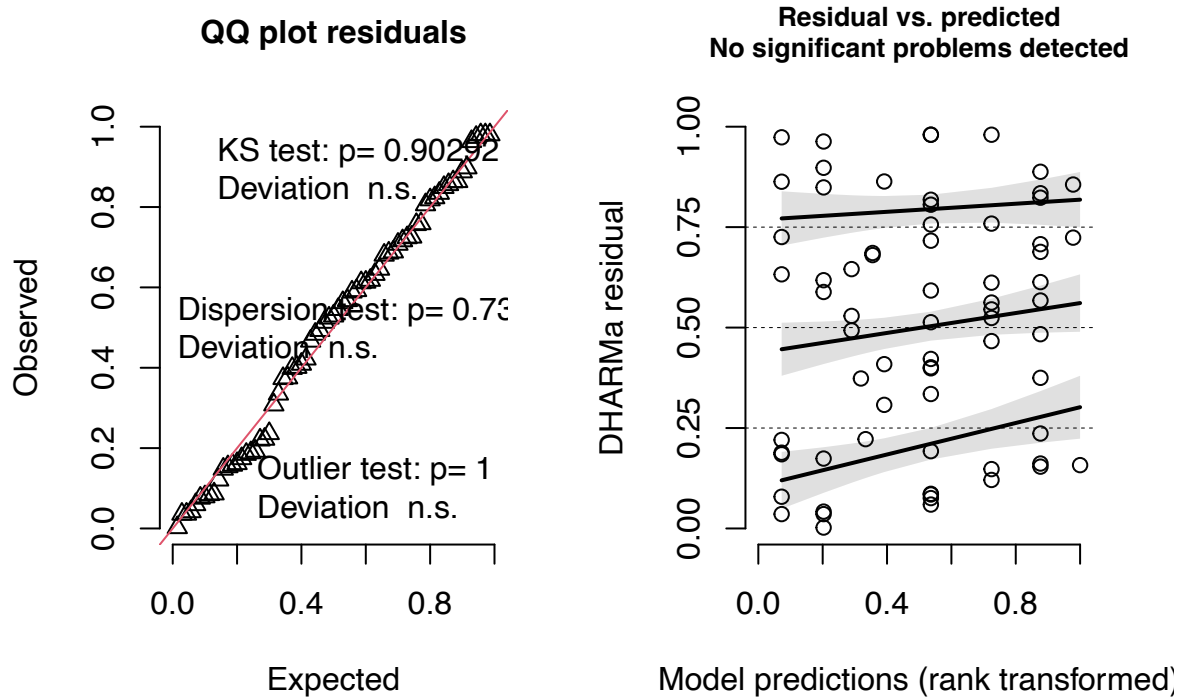
residuals(m4)

## [1] 0.174009546 0.041959342 0.035803856 0.078992320 0.084590621 0.059502183
## [7] 0.035702519 0.002258117 0.075588634 0.220347624 0.086229807 0.153843648
## [13] 0.222318796 0.147748760 0.157648770 0.161613703 0.184668082 0.529018961
## [19] 0.192416294 0.373321145 0.307672608 0.588625374 0.408913256 0.632526893
## [25] 0.374972914 0.483521704 0.466382643 0.120738497 0.523920086 0.617903940
## [31] 0.611786900 0.545293838 0.567388002 0.592246388 0.421936778 0.716255667
## [37] 0.707564665 0.848828882 0.398556665 0.680383126 0.513351009 0.685624130
## [43] 0.973555666 0.756560127 0.562297220 0.613157033 0.334635328 0.863045705
## [49] 0.236295245 0.897712914 0.688400581 0.823315927 0.758793394 0.401185360
## [55] 0.834364344 0.805822717 0.188647478 0.963170315 0.645509598 0.887817406
## [61] 0.725391469 0.980000000 0.980000000 0.980000000 0.856000000 0.723896576
## [67] 0.493054102 0.863553559 0.818158235

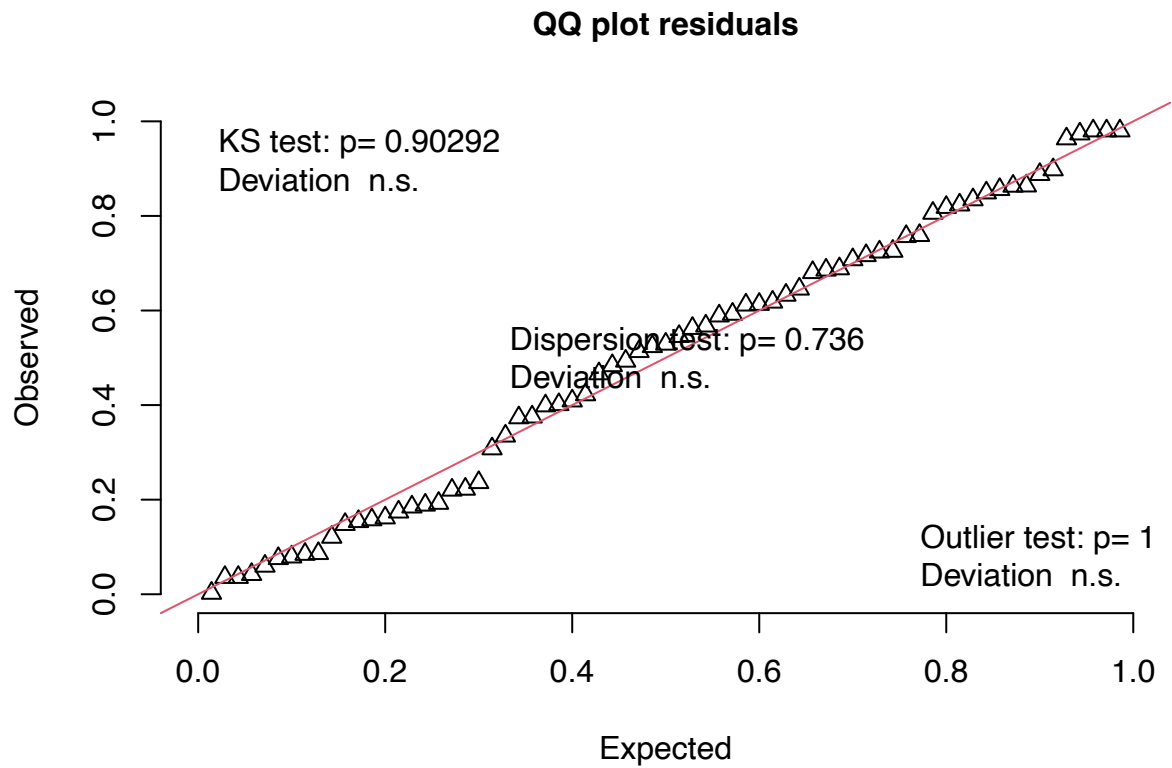
```

```
#plot simulation output
plot(m4)
```

DHARMA residual

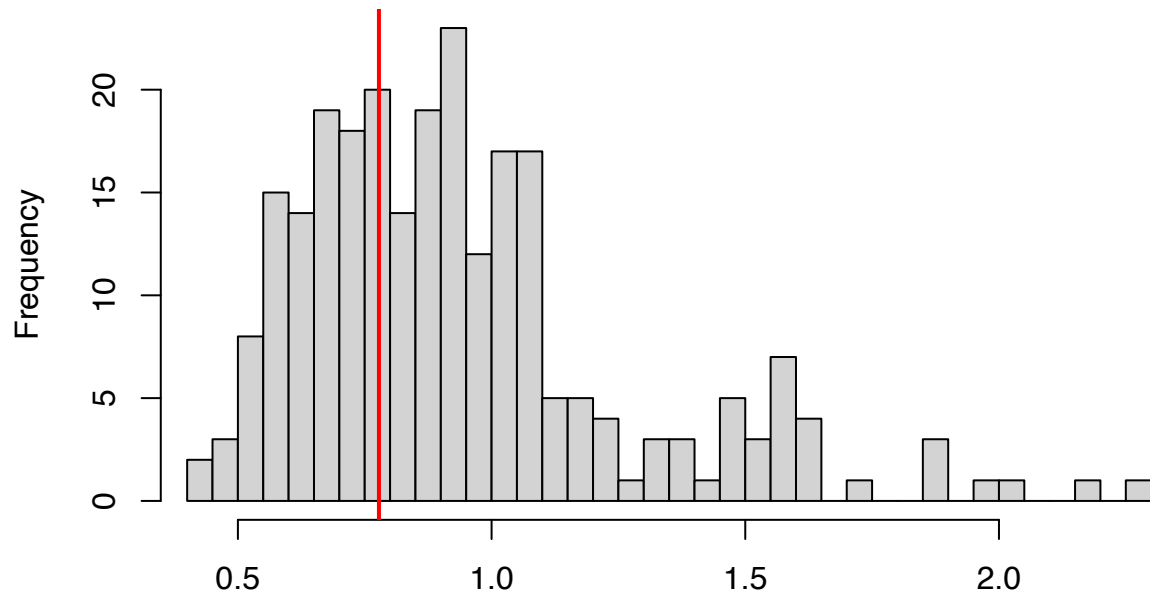


```
#tests if the overall distribution conforms to expectations.
testUniformity(m4)
```



```
##  
## Asymptotic one-sample Kolmogorov-Smirnov test  
##  
## data: simulationOutput$scaledResiduals  
## D = 0.068453, p-value = 0.9029  
## alternative hypothesis: two-sided  
testDispersion(m4)
```

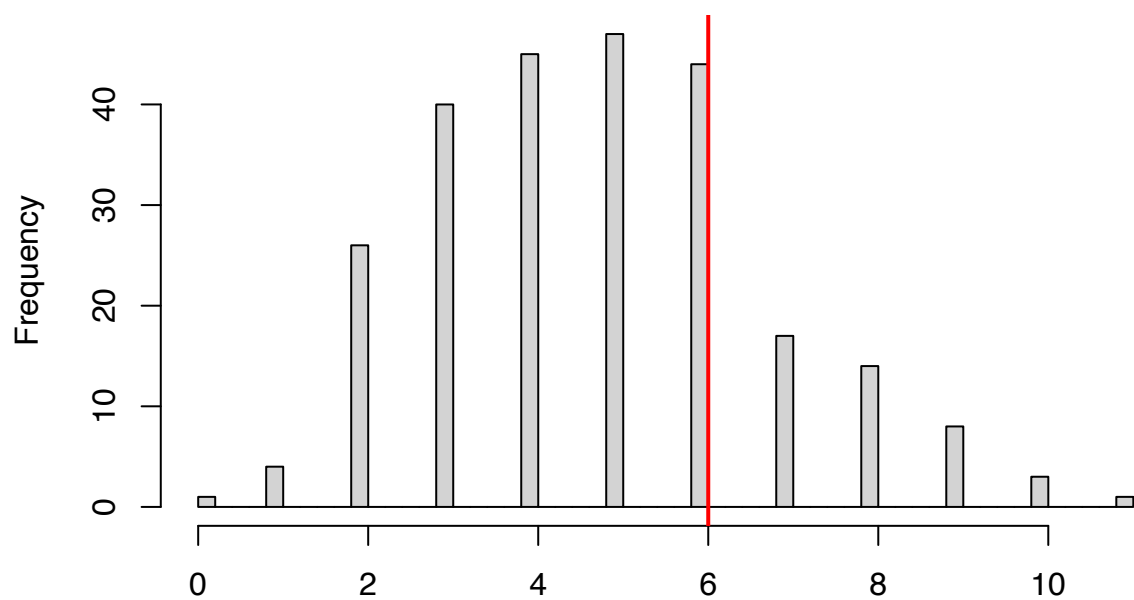
**DHARMA nonparametric dispersion test via sd of
residuals fitted vs. simulated**



Simulated values, red line = fitted model. p-value (two.sided) = 0.736

```
##  
## DHARMA nonparametric dispersion test via sd of residuals fitted vs.  
## simulated  
##  
## data: simulationOutput  
## dispersion = 0.82933, p-value = 0.736  
## alternative hypothesis: two.sided  
testZeroInflation(m4)
```

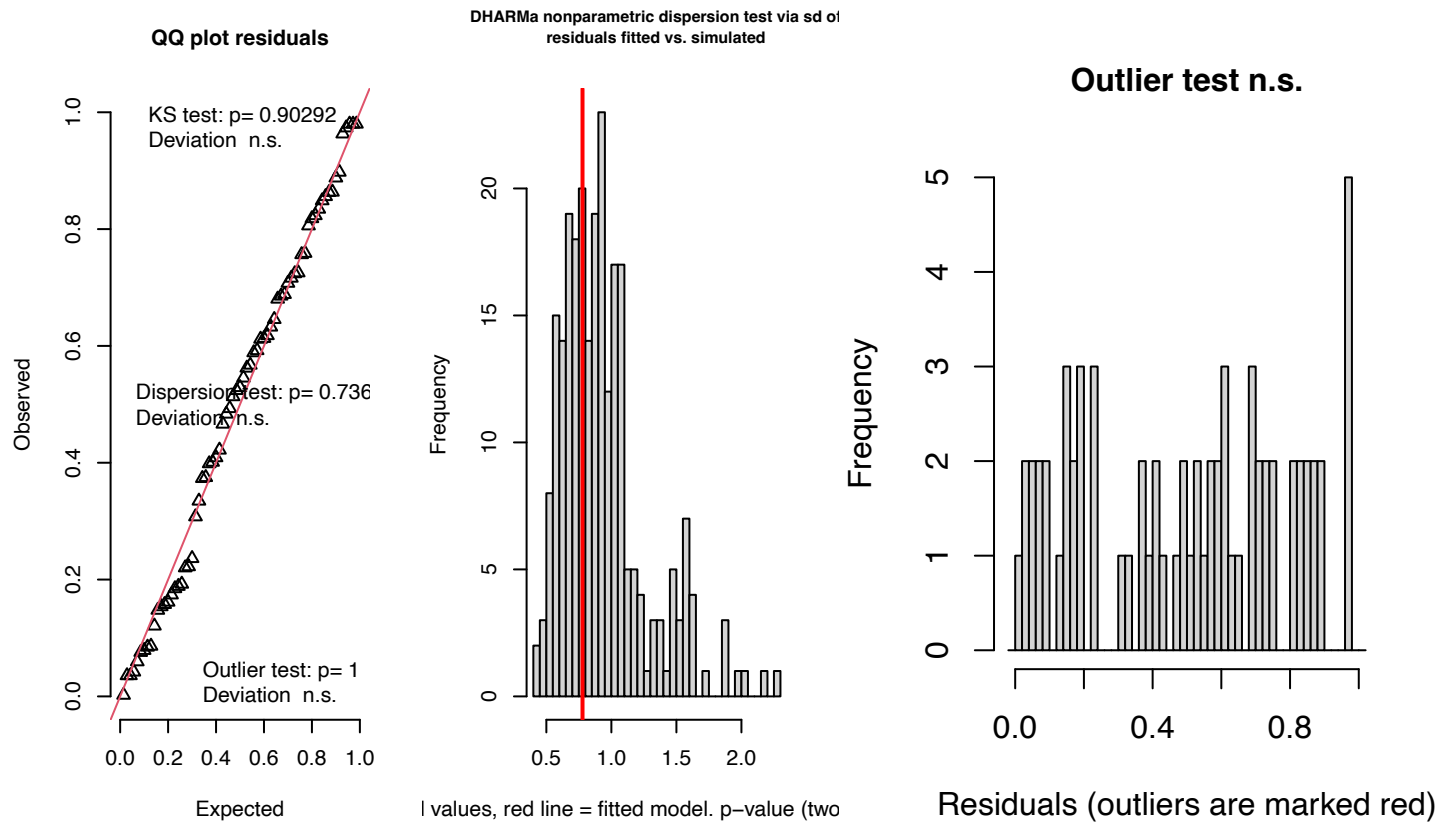
DHARMA zero-inflation test via comparison to expected zeros with simulation under H0 = fitted model



Simulated values, red line = fitted model. p-value (two.sided) = 0.696

```
##
## DHARMA zero-inflation test via comparison to expected zeros with
## simulation under H0 = fitted model
##
## data: simulationOutput
## ratioObsSim = 1.251, p-value = 0.696
## alternative hypothesis: two.sided
```

```
testResiduals(m4)
```



```
## $uniformity
##
## Asymptotic one-sample Kolmogorov-Smirnov test
##
## data: simulationOutput$scaledResiduals
## D = 0.068453, p-value = 0.9029
## alternative hypothesis: two-sided
##
##
## $dispersion
##
## DHARMA nonparametric dispersion test via sd of residuals fitted vs.
## simulated
##
## data: simulationOutput
## dispersion = 0.82933, p-value = 0.736
## alternative hypothesis: two.sided
##
##
## $outliers
##
## DHARMA bootstrapped outlier test
##
## data: simulationOutput
## outliers at both margin(s) = 0, observations = 69, p-value = 1
## alternative hypothesis: two.sided
## percent confidence interval:
## 0.00000000 0.02898551
```

```

## sample estimates:
## outlier frequency (expected: 0.00608695652173913 )
##
## $uniformity
##
## Asymptotic one-sample Kolmogorov-Smirnov test
##
## data: simulationOutput$scaledResiduals
## D = 0.068453, p-value = 0.9029
## alternative hypothesis: two-sided
##
##
## $dispersion
##
## DHARMA nonparametric dispersion test via sd of residuals fitted vs.
## simulated
##
## data: simulationOutput
## dispersion = 0.82933, p-value = 0.736
## alternative hypothesis: two.sided
##
##
## $outliers
##
## DHARMA bootstrapped outlier test
##
## data: simulationOutput
## outliers at both margin(s) = 0, observations = 69, p-value = 1
## alternative hypothesis: two.sided
## percent confidence interval:
## 0.00000000 0.02898551
## sample estimates:
## outlier frequency (expected: 0.00608695652173913 )
##
##
# calculate the residuals for the model
m9 <- simulateResiduals(fittedModel = additive_direct_brumol, plot = F)

residuals(m9)

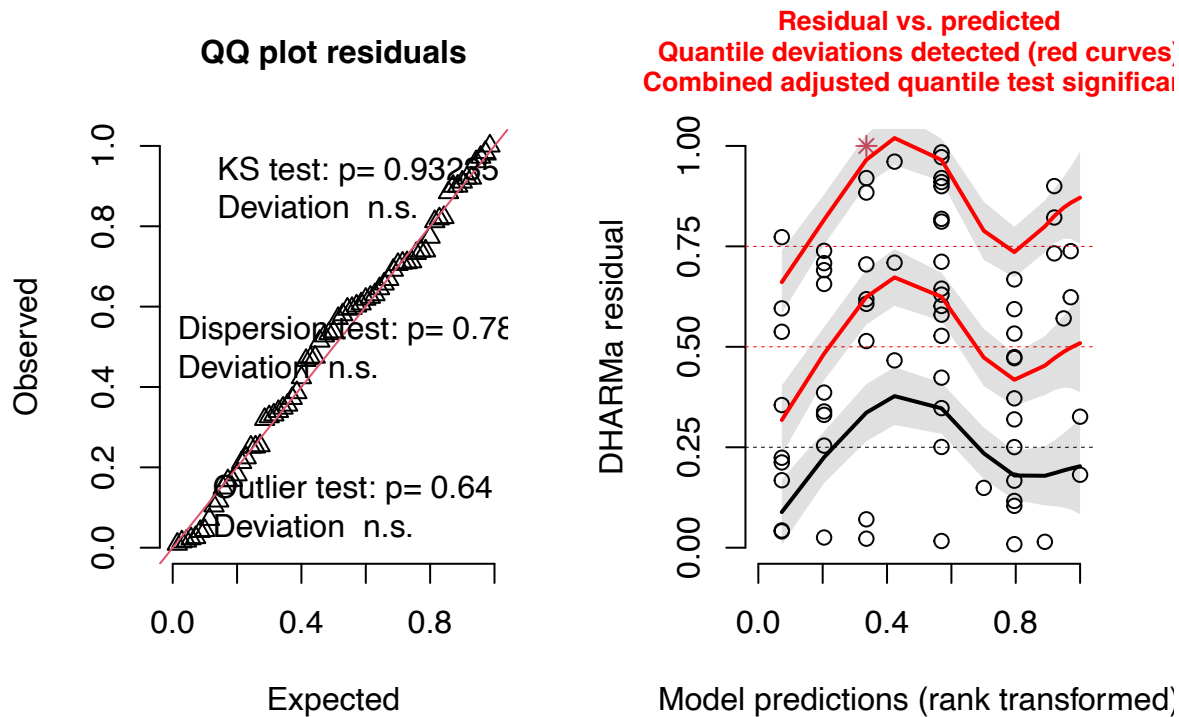
## [1] 0.223921495 0.040633140 0.022464019 0.070656671 0.423301931 0.016691917
## [7] 0.167916320 0.212216338 0.347495614 0.606942533 0.527277506 0.104160862
## [13] 0.014674290 0.254079183 0.148848661 0.116367504 0.618506186 0.821830019
## [19] 0.579995017 0.570653874 0.709455605 0.355053830 0.466248660 0.514303679
## [25] 0.250385619 0.593766758 0.339118927 0.708258901 0.386119095 0.595626165
## [31] 0.025385441 0.330592605 0.008785363 0.811637002 0.601694817 0.644779573
## [37] 0.166837967 0.042975837 0.711719967 0.181572313 0.629818366 0.326125744
## [43] 0.883918179 0.250795914 0.656421456 0.371729887 0.818550478 0.705135757
## [49] 0.667763378 0.537444365 0.532959902 0.319393142 0.690722111 0.910110054
## [55] 0.474452320 0.920000000 0.920000000 0.772925717 0.732499232 0.471758873
## [61] 1.000000000 0.972000000 0.899724977 0.738871320 0.623426393 0.738124536
## [67] 0.900000000 0.960906675 0.984000000

#plot simulation output
plot(m9)

```

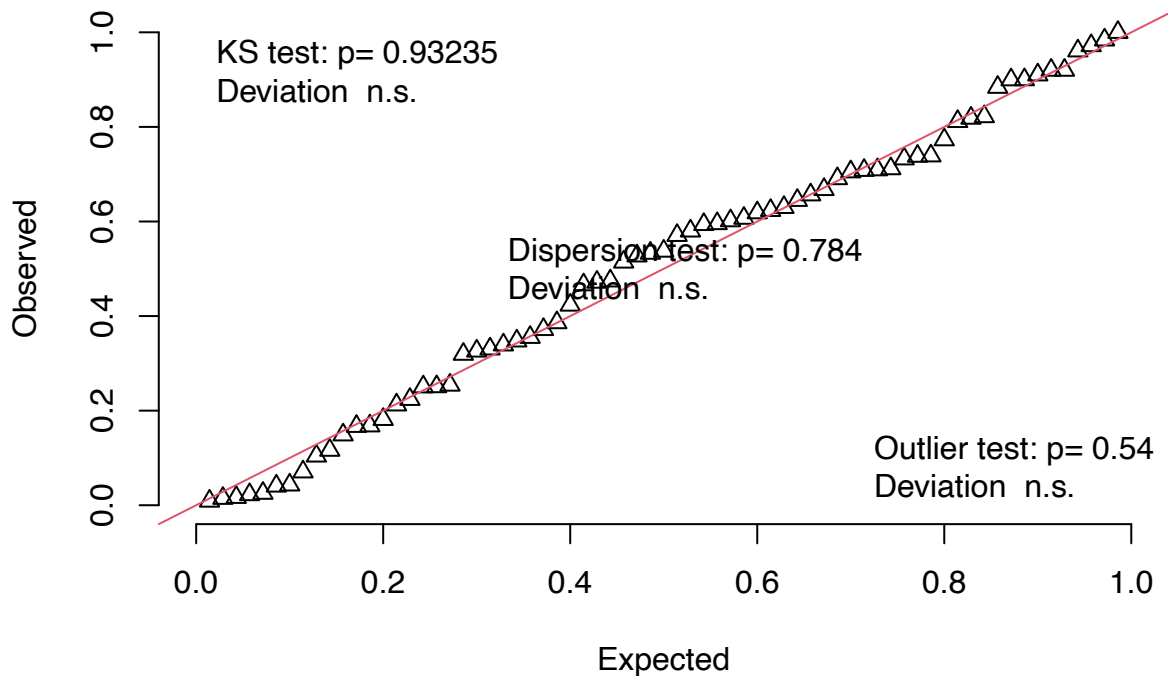
```
## Warning in newton(lsp = lsp, X = G$X, y = G$y, Eb = G$Eb, UrS = G$UrS, L = G$L,
## : Fitting terminated with step failure - check results carefully
```

DHARMA residual



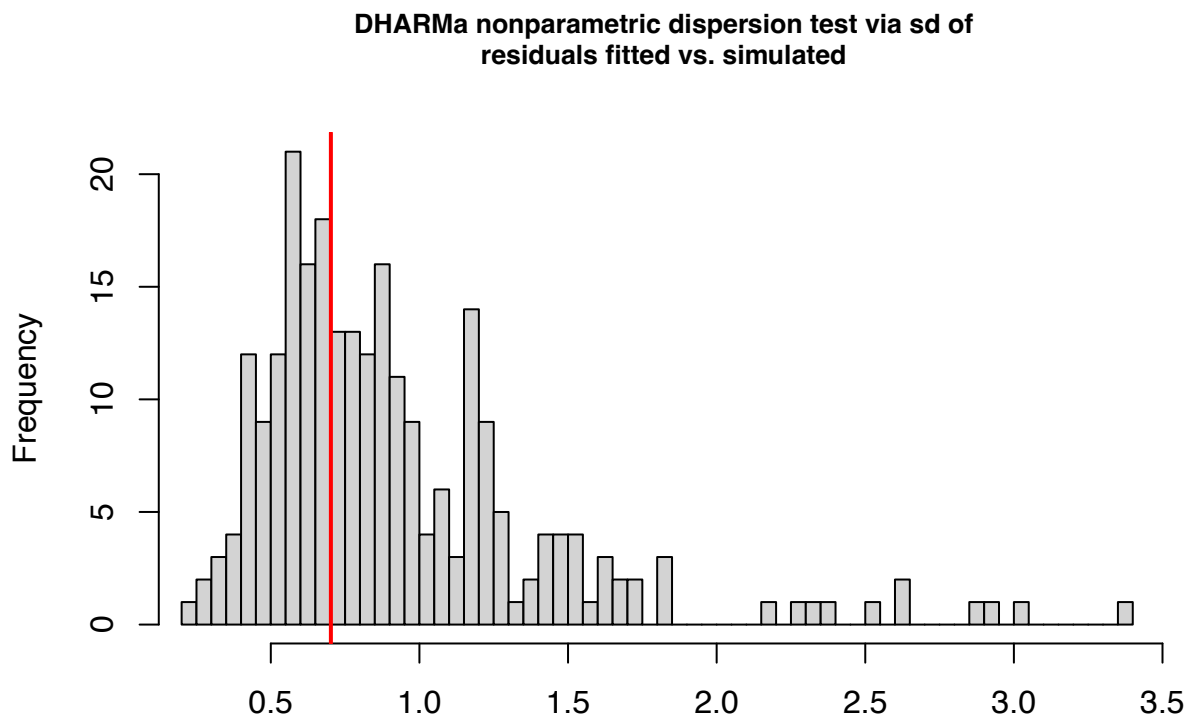
```
#tests if the overall distribution conforms to expectations.
testUniformity(m9)
```

QQ plot residuals



```
##
## Asymptotic one-sample Kolmogorov-Smirnov test
##
## data: simulationOutput$scaledResiduals
## D = 0.065028, p-value = 0.9323
## alternative hypothesis: two-sided
```

```
testDispersion(m9)
```

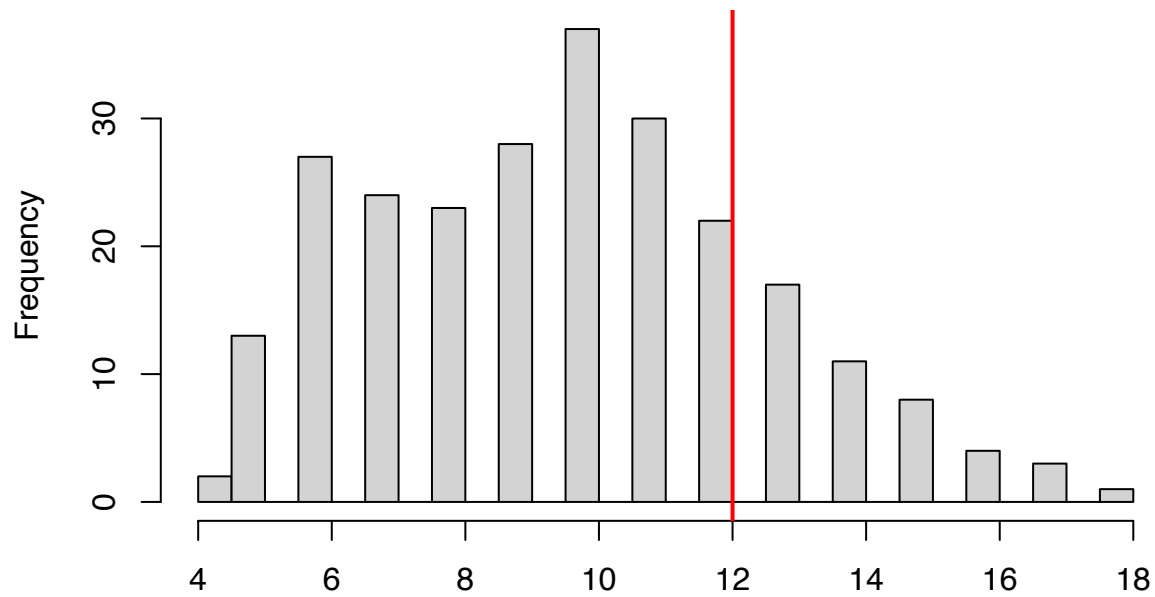


Simulated values, red line = fitted model. p-value (two.sided) = 0.784

```
##
## DHARMA nonparametric dispersion test via sd of residuals fitted vs.
## simulated
##
## data: simulationOutput
## dispersion = 0.75387, p-value = 0.784
## alternative hypothesis: two.sided
```

```
testZeroInflation(m9)
```

DHARMA zero-inflation test via comparison to expected zeros with simulation under H0 = fitted model



Simulated values, red line = fitted model. p-value (two.sided) = 0.528

```
##
## DHARMA zero-inflation test via comparison to expected zeros with
## simulation under H0 = fitted model
##
## data: simulationOutput
## ratioObsSim = 1.2341, p-value = 0.528
## alternative hypothesis: two.sided
```

```
testResiduals(m9)
```



```

## sample estimates:
## outlier frequency (expected: 0.00405797101449275 )
##                                0.01449275

## $uniformity
##
## Asymptotic one-sample Kolmogorov-Smirnov test
##
## data: simulationOutput$scaledResiduals
## D = 0.065028, p-value = 0.9323
## alternative hypothesis: two-sided
##
##
## $dispersion
##
## DHARMA nonparametric dispersion test via sd of residuals fitted vs.
## simulated
##
## data: simulationOutput
## dispersion = 0.75387, p-value = 0.784
## alternative hypothesis: two.sided
##
##
## $outliers
##
## DHARMA bootstrapped outlier test
##
## data: simulationOutput
## outliers at both margin(s) = 1, observations = 69, p-value = 0.48
## alternative hypothesis: two.sided
## percent confidence interval:
## 0.00000000 0.02898551
## sample estimates:
## outlier frequency (expected: 0.00405797101449275 )
##                                0.01449275

# calculate the residuals for the model
m10 <- simulateResiduals(fittedModel = interaction_direct_brumol, plot = F)

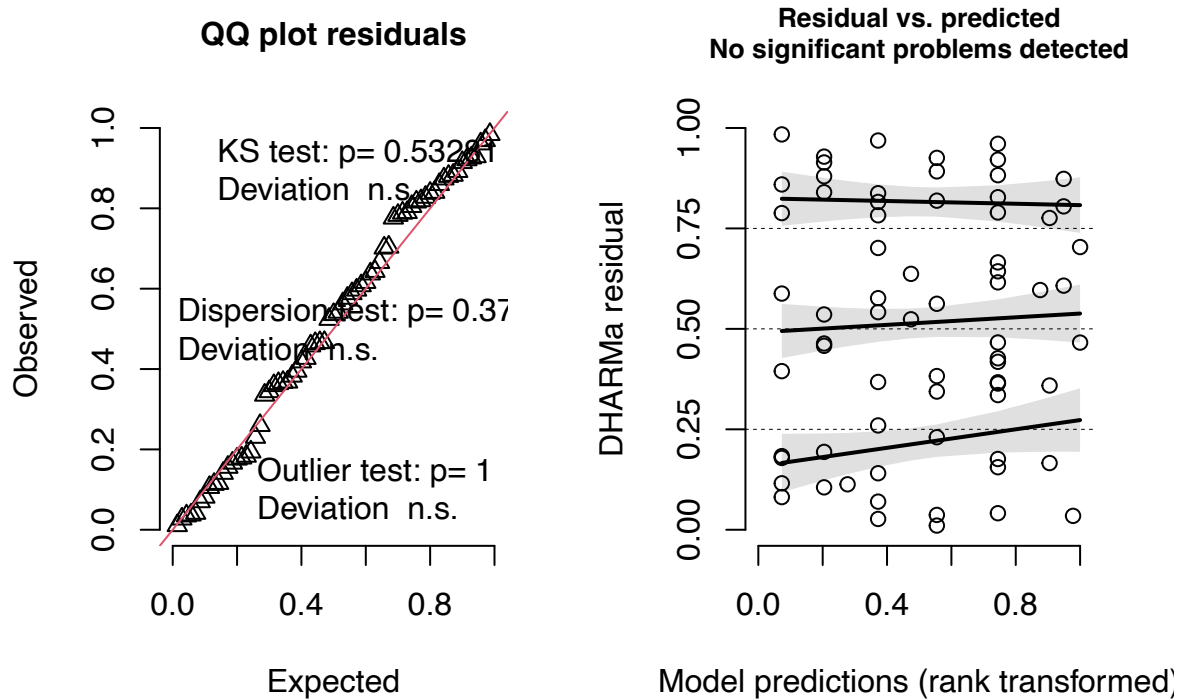
residuals(m10)

## [1] 0.39483232 0.11566192 0.03689696 0.01047204 0.33522734 0.04119060
## [7] 0.08131108 0.18325915 0.17613273 0.34395812 0.36396287 0.07000782
## [13] 0.11312055 0.19373707 0.03461264 0.14068616 0.38255308 0.80481221
## [19] 0.46612933 0.59665799 0.35899832 0.58789572 0.16603546 0.23030397
## [25] 0.36779550 0.81594524 0.45782850 0.91415483 0.53583818 0.85946108
## [31] 0.10552357 0.46357412 0.02689741 0.66547153 0.36698399 0.41800892
## [37] 0.25965035 0.17873939 0.61573649 0.52384506 0.42679632 0.63693772
## [43] 0.81914267 0.15567248 0.84000031 0.57637334 0.64320492 0.56244437
## [49] 0.96880315 0.78818049 0.78265525 0.54158705 0.87974650 0.82770687
## [55] 0.83763492 0.88249998 0.89200000 0.98400000 0.60755659 0.70135413
## [61] 0.92560825 0.92106085 0.78972390 0.92871553 0.46590494 0.70365093
## [67] 0.87335552 0.77600000 0.96054764

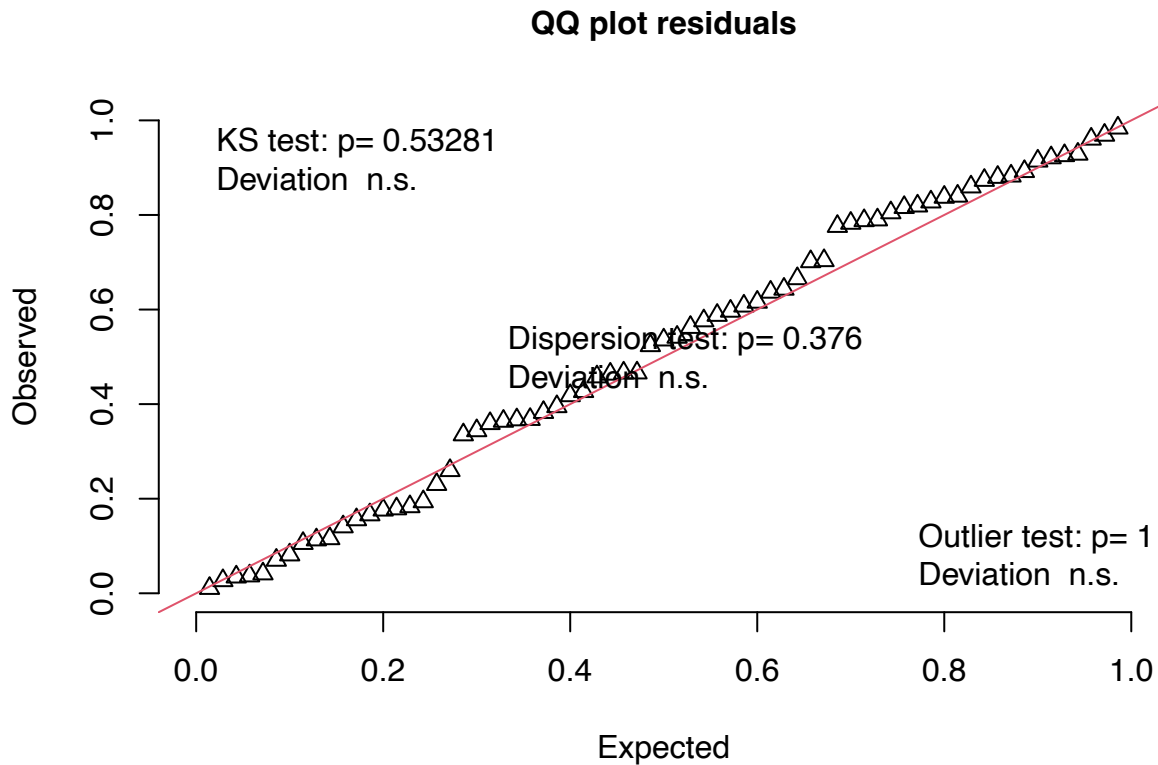
#plot simulation output
plot(m10)

```

DHARMA residual



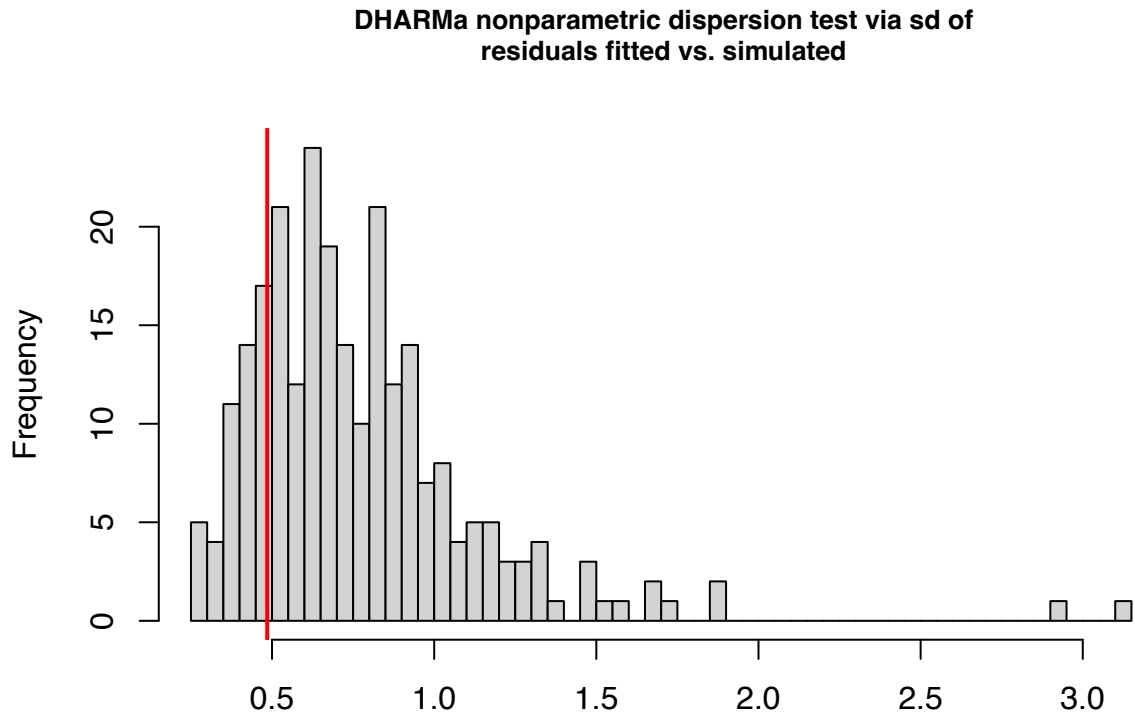
#tests if the overall distribution conforms to expectations.
`testUniformity(m10)`



 ## Exact one-sample Kolmogorov-Smirnov test

```
##
## data:  simulationOutput$scaledResiduals
## D = 0.094841, p-value = 0.5328
## alternative hypothesis: two-sided
```

```
testDispersion(m10)
```

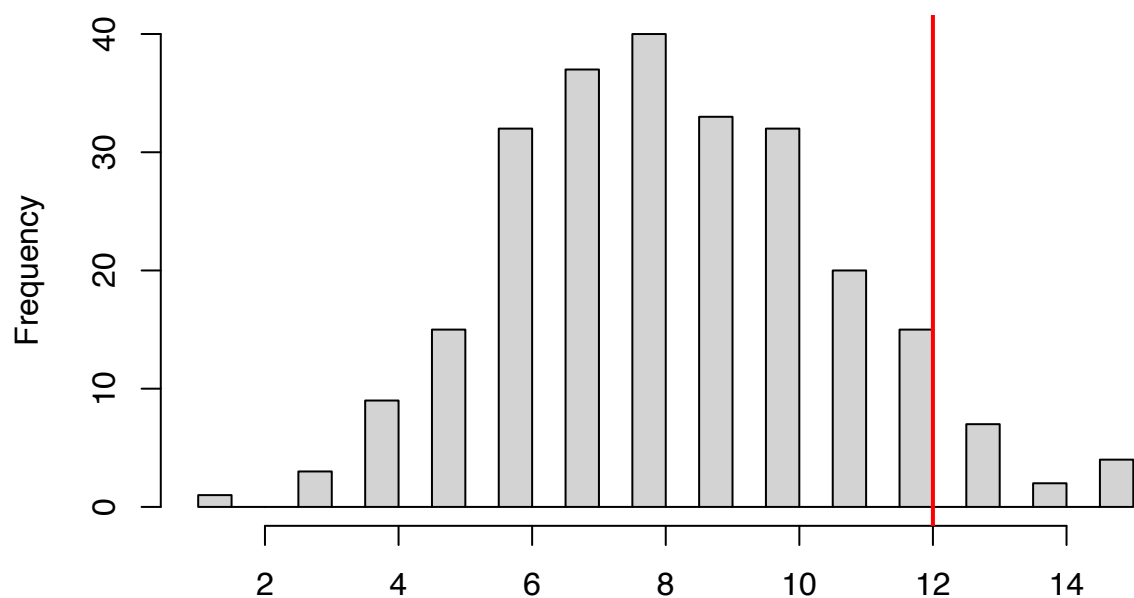


Simulated values, red line = fitted model. p-value (two.sided) = 0.376

```
##
## DHARMa nonparametric dispersion test via sd of residuals fitted vs.
## simulated
##
## data:  simulationOutput
## dispersion = 0.62877, p-value = 0.376
## alternative hypothesis: two.sided
```

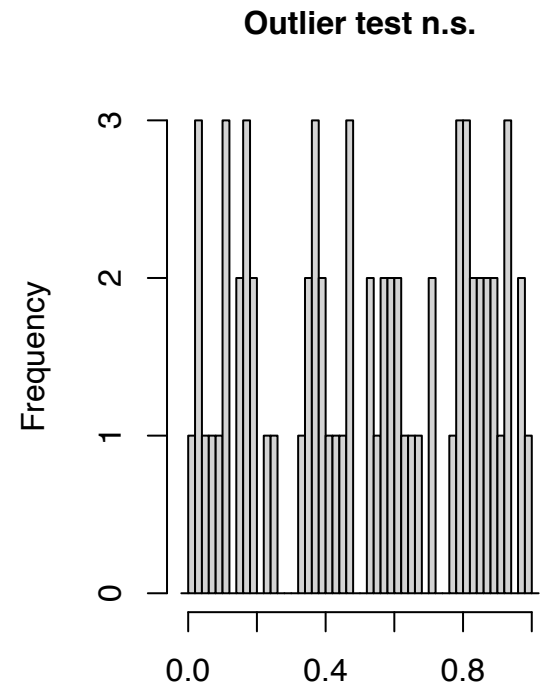
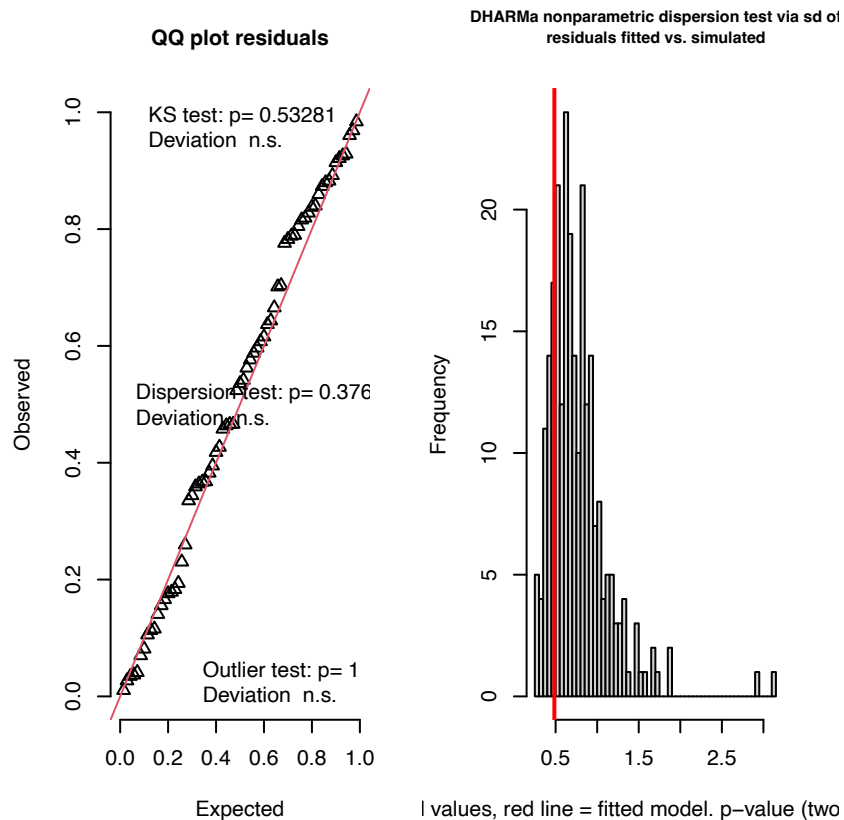
```
testZeroInflation(m10)
```

DHARMA zero-inflation test via comparison to expected zeros with simulation under H0 = fitted model



Simulated values, red line = fitted model. p-value (two.sided) = 0.224

```
##
## DHARMA zero-inflation test via comparison to expected zeros with
## simulation under H0 = fitted model
##
## data: simulationOutput
## ratioObsSim = 1.4368, p-value = 0.224
## alternative hypothesis: two.sided
testResiduals(m10)
```



Residuals (outliers are marked red)

```
## $uniformity
##
##   Exact one-sample Kolmogorov-Smirnov test
##
## data:  simulationOutput$scaledResiduals
## D = 0.094841, p-value = 0.5328
## alternative hypothesis: two-sided
##
##
## $dispersion
##
##   DHARMa nonparametric dispersion test via sd of residuals fitted vs.
##   simulated
##
## data:  simulationOutput
## dispersion = 0.62877, p-value = 0.376
## alternative hypothesis: two.sided
##
##
## $outliers
##
##   DHARMa bootstrapped outlier test
##
## data:  simulationOutput
## outliers at both margin(s) = 0, observations = 69, p-value = 1
## alternative hypothesis: two.sided
##   percent confidence interval:
##   0.00000000 0.02210145
```

```

## sample estimates:
## outlier frequency (expected: 0.00550724637681159 )
##
## $uniformity
##
## Exact one-sample Kolmogorov-Smirnov test
##
## data: simulationOutput$scaledResiduals
## D = 0.094841, p-value = 0.5328
## alternative hypothesis: two-sided
##
##
## $dispersion
##
## DHARMA nonparametric dispersion test via sd of residuals fitted vs.
## simulated
##
## data: simulationOutput
## dispersion = 0.62877, p-value = 0.376
## alternative hypothesis: two.sided
##
##
## $outliers
##
## DHARMA bootstrapped outlier test
##
## data: simulationOutput
## outliers at both margin(s) = 0, observations = 69, p-value = 1
## alternative hypothesis: two.sided
## percent confidence interval:
## 0.00000000 0.02210145
## sample estimates:
## outlier frequency (expected: 0.00550724637681159 )
##
##
# calculate the residuals for the model
m21 <- simulateResiduals(fittedModel = additive_harass_brumol, plot = F)

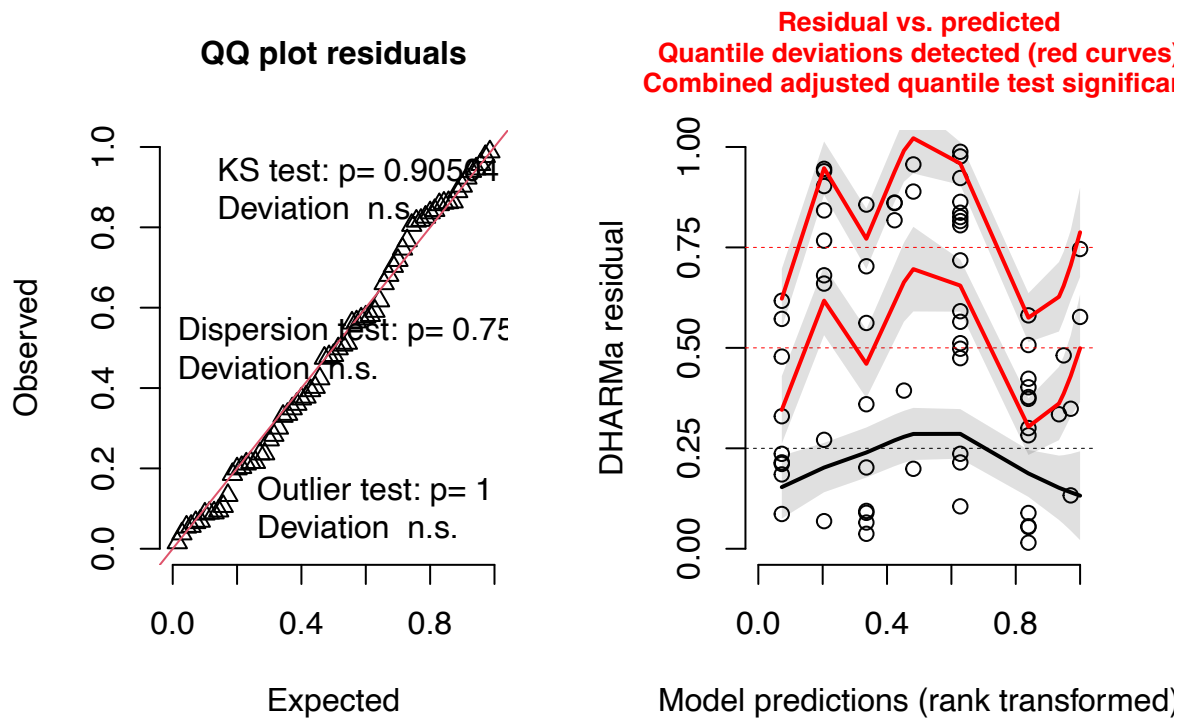
residuals(m21)

## [1] 0.32984950 0.23625109 0.27108364 0.06871264 0.51167623 0.23637763
## [7] 0.61715774 0.21496792 0.49780055 0.76679842 0.71774999 0.28252443
## [13] 0.39368454 0.06527914 0.33471768 0.01515651 0.68059248 0.95659888
## [19] 0.56483292 0.48130022 0.81759076 0.18507827 0.86033758 0.66001317
## [25] 0.05596276 0.40189474 0.35970074 0.85660571 0.09423629 0.47823583
## [31] 0.03726733 0.08944538 0.08905087 0.83635821 0.10554960 0.47440645
## [37] 0.37303767 0.21110312 0.86337619 0.34844229 0.59120872 0.13284112
## [43] 0.94559402 0.21479881 0.56197538 0.30044549 0.82549979 0.90264666
## [49] 0.37760860 0.57208729 0.50702218 0.42314498 0.20233143 0.92272519
## [55] 0.58088647 0.81556598 0.93952778 0.08653569 0.19900308 0.05405670
## [61] 0.84208331 0.97645122 0.80513487 0.70302284 0.57697505 0.74603033
## [67] 0.88901974 0.86164830 0.98800000

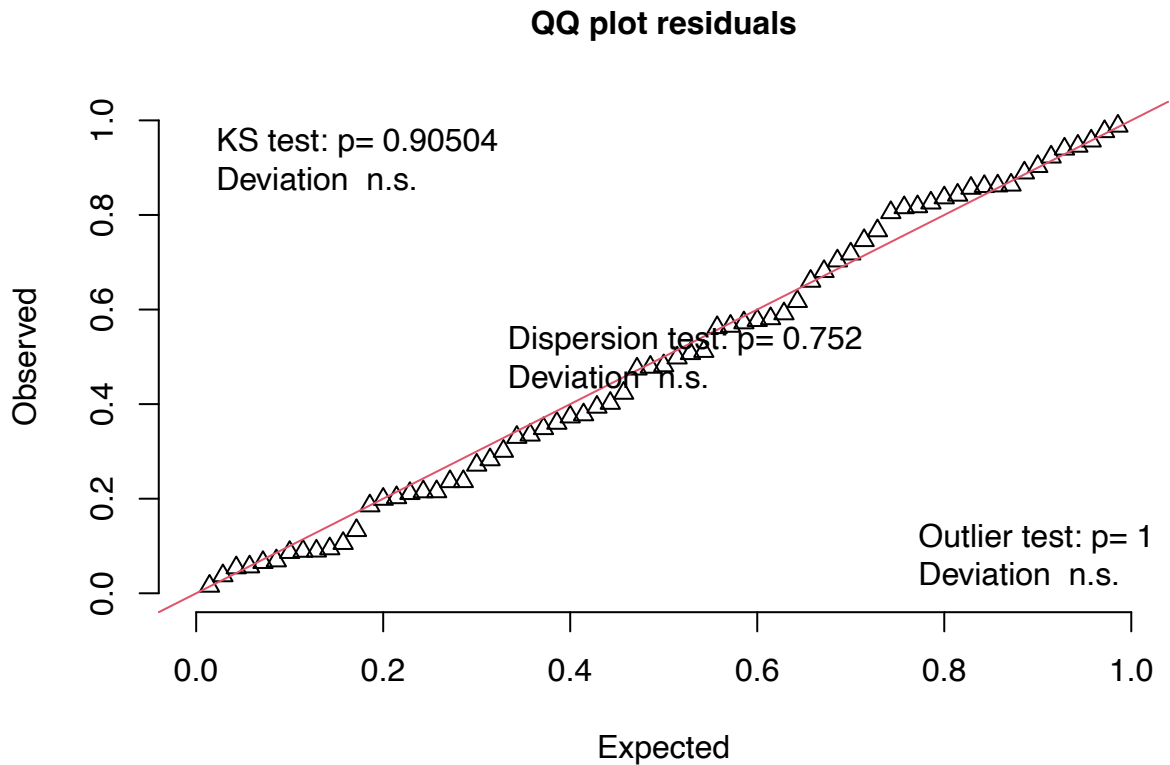
#plot simulation output
plot(m21)

```

DHARMA residual



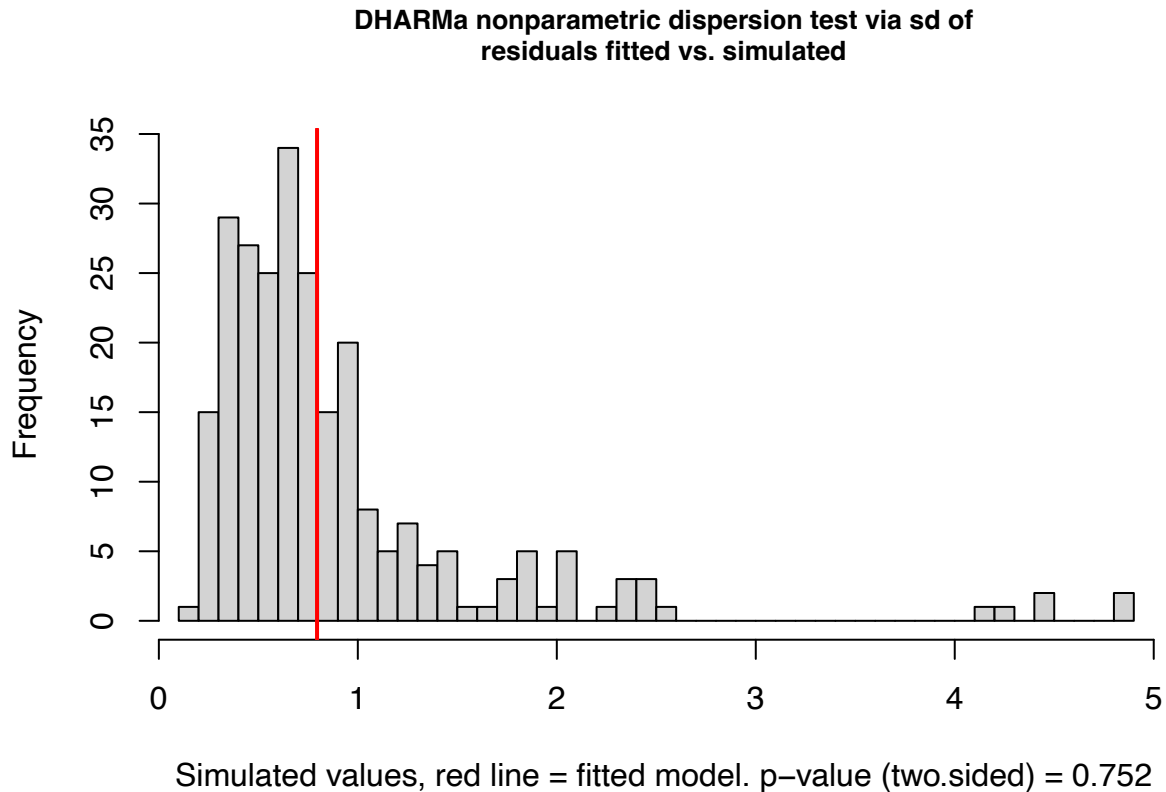
#tests if the overall distribution conforms to expectations.
`testUniformity(m21)`



 ## Exact one-sample Kolmogorov-Smirnov test

```
##
## data: simulationOutput$scaledResiduals
## D = 0.066004, p-value = 0.905
## alternative hypothesis: two-sided
```

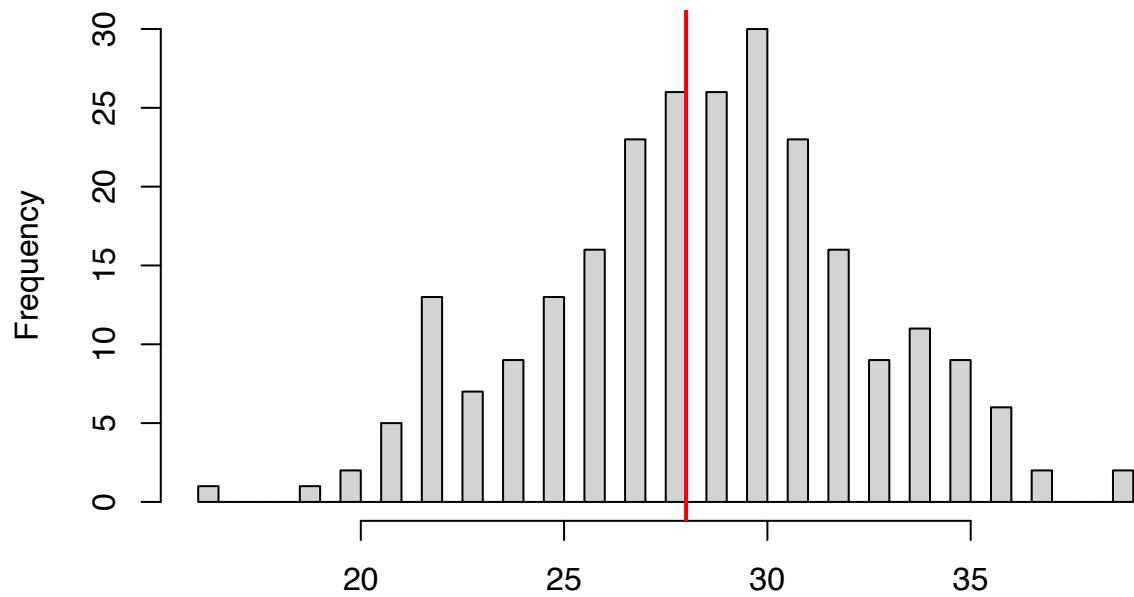
```
testDispersion(m21)
```



```
##
## DHARMa nonparametric dispersion test via sd of residuals fitted vs.
## simulated
##
## data: simulationOutput
## dispersion = 0.88607, p-value = 0.752
## alternative hypothesis: two.sided
```

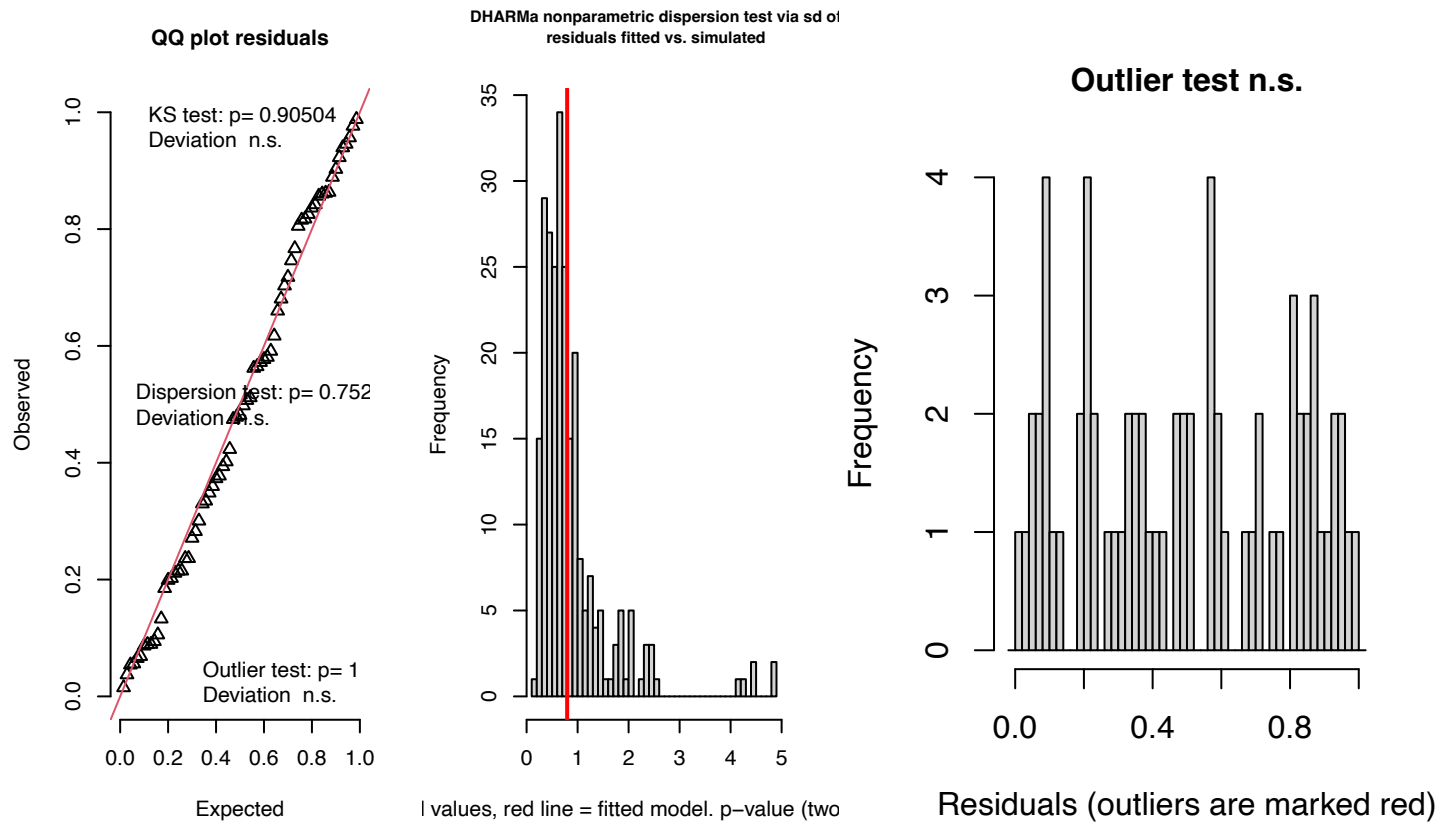
```
testZeroInflation(m21)
```

DHARMA zero-inflation test via comparison to expected zeros with simulation under H0 = fitted model



Simulated values, red line = fitted model. p-value (two.sided) = 0.928

```
##
## DHARMA zero-inflation test via comparison to expected zeros with
## simulation under H0 = fitted model
##
## data: simulationOutput
## ratioObsSim = 0.97684, p-value = 0.928
## alternative hypothesis: two.sided
testResiduals(m21)
```



```
## $uniformity
##
## Exact one-sample Kolmogorov-Smirnov test
##
## data: simulationOutput$scaledResiduals
## D = 0.066004, p-value = 0.905
## alternative hypothesis: two-sided
##
##
## $dispersion
##
## DHARMA nonparametric dispersion test via sd of residuals fitted vs.
## simulated
##
## data: simulationOutput
## dispersion = 0.88607, p-value = 0.752
## alternative hypothesis: two.sided
##
##
## $outliers
##
## DHARMA bootstrapped outlier test
##
## data: simulationOutput
## outliers at both margin(s) = 0, observations = 69, p-value = 1
## alternative hypothesis: two.sided
## percent confidence interval:
## 0.00000000 0.02898551
```

```

## sample estimates:
## outlier frequency (expected: 0.00594202898550725 )
##
## $uniformity
##
## Exact one-sample Kolmogorov-Smirnov test
##
## data: simulationOutput$scaledResiduals
## D = 0.066004, p-value = 0.905
## alternative hypothesis: two-sided
##
##
## $dispersion
##
## DHARMA nonparametric dispersion test via sd of residuals fitted vs.
## simulated
##
## data: simulationOutput
## dispersion = 0.88607, p-value = 0.752
## alternative hypothesis: two.sided
##
##
## $outliers
##
## DHARMA bootstrapped outlier test
##
## data: simulationOutput
## outliers at both margin(s) = 0, observations = 69, p-value = 1
## alternative hypothesis: two.sided
## percent confidence interval:
## 0.00000000 0.02898551
## sample estimates:
## outlier frequency (expected: 0.00594202898550725 )
##
0

# calculate the residuals for the model
m22 <- simulateResiduals(fittedModel = interaction_harass_brumol, plot = F)

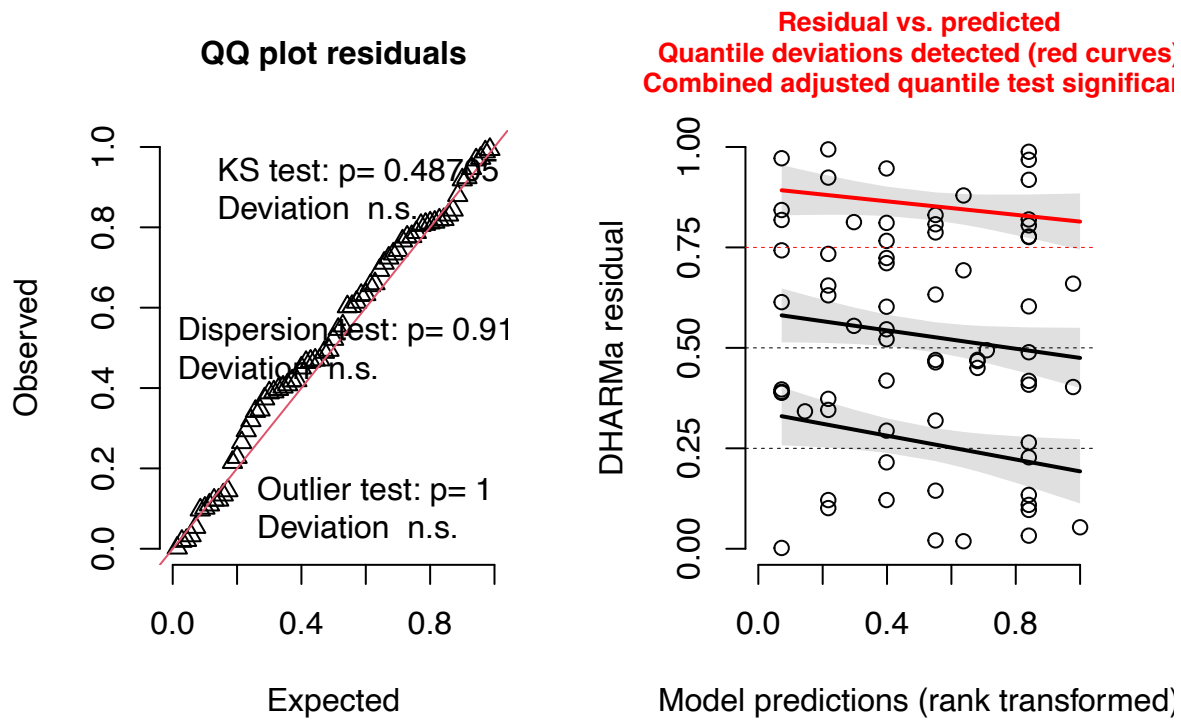
residuals(m22)

## [1] 0.390601127 0.388160677 0.144602728 0.020835556 0.264349755 0.096658062
## [7] 0.742715105 0.613829274 0.227618018 0.463862820 0.489140827 0.521445887
## [13] 0.342013392 0.630742474 0.053465283 0.546217792 0.469773706 0.879283562
## [19] 0.408318939 0.494119364 0.450068022 0.001990125 0.470210989 0.319048834
## [25] 0.418525678 0.766253657 0.121187210 0.993992152 0.655128070 0.818338168
## [31] 0.345571313 0.101186417 0.215066501 0.819568627 0.032638299 0.134033114
## [37] 0.602439456 0.396620037 0.775751607 0.812921655 0.417809032 0.554771594
## [43] 0.830711517 0.109169445 0.733869108 0.293952793 0.777094700 0.787352403
## [49] 0.723690186 0.971952143 0.710829103 0.811051285 0.373088242 0.918293328
## [55] 0.946554730 0.804989528 0.807636906 0.843076080 0.018679263 0.121222110
## [61] 0.632760957 0.967982203 0.603301633 0.923762723 0.402573038 0.659847382
## [67] 0.693015631 0.466959014 0.988000000

#plot simulation output
plot(m22)

```

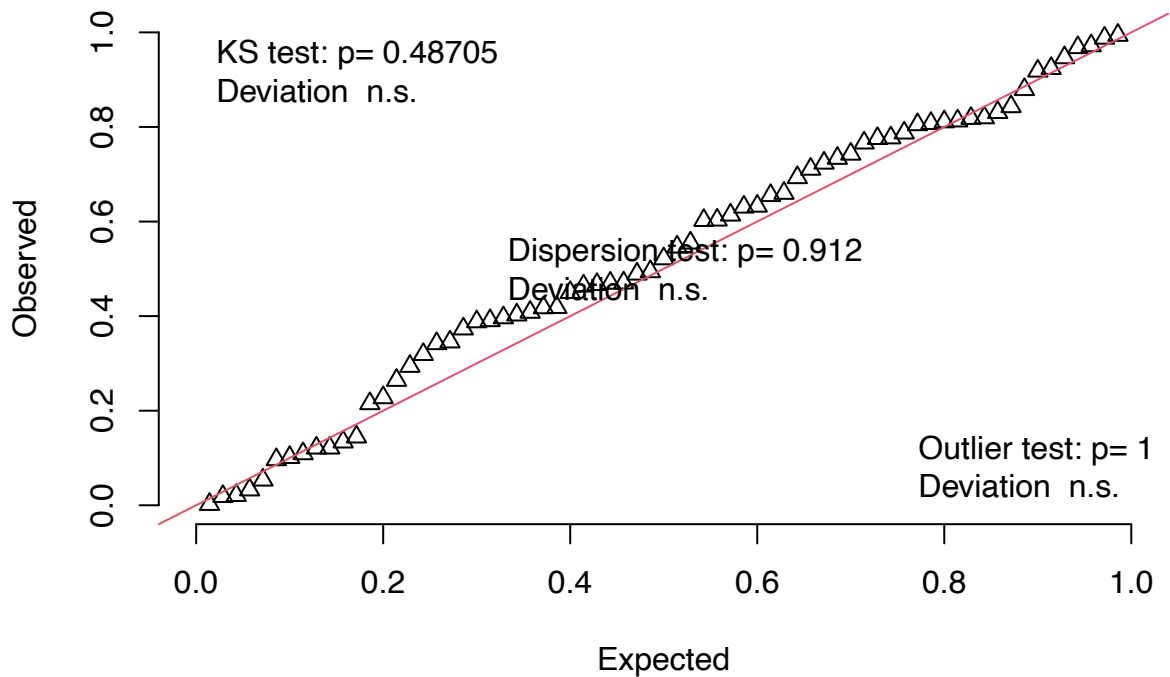
DHARMa residual



#tests if the overall distribution conforms to expectations.

`testUniformity(m22)`

QQ plot residuals

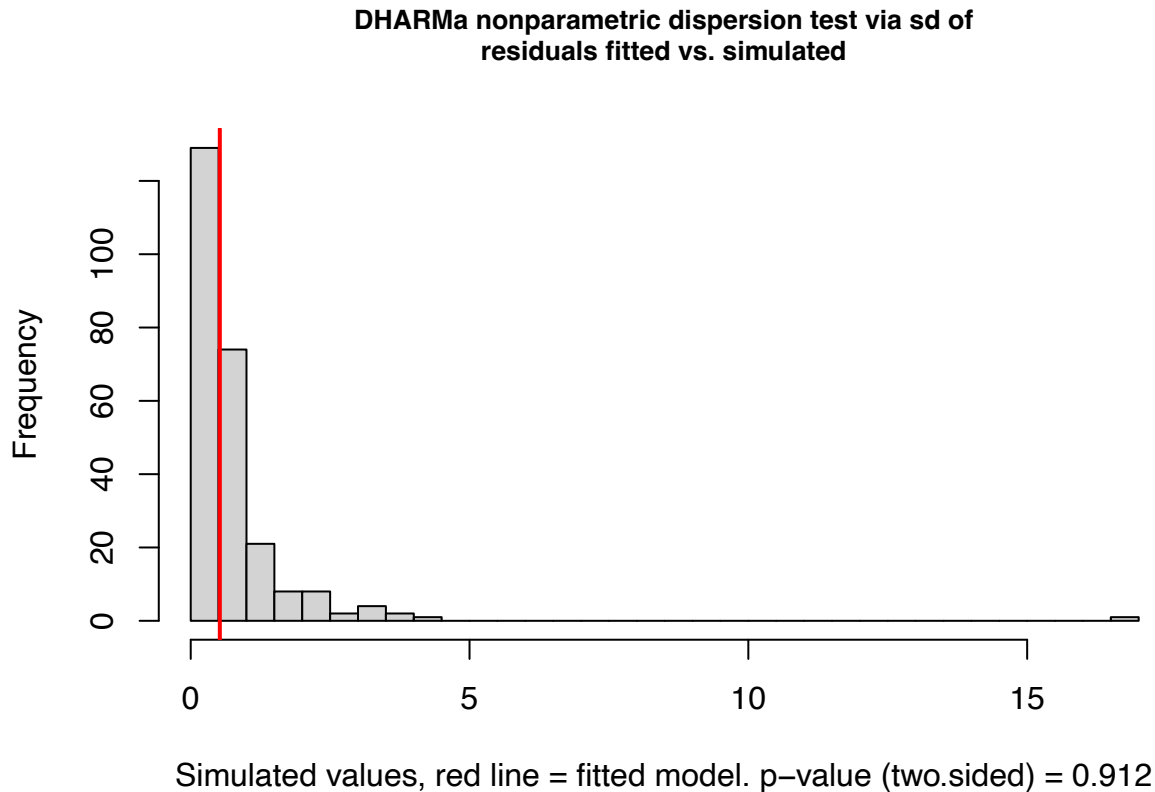


##

Exact one-sample Kolmogorov-Smirnov test

```
##
## data: simulationOutput$scaledResiduals
## D = 0.098306, p-value = 0.487
## alternative hypothesis: two-sided
```

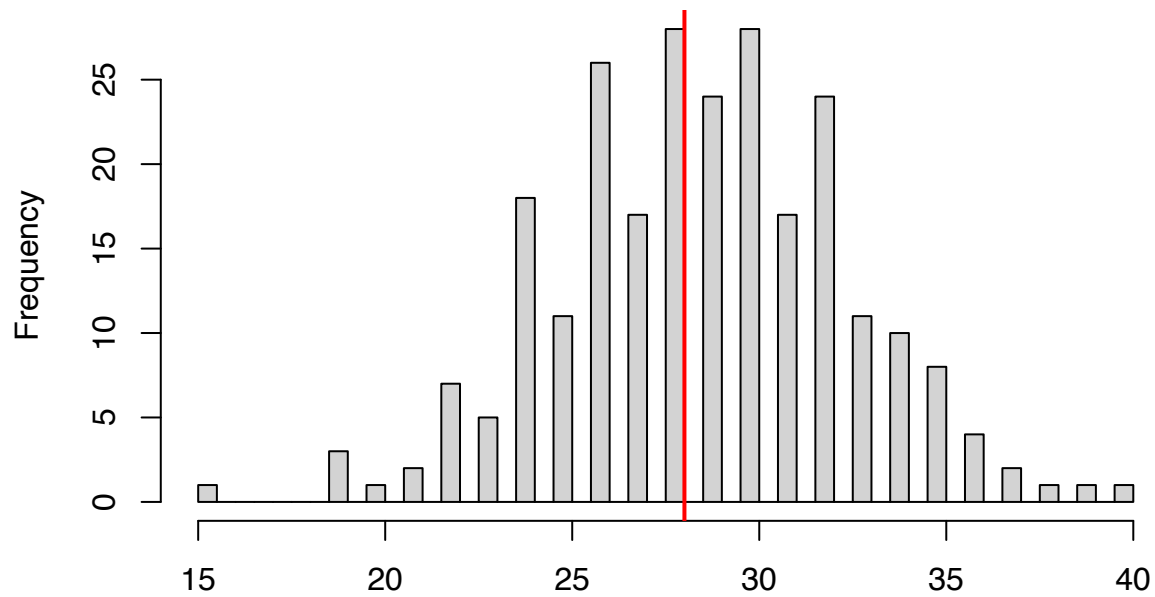
```
testDispersion(m22)
```



```
##
## DHARMa nonparametric dispersion test via sd of residuals fitted vs.
## simulated
##
## data: simulationOutput
## dispersion = 0.66919, p-value = 0.912
## alternative hypothesis: two.sided
```

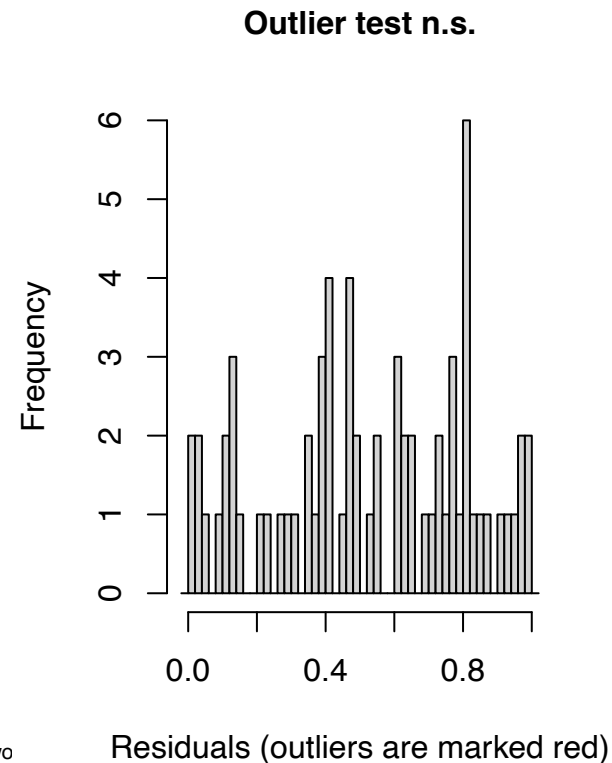
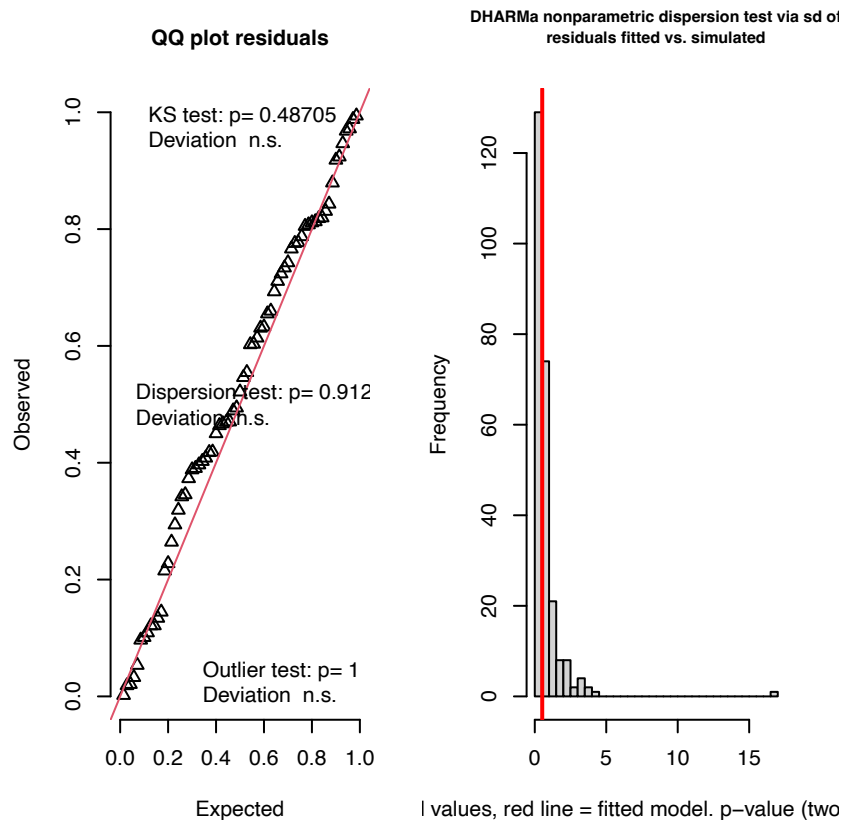
```
testZeroInflation(m22)
```

DHARMA zero-inflation test via comparison to expected zeros with simulation under H0 = fitted model



Simulated values, red line = fitted model. p-value (two.sided) = 0.952

```
##
## DHARMA zero-inflation test via comparison to expected zeros with
## simulation under H0 = fitted model
##
## data: simulationOutput
## ratioObsSim = 0.9752, p-value = 0.952
## alternative hypothesis: two.sided
testResiduals(m22)
```



```
## $uniformity
##
## Exact one-sample Kolmogorov-Smirnov test
##
## data: simulationOutput$scaledResiduals
## D = 0.098306, p-value = 0.487
## alternative hypothesis: two-sided
##
##
## $dispersion
##
## DHARMa nonparametric dispersion test via sd of residuals fitted vs.
## simulated
##
## data: simulationOutput
## dispersion = 0.66919, p-value = 0.912
## alternative hypothesis: two.sided
##
##
## $outliers
##
## DHARMa bootstrapped outlier test
##
## data: simulationOutput
## outliers at both margin(s) = 0, observations = 69, p-value = 1
## alternative hypothesis: two.sided
## percent confidence interval:
## 0.00000000 0.02898551
```

```

## sample estimates:
## outlier frequency (expected: 0.00550724637681159 )
##
## $uniformity
##
## Exact one-sample Kolmogorov-Smirnov test
##
## data: simulationOutput$scaledResiduals
## D = 0.098306, p-value = 0.487
## alternative hypothesis: two-sided
##
##
## $dispersion
##
## DHARMA nonparametric dispersion test via sd of residuals fitted vs.
## simulated
##
## data: simulationOutput
## dispersion = 0.66919, p-value = 0.912
## alternative hypothesis: two.sided
##
##
## $outliers
##
## DHARMA bootstrapped outlier test
##
## data: simulationOutput
## outliers at both margin(s) = 0, observations = 69, p-value = 1
## alternative hypothesis: two.sided
## percent confidence interval:
## 0.00000000 0.02898551
## sample estimates:
## outlier frequency (expected: 0.00550724637681159 )
##
##
# calculate the residuals for the model
m15 <- simulateResiduals(fittedModel = additive_resp_brumol, plot = F)

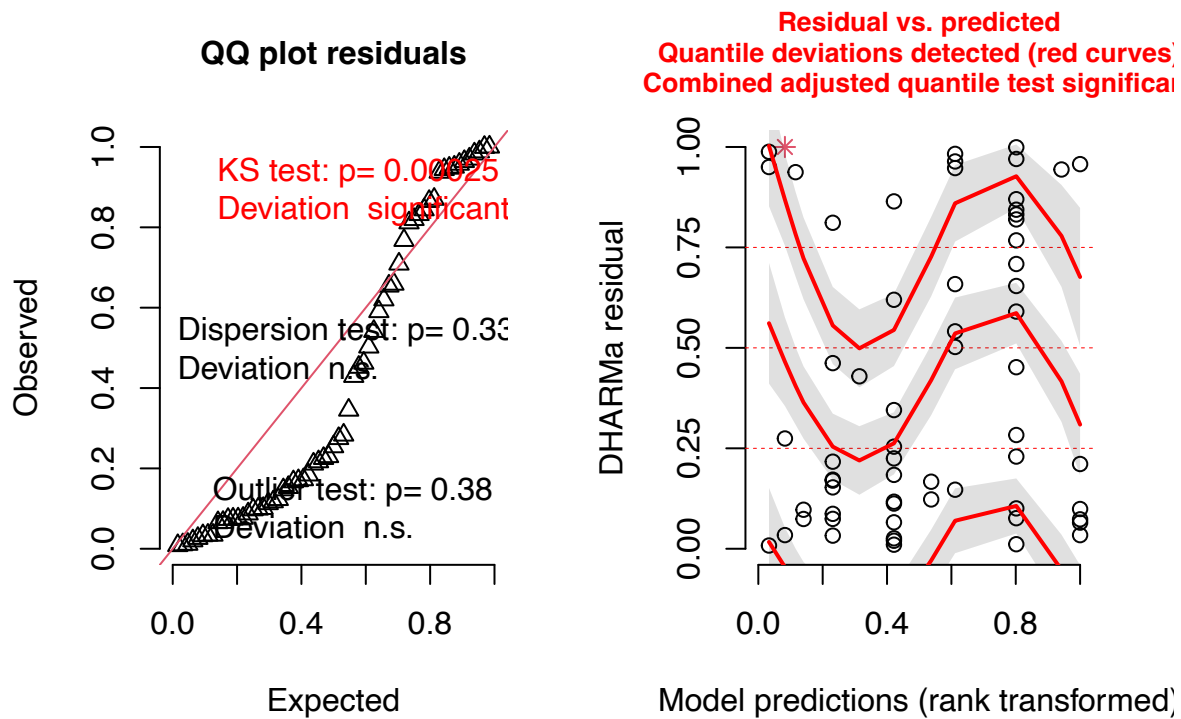
residuals(m15)

## [1] 0.957313424 0.999190180 0.870010933 0.982811866 0.970091437 0.619645326
## [7] 0.429192539 0.152757239 0.937294552 0.254085595 0.947496573 1.000000000
## [13] 0.819665328 0.943993026 0.988000000 0.210949583 0.950062796 0.658714899
## [19] 0.345172642 0.864556348 0.086904320 0.074389786 0.462060196 0.065091793
## [25] 0.032690500 0.216731430 0.183881329 0.843885227 0.100451407 0.767527834
## [31] 0.010109730 0.073221747 0.590364597 0.122770498 0.229500616 0.166857016
## [37] 0.963959311 0.076064096 0.172009579 0.224906766 0.283083174 0.541178443
## [43] 0.019951233 0.098909816 0.117070065 0.025997330 0.169595915 0.451897178
## [49] 0.112479630 0.708744528 0.146963057 0.033949458 0.274484037 0.065791931
## [55] 0.502272533 0.832052117 0.653998541 0.811318274 0.097118454 0.074077231
## [61] 0.034004596 0.008066622 0.011337570

#plot simulation output
plot(m15)

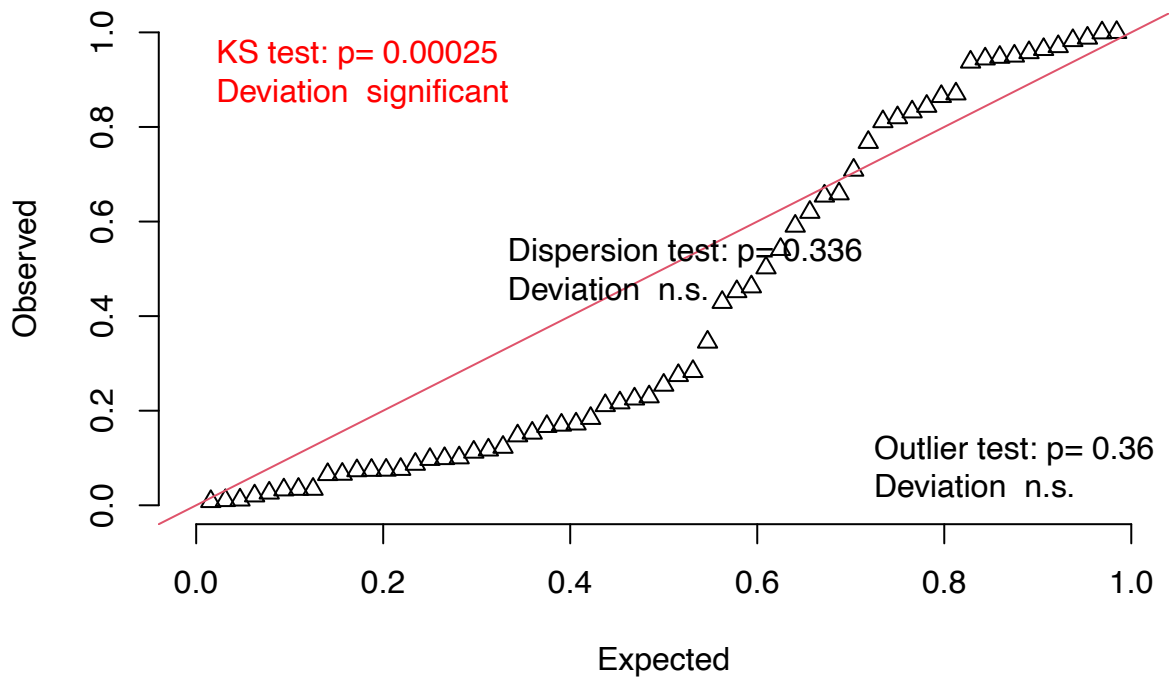
```

DHARMa residual



#tests if the overall distribution conforms to expectations.
`testUniformity(m15)`

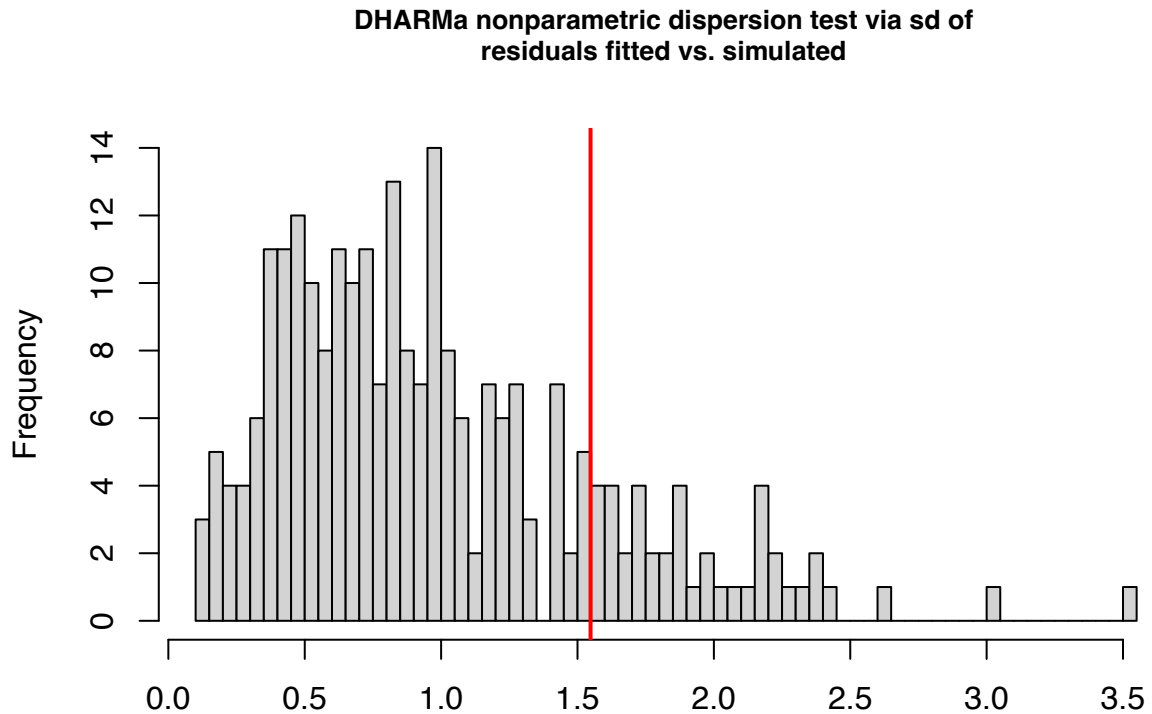
QQ plot residuals



 ## Exact one-sample Kolmogorov-Smirnov test

```
##
## data:  simulationOutput$scaledResiduals
## D = 0.26256, p-value = 0.000254
## alternative hypothesis: two-sided
```

```
testDispersion(m15)
```

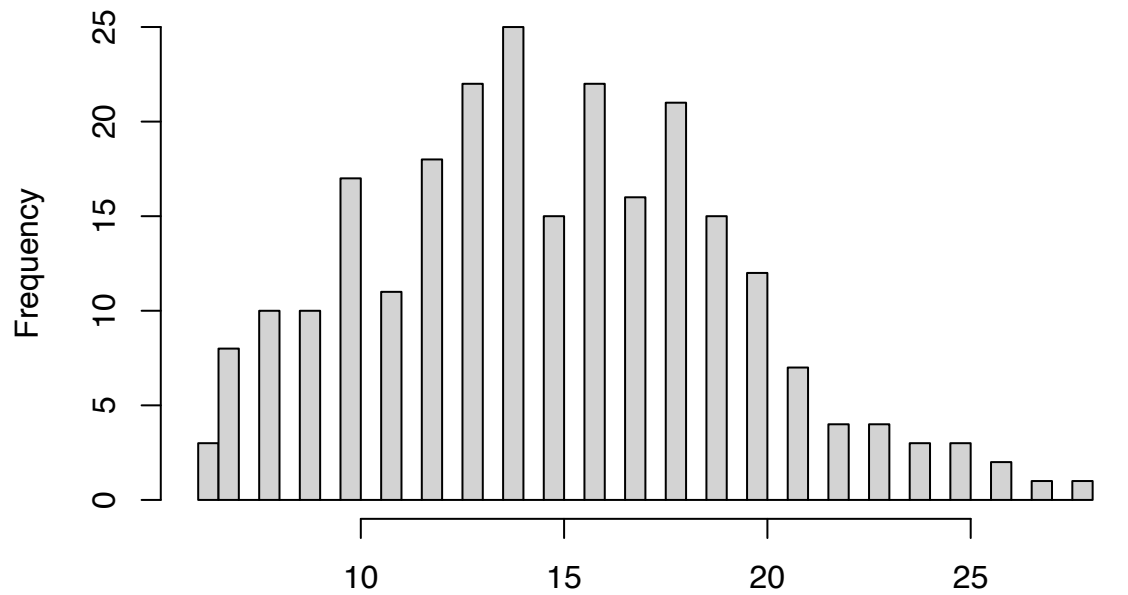


Simulated values, red line = fitted model. p-value (two.sided) = 0.336

```
##
## DHARMa nonparametric dispersion test via sd of residuals fitted vs.
## simulated
##
## data:  simulationOutput
## dispersion = 1.5839, p-value = 0.336
## alternative hypothesis: two.sided
```

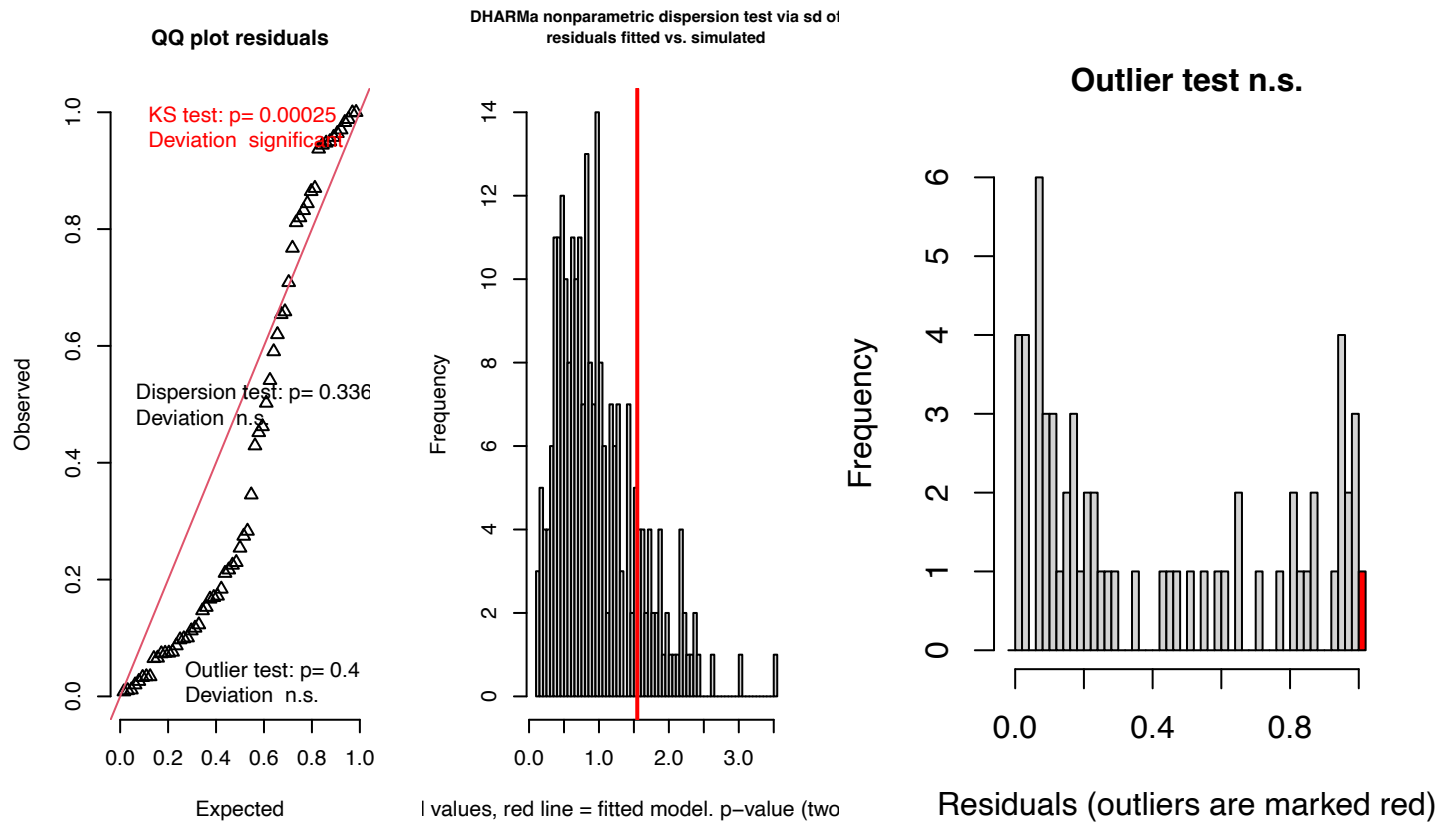
```
testZeroInflation(m15)
```

DHARMA zero-inflation test via comparison to expected zeros with simulation under H0 = fitted model



Simulated values, red line = fitted model. p-value (two.sided) = 0

```
##
## DHARMA zero-inflation test via comparison to expected zeros with
## simulation under H0 = fitted model
##
## data: simulationOutput
## ratioObsSim = 1.9489, p-value < 2.2e-16
## alternative hypothesis: two.sided
testResiduals(m15)
```



```
## $uniformity
##
## Exact one-sample Kolmogorov-Smirnov test
##
## data: simulationOutput$scaledResiduals
## D = 0.26256, p-value = 0.000254
## alternative hypothesis: two-sided
##
##
## $dispersion
##
## DHARMA nonparametric dispersion test via sd of residuals fitted vs.
## simulated
##
## data: simulationOutput
## dispersion = 1.5839, p-value = 0.336
## alternative hypothesis: two.sided
##
##
## $outliers
##
## DHARMA bootstrapped outlier test
##
## data: simulationOutput
## outliers at both margin(s) = 1, observations = 63, p-value = 0.38
## alternative hypothesis: two.sided
## percent confidence interval:
## 0.00000000 0.04007937
```

```

## sample estimates:
## outlier frequency (expected: 0.0046031746031746 )
##                                0.01587302

## $uniformity
##
## Exact one-sample Kolmogorov-Smirnov test
##
## data: simulationOutput$scaledResiduals
## D = 0.26256, p-value = 0.000254
## alternative hypothesis: two-sided
##
##
## $dispersion
##
## DHARMA nonparametric dispersion test via sd of residuals fitted vs.
## simulated
##
## data: simulationOutput
## dispersion = 1.5839, p-value = 0.336
## alternative hypothesis: two.sided
##
##
## $outliers
##
## DHARMA bootstrapped outlier test
##
## data: simulationOutput
## outliers at both margin(s) = 1, observations = 63, p-value = 0.38
## alternative hypothesis: two.sided
## percent confidence interval:
## 0.00000000 0.04007937
## sample estimates:
## outlier frequency (expected: 0.0046031746031746 )
##                                0.01587302

# calculate the residuals for the model
m16 <- simulateResiduals(fittedModel = interaction_resp_brumol, plot = F)

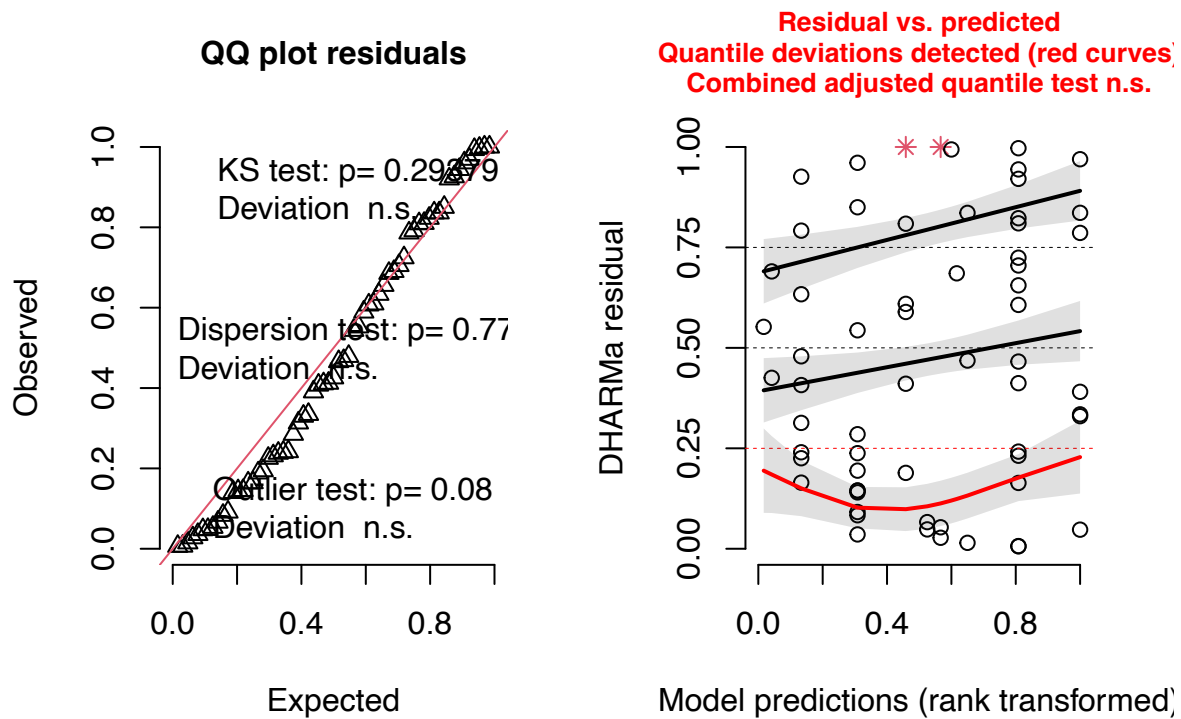
residuals(m16)

## [1] 1.000000000 0.996953094 0.920652450 0.969495709 0.944138978 0.849989789
## [7] 0.552331772 0.478970558 0.685529858 0.144519531 0.786151099 1.000000000
## [13] 0.822675140 0.993784148 0.835950898 0.809034792 0.468451039 0.390619738
## [19] 0.091461910 0.960622488 0.633447889 0.313307626 0.791773470 0.589332849
## [25] 0.240168906 0.164542728 0.145167738 0.809962063 0.163823314 0.705072577
## [31] 0.083382200 0.410717245 0.465907468 0.690301677 0.241800110 0.425448202
## [37] 0.836089768 0.006097805 0.407427405 0.543739392 0.231548191 0.330011984
## [43] 0.237899259 0.609481265 0.285086057 0.035649081 0.225264139 0.412149647
## [49] 0.140237052 0.655673224 0.047568247 0.188846099 0.027399383 0.194400242
## [55] 0.334521163 0.724404086 0.606893764 0.926031213 0.048156555 0.066326396
## [61] 0.053479779 0.014789164 0.006665046

#plot simulation output
plot(m16)

```

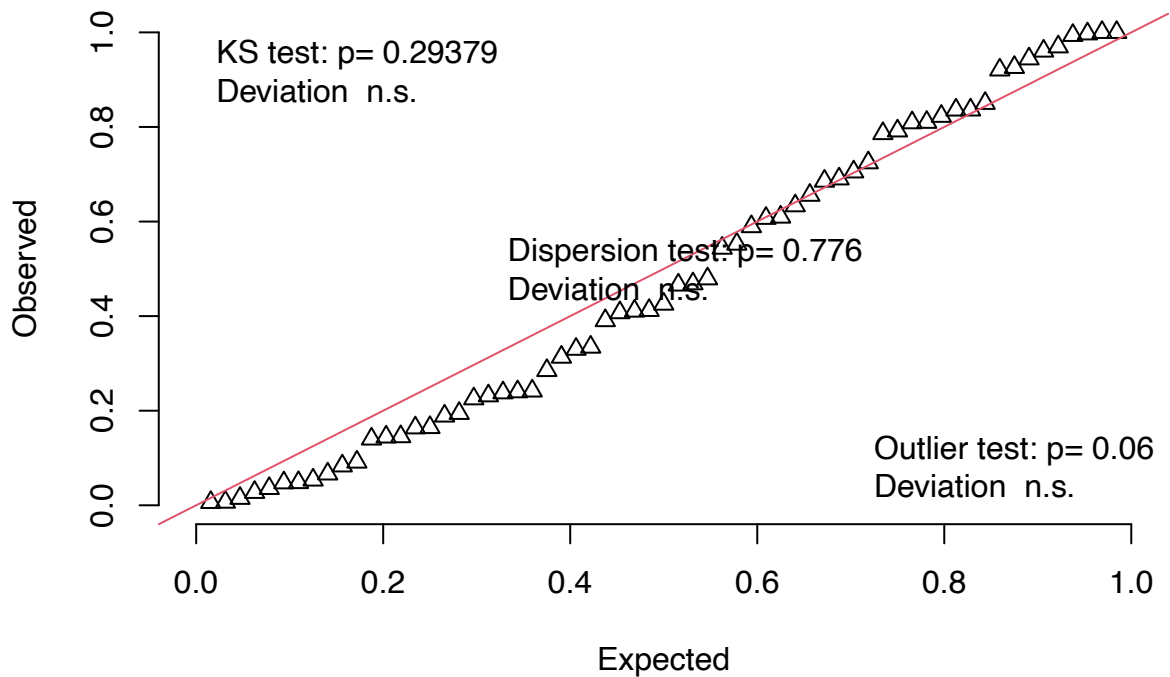
DHARMA residual



#tests if the overall distribution conforms to expectations.

`testUniformity(m16)`

QQ plot residuals

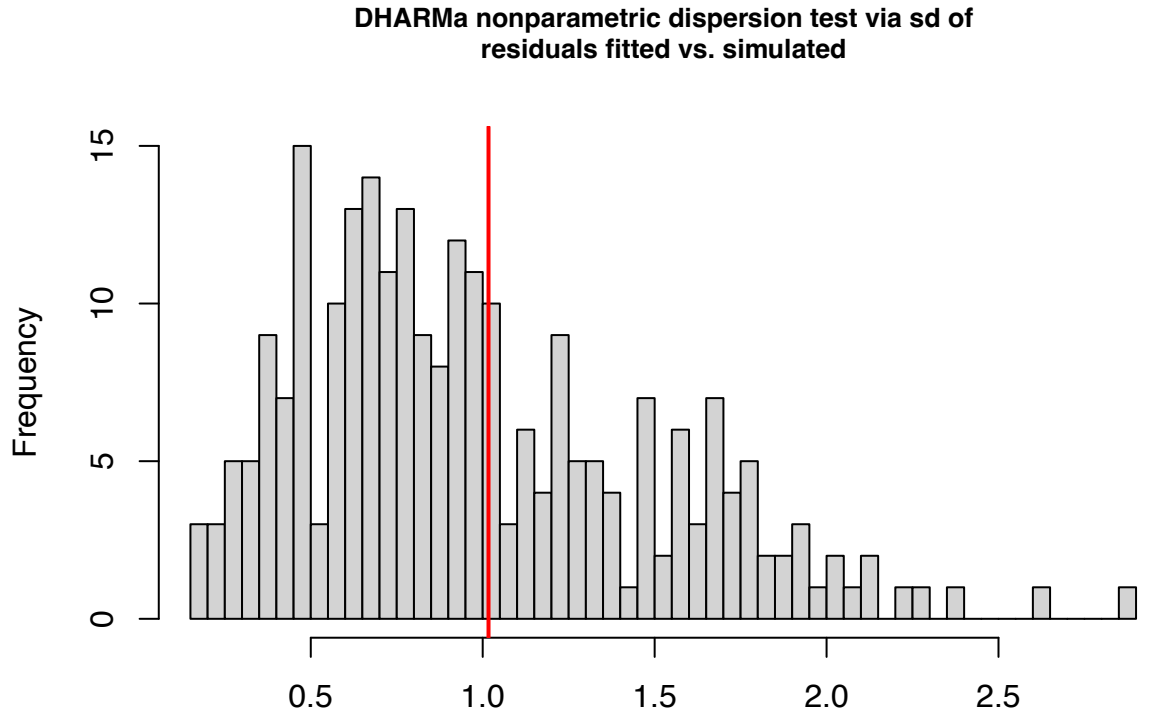


##

Asymptotic one-sample Kolmogorov-Smirnov test

```
##
## data: simulationOutput$scaledResiduals
## D = 0.12328, p-value = 0.2938
## alternative hypothesis: two-sided
```

```
testDispersion(m16)
```

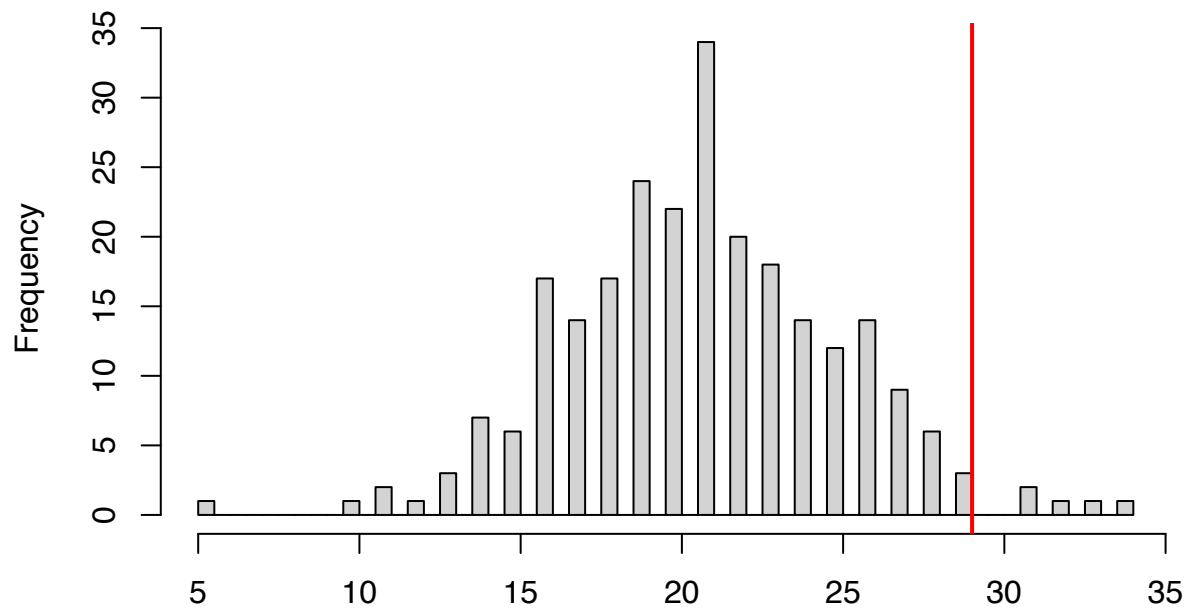


Simulated values, red line = fitted model. p-value (two.sided) = 0.776

```
##
## DHARMa nonparametric dispersion test via sd of residuals fitted vs.
## simulated
##
## data: simulationOutput
## dispersion = 1.0334, p-value = 0.776
## alternative hypothesis: two.sided
```

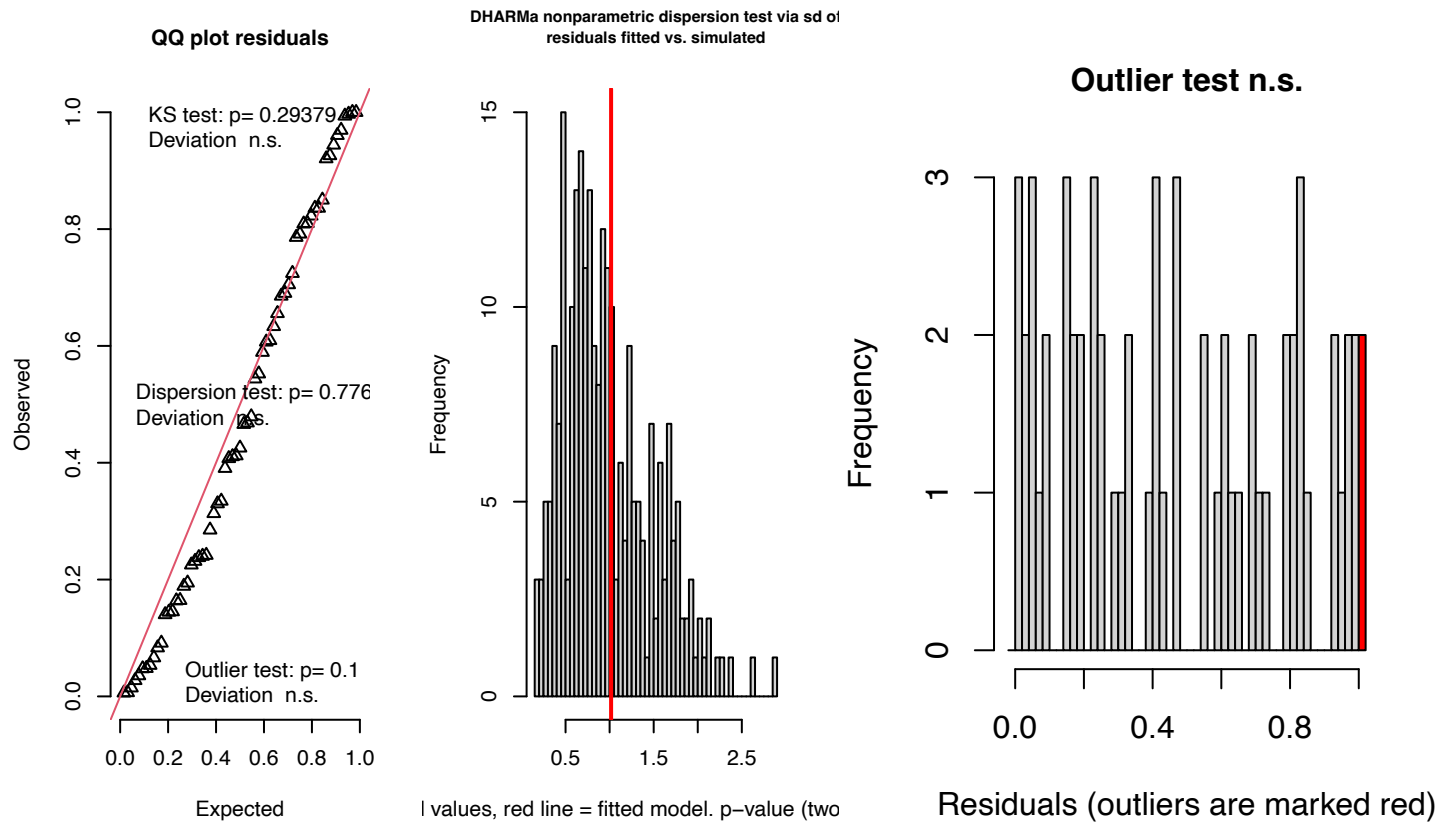
```
testZeroInflation(m16)
```

**DHARMA zero-inflation test via comparison to
expected zeros with simulation under H0 = fitted
model**



Simulated values, red line = fitted model. p-value (two.sided) = 0.064

```
##
## DHARMA zero-inflation test via comparison to expected zeros with
## simulation under H0 = fitted model
##
## data: simulationOutput
## ratioObsSim = 1.3902, p-value = 0.064
## alternative hypothesis: two.sided
testResiduals(m16)
```



```
## $uniformity
##
## Asymptotic one-sample Kolmogorov-Smirnov test
##
## data: simulationOutput$scaledResiduals
## D = 0.12328, p-value = 0.2938
## alternative hypothesis: two-sided
##
##
## $dispersion
##
## DHARMA nonparametric dispersion test via sd of residuals fitted vs.
## simulated
##
## data: simulationOutput
## dispersion = 1.0334, p-value = 0.776
## alternative hypothesis: two.sided
##
##
## $outliers
##
## DHARMA bootstrapped outlier test
##
## data: simulationOutput
## outliers at both margin(s) = 2, observations = 63, p-value = 0.08
## alternative hypothesis: two.sided
## percent confidence interval:
## 0.00000000 0.03174603
```

```

## sample estimates:
## outlier frequency (expected: 0.00396825396825397 )
##                                0.03174603

## $uniformity
##
## Asymptotic one-sample Kolmogorov-Smirnov test
##
## data: simulationOutput$scaledResiduals
## D = 0.12328, p-value = 0.2938
## alternative hypothesis: two-sided
##
##
## $dispersion
##
## DHARMA nonparametric dispersion test via sd of residuals fitted vs.
## simulated
##
## data: simulationOutput
## dispersion = 1.0334, p-value = 0.776
## alternative hypothesis: two.sided
##
##
## $outliers
##
## DHARMA bootstrapped outlier test
##
## data: simulationOutput
## outliers at both margin(s) = 2, observations = 63, p-value = 0.08
## alternative hypothesis: two.sided
## percent confidence interval:
## 0.00000000 0.03174603
## sample estimates:
## outlier frequency (expected: 0.00396825396825397 )
##                                0.03174603

# calculate the residuals for the model
m27 <- simulateResiduals(fittedModel = additive_rej_brumol, plot = F)

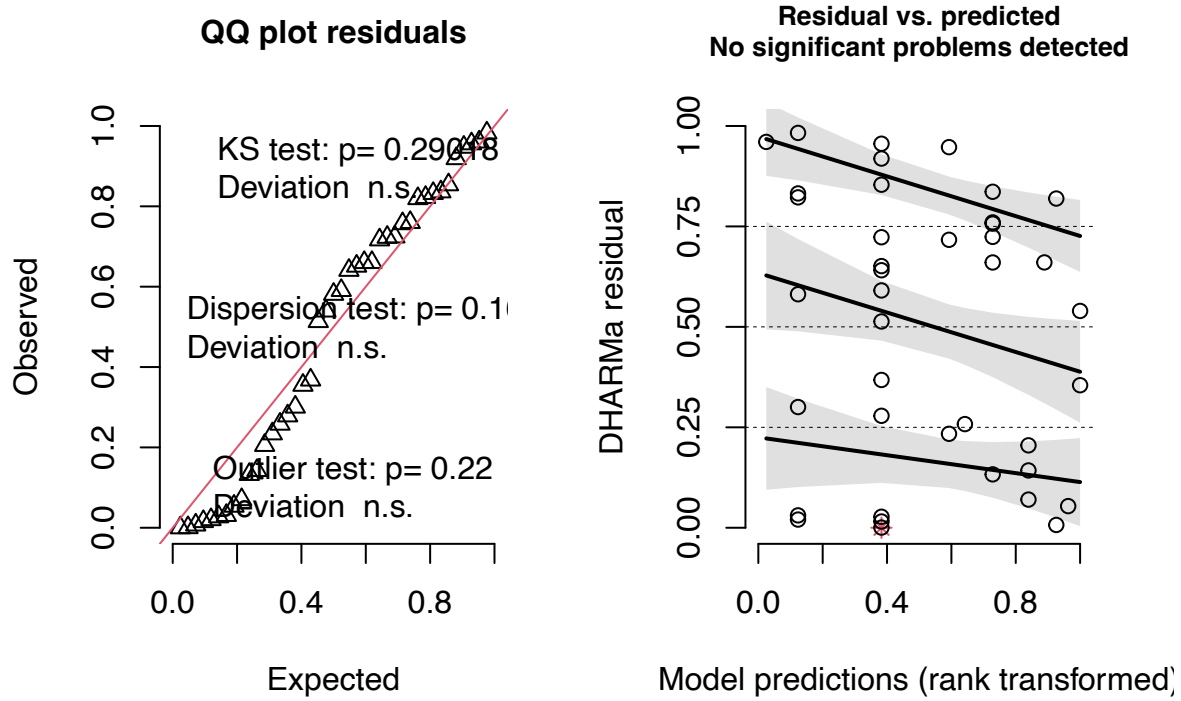
residuals(m27)

## [1] 0.2786600919 0.5133876578 0.3004530984 0.5906949527 0.0538566952
## [6] 0.5812031831 0.0070050725 0.6409038302 0.2580372806 0.1426346915
## [11] 0.0701839851 0.8222792874 0.6602753099 0.9472796179 0.0000000000
## [16] 0.0273862356 0.7599457208 0.9558150569 0.6604430219 0.6512570997
## [21] 0.0203934066 0.2338812769 0.0004721712 0.8321489438 0.1328772252
## [26] 0.9603448591 0.7239933345 0.8362491972 0.7230562079 0.7569603162
## [31] 0.3678643747 0.9827265516 0.0308318941 0.8538364192 0.0155469400
## [36] 0.7167187445 0.5396368934 0.3547707441 0.8193809929 0.2051192928
## [41] 0.9188393815

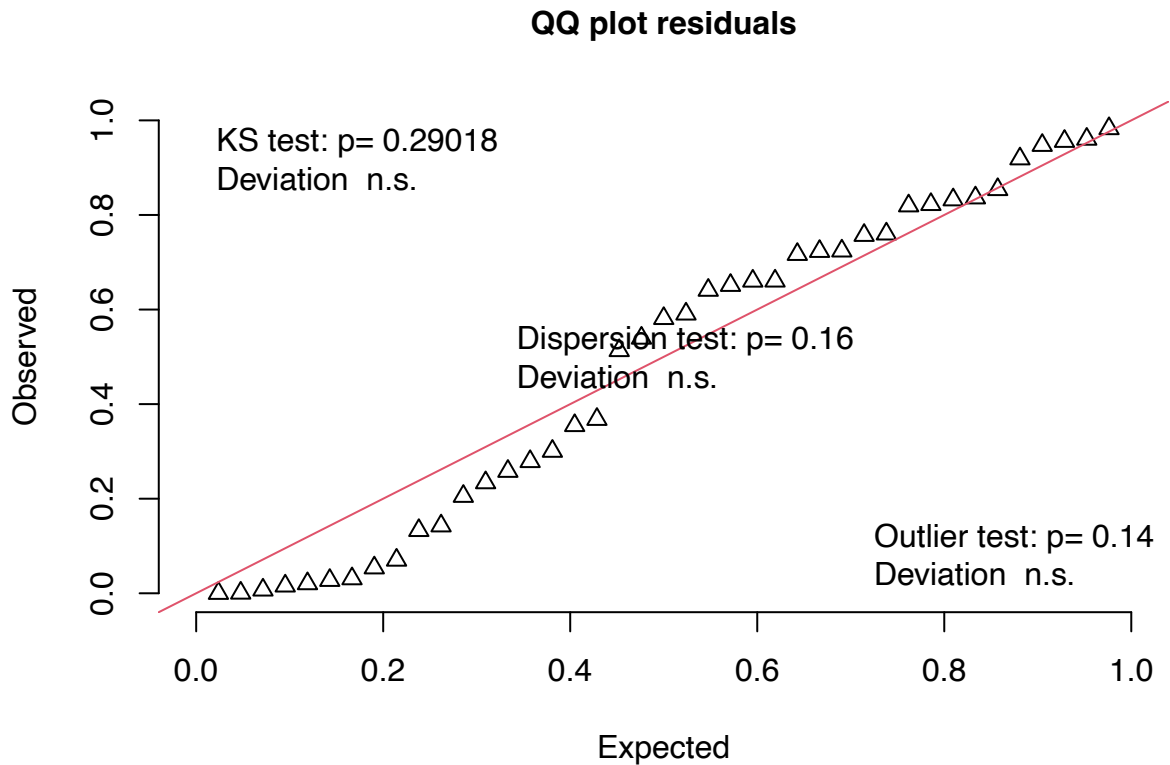
#plot simulation output
plot(m27)

```

DHARMA residual



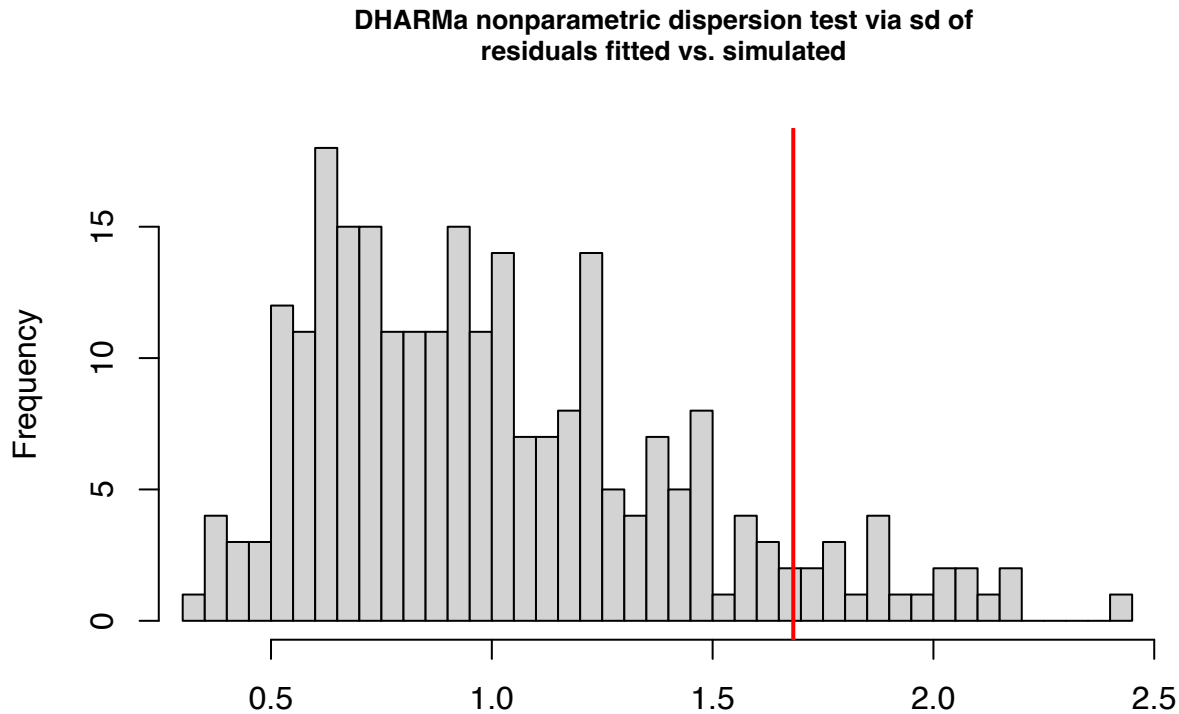
#tests if the overall distribution conforms to expectations.
`testUniformity(m27)`



 ## Exact one-sample Kolmogorov-Smirnov test

```
##
## data: simulationOutput$scaledResiduals
## D = 0.14933, p-value = 0.2902
## alternative hypothesis: two-sided
```

```
testDispersion(m27)
```

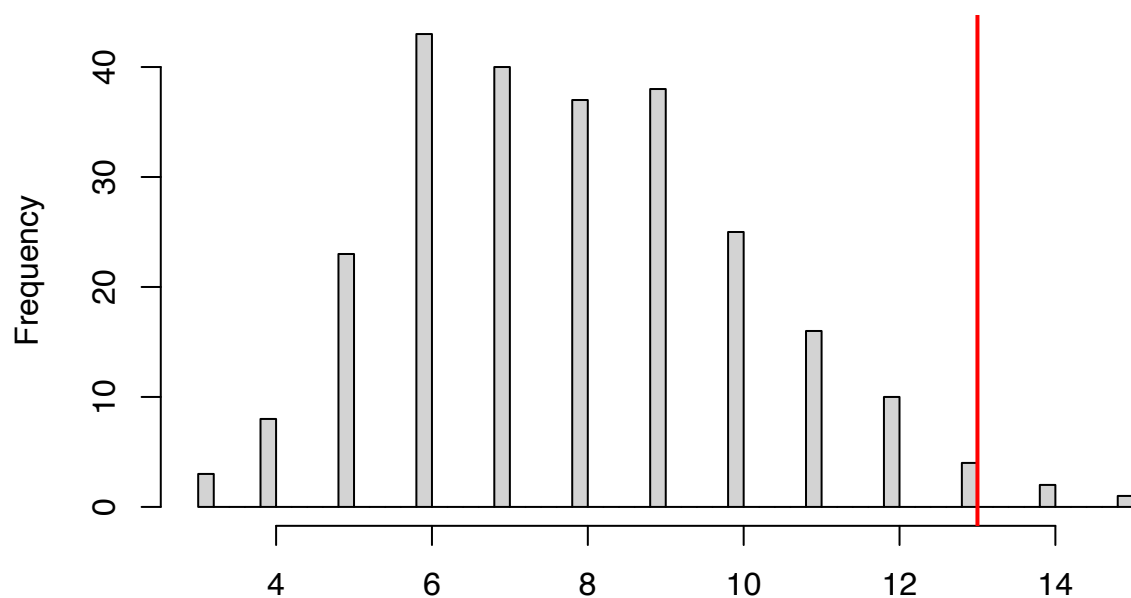


Simulated values, red line = fitted model. p-value (two.sided) = 0.16

```
##
## DHARMa nonparametric dispersion test via sd of residuals fitted vs.
## simulated
##
## data: simulationOutput
## dispersion = 1.6628, p-value = 0.16
## alternative hypothesis: two.sided
```

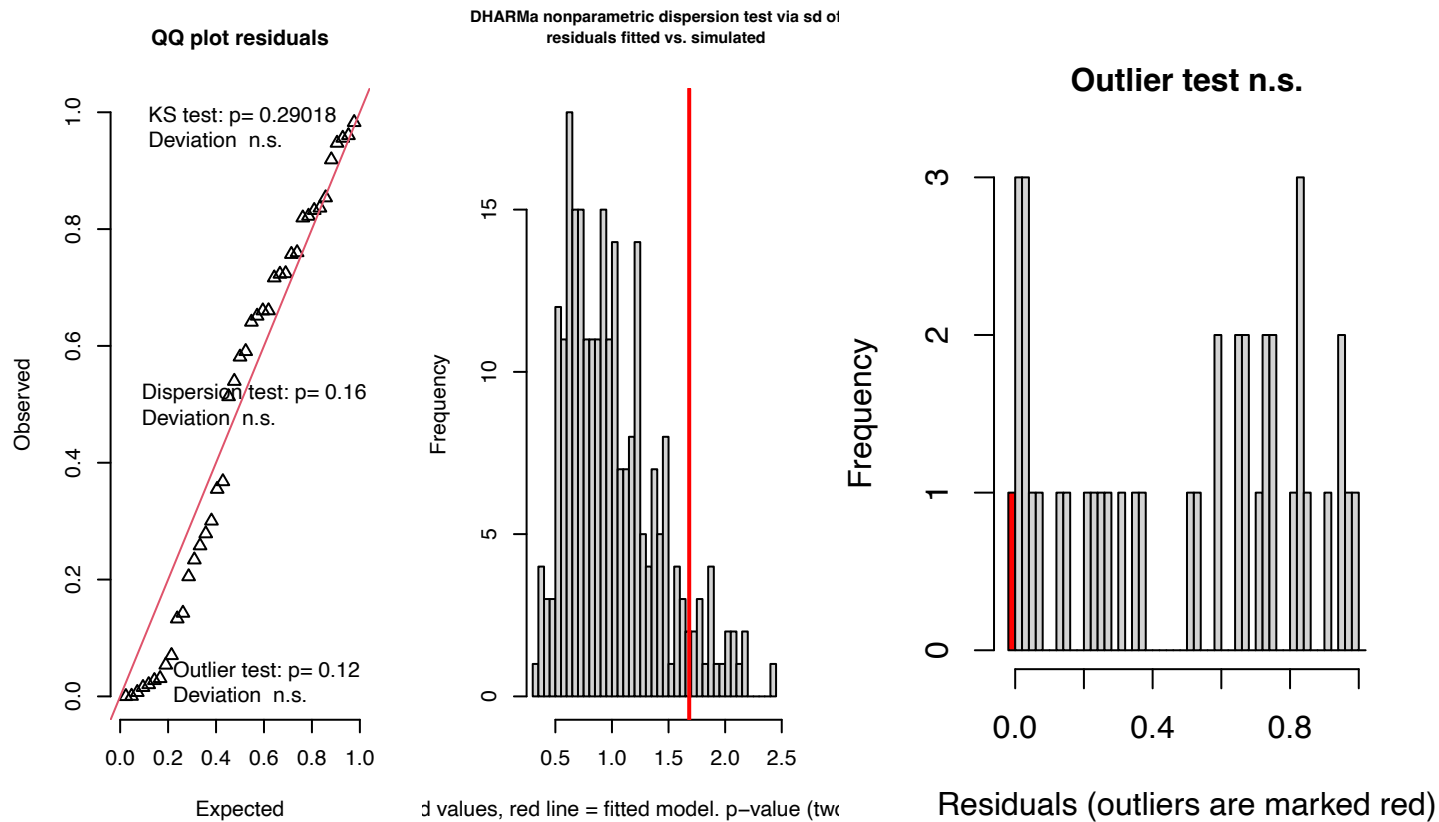
```
testZeroInflation(m27)
```

DHARMA zero-inflation test via comparison to expected zeros with simulation under H0 = fitted model



Simulated values, red line = fitted model. p-value (two.sided) = 0.056

```
##
## DHARMA zero-inflation test via comparison to expected zeros with
## simulation under H0 = fitted model
##
## data: simulationOutput
## ratioObsSim = 1.6472, p-value = 0.056
## alternative hypothesis: two.sided
testResiduals(m27)
```



```
## $uniformity
##
## Exact one-sample Kolmogorov-Smirnov test
##
## data: simulationOutput$scaledResiduals
## D = 0.14933, p-value = 0.2902
## alternative hypothesis: two-sided
##
##
## $dispersion
##
## DHARMA nonparametric dispersion test via sd of residuals fitted vs.
## simulated
##
## data: simulationOutput
## dispersion = 1.6628, p-value = 0.16
## alternative hypothesis: two.sided
##
##
## $outliers
##
## DHARMA bootstrapped outlier test
##
## data: simulationOutput
## outliers at both margin(s) = 1, observations = 41, p-value = 0.18
## alternative hypothesis: two.sided
## percent confidence interval:
## 0.00000000 0.02439024
```

```

## sample estimates:
## outlier frequency (expected: 0.0024390243902439 )
##                                0.02439024

## $uniformity
##
## Exact one-sample Kolmogorov-Smirnov test
##
## data: simulationOutput$scaledResiduals
## D = 0.14933, p-value = 0.2902
## alternative hypothesis: two-sided
##
##
## $dispersion
##
## DHARMA nonparametric dispersion test via sd of residuals fitted vs.
## simulated
##
## data: simulationOutput
## dispersion = 1.6628, p-value = 0.16
## alternative hypothesis: two.sided
##
##
## $outliers
##
## DHARMA bootstrapped outlier test
##
## data: simulationOutput
## outliers at both margin(s) = 1, observations = 41, p-value = 0.18
## alternative hypothesis: two.sided
## percent confidence interval:
## 0.00000000 0.02439024
## sample estimates:
## outlier frequency (expected: 0.0024390243902439 )
##                                0.02439024

# calculate the residuals for the model
m28 <- simulateResiduals(fittedModel = interaction_rej_brumol, plot = F)

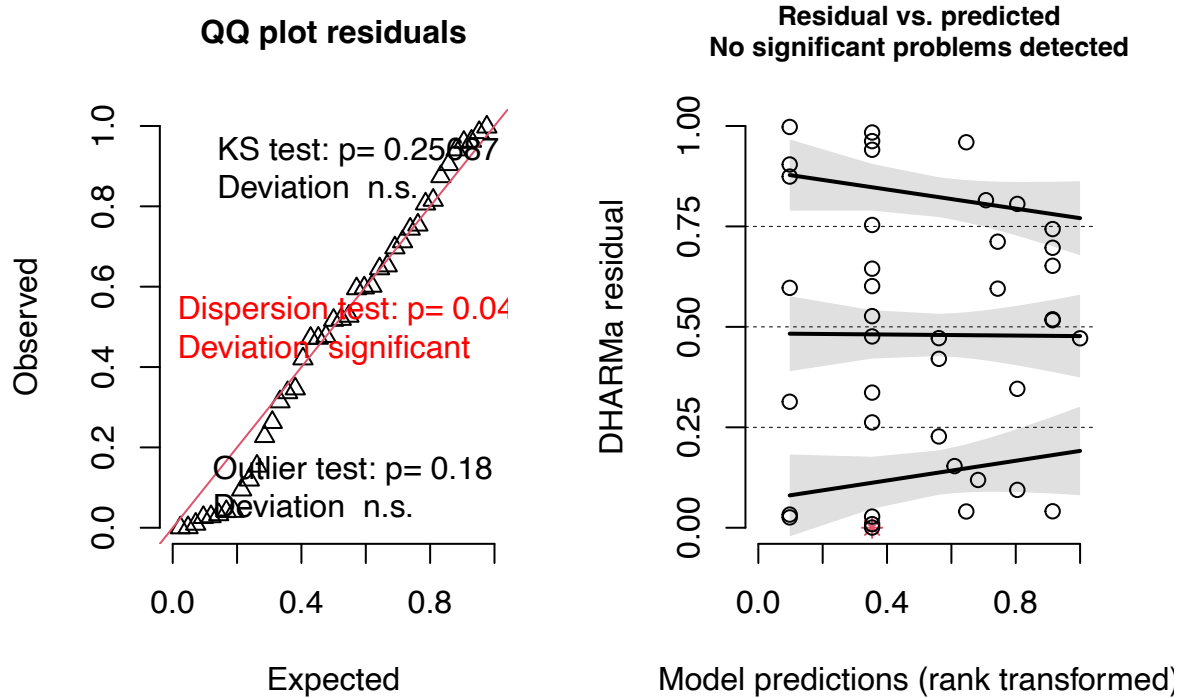
residuals(m28)

## [1] 0.2622683218 0.4761856536 0.3136308659 0.5266949527 0.1189857220
## [6] 0.5971265580 0.0405177539 0.6447914026 0.1535033063 0.4205108784
## [11] 0.2272783613 0.8740426017 0.5188991966 0.8060643089 0.0000000000
## [16] 0.0277831376 0.6968457388 0.9838034980 0.4715165829 0.6014269780
## [21] 0.0257600926 0.0942506638 0.0004721712 0.9041117079 0.0410114893
## [26] 0.8152650752 0.6519915956 0.7433756075 0.7535977767 0.5165061610
## [31] 0.3363970531 0.9975323645 0.0327350974 0.9407651747 0.0090690484
## [36] 0.3458376074 0.7120004163 0.5951215110 0.9594368671 0.4719383471
## [41] 0.9622590722

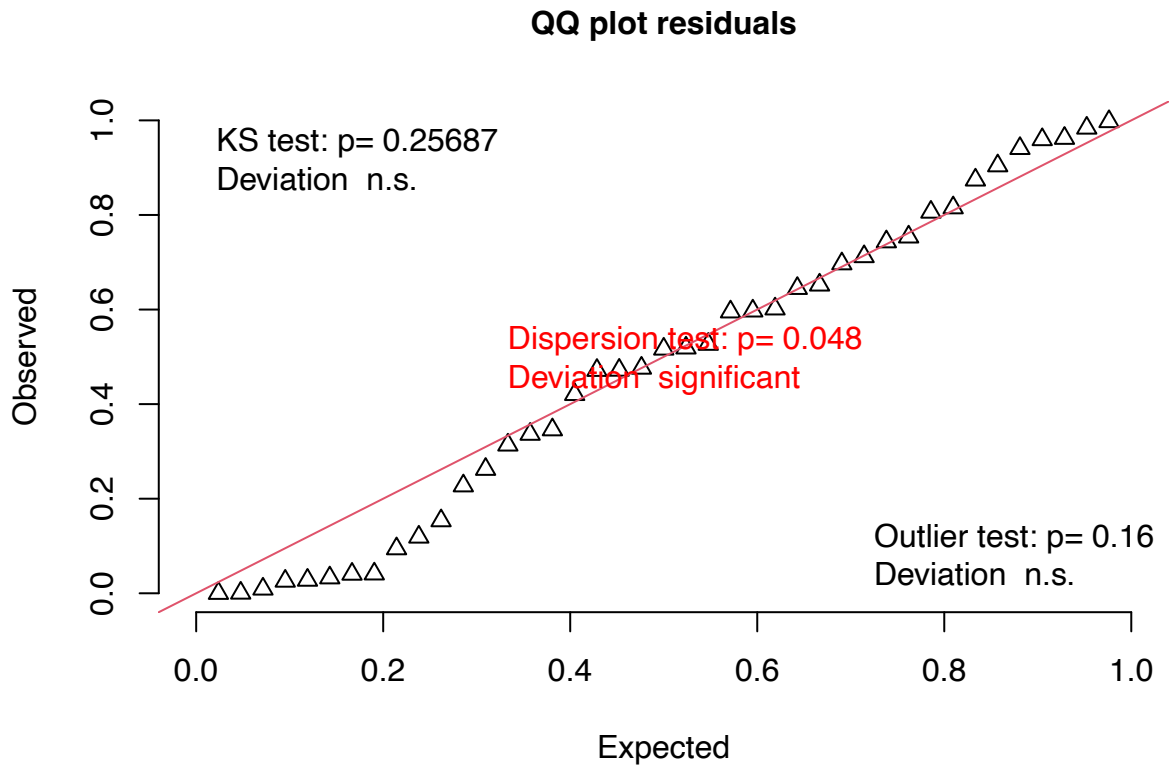
#plot simulation output
plot(m28)

```

DHARMA residual



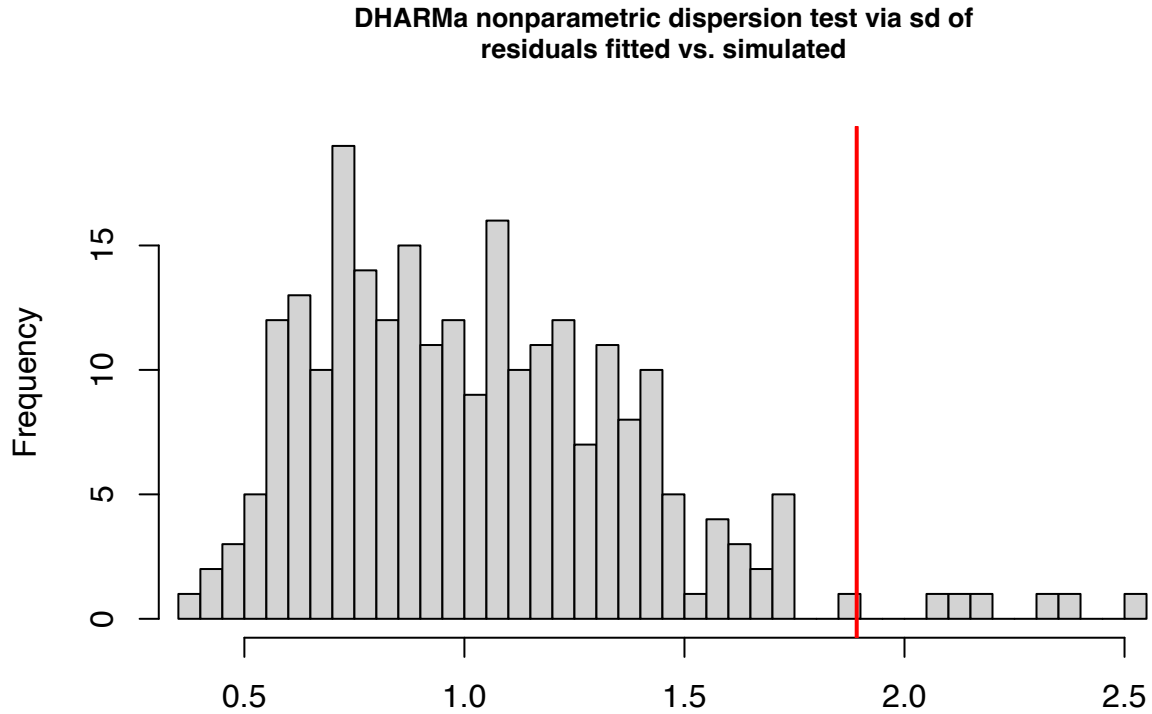
#tests if the overall distribution conforms to expectations.
`testUniformity(m28)`



 ## Exact one-sample Kolmogorov-Smirnov test

```
##
## data:  simulationOutput$scaledResiduals
## D = 0.15411, p-value = 0.2569
## alternative hypothesis: two-sided
```

```
testDispersion(m28)
```

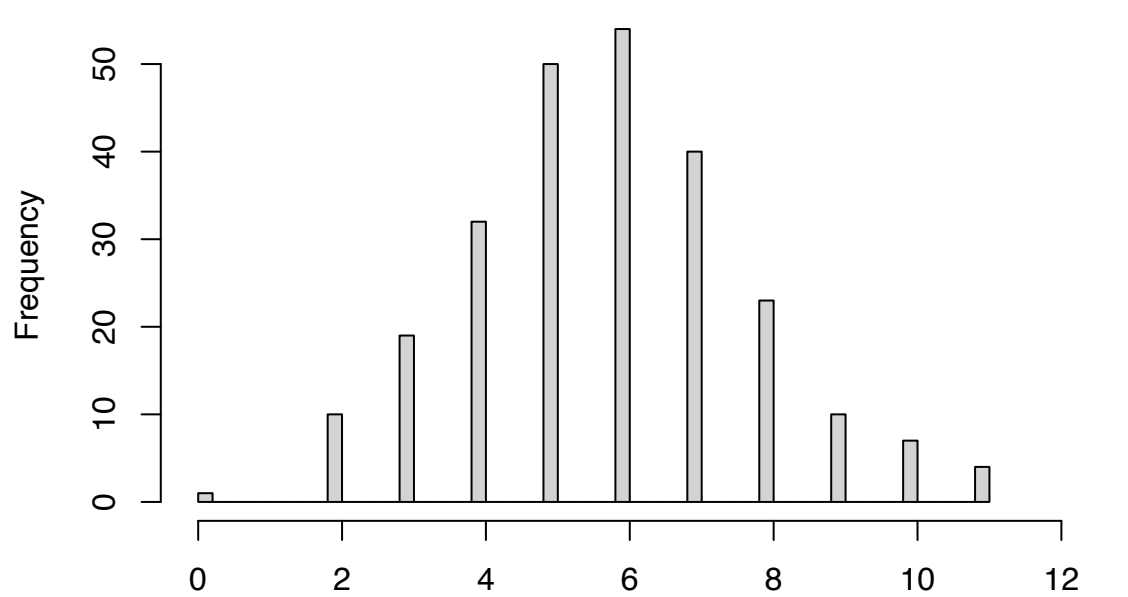


Simulated values, red line = fitted model. p-value (two.sided) = 0.048

```
##
## DHARMa nonparametric dispersion test via sd of residuals fitted vs.
## simulated
##
## data:  simulationOutput
## dispersion = 1.835, p-value = 0.048
## alternative hypothesis: two.sided
```

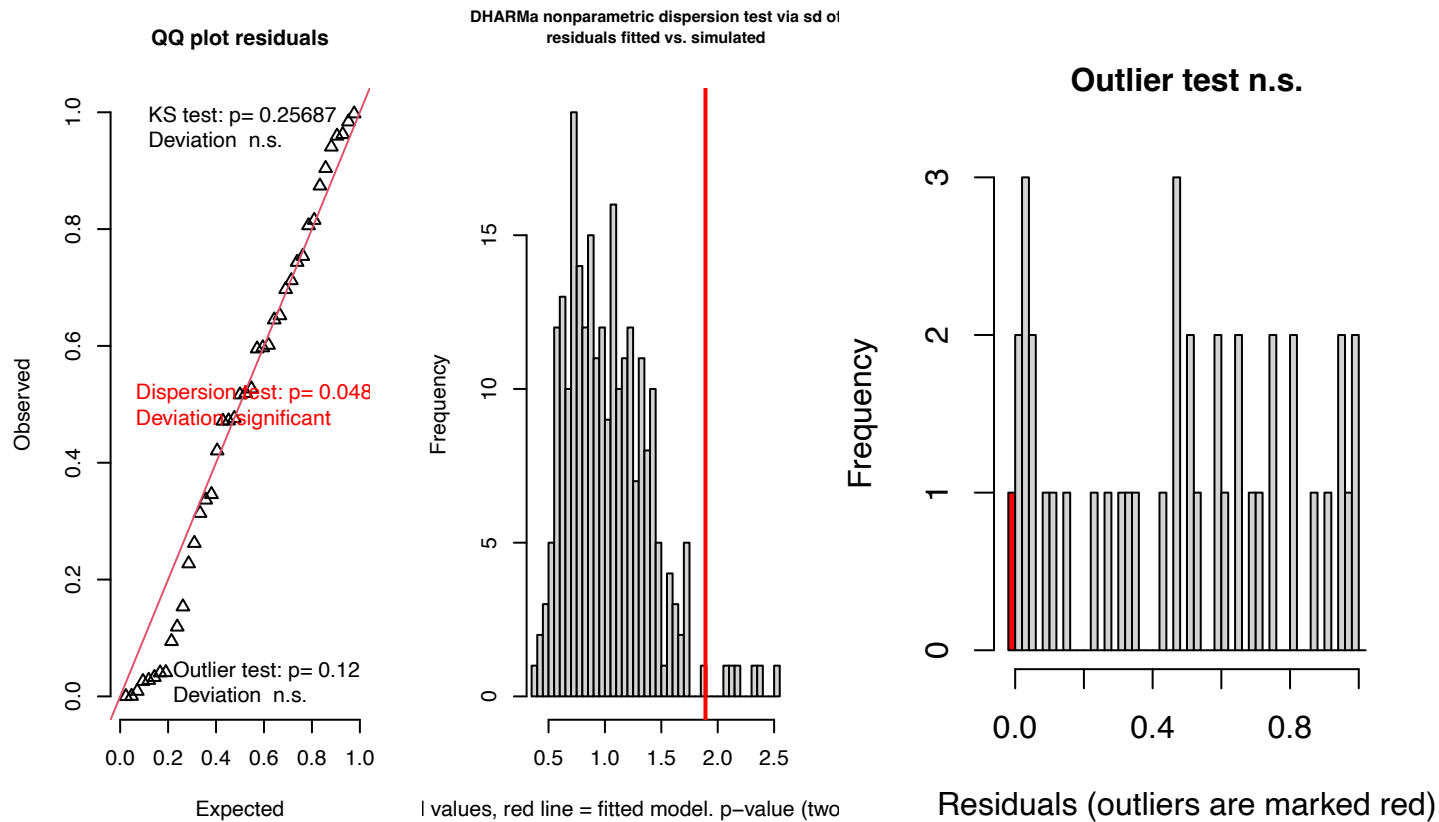
```
testZeroInflation(m28)
```

DHARMA zero-inflation test via comparison to expected zeros with simulation under H0 = fitted model



Simulated values, red line = fitted model. p-value (two.sided) = 0

```
##
## DHARMA zero-inflation test via comparison to expected zeros with
## simulation under H0 = fitted model
##
## data: simulationOutput
## ratioObsSim = 2.246, p-value < 2.2e-16
## alternative hypothesis: two.sided
testResiduals(m28)
```



```
## $uniformity
##
## Exact one-sample Kolmogorov-Smirnov test
##
## data: simulationOutput$scaledResiduals
## D = 0.15411, p-value = 0.2569
## alternative hypothesis: two-sided
##
##
## $dispersion
##
## DHARMA nonparametric dispersion test via sd of residuals fitted vs.
## simulated
##
## data: simulationOutput
## dispersion = 1.835, p-value = 0.048
## alternative hypothesis: two.sided
##
##
## $outliers
##
## DHARMA bootstrapped outlier test
##
## data: simulationOutput
## outliers at both margin(s) = 1, observations = 41, p-value = 0.14
## alternative hypothesis: two.sided
## percent confidence interval:
## 0.00000000 0.02439024
```

```

## sample estimates:
## outlier frequency (expected: 0.00195121951219512 )
##                                0.02439024

## $uniformity
##
## Exact one-sample Kolmogorov-Smirnov test
##
## data: simulationOutput$scaledResiduals
## D = 0.15411, p-value = 0.2569
## alternative hypothesis: two-sided
##
##
## $dispersion
##
## DHARMA nonparametric dispersion test via sd of residuals fitted vs.
## simulated
##
## data: simulationOutput
## dispersion = 1.835, p-value = 0.048
## alternative hypothesis: two.sided
##
##
## $outliers
##
## DHARMA bootstrapped outlier test
##
## data: simulationOutput
## outliers at both margin(s) = 1, observations = 41, p-value = 0.14
## alternative hypothesis: two.sided
## percent confidence interval:
## 0.00000000 0.02439024
## sample estimates:
## outlier frequency (expected: 0.00195121951219512 )
##                                0.02439024

# calculate the residuals for the model
m33 <- simulateResiduals(fittedModel = additive_mating_brumol, plot = F)

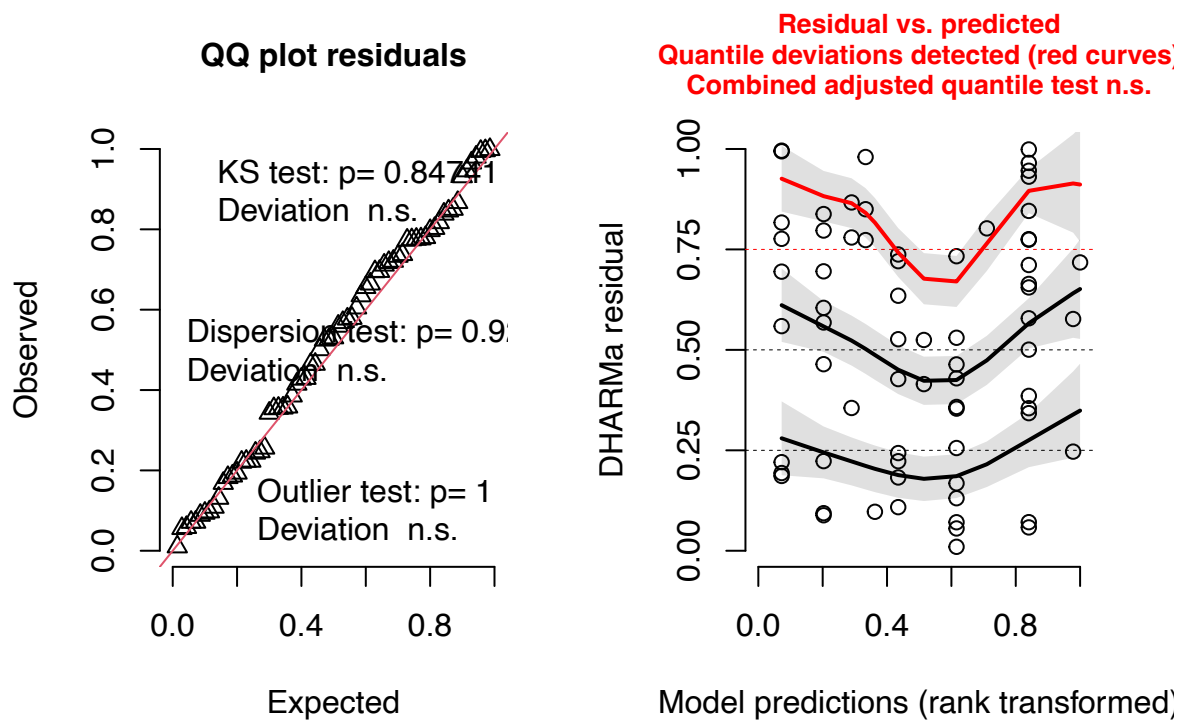
residuals(m33)

## [1] 0.60453472 0.79703829 0.81688224 0.69467627 0.77416494 0.35484691
## [7] 0.09355361 0.56816747 0.93106942 0.99578382 0.99876362 0.05547826
## [13] 0.09701251 0.63473809 0.71762854 0.25568494 0.99411942 0.86687886
## [19] 0.05809379 0.80241962 0.98001659 0.08839511 0.85004600 0.22051152
## [25] 0.00985215 0.53012467 0.22282417 0.24311446 0.72074956 0.83801884
## [31] 0.42722432 0.52649117 0.35378718 0.66407393 0.50062688 0.57856487
## [37] 0.35799207 0.69547450 0.77547825 0.52459963 0.94583522 0.41476247
## [43] 0.55906878 0.38545581 0.10851220 0.46378575 0.84587720 0.19345153
## [49] 0.07095488 0.46438723 0.16788376 0.42920459 0.18242099 0.07117886
## [55] 0.13078405 0.96529091 0.18616764 0.22308704 0.35539068 0.73311932
## [61] 0.77636952 0.34251975 0.71116269 0.73742834 0.24672139 0.57679981
## [67] 0.77971346 0.77326634 0.65536925

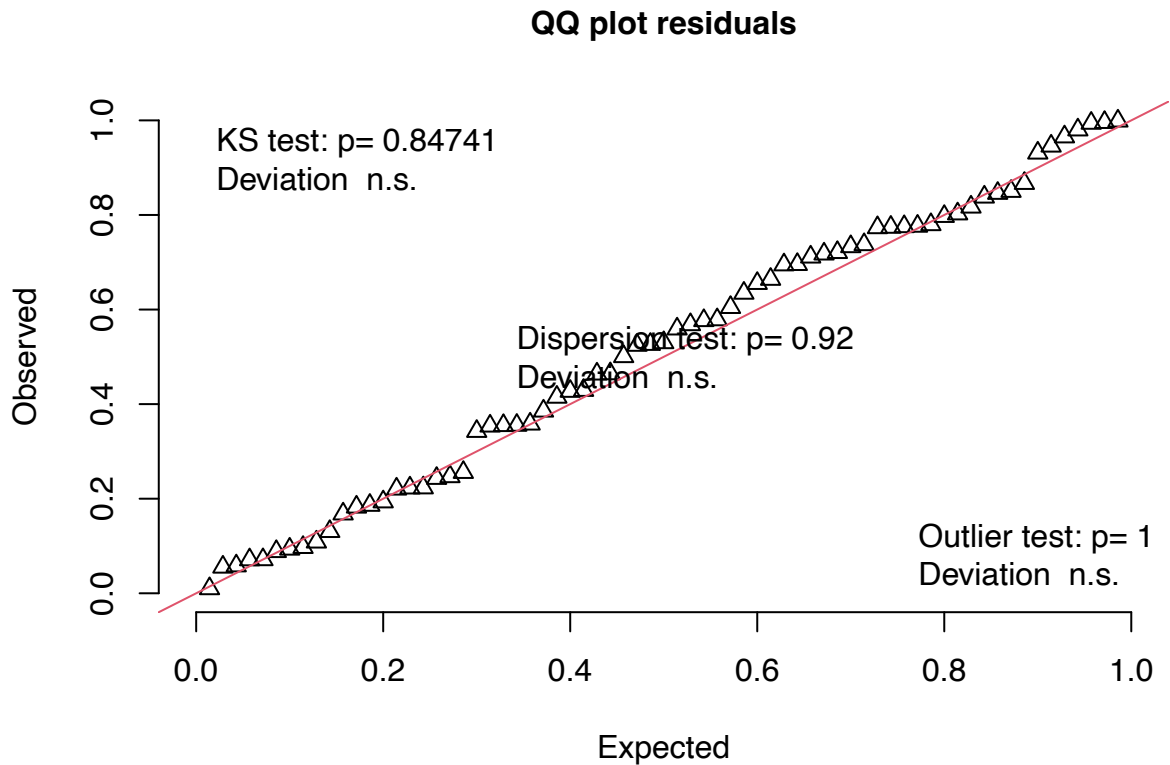
#plot simulation output
plot(m33)

```

DHARMA residual



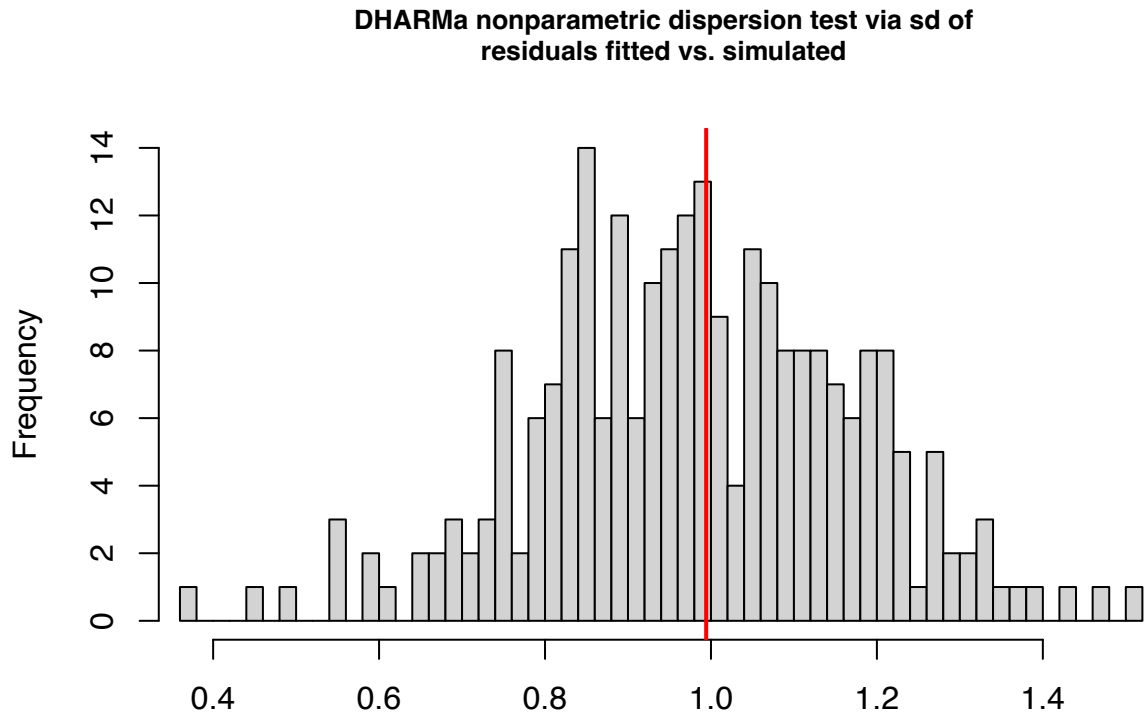
#tests if the overall distribution conforms to expectations.
`testUniformity(m33)`



 ## Exact one-sample Kolmogorov-Smirnov test

```
##
## data: simulationOutput$scaledResiduals
## D = 0.071488, p-value = 0.8474
## alternative hypothesis: two-sided
```

```
testDispersion(m33)
```

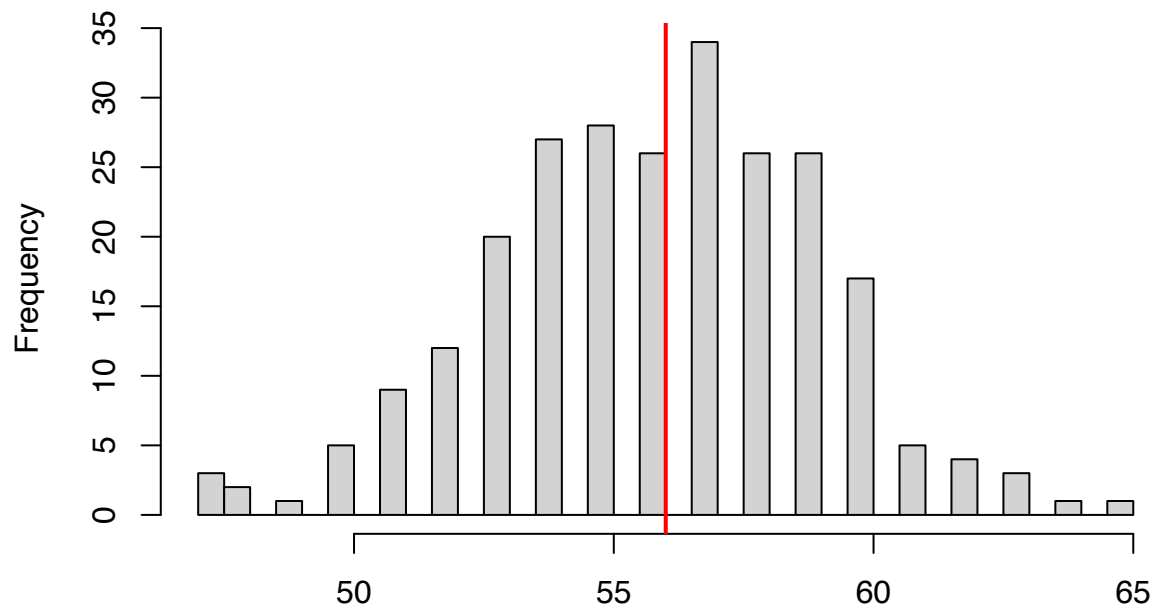


Simulated values, red line = fitted model. p-value (two.sided) = 0.92

```
##
## DHARMA nonparametric dispersion test via sd of residuals fitted vs.
## simulated
##
## data: simulationOutput
## dispersion = 1.0123, p-value = 0.92
## alternative hypothesis: two.sided
```

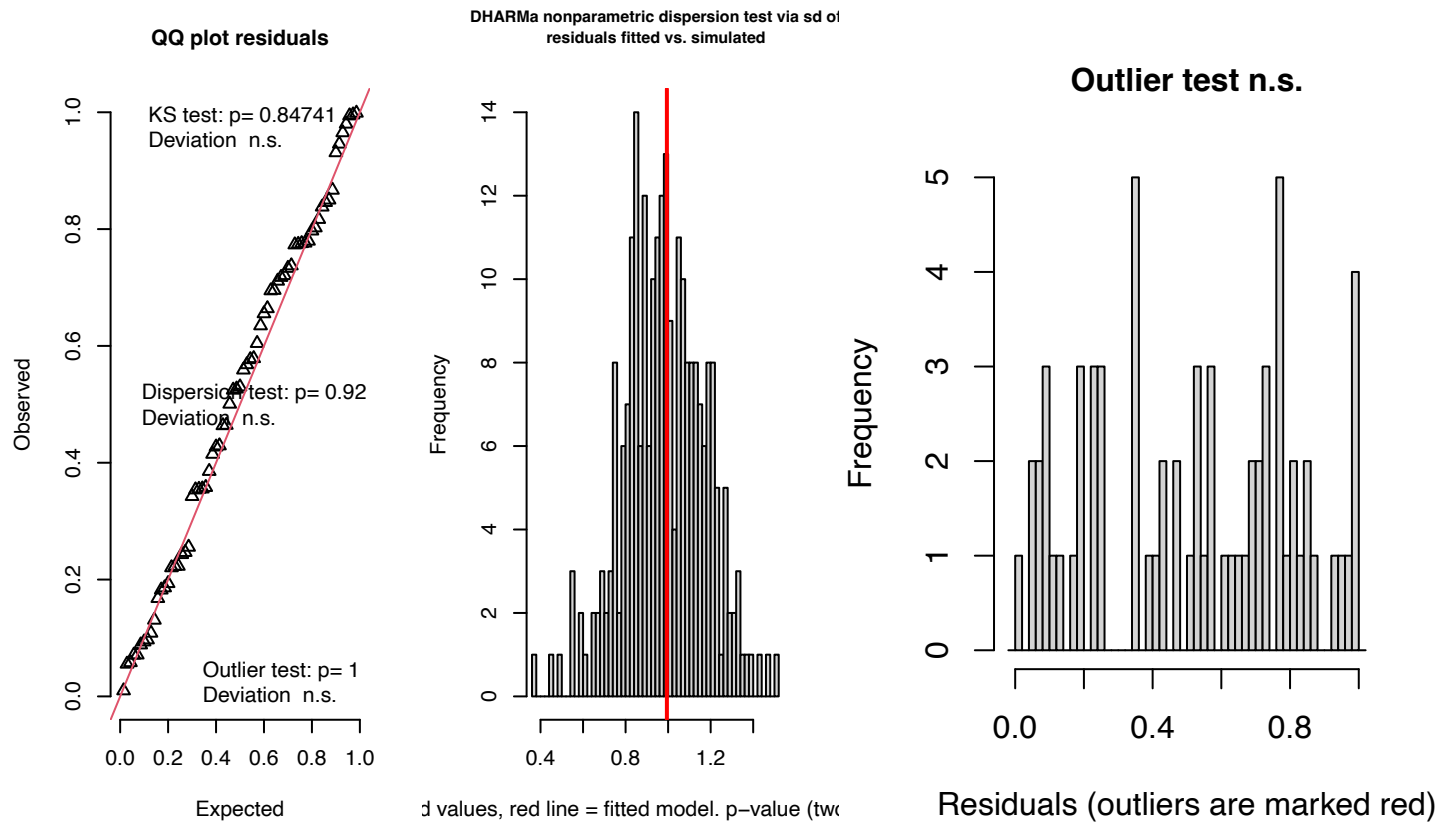
```
testZeroInflation(m33)
```

DHARMA zero-inflation test via comparison to expected zeros with simulation under H0 = fitted model



Simulated values, red line = fitted model. p-value (two.sided) = 1

```
##
## DHARMA zero-inflation test via comparison to expected zeros with
## simulation under H0 = fitted model
##
## data: simulationOutput
## ratioObsSim = 0.99971, p-value = 1
## alternative hypothesis: two.sided
testResiduals(m33)
```



```
## $uniformity
##
## Exact one-sample Kolmogorov-Smirnov test
##
## data: simulationOutput$scaledResiduals
## D = 0.071488, p-value = 0.8474
## alternative hypothesis: two-sided
##
##
## $dispersion
##
## DHARMA nonparametric dispersion test via sd of residuals fitted vs.
## simulated
##
## data: simulationOutput
## dispersion = 1.0123, p-value = 0.92
## alternative hypothesis: two.sided
##
##
## $outliers
##
## DHARMA bootstrapped outlier test
##
## data: simulationOutput
## outliers at both margin(s) = 0, observations = 69, p-value = 1
## alternative hypothesis: two.sided
## percent confidence interval:
## 0 0
```

```

## sample estimates:
## outlier frequency (expected: 0 )
##                                0

## $uniformity
##
## Exact one-sample Kolmogorov-Smirnov test
##
## data: simulationOutput$scaledResiduals
## D = 0.071488, p-value = 0.8474
## alternative hypothesis: two-sided
##
##
## $dispersion
##
## DHARMA nonparametric dispersion test via sd of residuals fitted vs.
## simulated
##
## data: simulationOutput
## dispersion = 1.0123, p-value = 0.92
## alternative hypothesis: two.sided
##
##
## $outliers
##
## DHARMA bootstrapped outlier test
##
## data: simulationOutput
## outliers at both margin(s) = 0, observations = 69, p-value = 1
## alternative hypothesis: two.sided
## percent confidence interval:
## 0 0
## sample estimates:
## outlier frequency (expected: 0 )
##                                0

# calculate the residuals for the model
m34 <- simulateResiduals(fittedModel = interaction_mating_brumol, plot = F)

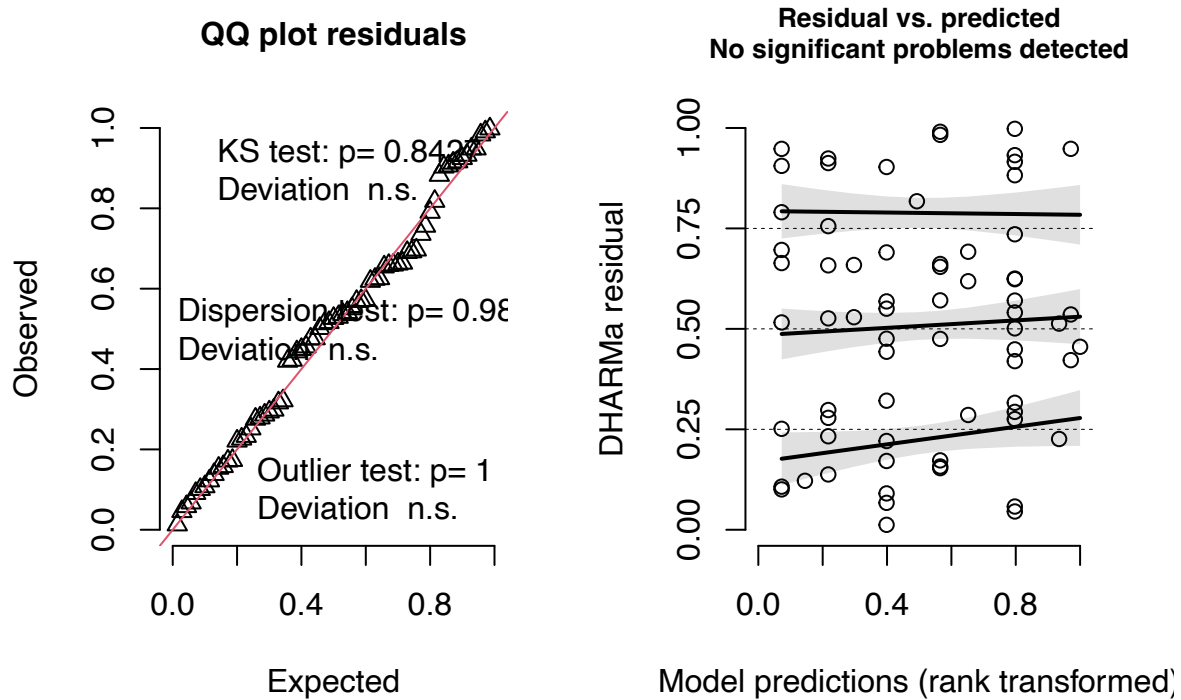
residuals(m34)

## [1] 0.69646857 0.90572533 0.66145774 0.57096680 0.62360823 0.29336354
## [7] 0.10728625 0.66374705 0.88234264 0.99054553 0.99787664 0.06700273
## [13] 0.12187501 0.75564058 0.45542647 0.32121224 0.98284831 0.69191965
## [19] 0.04547746 0.81761811 0.94804312 0.10044899 0.53585667 0.17270150
## [25] 0.01231519 0.69026649 0.27853021 0.29793438 0.92403790 0.94798511
## [31] 0.52613832 0.65811396 0.47552040 0.54084372 0.41971748 0.44899044
## [37] 0.44305949 0.79031194 0.62513043 0.65904476 0.91585114 0.52903377
## [43] 0.47496109 0.31594739 0.13770584 0.56836489 0.73542252 0.15336698
## [49] 0.09050367 0.51598581 0.22089969 0.55026229 0.23267984 0.05781024
## [55] 0.17118331 0.93231727 0.15745955 0.25122415 0.28570623 0.90285630
## [61] 0.65432017 0.27539277 0.57041174 0.91265884 0.22591357 0.51310413
## [67] 0.61852270 0.42212124 0.50116472

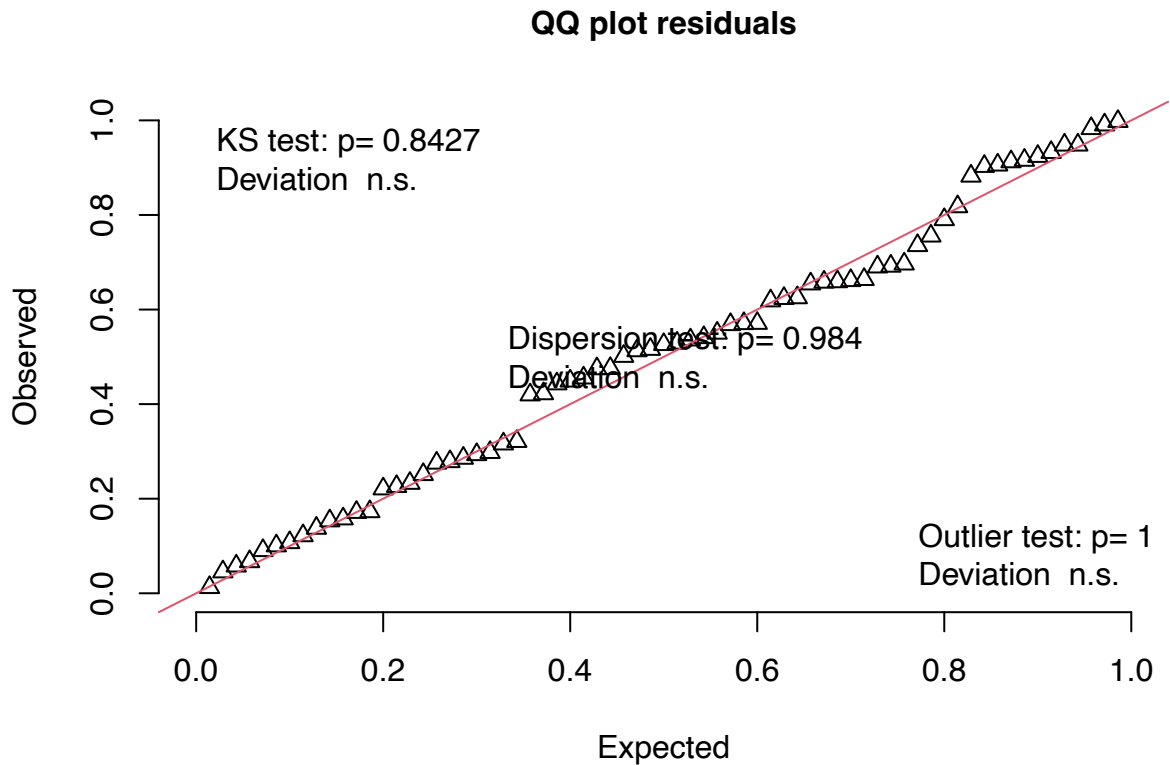
#plot simulation output
plot(m34)

```

DHARMA residual



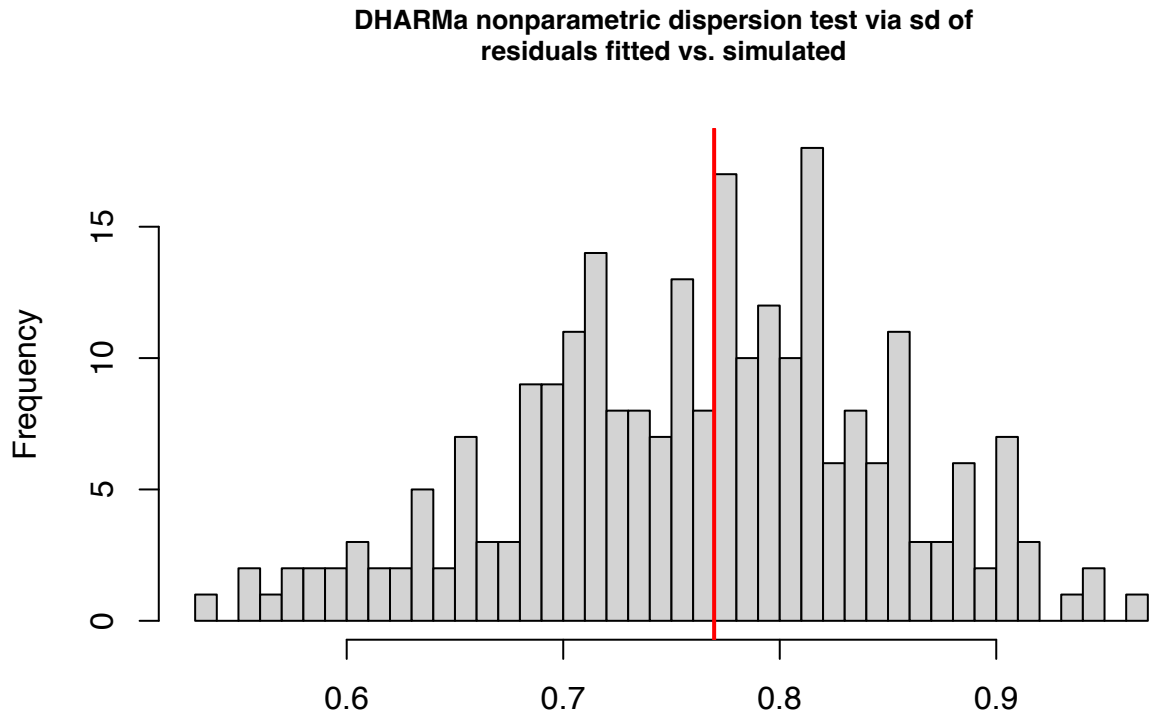
#tests if the overall distribution conforms to expectations.
`testUniformity(m34)`



 ## Exact one-sample Kolmogorov-Smirnov test

```
##
## data: simulationOutput$scaledResiduals
## D = 0.071891, p-value = 0.8427
## alternative hypothesis: two-sided
```

```
testDispersion(m34)
```

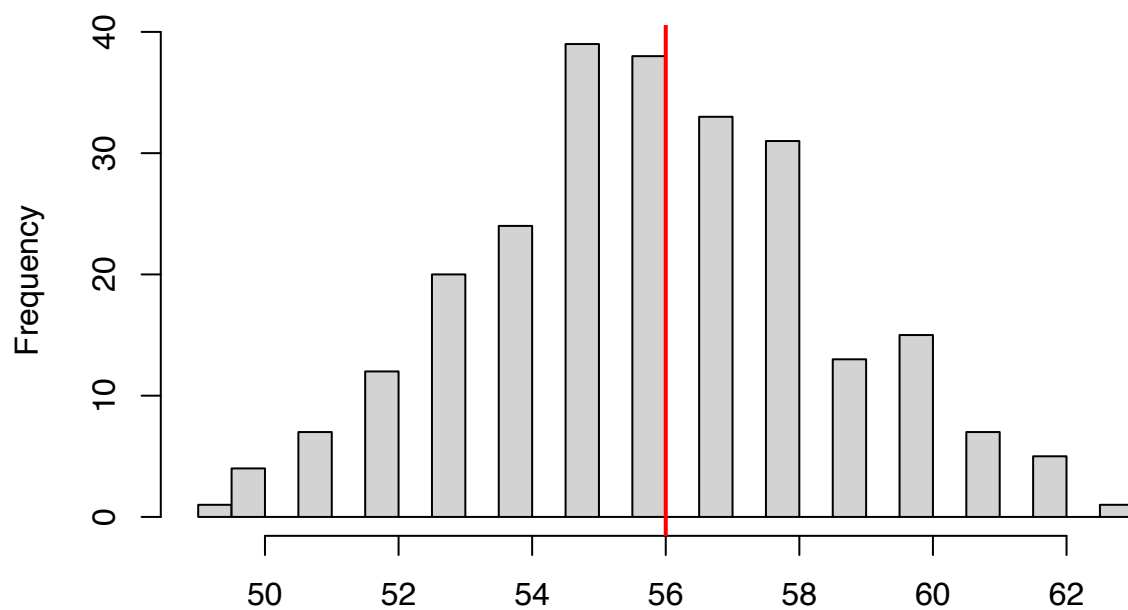


Simulated values, red line = fitted model. p-value (two.sided) = 0.984

```
##
## DHARMa nonparametric dispersion test via sd of residuals fitted vs.
## simulated
##
## data: simulationOutput
## dispersion = 1.0089, p-value = 0.984
## alternative hypothesis: two.sided
```

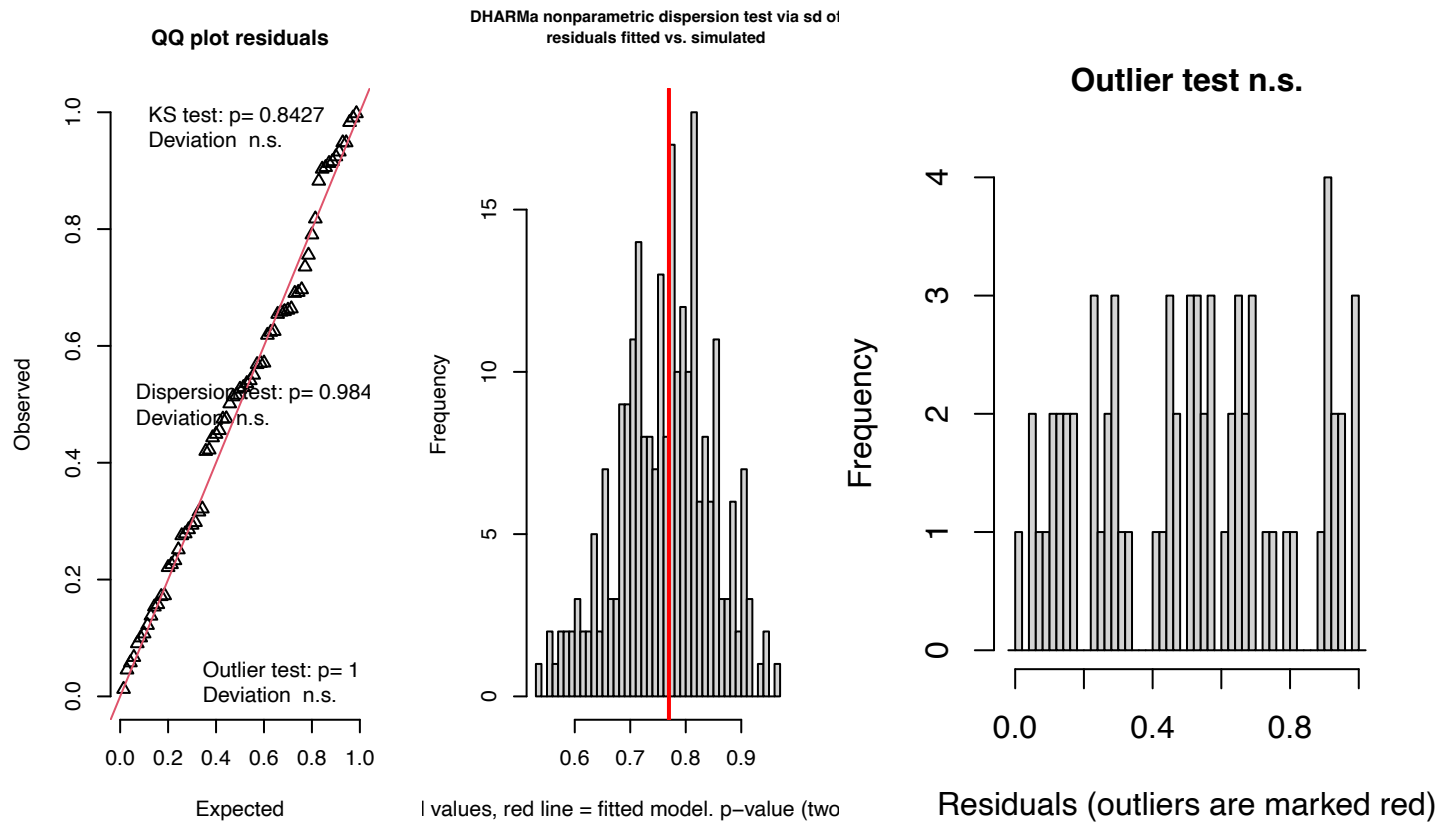
```
testZeroInflation(m34)
```

DHARMA zero-inflation test via comparison to expected zeros with simulation under H0 = fitted model



Simulated values, red line = fitted model. p-value (two.sided) = 1

```
##
## DHARMA zero-inflation test via comparison to expected zeros with
## simulation under H0 = fitted model
##
## data: simulationOutput
## ratioObsSim = 0.99964, p-value = 1
## alternative hypothesis: two.sided
testResiduals(m34)
```



```
## $uniformity
##
## Exact one-sample Kolmogorov-Smirnov test
##
## data: simulationOutput$scaledResiduals
## D = 0.071891, p-value = 0.8427
## alternative hypothesis: two-sided
##
##
## $dispersion
##
## DHARMA nonparametric dispersion test via sd of residuals fitted vs.
## simulated
##
## data: simulationOutput
## dispersion = 1.0089, p-value = 0.984
## alternative hypothesis: two.sided
##
##
## $outliers
##
## DHARMA bootstrapped outlier test
##
## data: simulationOutput
## outliers at both margin(s) = 0, observations = 69, p-value = 1
## alternative hypothesis: two.sided
## percent confidence interval:
## 0 0
```

```
## sample estimates:
## outlier frequency (expected: 0 )
##
## $uniformity
##
## Exact one-sample Kolmogorov-Smirnov test
##
## data: simulationOutput$scaledResiduals
## D = 0.071891, p-value = 0.8427
## alternative hypothesis: two-sided
##
##
## $dispersion
##
## DHARMA nonparametric dispersion test via sd of residuals fitted vs.
## simulated
##
## data: simulationOutput
## dispersion = 1.0089, p-value = 0.984
## alternative hypothesis: two.sided
##
##
## $outliers
##
## DHARMA bootstrapped outlier test
##
## data: simulationOutput
## outliers at both margin(s) = 0, observations = 69, p-value = 1
## alternative hypothesis: two.sided
## percent confidence interval:
## 0 0
## sample estimates:
## outlier frequency (expected: 0 )
##
## 0
```

Mol-Big combination

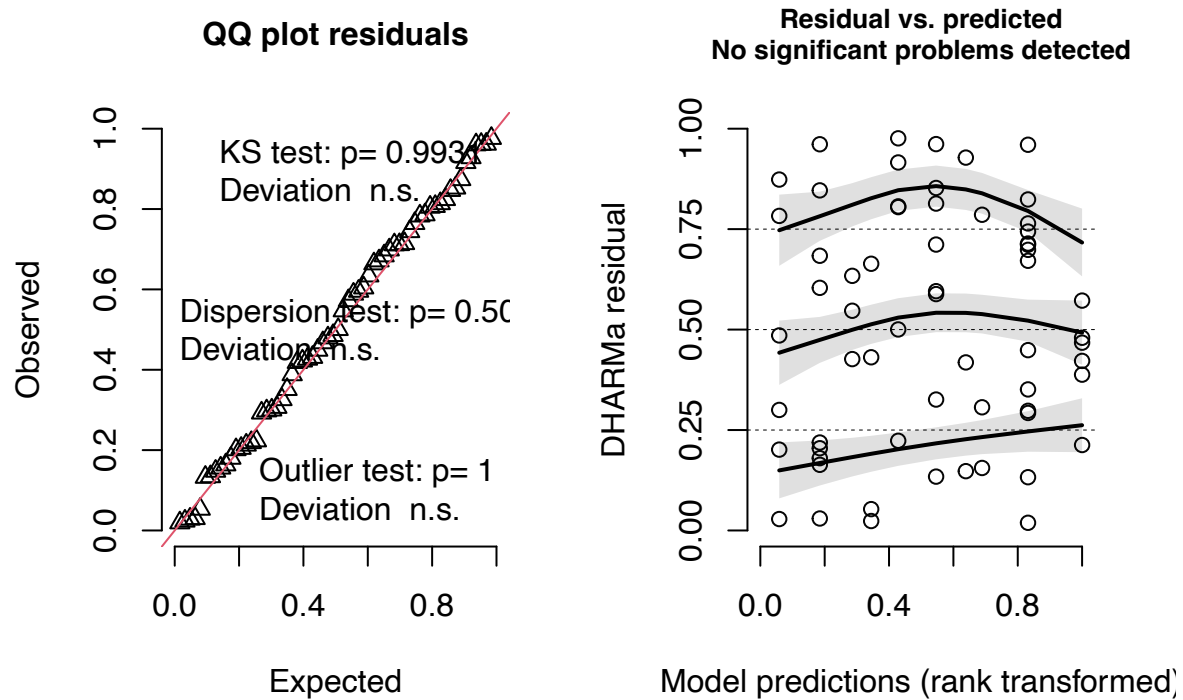
```
# calculate the residuals for the model
m5 <- simulateResiduals(fittedModel = additive_indirect_molbig, plot = F)

residuals(m5)

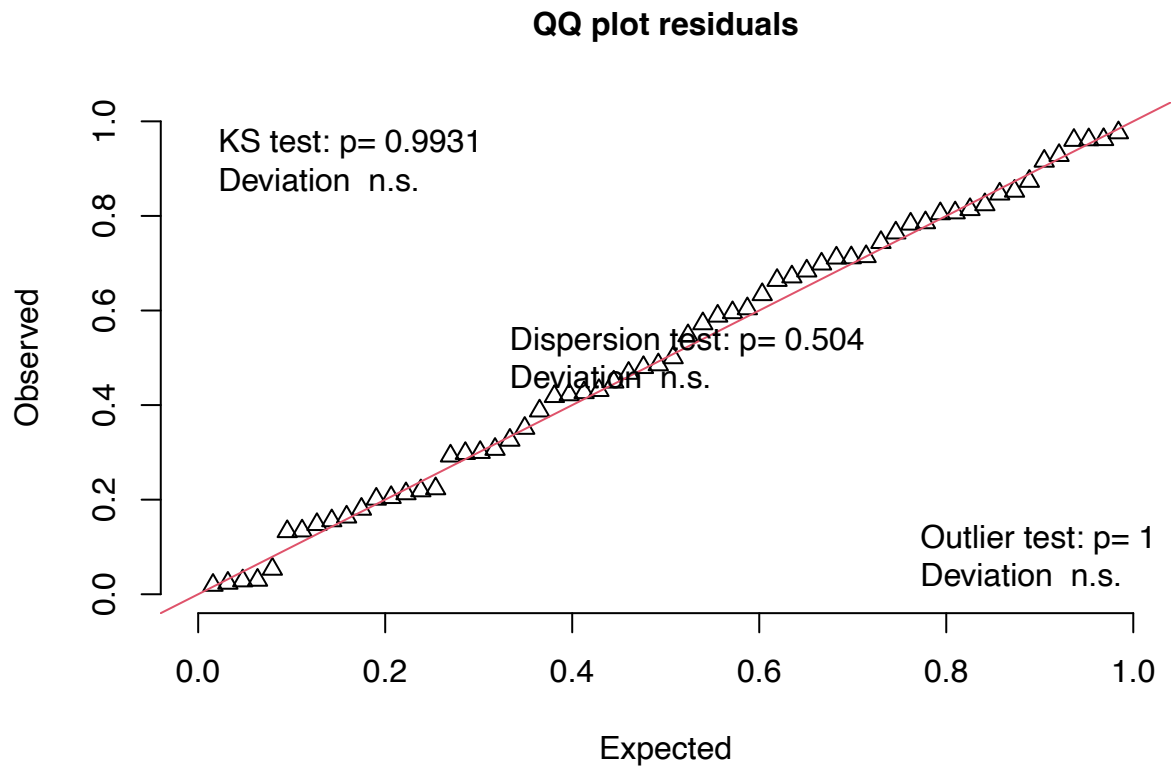
## [1] 0.02843881 0.29784075 0.35104955 0.05310125 0.16343011 0.02982276
## [7] 0.17995134 0.13258333 0.30001955 0.20123132 0.44885221 0.22323065
## [13] 0.21889616 0.14760127 0.54684234 0.21267038 0.48536639 0.43104289
## [19] 0.29247069 0.15553597 0.50034991 0.02352106 0.13404902 0.30658083
## [25] 0.60389062 0.41827771 0.01911095 0.66382859 0.78292263 0.58832948
## [31] 0.32584965 0.46741117 0.80665707 0.96122867 0.71162497 0.80479805
## [37] 0.84643813 0.42199478 0.57216928 0.87338843 0.91594810 0.85242858
## [43] 0.81358605 0.20467341 0.71114902 0.63375094 0.38780254 0.97600000
## [49] 0.68372296 0.59558538 0.69847191 0.47959782 0.96164060 0.92800000
## [55] 0.42652969 0.78576472 0.71414980 0.76420037 0.67124270 0.74400000
## [61] 0.96000000 0.82383948
```

```
#plot simulation output
plot(m5)
```

DHARMA residual

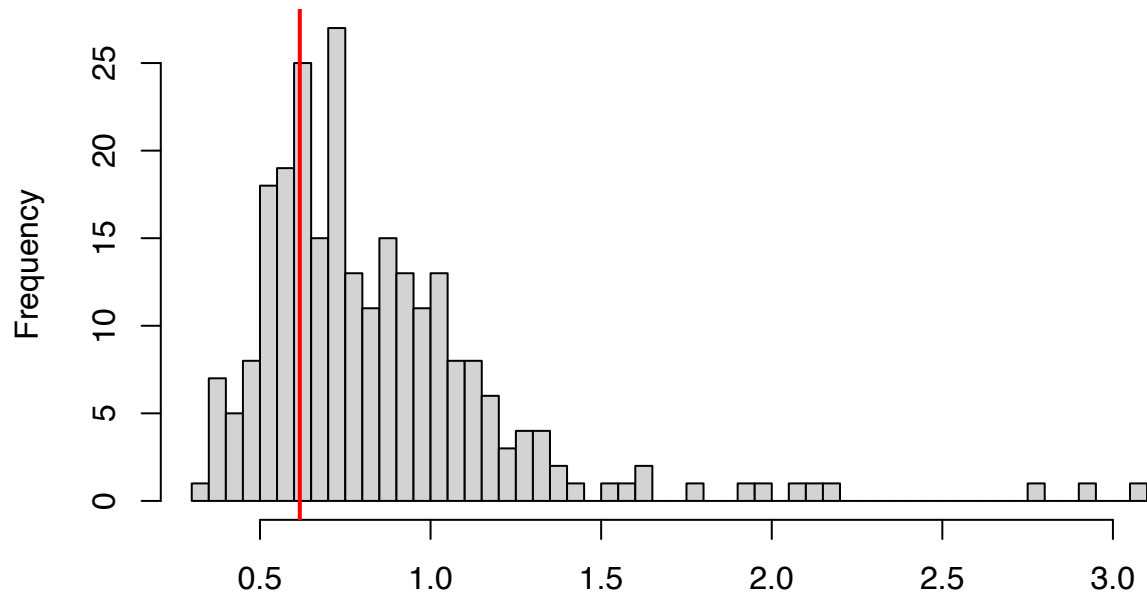


```
#tests if the overall distribution conforms to expectations.
testUniformity(m5)
```



```
##  
## Exact one-sample Kolmogorov-Smirnov test  
##  
## data: simulationOutput$scaledResiduals  
## D = 0.051938, p-value = 0.9931  
## alternative hypothesis: two-sided  
testDispersion(m5)
```

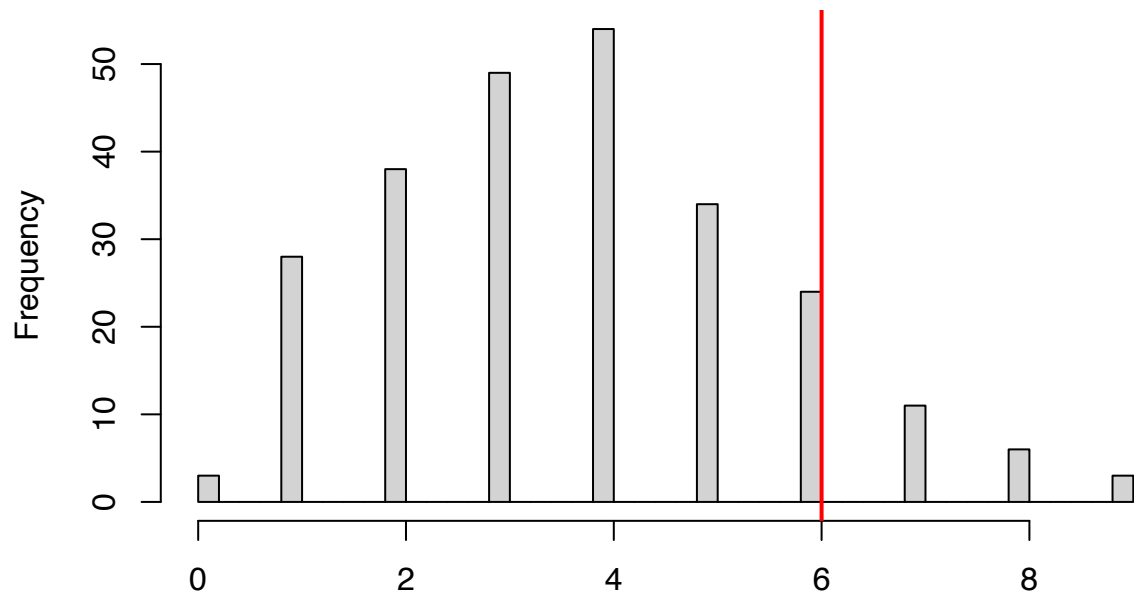
**DHARMa nonparametric dispersion test via sd of
residuals fitted vs. simulated**



Simulated values, red line = fitted model. p-value (two.sided) = 0.504

```
##
## DHARMa nonparametric dispersion test via sd of residuals fitted vs.
## simulated
##
## data:  simulationOutput
## dispersion = 0.72322, p-value = 0.504
## alternative hypothesis: two.sided
testZeroInflation(m5)
```

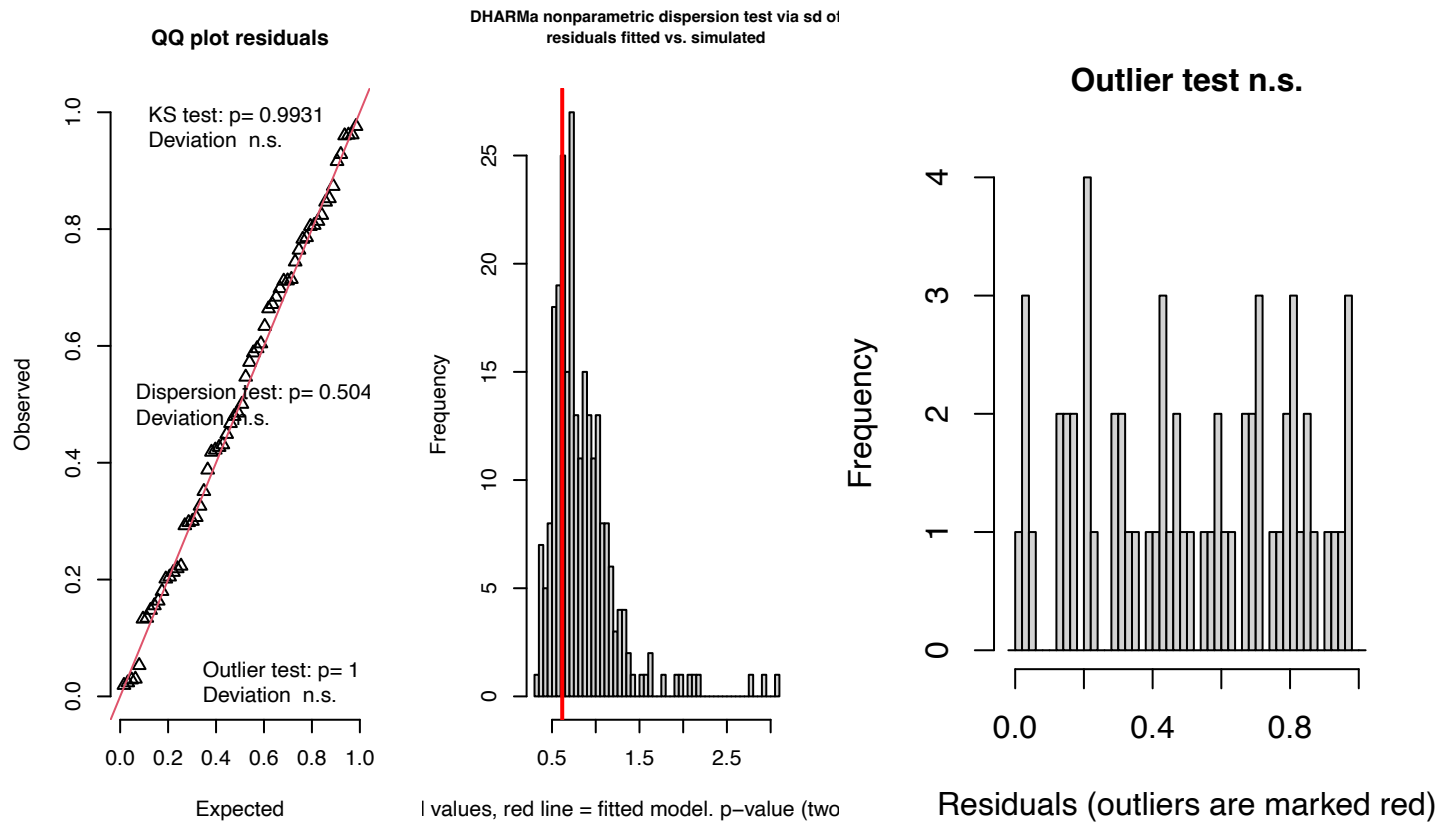
DHARMA zero-inflation test via comparison to expected zeros with simulation under H0 = fitted model



Simulated values, red line = fitted model. p-value (two.sided) = 0.352

```
##
## DHARMA zero-inflation test via comparison to expected zeros with
## simulation under H0 = fitted model
##
## data: simulationOutput
## ratioObsSim = 1.6077, p-value = 0.352
## alternative hypothesis: two.sided
```

```
testResiduals(m5)
```



```
## $uniformity
##
## Exact one-sample Kolmogorov-Smirnov test
##
## data: simulationOutput$scaledResiduals
## D = 0.051938, p-value = 0.9931
## alternative hypothesis: two-sided
##
##
## $dispersion
##
## DHARMA nonparametric dispersion test via sd of residuals fitted vs.
## simulated
##
## data: simulationOutput
## dispersion = 0.72322, p-value = 0.504
## alternative hypothesis: two.sided
##
##
## $outliers
##
## DHARMA bootstrapped outlier test
##
## data: simulationOutput
## outliers at both margin(s) = 0, observations = 62, p-value = 1
## alternative hypothesis: two.sided
## percent confidence interval:
## 0.00000000 0.03225806
```

```

## sample estimates:
## outlier frequency (expected: 0.00516129032258065 )
##
## $uniformity
##
## Exact one-sample Kolmogorov-Smirnov test
##
## data: simulationOutput$scaledResiduals
## D = 0.051938, p-value = 0.9931
## alternative hypothesis: two-sided
##
##
## $dispersion
##
## DHARMA nonparametric dispersion test via sd of residuals fitted vs.
## simulated
##
## data: simulationOutput
## dispersion = 0.72322, p-value = 0.504
## alternative hypothesis: two.sided
##
##
## $outliers
##
## DHARMA bootstrapped outlier test
##
## data: simulationOutput
## outliers at both margin(s) = 0, observations = 62, p-value = 1
## alternative hypothesis: two.sided
## percent confidence interval:
## 0.00000000 0.03225806
## sample estimates:
## outlier frequency (expected: 0.00516129032258065 )
##
##
# calculate the residuals for the model
m6 <- simulateResiduals(fittedModel = interaction_indirect_molbig, plot = F)

residuals(m6)

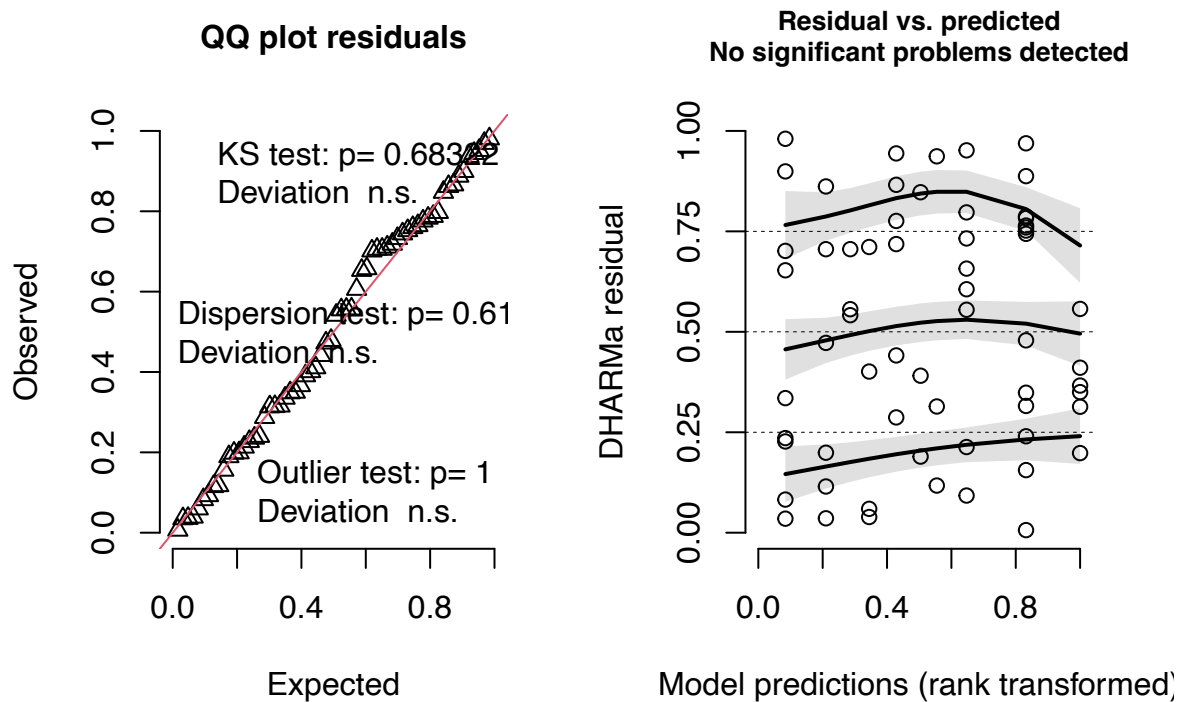
## [1] 0.035839344 0.348431891 0.315502588 0.059395277 0.082475428 0.034956059
## [7] 0.227083146 0.155804008 0.199068810 0.115029702 0.478765456 0.286961464
## [13] 0.235529907 0.117384552 0.556264141 0.198159597 0.472364573 0.401110624
## [19] 0.239991228 0.189434678 0.441257338 0.039025335 0.092461880 0.390734533
## [25] 0.653143120 0.314162778 0.006447799 0.711053665 0.705633000 0.554890821
## [31] 0.213199900 0.366251036 0.775947764 0.980405431 0.657325749 0.718312937
## [37] 0.899270367 0.349878095 0.556928272 0.861780171 0.865821306 0.797238705
## [43] 0.732350619 0.334988296 0.759561152 0.705709483 0.313478425 0.944000000
## [49] 0.701402124 0.605926352 0.751997090 0.410661675 0.951667322 0.936802633
## [55] 0.541365623 0.847418425 0.782994678 0.764479514 0.743832675 0.787447278
## [61] 0.969162982 0.887505717

#plot simulation output
plot(m6)

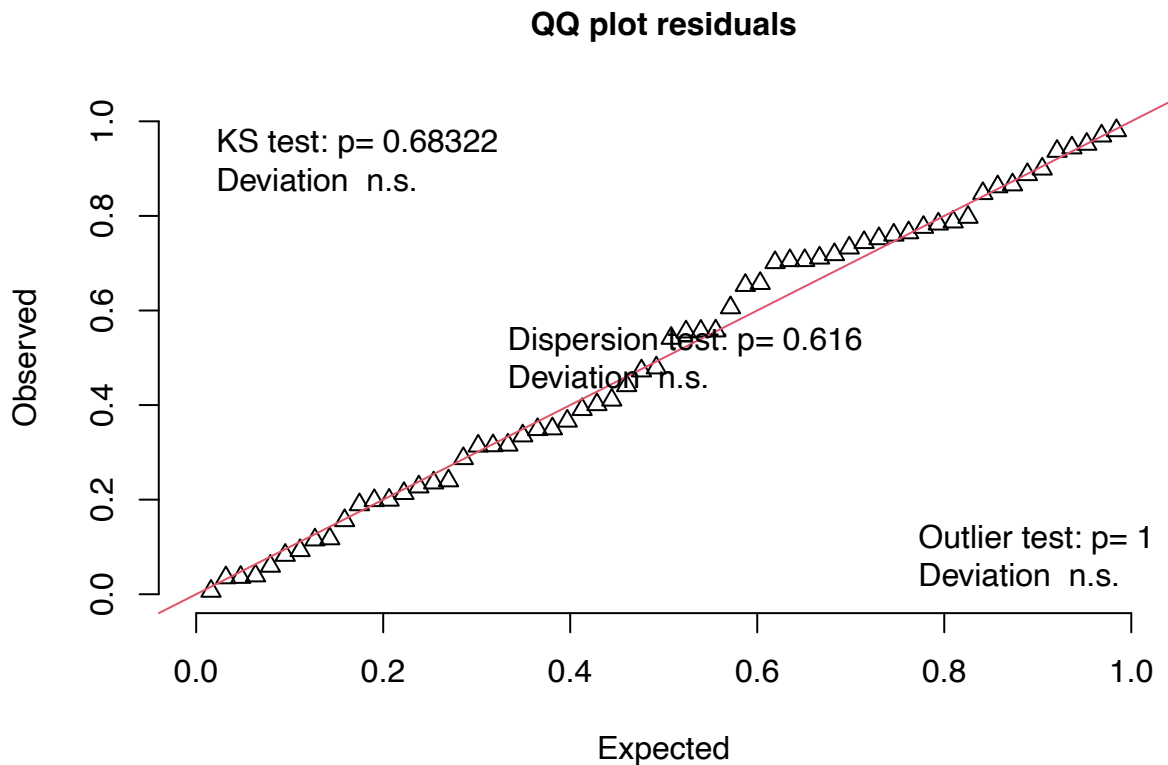
```

```
## Warning in newton(lsp = lsp, X = G$X, y = G$y, Eb = G$Eb, UrS = G$UrS, L = G$L,
## : Fitting terminated with step failure - check results carefully
```

DHARMA residual

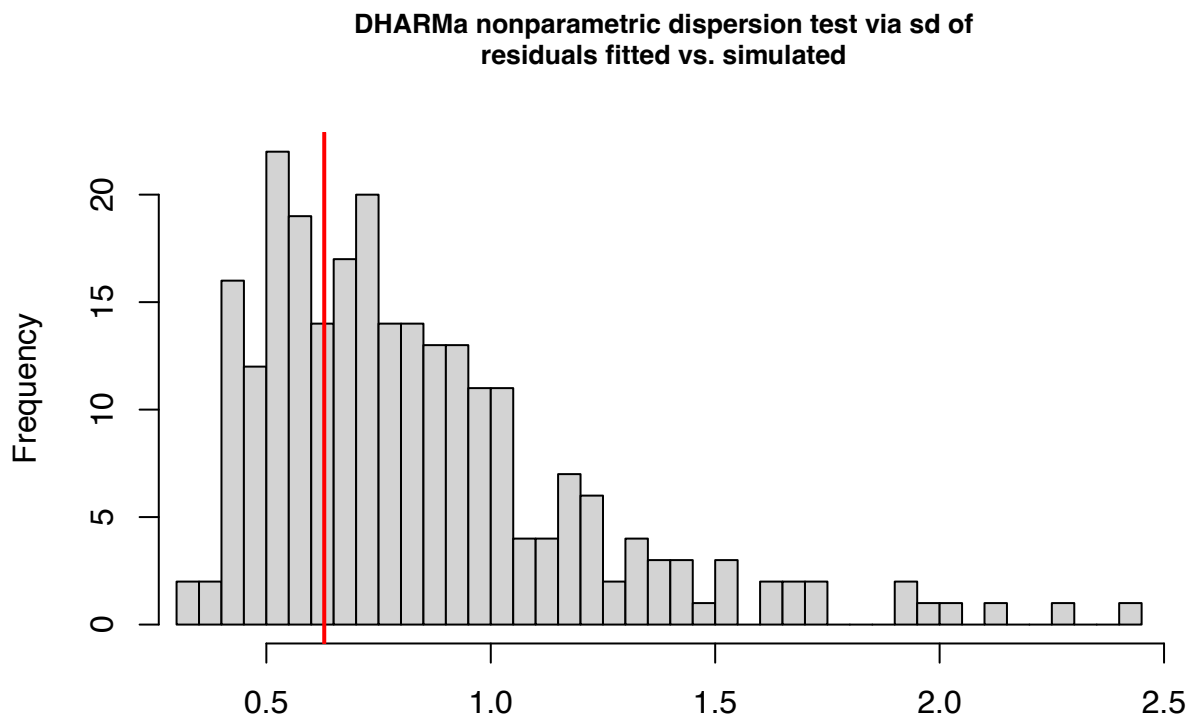


```
#tests if the overall distribution conforms to expectations.
testUniformity(m6)
```



```
##
## Exact one-sample Kolmogorov-Smirnov test
##
## data: simulationOutput$scaledResiduals
## D = 0.088499, p-value = 0.6832
## alternative hypothesis: two-sided
```

```
testDispersion(m6)
```

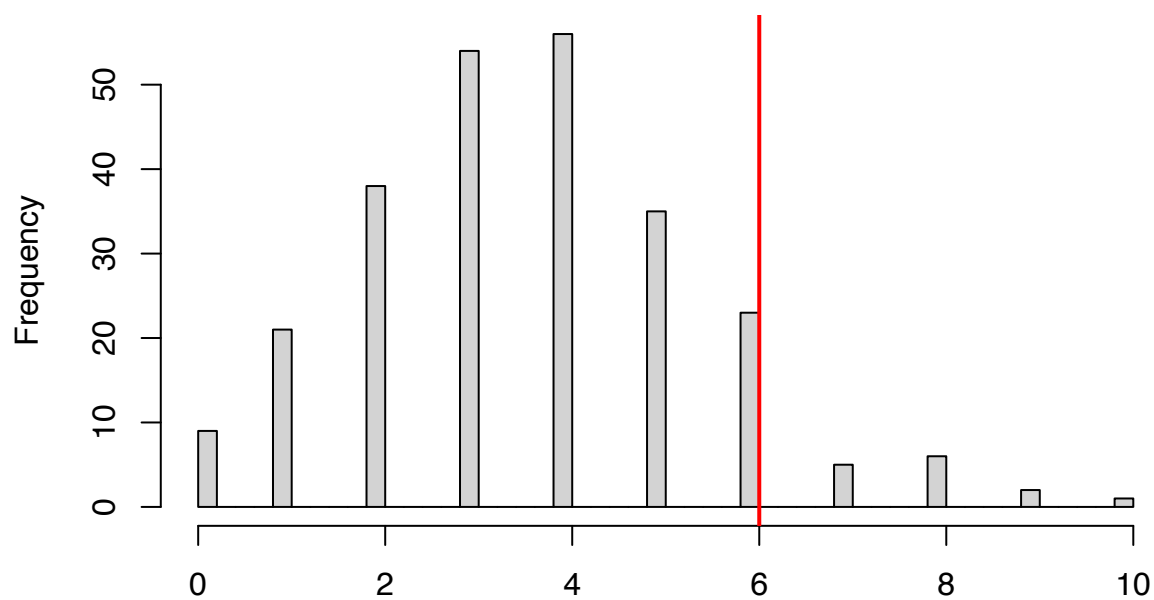


Simulated values, red line = fitted model. p-value (two.sided) = 0.616

```
##
## DHARMa nonparametric dispersion test via sd of residuals fitted vs.
## simulated
##
## data: simulationOutput
## dispersion = 0.74628, p-value = 0.616
## alternative hypothesis: two.sided
```

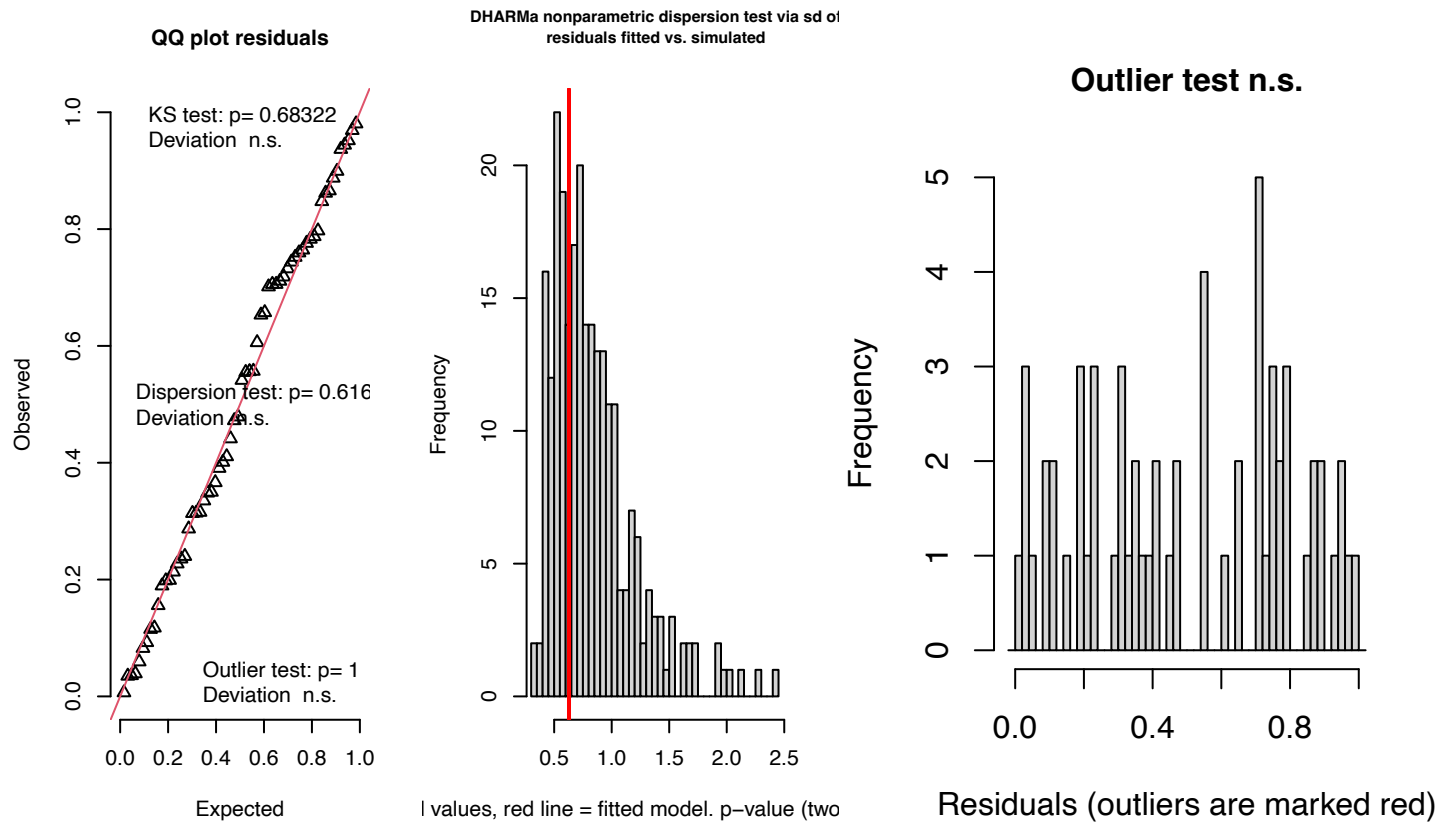
```
testZeroInflation(m6)
```

**DHARMA zero-inflation test via comparison to
expected zeros with simulation under H0 = fitted
model**



Simulated values, red line = fitted model. p-value (two.sided) = 0.296

```
##  
## DHARMA zero-inflation test via comparison to expected zeros with  
## simulation under H0 = fitted model  
##  
## data: simulationOutput  
## ratioObsSim = 1.6538, p-value = 0.296  
## alternative hypothesis: two.sided  
testResiduals(m6)
```



```
## $uniformity
##
## Exact one-sample Kolmogorov-Smirnov test
##
## data: simulationOutput$scaledResiduals
## D = 0.088499, p-value = 0.6832
## alternative hypothesis: two-sided
##
##
## $dispersion
##
## DHARMA nonparametric dispersion test via sd of residuals fitted vs.
## simulated
##
## data: simulationOutput
## dispersion = 0.74628, p-value = 0.616
## alternative hypothesis: two.sided
##
##
## $outliers
##
## DHARMA bootstrapped outlier test
##
## data: simulationOutput
## outliers at both margin(s) = 0, observations = 62, p-value = 1
## alternative hypothesis: two.sided
## percent confidence interval:
## 0.00000000 0.02459677
```

```

## sample estimates:
## outlier frequency (expected: 0.00596774193548387 )
##
## $uniformity
##
## Exact one-sample Kolmogorov-Smirnov test
##
## data: simulationOutput$scaledResiduals
## D = 0.088499, p-value = 0.6832
## alternative hypothesis: two-sided
##
##
## $dispersion
##
## DHARMA nonparametric dispersion test via sd of residuals fitted vs.
## simulated
##
## data: simulationOutput
## dispersion = 0.74628, p-value = 0.616
## alternative hypothesis: two.sided
##
##
## $outliers
##
## DHARMA bootstrapped outlier test
##
## data: simulationOutput
## outliers at both margin(s) = 0, observations = 62, p-value = 1
## alternative hypothesis: two.sided
## percent confidence interval:
## 0.00000000 0.02459677
## sample estimates:
## outlier frequency (expected: 0.00596774193548387 )
##
##
# calculate the residuals for the model
m11 <- simulateResiduals(fittedModel = additive_direct_molbig, plot = F)

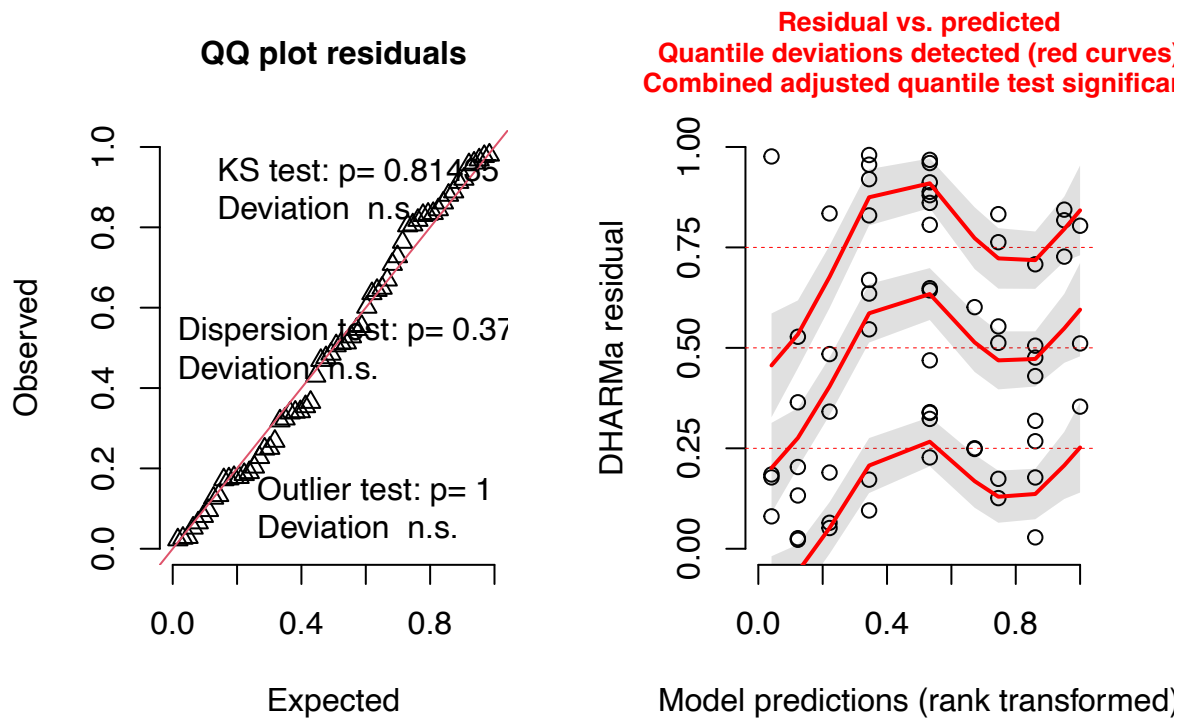
residuals(m11)

## [1] 0.06464251 0.64309885 0.34007336 0.17762268 0.09535389 0.17195721
## [7] 0.63510232 0.22730413 0.05159969 0.48454648 0.32321834 0.02634497
## [13] 0.66914253 0.25016480 0.81757109 0.17407662 0.18974377 0.08065868
## [19] 0.33813005 0.51104168 0.02235430 0.97641105 0.42980895 0.35380024
## [25] 0.54609574 0.24828988 0.64858788 0.18485406 0.34131451 0.02838080
## [31] 0.50539698 0.51249926 0.13261828 0.91984750 0.31837598 0.52758108
## [37] 0.82975326 0.55335682 0.12608616 0.83456650 0.20344910 0.17742718
## [43] 0.47568674 0.95600000 0.46891412 0.72713254 0.76299015 0.36452533
## [49] 0.98000000 0.70800000 0.80625526 0.83278739 0.26760527 0.60136773
## [55] 0.84400000 0.80400000 0.86083497 0.88000000 0.91200000 0.96800000
## [61] 0.88831136 0.96000000

#plot simulation output
plot(m11)

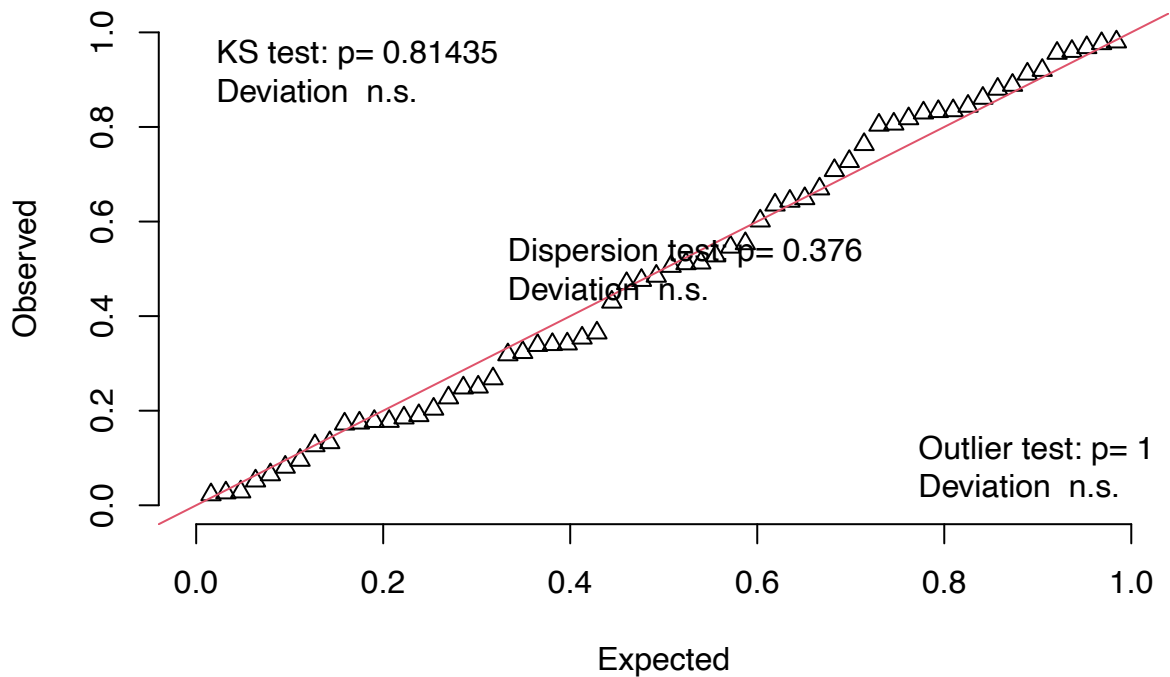
```

DHARMa residual



#tests if the overall distribution conforms to expectations.
`testUniformity(m11)`

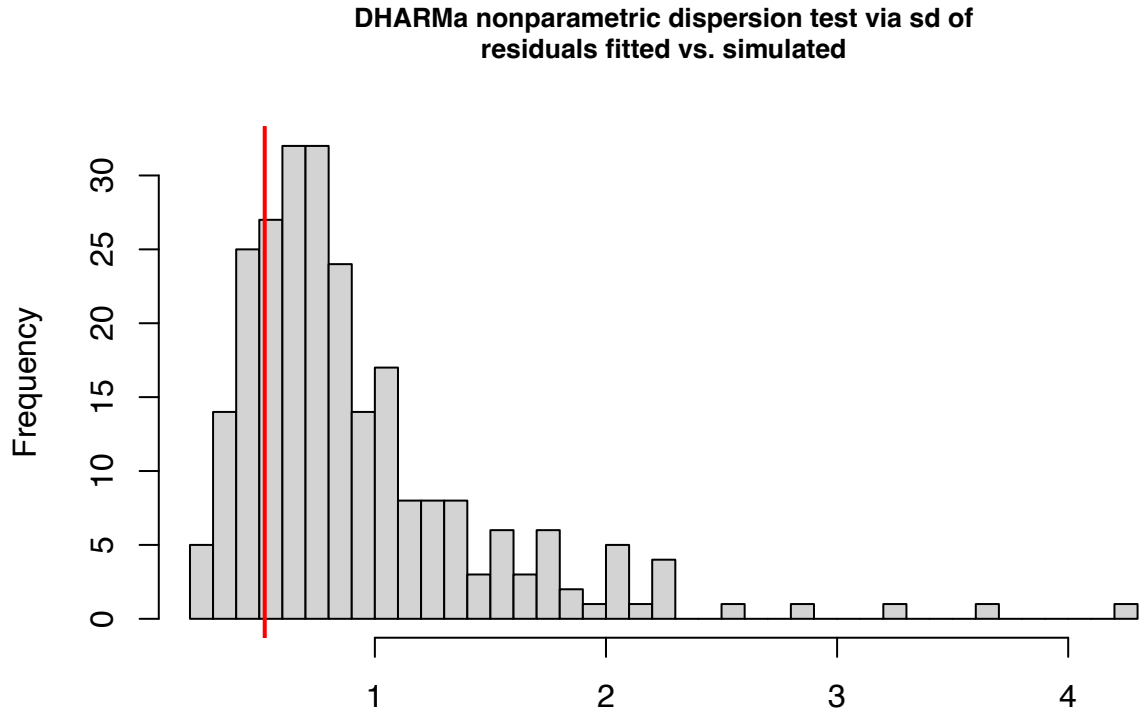
QQ plot residuals



 ## Exact one-sample Kolmogorov-Smirnov test

```
##
## data: simulationOutput$scaledResiduals
## D = 0.078194, p-value = 0.8144
## alternative hypothesis: two-sided
```

```
testDispersion(m11)
```

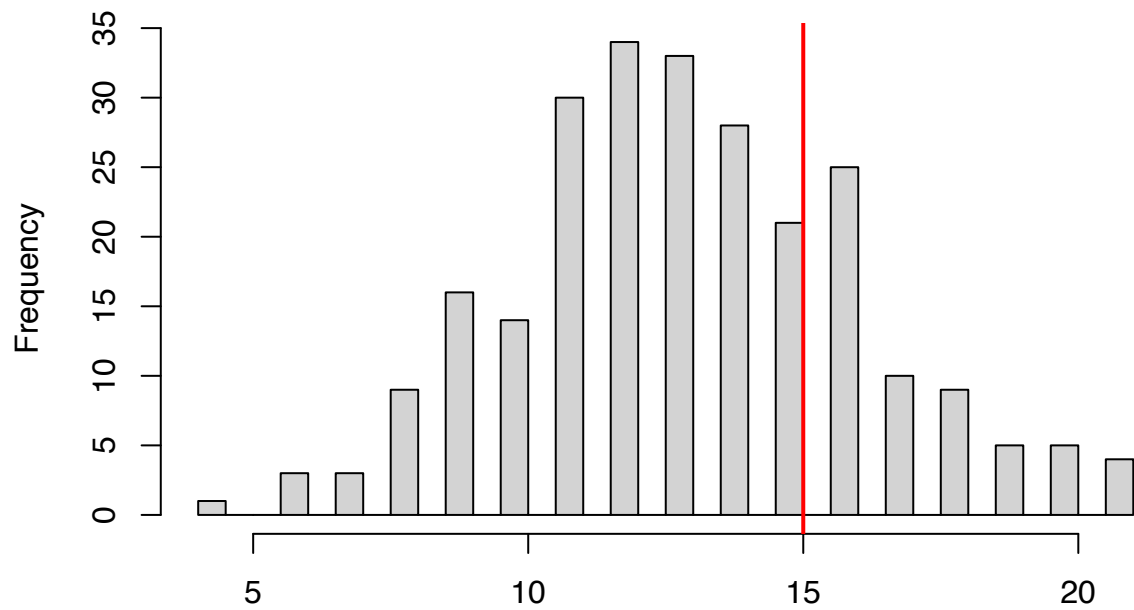


Simulated values, red line = fitted model. p-value (two.sided) = 0.376

```
##
## DHARMA nonparametric dispersion test via sd of residuals fitted vs.
## simulated
##
## data: simulationOutput
## dispersion = 0.56701, p-value = 0.376
## alternative hypothesis: two.sided
```

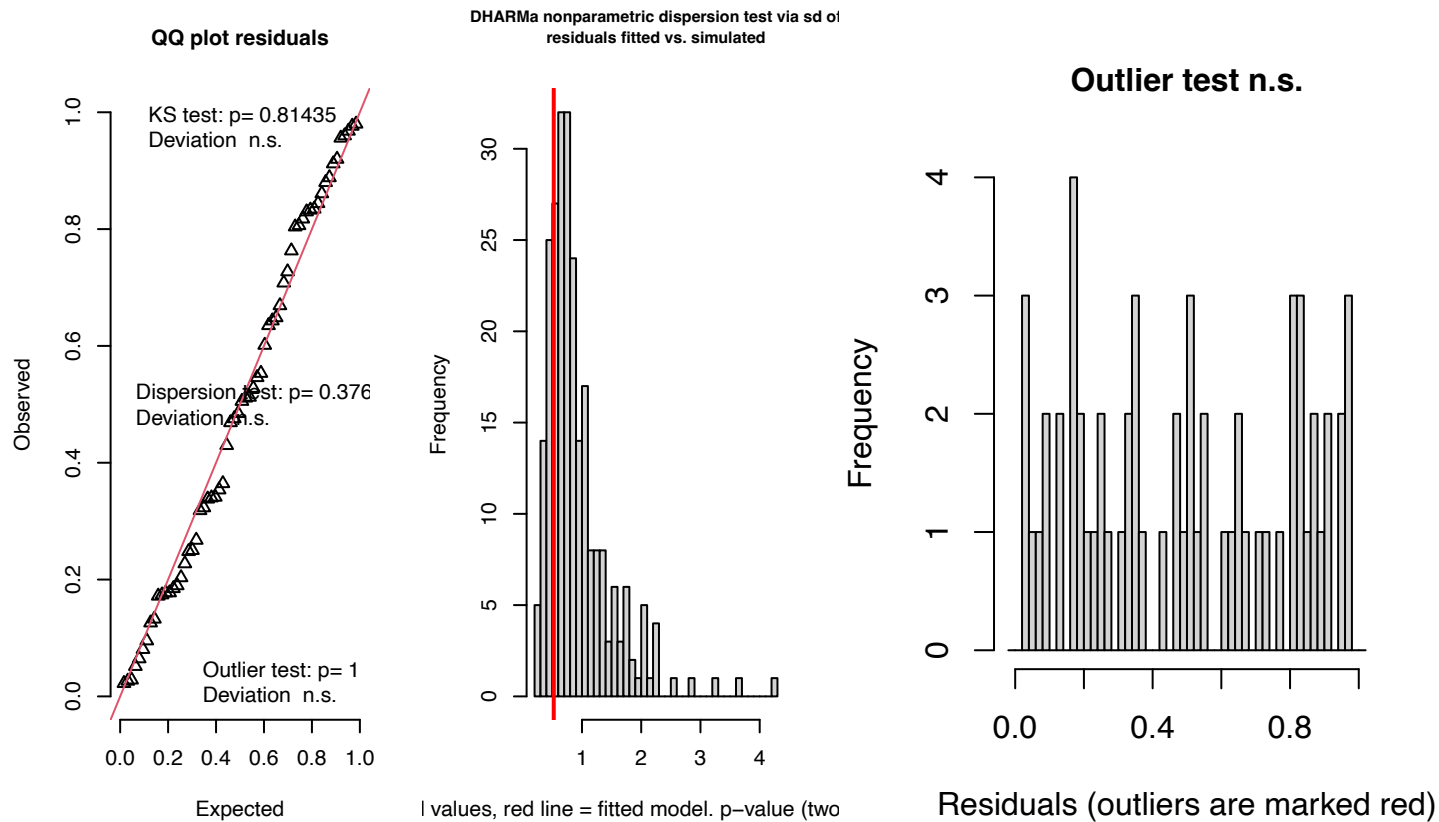
```
testZeroInflation(m11)
```

DHARMA zero-inflation test via comparison to expected zeros with simulation under H0 = fitted model



Simulated values, red line = fitted model. p-value (two.sided) = 0.632

```
##
## DHARMA zero-inflation test via comparison to expected zeros with
## simulation under H0 = fitted model
##
## data: simulationOutput
## ratioObsSim = 1.1419, p-value = 0.632
## alternative hypothesis: two.sided
testResiduals(m11)
```



```
## $uniformity
##
##   Exact one-sample Kolmogorov-Smirnov test
##
## data:  simulationOutput$scaledResiduals
## D = 0.078194, p-value = 0.8144
## alternative hypothesis: two-sided
##
##
## $dispersion
##
##   DHARMA nonparametric dispersion test via sd of residuals fitted vs.
##   simulated
##
## data:  simulationOutput
## dispersion = 0.56701, p-value = 0.376
## alternative hypothesis: two.sided
##
##
## $outliers
##
##   DHARMA bootstrapped outlier test
##
## data:  simulationOutput
## outliers at both margin(s) = 0, observations = 62, p-value = 1
## alternative hypothesis: two.sided
## percent confidence interval:
## 0.00000000 0.02459677
```

```

## sample estimates:
## outlier frequency (expected: 0.00564516129032258 )
##
## 0

## $uniformity
##
## Exact one-sample Kolmogorov-Smirnov test
##
## data: simulationOutput$scaledResiduals
## D = 0.078194, p-value = 0.8144
## alternative hypothesis: two-sided
##
##
## $dispersion
##
## DHARMA nonparametric dispersion test via sd of residuals fitted vs.
## simulated
##
## data: simulationOutput
## dispersion = 0.56701, p-value = 0.376
## alternative hypothesis: two.sided
##
##
## $outliers
##
## DHARMA bootstrapped outlier test
##
## data: simulationOutput
## outliers at both margin(s) = 0, observations = 62, p-value = 1
## alternative hypothesis: two.sided
## percent confidence interval:
## 0.00000000 0.02459677
## sample estimates:
## outlier frequency (expected: 0.00564516129032258 )
##
## 0

# calculate the residuals for the model
m12 <- simulateResiduals(fittedModel = interaction_direct_molbig, plot = F)

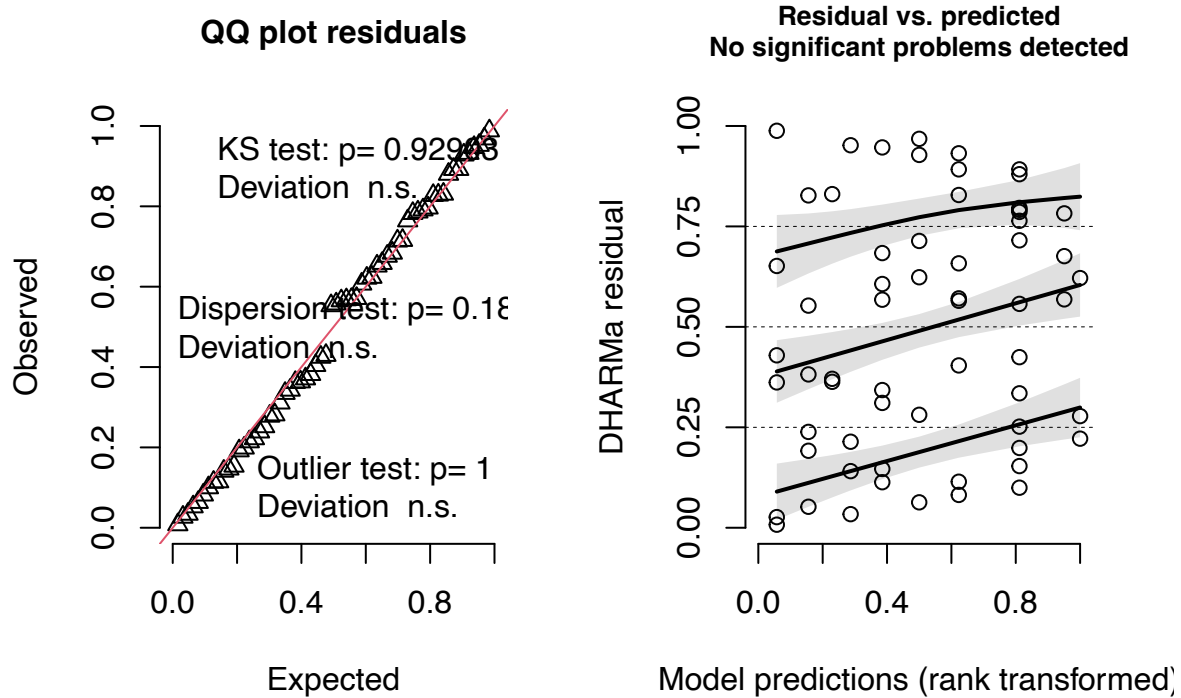
residuals(m12)

## [1] 0.007631732 0.557237344 0.251710674 0.214146536 0.114571160 0.081731229
## [7] 0.565009628 0.099704783 0.361773795 0.651614455 0.153509375 0.381674578
## [13] 0.570977818 0.363259421 0.676000000 0.281162808 0.026513389 0.140930494
## [19] 0.198605666 0.277942776 0.191638996 0.952000000 0.606682191 0.221786496
## [25] 0.404024468 0.370713998 0.424572309 0.033858344 0.429209395 0.113386092
## [31] 0.683634857 0.623499280 0.238592373 0.828233281 0.342672641 0.827203004
## [37] 0.658083744 0.713847819 0.063303290 0.988000000 0.052228322 0.146371304
## [43] 0.567599905 0.892000000 0.334614944 0.568453384 0.927956595 0.552917562
## [49] 0.932000000 0.946637925 0.715357202 0.968000000 0.310694146 0.830428594
## [55] 0.782462923 0.621410110 0.791315999 0.764000000 0.796000000 0.892000000
## [61] 0.786067222 0.880000000

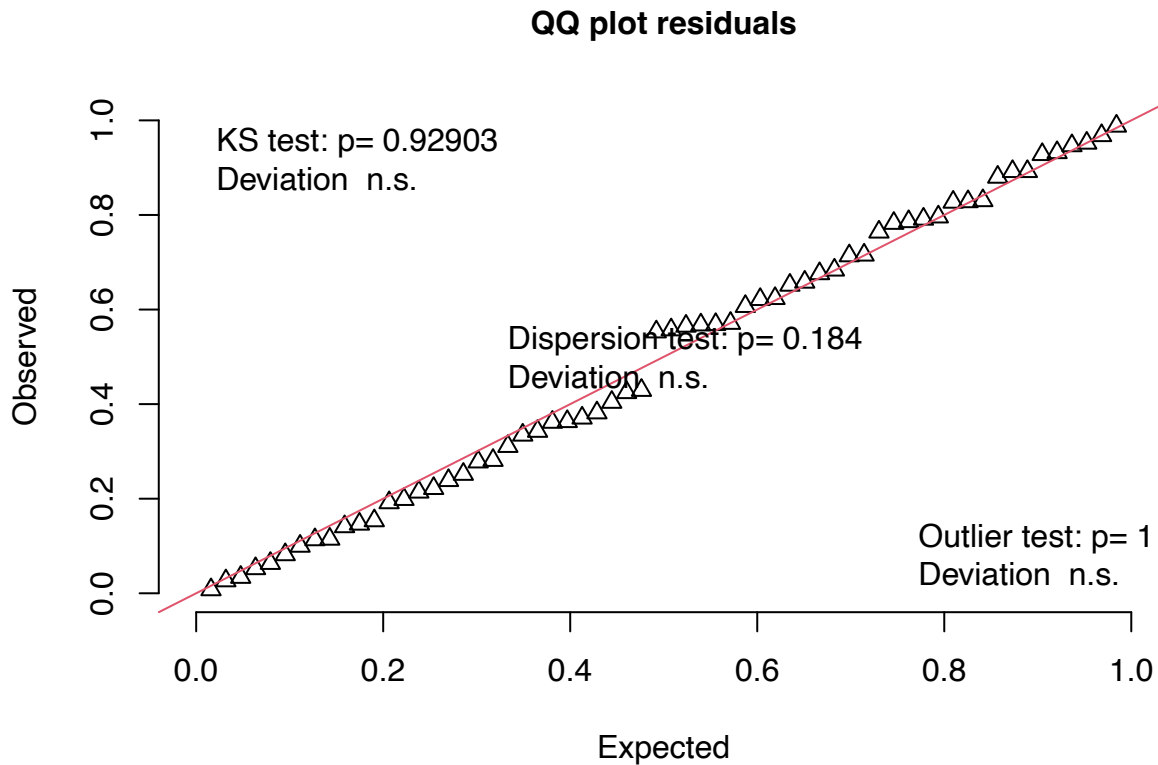
#plot simulation output
plot(m12)

```

DHARMA residual



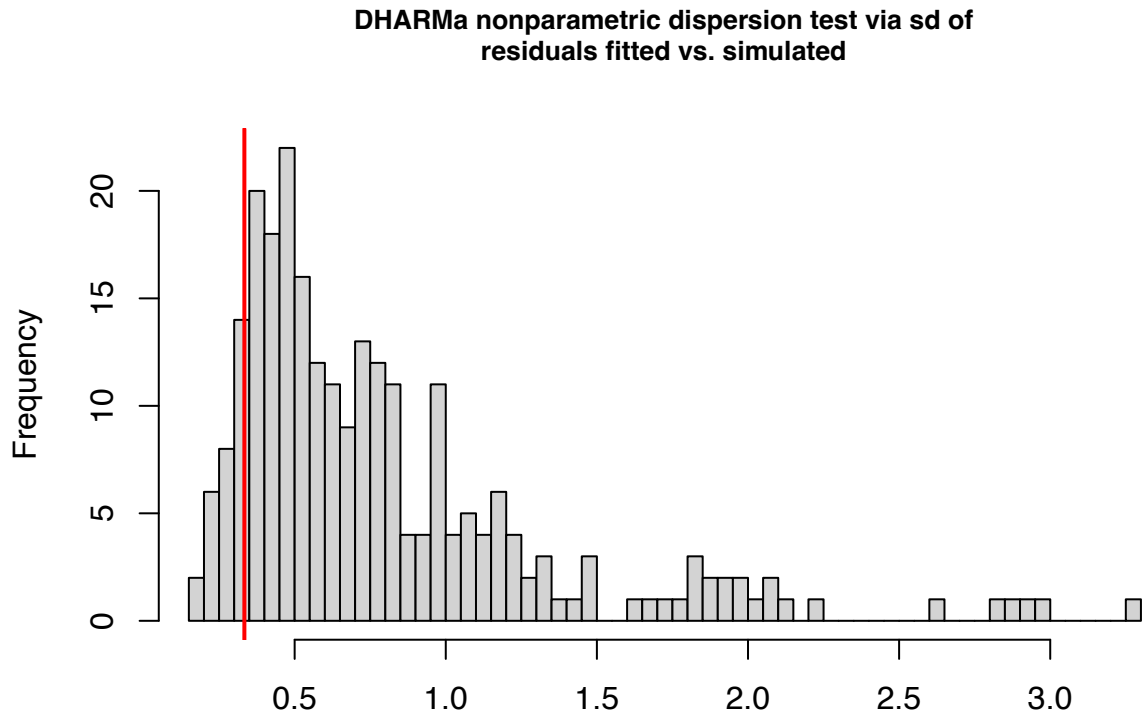
#tests if the overall distribution conforms to expectations.
`testUniformity(m12)`



 ## Asymptotic one-sample Kolmogorov-Smirnov test

```
##
## data:  simulationOutput$scaledResiduals
## D = 0.069047, p-value = 0.929
## alternative hypothesis: two-sided
```

```
testDispersion(m12)
```

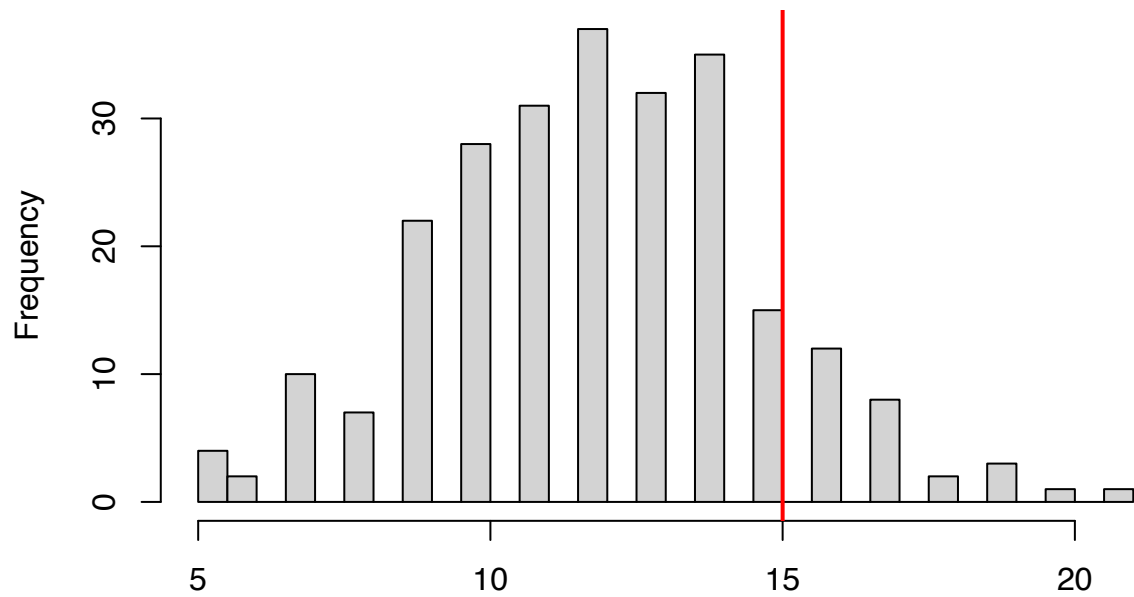


Simulated values, red line = fitted model. p-value (two.sided) = 0.184

```
##
## DHARMA nonparametric dispersion test via sd of residuals fitted vs.
## simulated
##
## data:  simulationOutput
## dispersion = 0.41794, p-value = 0.184
## alternative hypothesis: two.sided
```

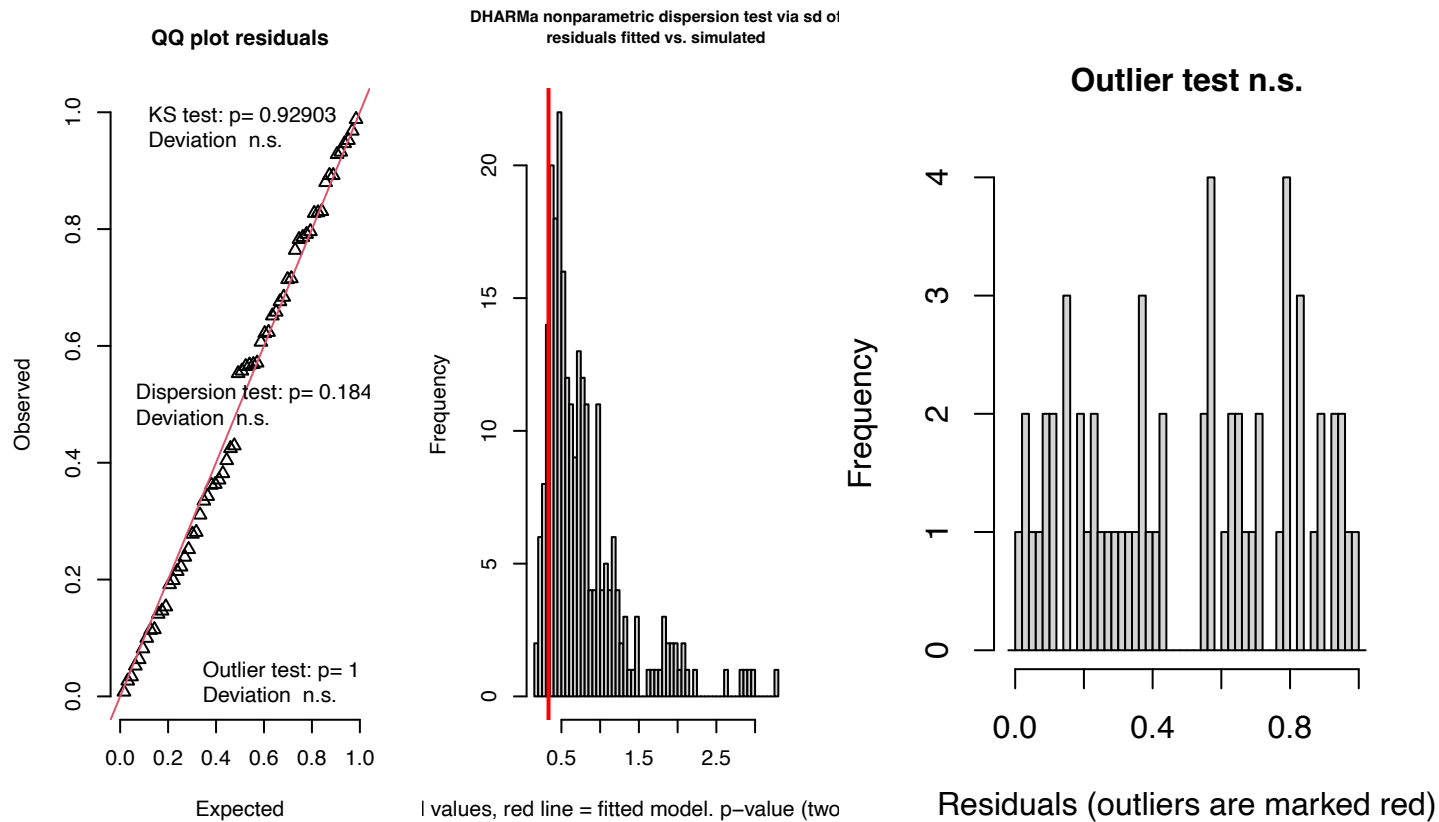
```
testZeroInflation(m12)
```

DHARMA zero-inflation test via comparison to expected zeros with simulation under H0 = fitted model



Simulated values, red line = fitted model. p-value (two.sided) = 0.336

```
##
## DHARMA zero-inflation test via comparison to expected zeros with
## simulation under H0 = fitted model
##
## data: simulationOutput
## ratioObsSim = 1.2442, p-value = 0.336
## alternative hypothesis: two.sided
testResiduals(m12)
```



```
## $uniformity
##
## Asymptotic one-sample Kolmogorov-Smirnov test
##
## data: simulationOutput$scaledResiduals
## D = 0.069047, p-value = 0.929
## alternative hypothesis: two-sided
##
##
## $dispersion
##
## DHARMA nonparametric dispersion test via sd of residuals fitted vs.
## simulated
##
## data: simulationOutput
## dispersion = 0.41794, p-value = 0.184
## alternative hypothesis: two.sided
##
##
## $outliers
##
## DHARMA bootstrapped outlier test
##
## data: simulationOutput
## outliers at both margin(s) = 0, observations = 62, p-value = 1
## alternative hypothesis: two.sided
## percent confidence interval:
## 0.00000000 0.02459677
```

```

## sample estimates:
## outlier frequency (expected: 0.00419354838709677 )
##
## $uniformity
##
## Asymptotic one-sample Kolmogorov-Smirnov test
##
## data: simulationOutput$scaledResiduals
## D = 0.069047, p-value = 0.929
## alternative hypothesis: two-sided
##
##
## $dispersion
##
## DHARMA nonparametric dispersion test via sd of residuals fitted vs.
## simulated
##
## data: simulationOutput
## dispersion = 0.41794, p-value = 0.184
## alternative hypothesis: two.sided
##
##
## $outliers
##
## DHARMA bootstrapped outlier test
##
## data: simulationOutput
## outliers at both margin(s) = 0, observations = 62, p-value = 1
## alternative hypothesis: two.sided
## percent confidence interval:
## 0.00000000 0.02459677
## sample estimates:
## outlier frequency (expected: 0.00419354838709677 )
##
##
# calculate the residuals for the model
m23 <- simulateResiduals(fittedModel = additive_harass_molbig, plot = F)

residuals(m23)

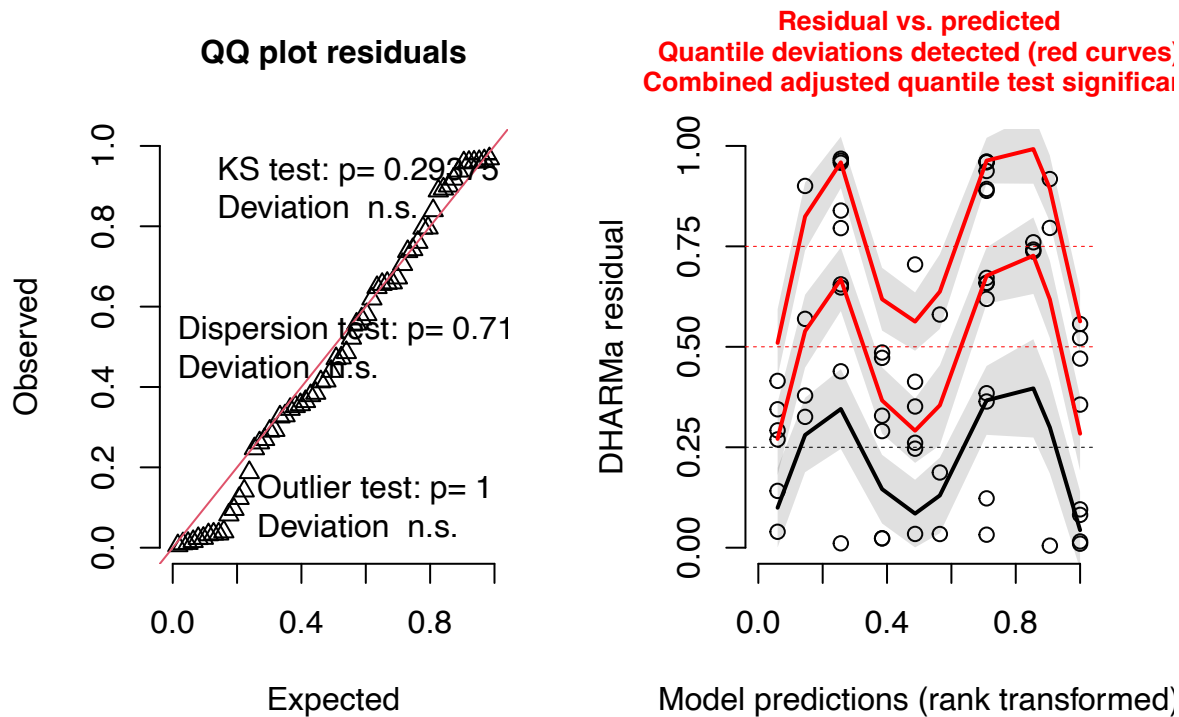
## [1] 0.351450126 0.384502244 0.363931954 0.569557295 0.010892322 0.439079460
## [7] 0.795440061 0.671578680 0.261053817 0.034240459 0.619288494 0.141415947
## [13] 0.647654040 0.187162655 0.917481513 0.472525098 0.246430229 0.378645393
## [19] 0.658174983 0.737215519 0.039661141 0.900447263 0.010002368 0.742045740
## [25] 0.655525143 0.580402804 0.888115955 0.325632107 0.413067293 0.356149966
## [31] 0.521639449 0.485785275 0.415170504 0.963224580 0.081703237 0.344560319
## [37] 0.958066297 0.290261349 0.022967637 0.705031258 0.292040890 0.015527509
## [43] 0.470586329 0.968199423 0.122822473 0.005163656 0.328464597 0.269504140
## [49] 0.839099844 0.556717014 0.959336448 0.023594657 0.095463715 0.034051870
## [55] 0.795918137 0.760165549 0.659287565 0.032406558 0.961207041 0.937673172
## [61] 0.892944021 0.960000000

#plot simulation output
plot(m23)

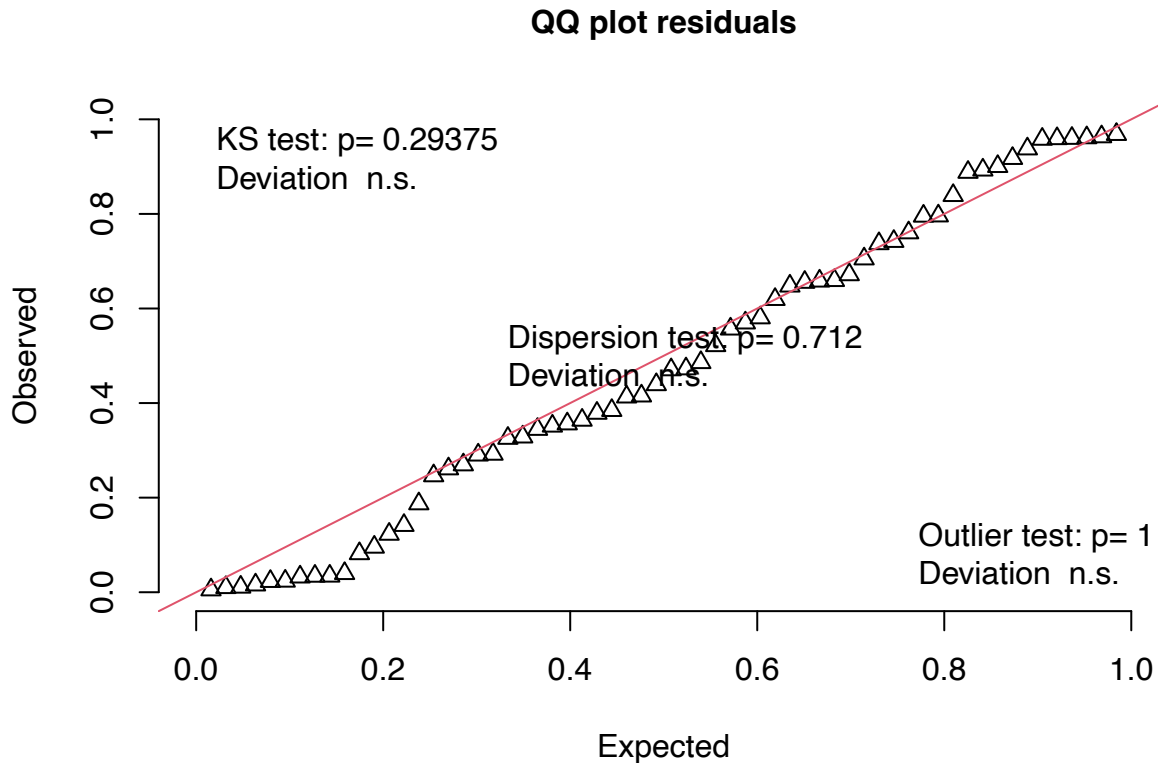
```

```
## Warning in newton(lsp = lsp, X = G$X, y = G$y, Eb = G$Eb, UrS = G$UrS, L = G$L,
## : Fitting terminated with step failure - check results carefully
```

DHARMA residual



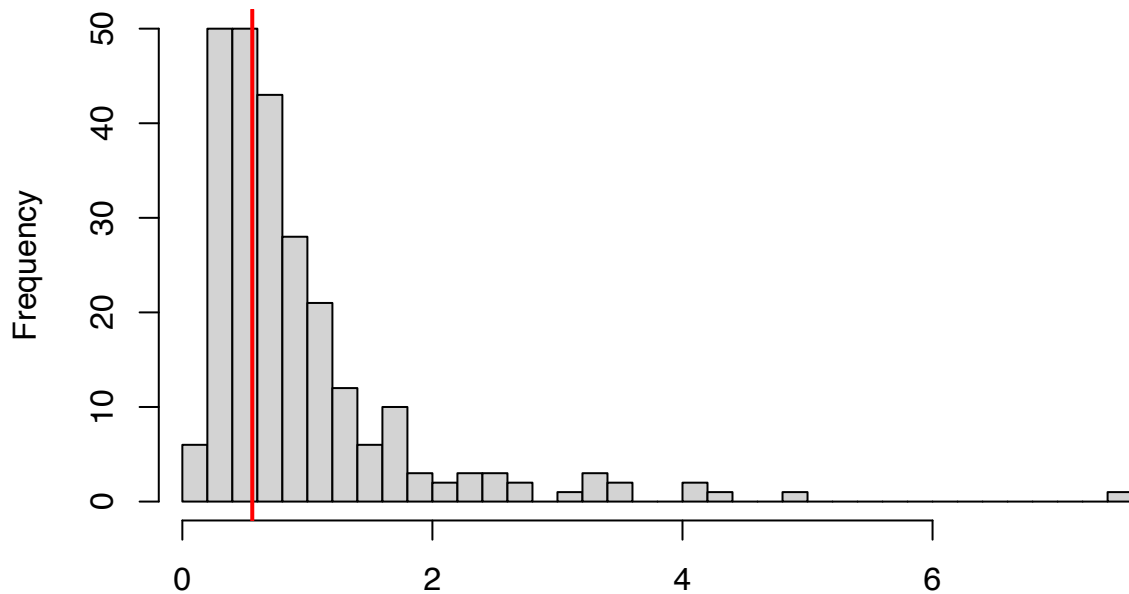
```
#tests if the overall distribution conforms to expectations.
testUniformity(m23)
```



```
##
## Exact one-sample Kolmogorov-Smirnov test
##
## data: simulationOutput$scaledResiduals
## D = 0.12163, p-value = 0.2938
## alternative hypothesis: two-sided
```

```
testDispersion(m23)
```

DHARMa nonparametric dispersion test via sd of residuals fitted vs. simulated

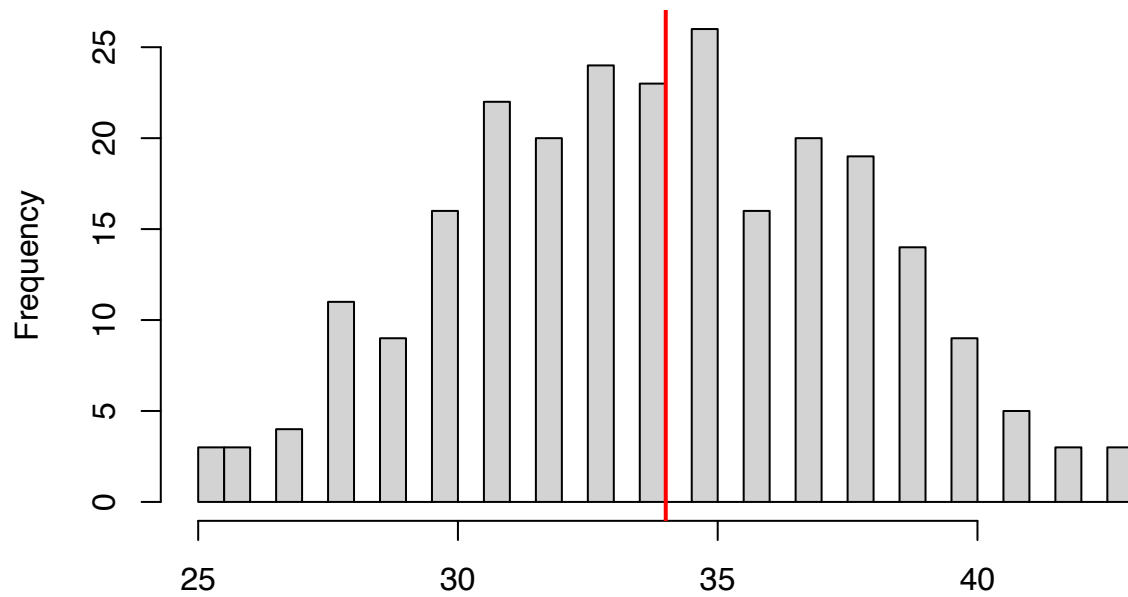


Simulated values, red line = fitted model. p-value (two.sided) = 0.712

```
##
## DHARMa nonparametric dispersion test via sd of residuals fitted vs.
## simulated
##
## data: simulationOutput
## dispersion = 0.59394, p-value = 0.712
## alternative hypothesis: two.sided
```

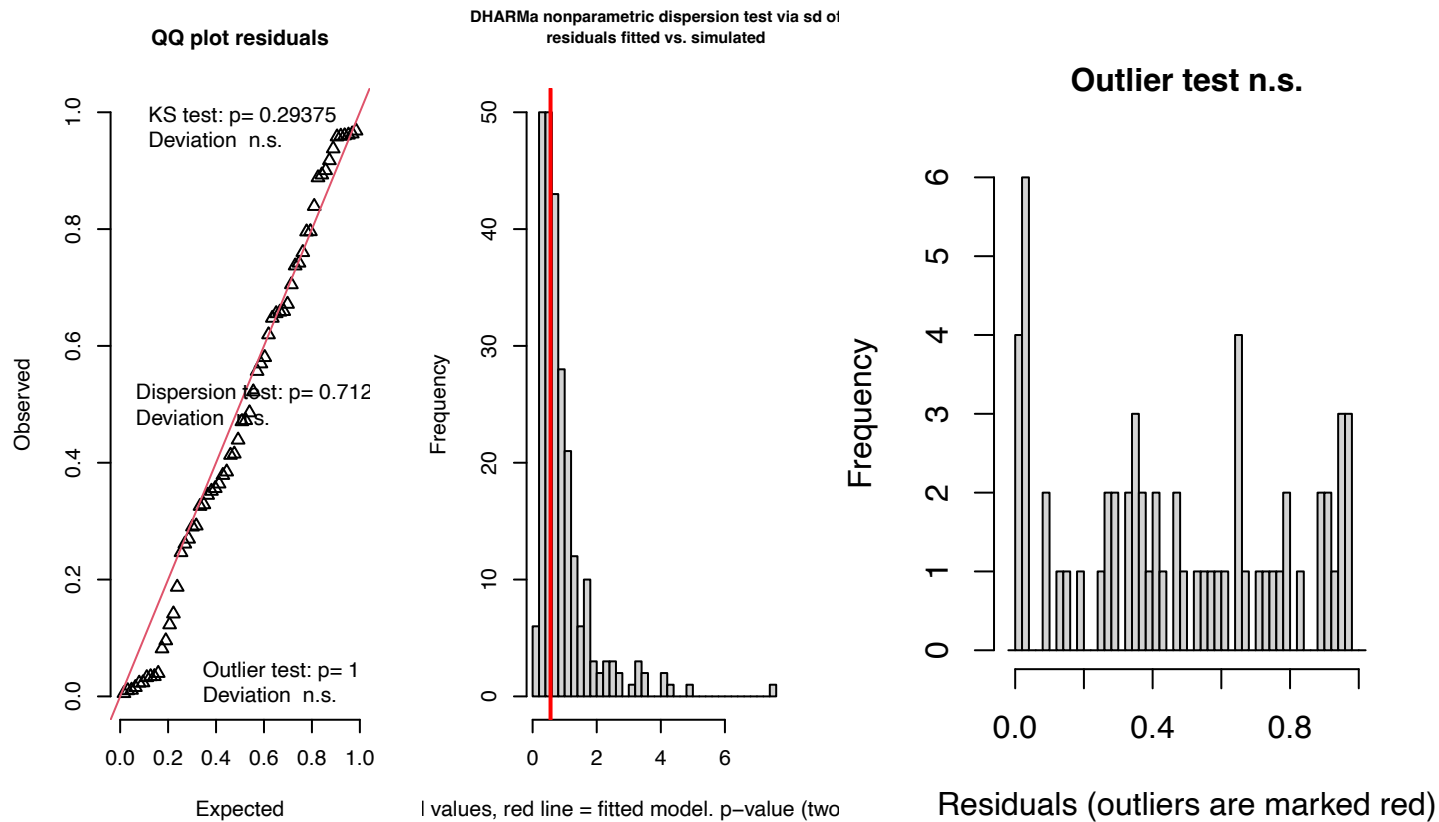
```
testZeroInflation(m23)
```

DHARMA zero-inflation test via comparison to expected zeros with simulation under H0 = fitted model



Simulated values, red line = fitted model. p-value (two.sided) = 1

```
##
## DHARMA zero-inflation test via comparison to expected zeros with
## simulation under H0 = fitted model
##
## data: simulationOutput
## ratioObsSim = 0.99765, p-value = 1
## alternative hypothesis: two.sided
testResiduals(m23)
```



```
## $uniformity
##
## Exact one-sample Kolmogorov-Smirnov test
##
## data: simulationOutput$scaledResiduals
## D = 0.12163, p-value = 0.2938
## alternative hypothesis: two-sided
##
##
## $dispersion
##
## DHARMA nonparametric dispersion test via sd of residuals fitted vs.
## simulated
##
## data: simulationOutput
## dispersion = 0.59394, p-value = 0.712
## alternative hypothesis: two.sided
##
##
## $outliers
##
## DHARMA bootstrapped outlier test
##
## data: simulationOutput
## outliers at both margin(s) = 0, observations = 62, p-value = 1
## alternative hypothesis: two.sided
## percent confidence interval:
## 0.00000000 0.03225806
```

```

## sample estimates:
## outlier frequency (expected: 0.00451612903225806 )
##
## 0

## $uniformity
##
## Exact one-sample Kolmogorov-Smirnov test
##
## data: simulationOutput$scaledResiduals
## D = 0.12163, p-value = 0.2938
## alternative hypothesis: two-sided
##
##
## $dispersion
##
## DHARMA nonparametric dispersion test via sd of residuals fitted vs.
## simulated
##
## data: simulationOutput
## dispersion = 0.59394, p-value = 0.712
## alternative hypothesis: two.sided
##
##
## $outliers
##
## DHARMA bootstrapped outlier test
##
## data: simulationOutput
## outliers at both margin(s) = 0, observations = 62, p-value = 1
## alternative hypothesis: two.sided
## percent confidence interval:
## 0.00000000 0.03225806
## sample estimates:
## outlier frequency (expected: 0.00451612903225806 )
##
## 0

# calculate the residuals for the model
m24 <- simulateResiduals(fittedModel = interaction_harass_molbig, plot = F)

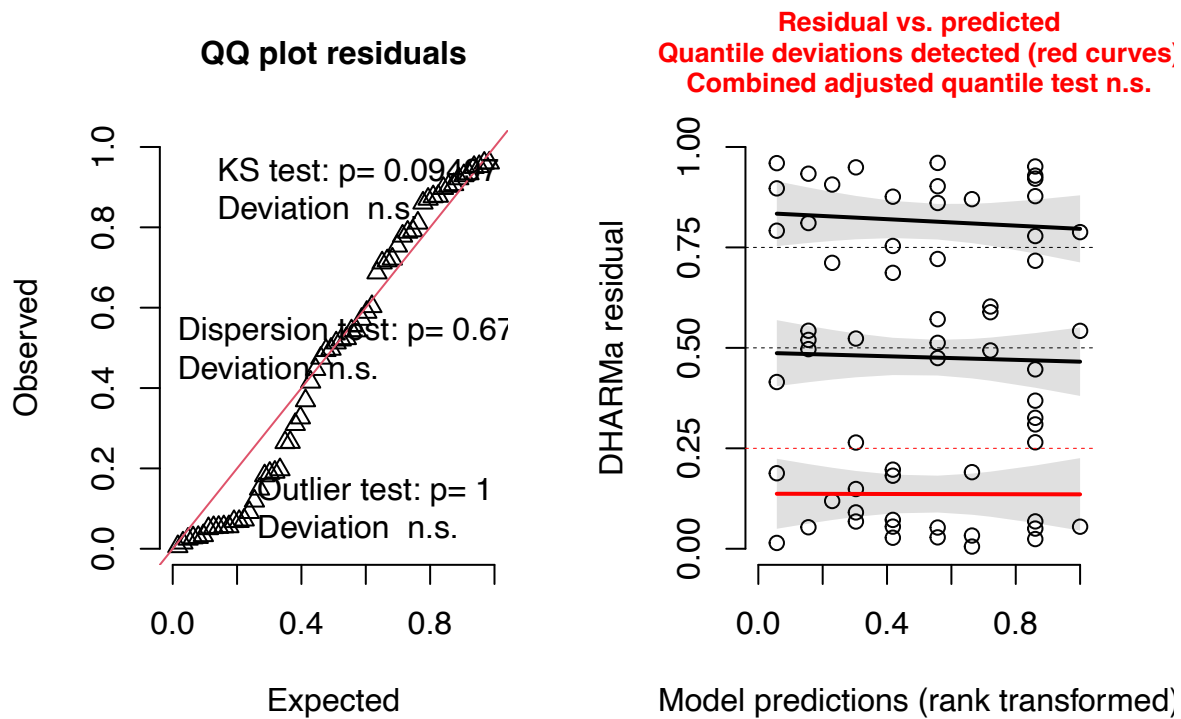
residuals(m24)

## [1] 0.264401483 0.050220597 0.264951692 0.005482038 0.052829103 0.028393441
## [7] 0.571307379 0.325916515 0.148709805 0.090505193 0.368605895 0.959788374
## [13] 0.512476637 0.118912485 0.788077689 0.810709360 0.067818328 0.190646529
## [19] 0.309526248 0.493561867 0.188008401 0.870118857 0.071852877 0.588709558
## [25] 0.474647335 0.906552567 0.716833588 0.032847824 0.523195856 0.027448381
## [31] 0.876291230 0.933486774 0.414978140 0.960357661 0.181788030 0.791544014
## [37] 0.860616988 0.542593278 0.519775703 0.949298867 0.014492947 0.055070861
## [43] 0.753604111 0.902269688 0.068086412 0.054922436 0.053498319 0.896958985
## [49] 0.721234417 0.686735500 0.928680764 0.496829460 0.196891122 0.711613331
## [55] 0.542377276 0.602952457 0.446495078 0.024098576 0.921101530 0.877581285
## [61] 0.778322761 0.951455518

#plot simulation output
plot(m24)

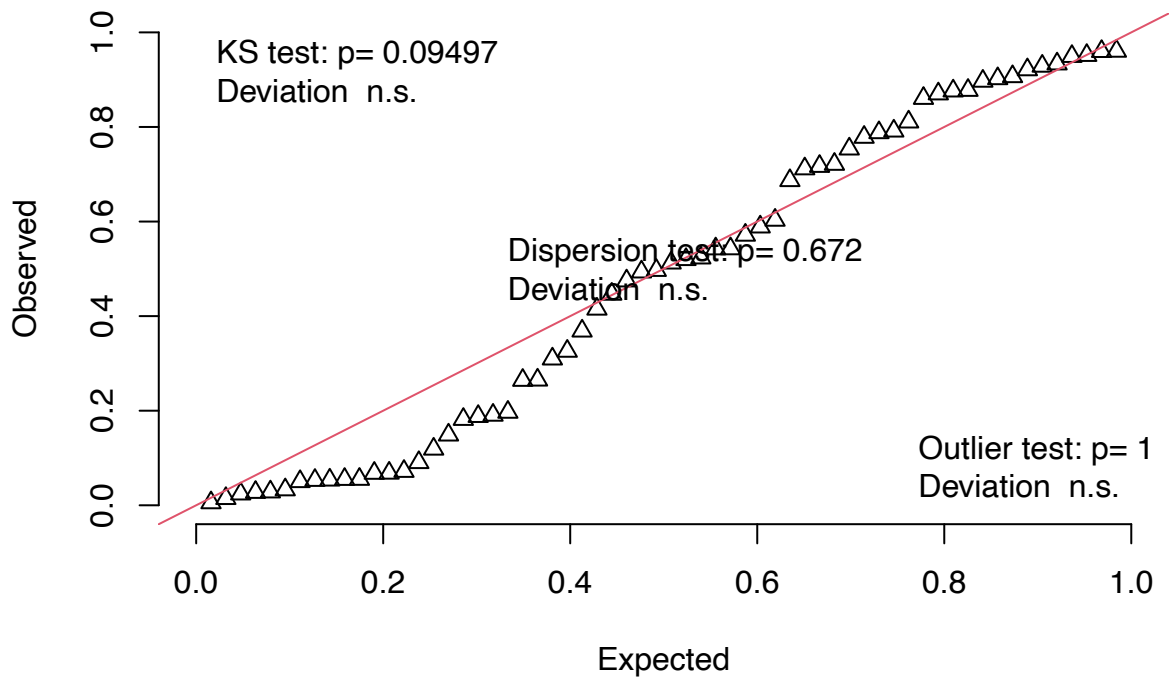
```

DHARMa residual



#tests if the overall distribution conforms to expectations.
`testUniformity(m24)`

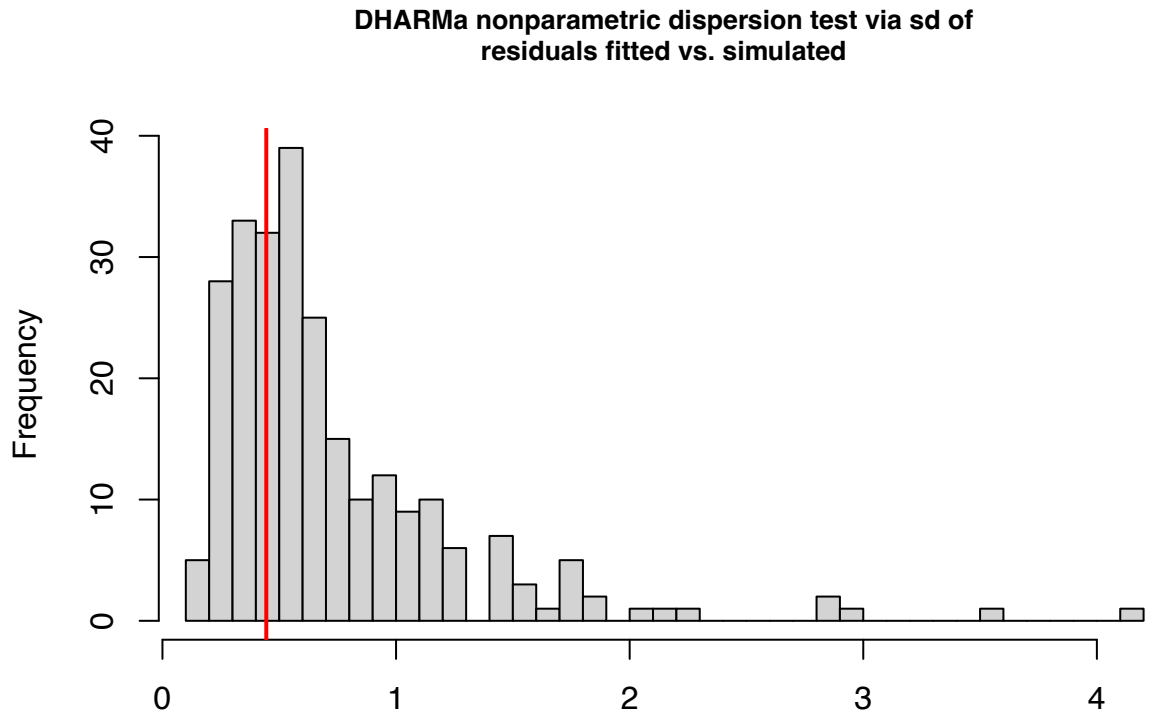
QQ plot residuals



 ## Exact one-sample Kolmogorov-Smirnov test

```
##
## data:  simulationOutput$scaledResiduals
## D = 0.15395, p-value = 0.09497
## alternative hypothesis: two-sided
```

```
testDispersion(m24)
```

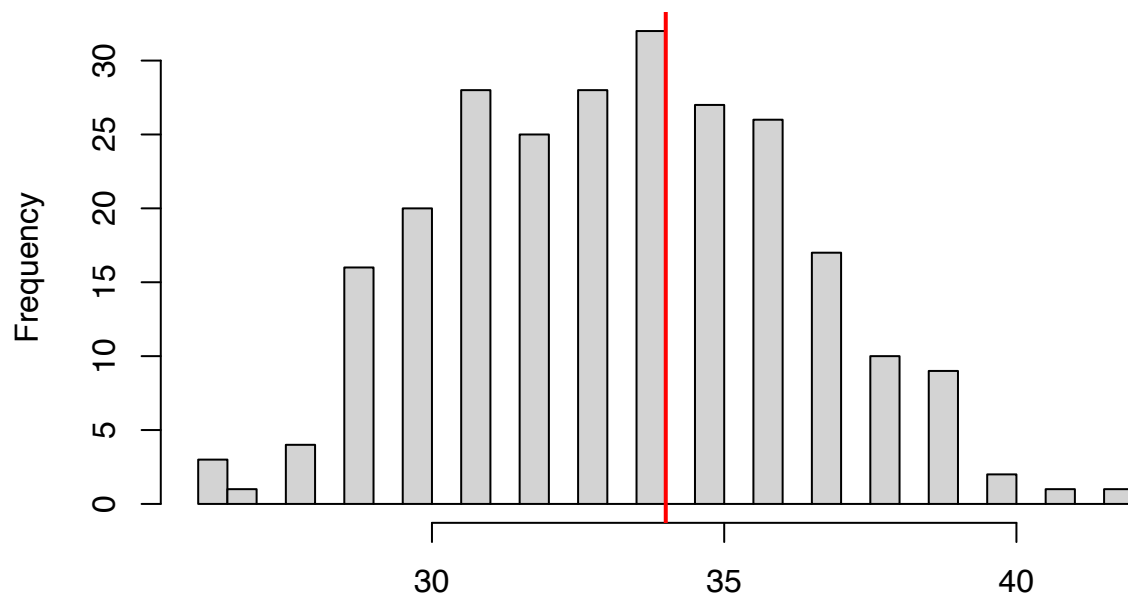


Simulated values, red line = fitted model. p-value (two.sided) = 0.672

```
##
## DHARMa nonparametric dispersion test via sd of residuals fitted vs.
## simulated
##
## data:  simulationOutput
## dispersion = 0.60801, p-value = 0.672
## alternative hypothesis: two.sided
```

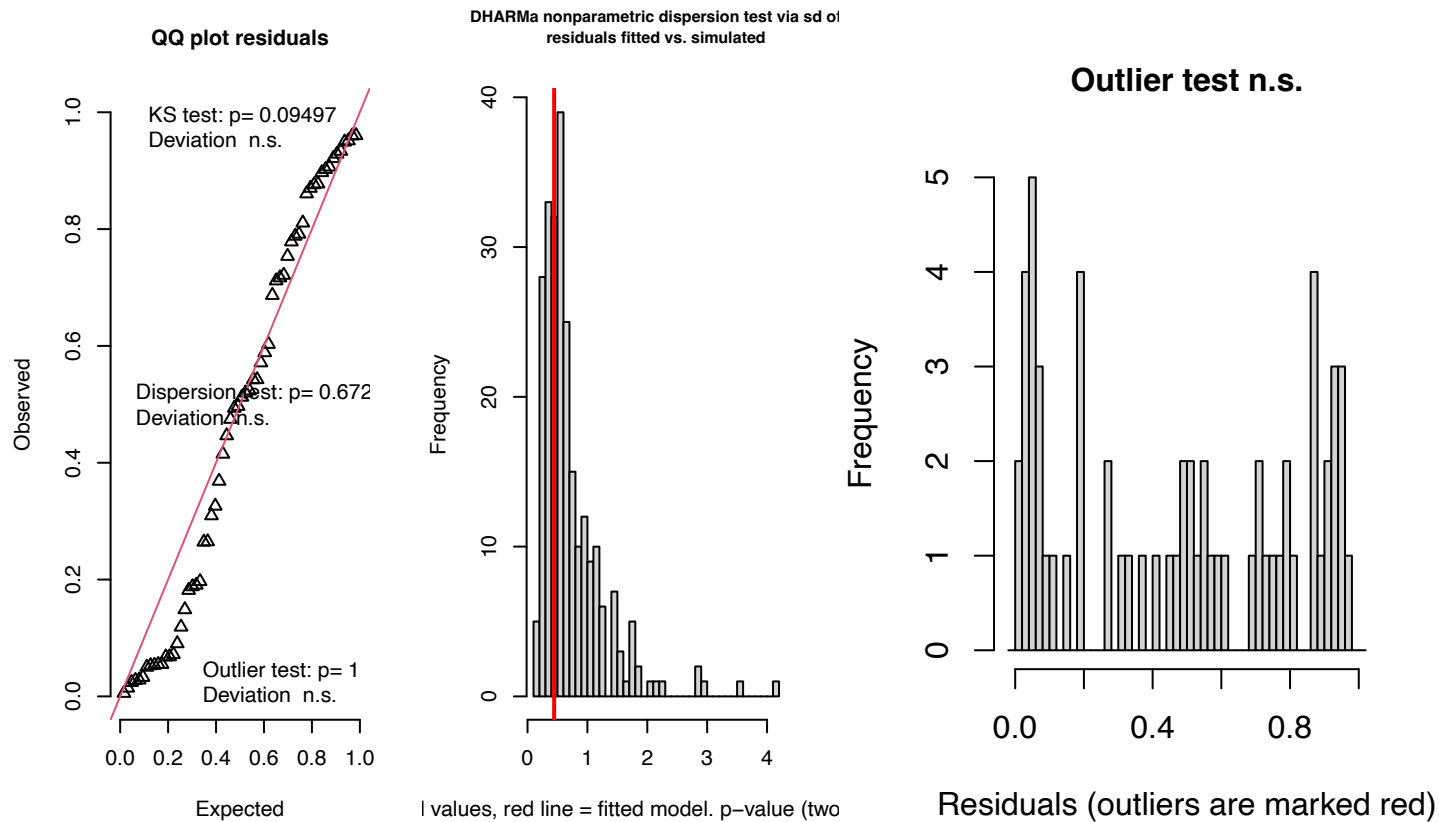
```
testZeroInflation(m24)
```

DHARMA zero-inflation test via comparison to expected zeros with simulation under H0 = fitted model



Simulated values, red line = fitted model. p-value (two.sided) = 1

```
##
## DHARMA zero-inflation test via comparison to expected zeros with
## simulation under H0 = fitted model
##
## data: simulationOutput
## ratioObsSim = 1.0161, p-value = 1
## alternative hypothesis: two.sided
testResiduals(m24)
```



```
## $uniformity
##
## Exact one-sample Kolmogorov-Smirnov test
##
## data: simulationOutput$scaledResiduals
## D = 0.15395, p-value = 0.09497
## alternative hypothesis: two-sided
##
##
## $dispersion
##
## DHARMA nonparametric dispersion test via sd of residuals fitted vs.
## simulated
##
## data: simulationOutput
## dispersion = 0.60801, p-value = 0.672
## alternative hypothesis: two.sided
##
##
## $outliers
##
## DHARMA bootstrapped outlier test
##
## data: simulationOutput
## outliers at both margin(s) = 0, observations = 62, p-value = 1
## alternative hypothesis: two.sided
## percent confidence interval:
## 0.00000000 0.03225806
```

```

## sample estimates:
## outlier frequency (expected: 0.00516129032258065 )
##
## $uniformity
##
## Exact one-sample Kolmogorov-Smirnov test
##
## data: simulationOutput$scaledResiduals
## D = 0.15395, p-value = 0.09497
## alternative hypothesis: two-sided
##
##
## $dispersion
##
## DHARMA nonparametric dispersion test via sd of residuals fitted vs.
## simulated
##
## data: simulationOutput
## dispersion = 0.60801, p-value = 0.672
## alternative hypothesis: two.sided
##
##
## $outliers
##
## DHARMA bootstrapped outlier test
##
## data: simulationOutput
## outliers at both margin(s) = 0, observations = 62, p-value = 1
## alternative hypothesis: two.sided
## percent confidence interval:
## 0.00000000 0.03225806
## sample estimates:
## outlier frequency (expected: 0.00516129032258065 )
##
##
# calculate the residuals for the model
m17 <- simulateResiduals(fittedModel = additive_resp_molbig, plot = F)

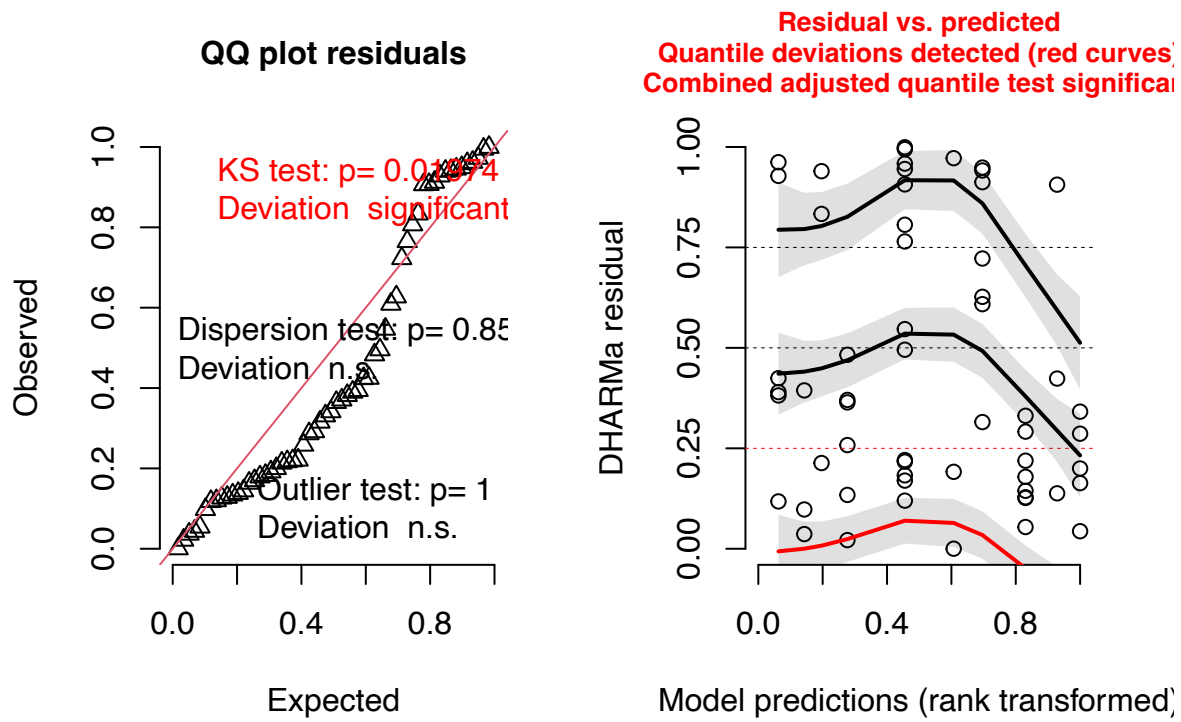
residuals(m17)

## [1] 0.994634859 0.999251685 0.948745332 0.945180725 0.199816380 0.286440676
## [7] 0.958653963 0.482779197 0.912357393 0.097817315 0.972110075 0.962105236
## [13] 0.341248172 0.423634241 0.906859250 0.939731695 0.258038940 0.906241700
## [19] 0.291508499 0.833417401 0.722372014 0.394161913 0.806436187 0.137976655
## [25] 0.163353077 0.126361523 0.330973669 0.928000000 0.370430613 0.941481946
## [31] 0.219170592 0.364253850 0.627016539 0.389869000 0.424501837 0.043685478
## [37] 0.133994776 0.144291941 0.128224560 0.315320299 0.546673967 0.191518187
## [43] 0.117801411 0.021526399 0.608798501 0.054098243 0.764841807 0.381503581
## [49] 0.179002398 0.036523593 0.000143534 0.213333891 0.119754161 0.169678030
## [55] 0.216876080 0.182773776 0.495396055 0.220840281

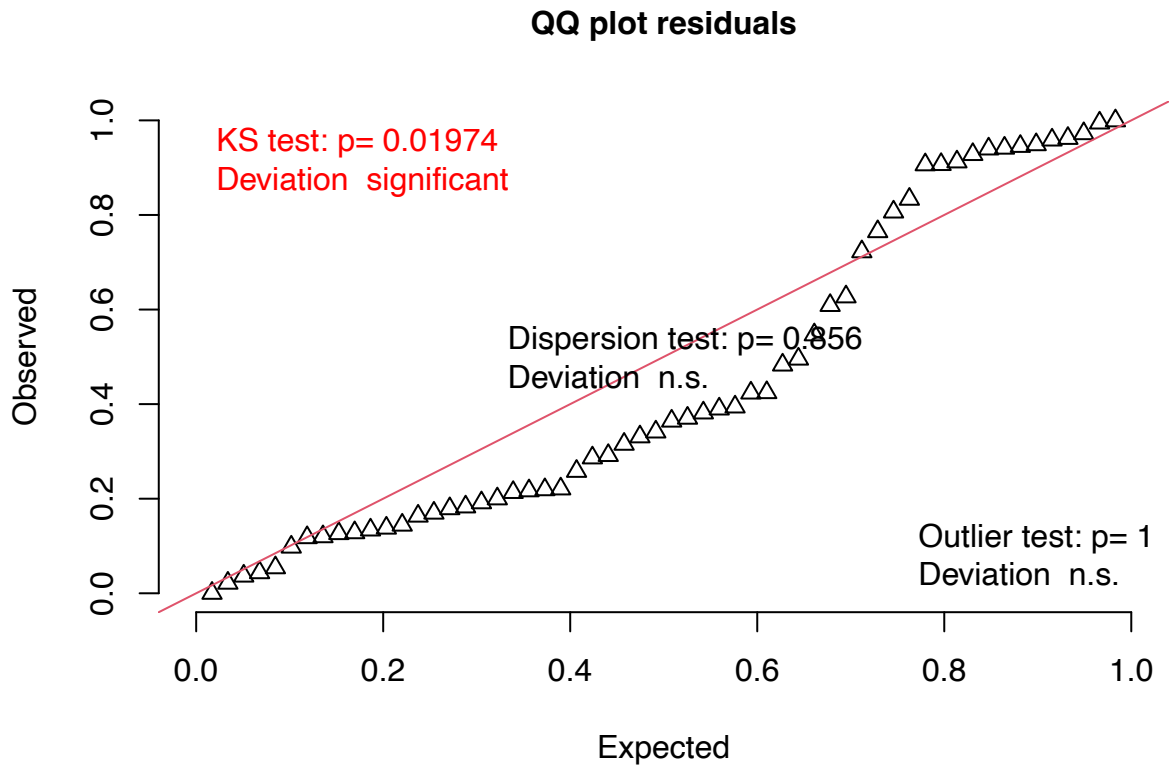
#plot simulation output
plot(m17)

```

DHARMA residual



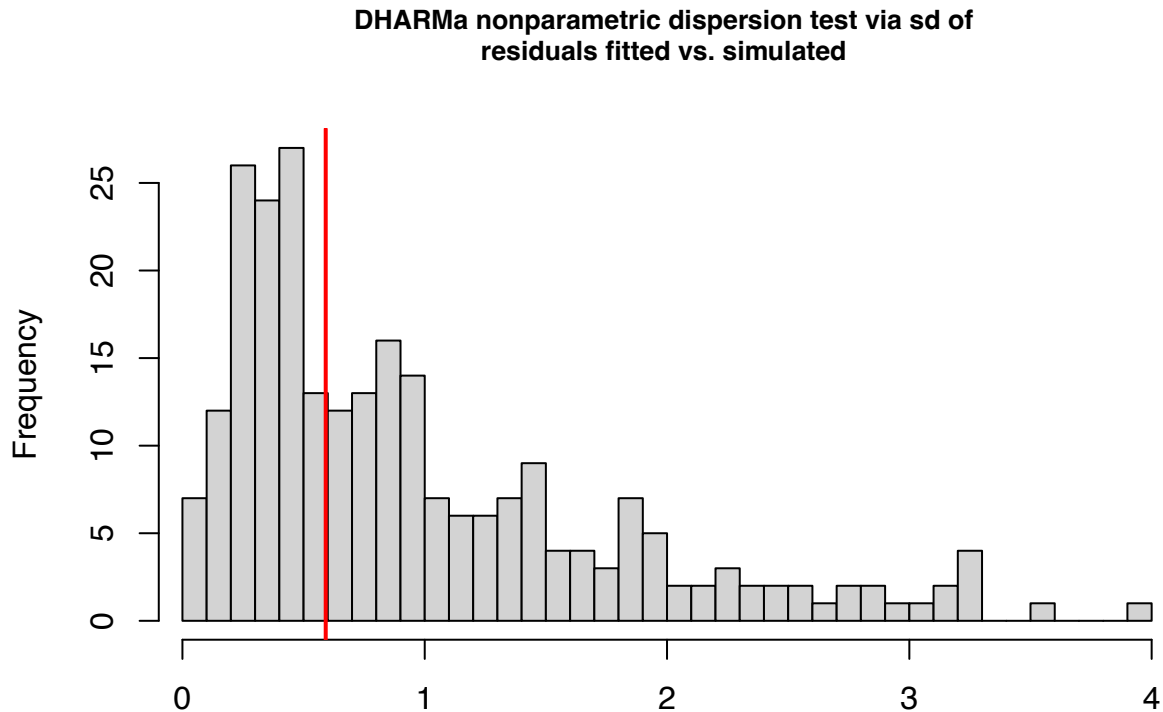
#tests if the overall distribution conforms to expectations.
`testUniformity(m17)`



 ## Exact one-sample Kolmogorov-Smirnov test

```
##
## data: simulationOutput$scaledResiduals
## D = 0.19619, p-value = 0.01974
## alternative hypothesis: two-sided
```

```
testDispersion(m17)
```

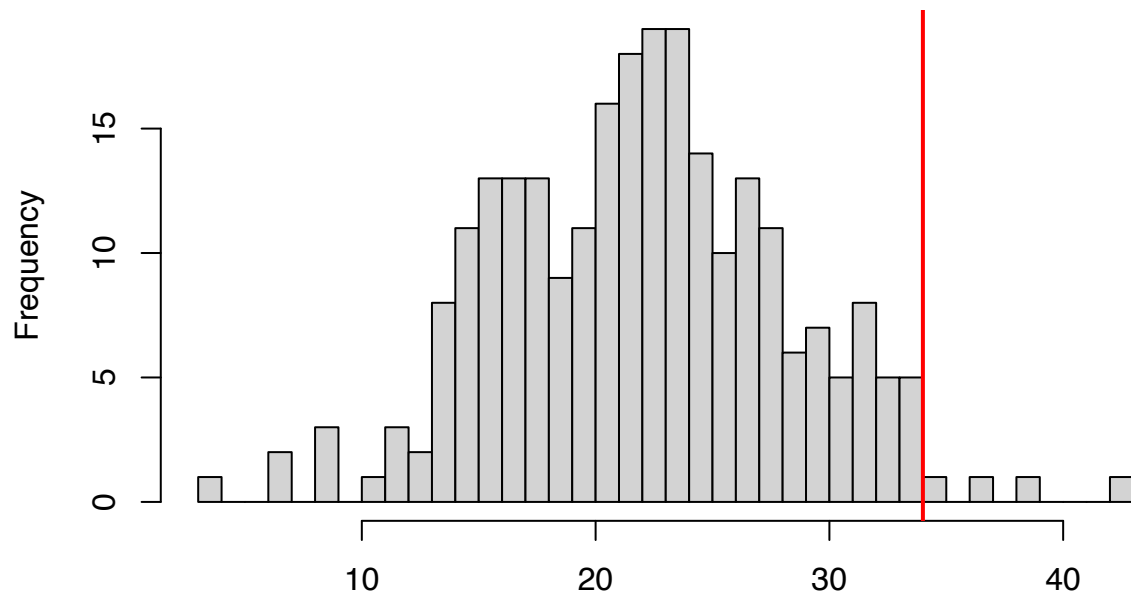


Simulated values, red line = fitted model. p-value (two.sided) = 0.856

```
##
## DHARMa nonparametric dispersion test via sd of residuals fitted vs.
## simulated
##
## data: simulationOutput
## dispersion = 0.60923, p-value = 0.856
## alternative hypothesis: two.sided
```

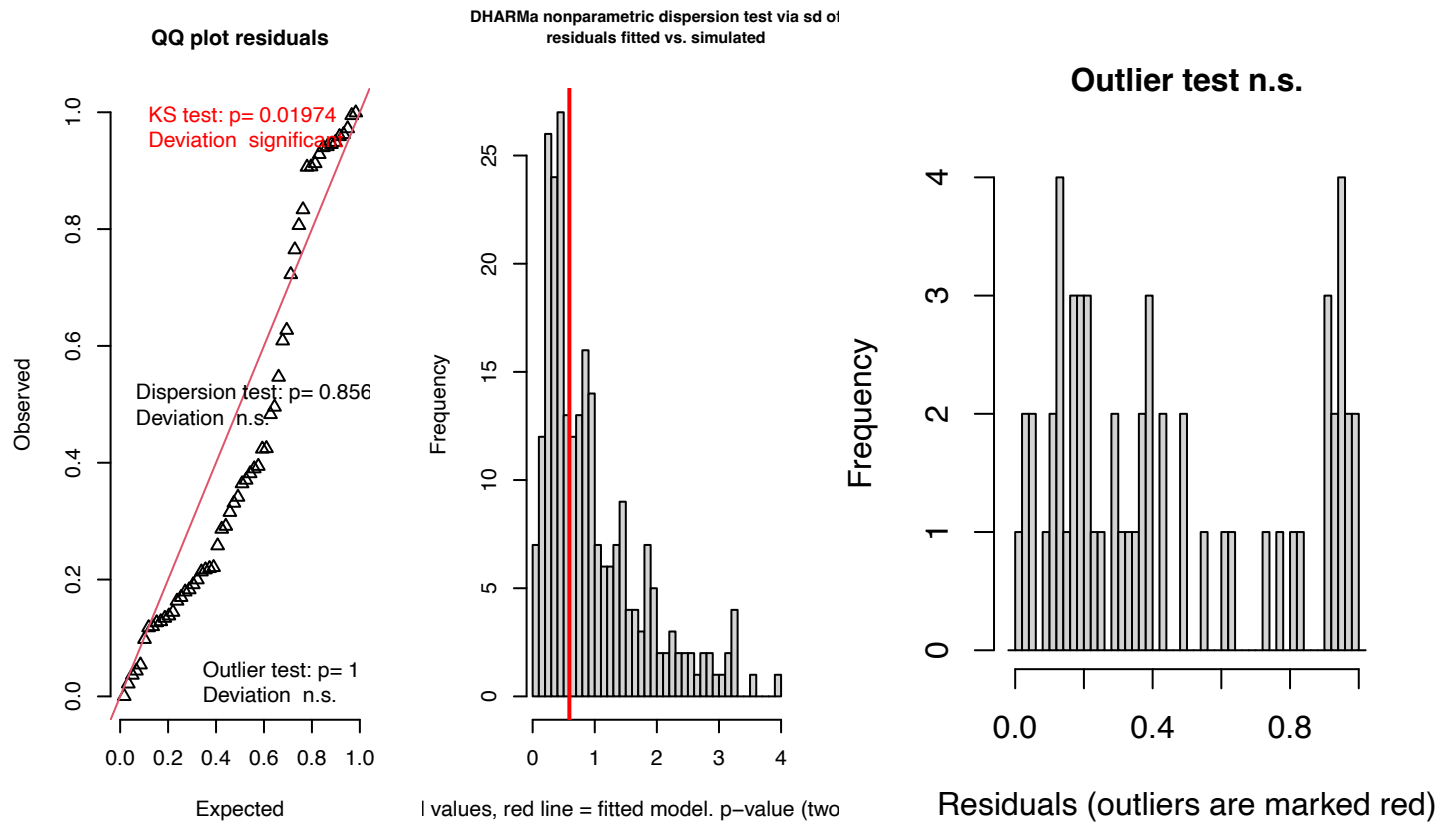
```
testZeroInflation(m17)
```

DHARMA zero-inflation test via comparison to expected zeros with simulation under H0 = fitted model



Simulated values, red line = fitted model. p-value (two.sided) = 0.072

```
##
## DHARMA zero-inflation test via comparison to expected zeros with
## simulation under H0 = fitted model
##
## data: simulationOutput
## ratioObsSim = 1.5108, p-value = 0.072
## alternative hypothesis: two.sided
testResiduals(m17)
```



```
## $uniformity
##
## Exact one-sample Kolmogorov-Smirnov test
##
## data: simulationOutput$scaledResiduals
## D = 0.19619, p-value = 0.01974
## alternative hypothesis: two-sided
##
##
## $dispersion
##
## DHARMA nonparametric dispersion test via sd of residuals fitted vs.
## simulated
##
## data: simulationOutput
## dispersion = 0.60923, p-value = 0.856
## alternative hypothesis: two.sided
##
##
## $outliers
##
## DHARMA bootstrapped outlier test
##
## data: simulationOutput
## outliers at both margin(s) = 0, observations = 58, p-value = 1
## alternative hypothesis: two.sided
## percent confidence interval:
## 0.00000000 0.03448276
```

```

## sample estimates:
## outlier frequency (expected: 0.00310344827586207 )
##
## $uniformity
##
## Exact one-sample Kolmogorov-Smirnov test
##
## data: simulationOutput$scaledResiduals
## D = 0.19619, p-value = 0.01974
## alternative hypothesis: two-sided
##
##
## $dispersion
##
## DHARMA nonparametric dispersion test via sd of residuals fitted vs.
## simulated
##
## data: simulationOutput
## dispersion = 0.60923, p-value = 0.856
## alternative hypothesis: two.sided
##
##
## $outliers
##
## DHARMA bootstrapped outlier test
##
## data: simulationOutput
## outliers at both margin(s) = 0, observations = 58, p-value = 1
## alternative hypothesis: two.sided
## percent confidence interval:
## 0.00000000 0.03448276
## sample estimates:
## outlier frequency (expected: 0.00310344827586207 )
##
##
# calculate the residuals for the model
m18 <- simulateResiduals(fittedModel = interaction_resp_molbig, plot = F)

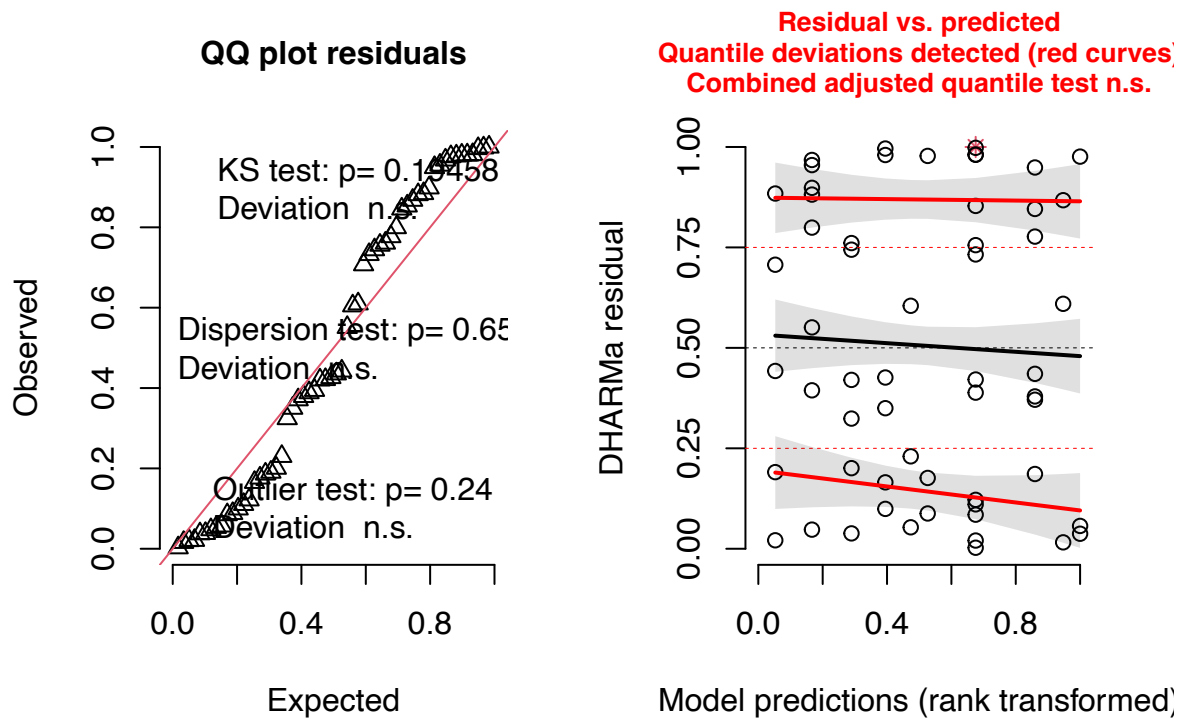
residuals(m18)

## [1] 1.000000000 0.997866324 0.948940564 0.981152844 0.021066642 0.706920567
## [7] 0.980447633 0.038221697 0.776972003 0.604705994 0.976000000 0.996125489
## [13] 0.190693235 0.176264738 0.853276190 0.867521716 0.744370065 0.977745907
## [19] 0.967669274 0.609625849 0.435364071 0.229984741 0.755665767 0.087863454
## [25] 0.884132636 0.881504285 0.047631457 0.980000000 0.200597123 0.845711758
## [31] 0.394308656 0.760818933 0.370895042 0.099274269 0.426191394 0.442493719
## [37] 0.324008302 0.550962969 0.799355613 0.186071724 0.421289583 0.056808827
## [43] 0.165291539 0.420266119 0.379138246 0.898323604 0.732468365 0.349857396
## [49] 0.954935418 0.053359042 0.037411382 0.015920528 0.084878301 0.121771032
## [55] 0.002553925 0.020454167 0.388549214 0.110491014

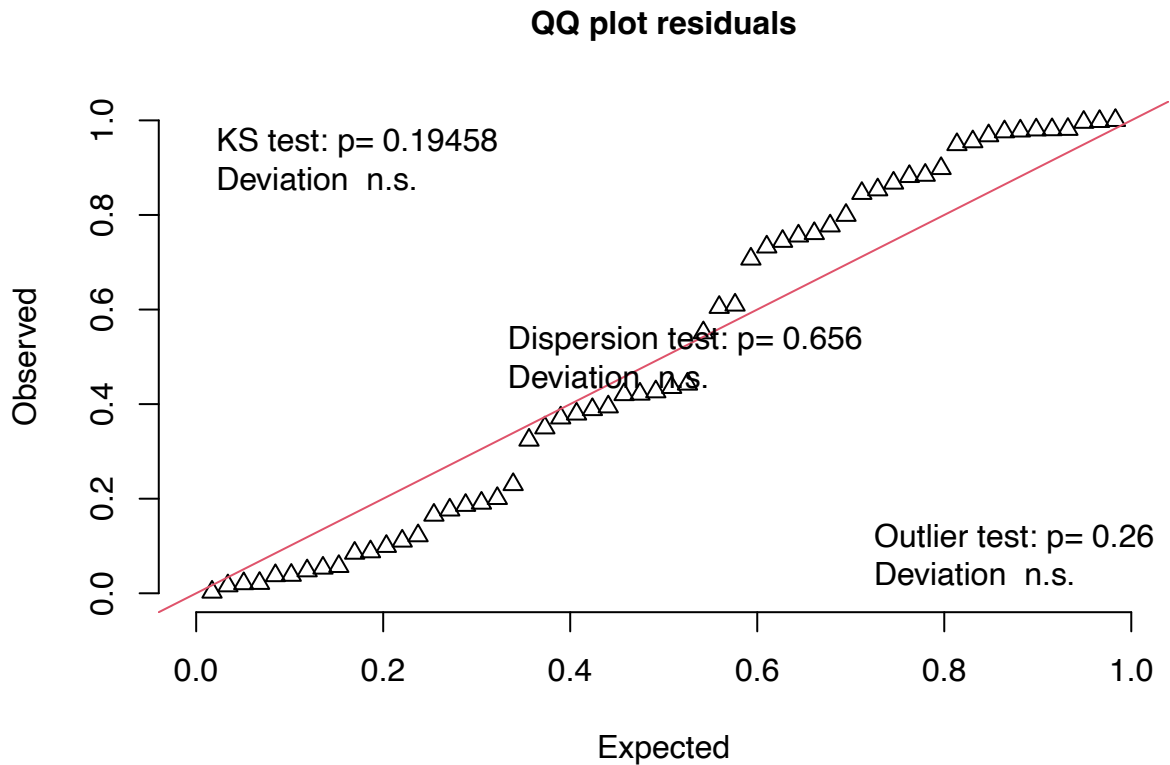
#plot simulation output
plot(m18)

```

DHARMA residual



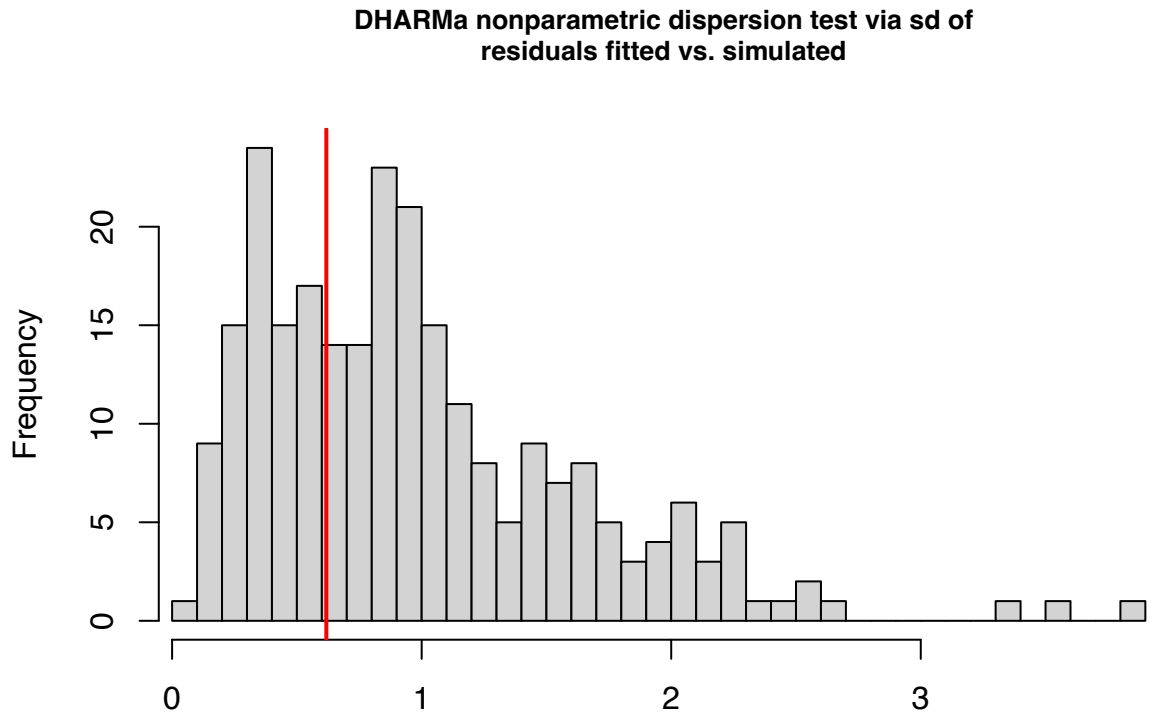
#tests if the overall distribution conforms to expectations.
`testUniformity(m18)`



 ## Exact one-sample Kolmogorov-Smirnov test

```
##
## data: simulationOutput$scaledResiduals
## D = 0.13882, p-value = 0.1946
## alternative hypothesis: two-sided
```

```
testDispersion(m18)
```

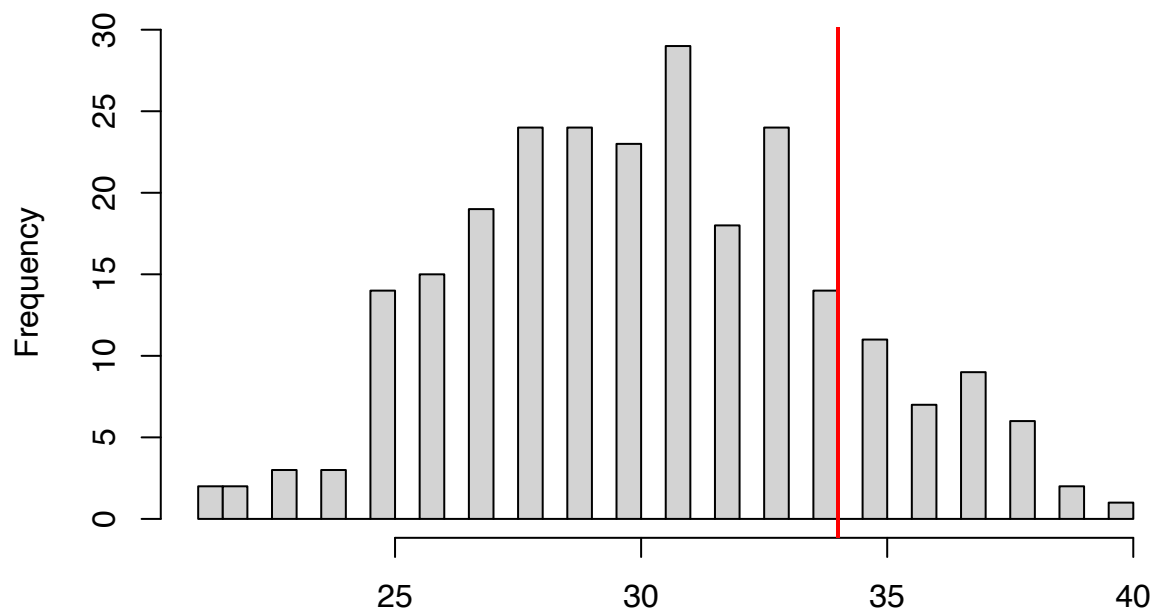


Simulated values, red line = fitted model. p-value (two.sided) = 0.656

```
##
## DHARMa nonparametric dispersion test via sd of residuals fitted vs.
## simulated
##
## data: simulationOutput
## dispersion = 0.62618, p-value = 0.656
## alternative hypothesis: two.sided
```

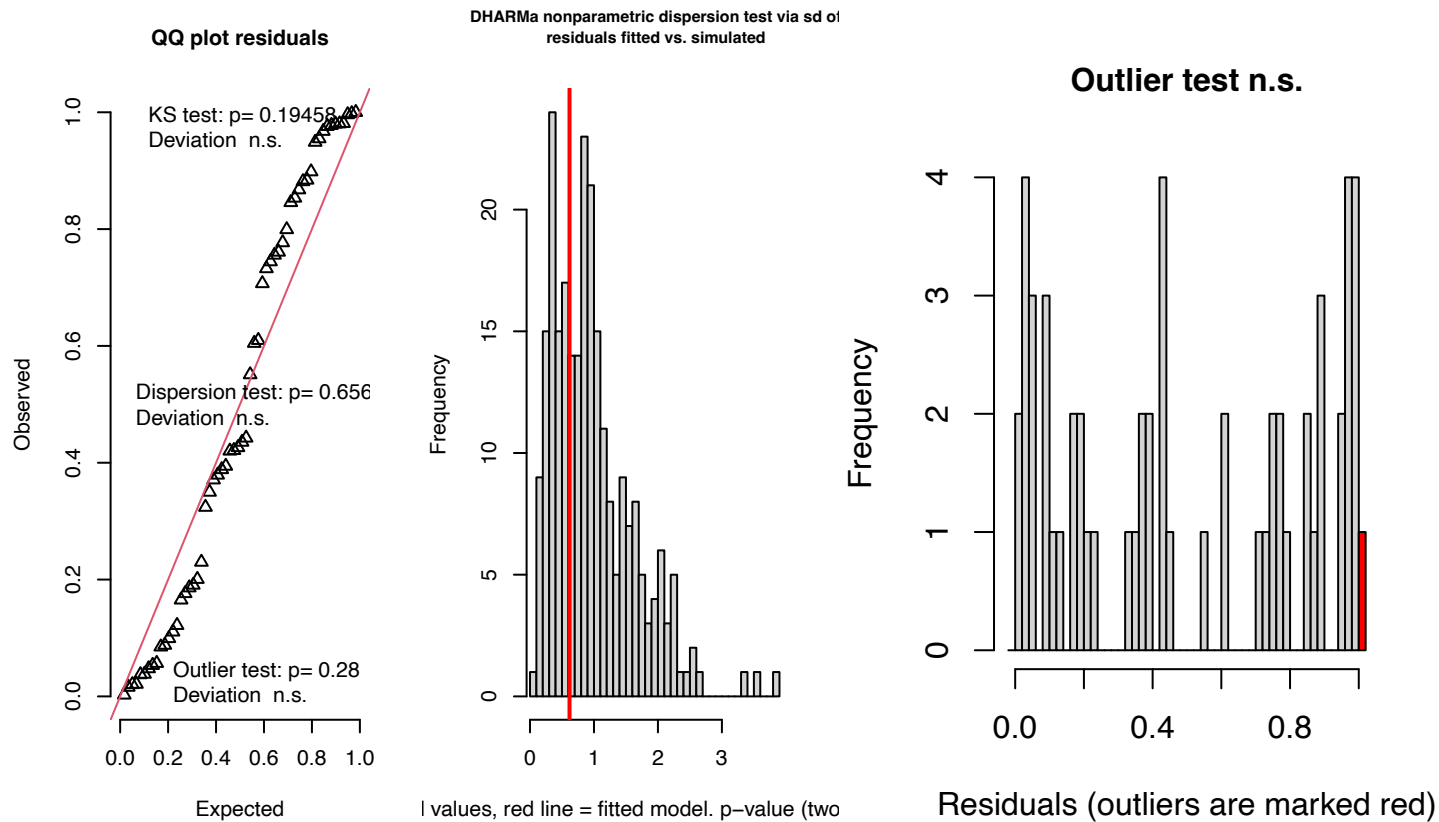
```
testZeroInflation(m18)
```

DHARMA zero-inflation test via comparison to expected zeros with simulation under H0 = fitted model



Simulated values, red line = fitted model. p-value (two.sided) = 0.4

```
##
## DHARMA zero-inflation test via comparison to expected zeros with
## simulation under H0 = fitted model
##
## data: simulationOutput
## ratioObsSim = 1.1189, p-value = 0.4
## alternative hypothesis: two.sided
testResiduals(m18)
```



```
## $uniformity
##
##   Exact one-sample Kolmogorov-Smirnov test
##
## data:  simulationOutput$scaledResiduals
## D = 0.13882, p-value = 0.1946
## alternative hypothesis: two-sided
##
##
## $dispersion
##
##   DHARMa nonparametric dispersion test via sd of residuals fitted vs.
##   simulated
##
## data:  simulationOutput
## dispersion = 0.62618, p-value = 0.656
## alternative hypothesis: two.sided
##
##
## $outliers
##
##   DHARMa bootstrapped outlier test
##
## data:  simulationOutput
## outliers at both margin(s) = 1, observations = 58, p-value = 0.22
## alternative hypothesis: two.sided
##   percent confidence interval:
##   0.00000000 0.03448276
```

```

## sample estimates:
## outlier frequency (expected: 0.00293103448275862 )
##                                0.01724138

## $uniformity
##
## Exact one-sample Kolmogorov-Smirnov test
##
## data: simulationOutput$scaledResiduals
## D = 0.13882, p-value = 0.1946
## alternative hypothesis: two-sided
##
##
## $dispersion
##
## DHARMA nonparametric dispersion test via sd of residuals fitted vs.
## simulated
##
## data: simulationOutput
## dispersion = 0.62618, p-value = 0.656
## alternative hypothesis: two.sided
##
##
## $outliers
##
## DHARMA bootstrapped outlier test
##
## data: simulationOutput
## outliers at both margin(s) = 1, observations = 58, p-value = 0.22
## alternative hypothesis: two.sided
## percent confidence interval:
## 0.00000000 0.03448276
## sample estimates:
## outlier frequency (expected: 0.00293103448275862 )
##                                0.01724138

# calculate the residuals for the model
m30 <- simulateResiduals(fittedModel = interaction_rej_molbig, plot = F)

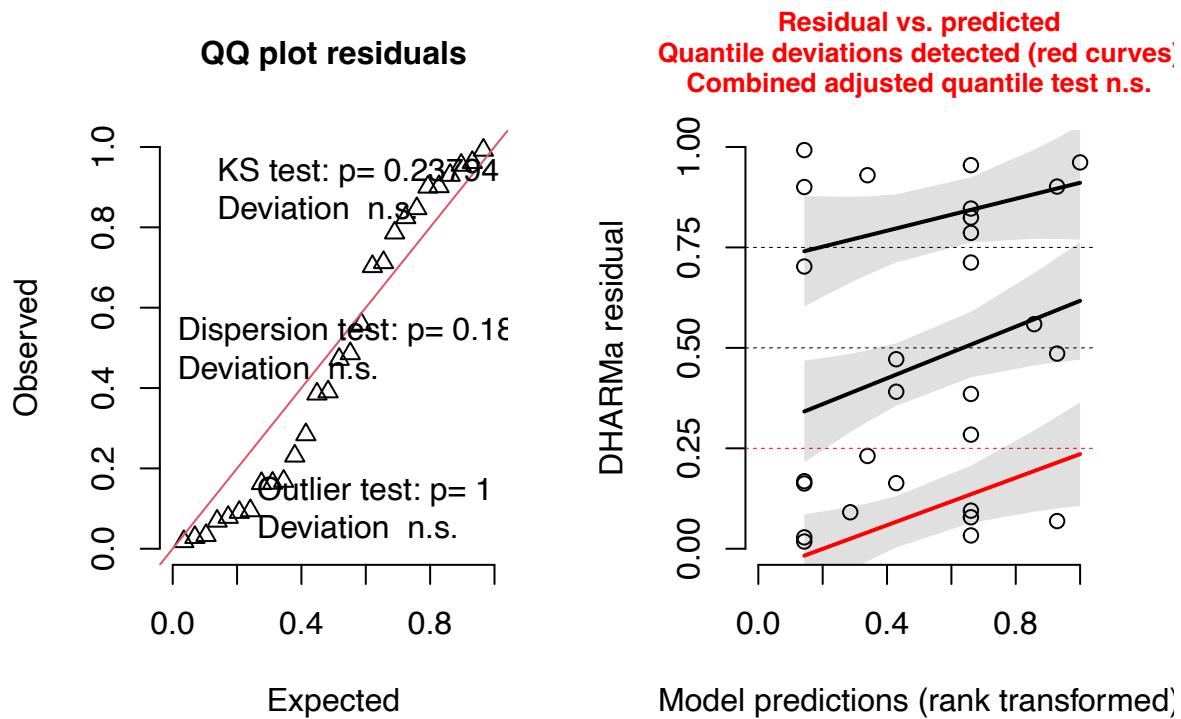
residuals(m30)

## [1] 0.16813357 0.28402805 0.38531230 0.16241866 0.23091928 0.09487309
## [7] 0.39080881 0.09080429 0.16336635 0.70251705 0.96153547 0.07838099
## [13] 0.06888305 0.02828293 0.90029568 0.55901121 0.90142674 0.99218892
## [19] 0.01798381 0.48573922 0.71238468 0.92968639 0.47174114 0.84671472
## [25] 0.95470641 0.82438473 0.03315419 0.78619368

#plot simulation output
plot(m30)

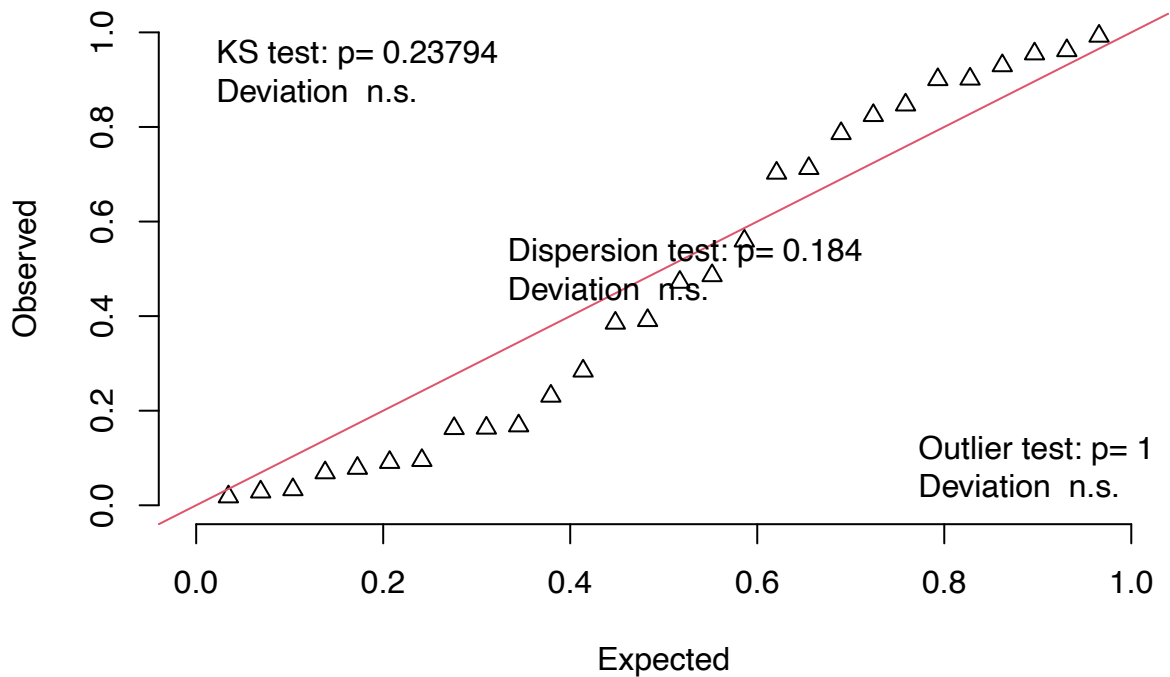
```

DHARMA residual



#tests if the overall distribution conforms to expectations.
`testUniformity(m30)`

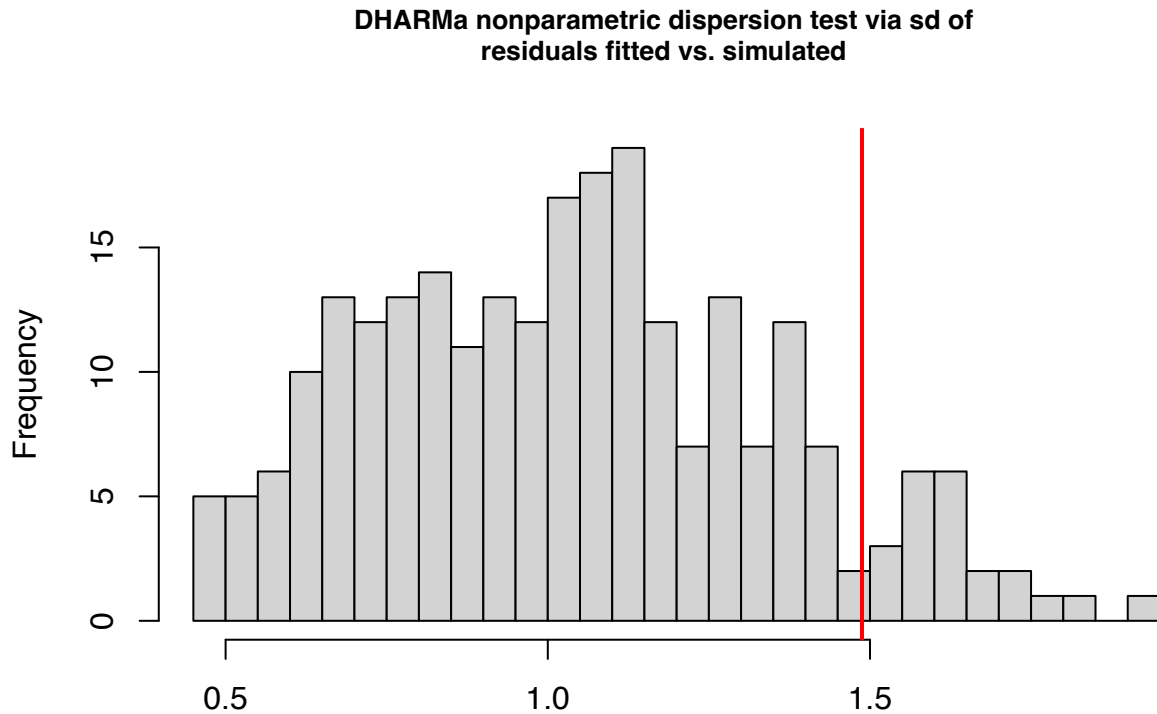
QQ plot residuals



 ## Exact one-sample Kolmogorov-Smirnov test

```
##
## data:  simulationOutput$scaledResiduals
## D = 0.18901, p-value = 0.2379
## alternative hypothesis: two-sided
```

```
testDispersion(m30)
```

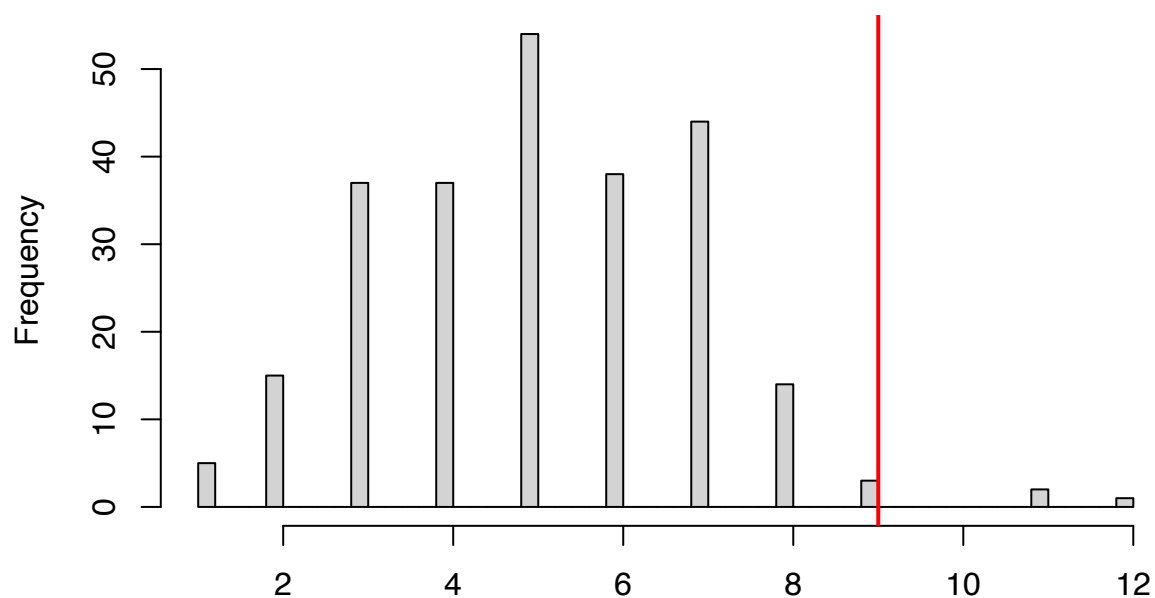


Simulated values, red line = fitted model. p-value (two.sided) = 0.184

```
##
## DHARMa nonparametric dispersion test via sd of residuals fitted vs.
## simulated
##
## data:  simulationOutput
## dispersion = 1.4335, p-value = 0.184
## alternative hypothesis: two.sided
```

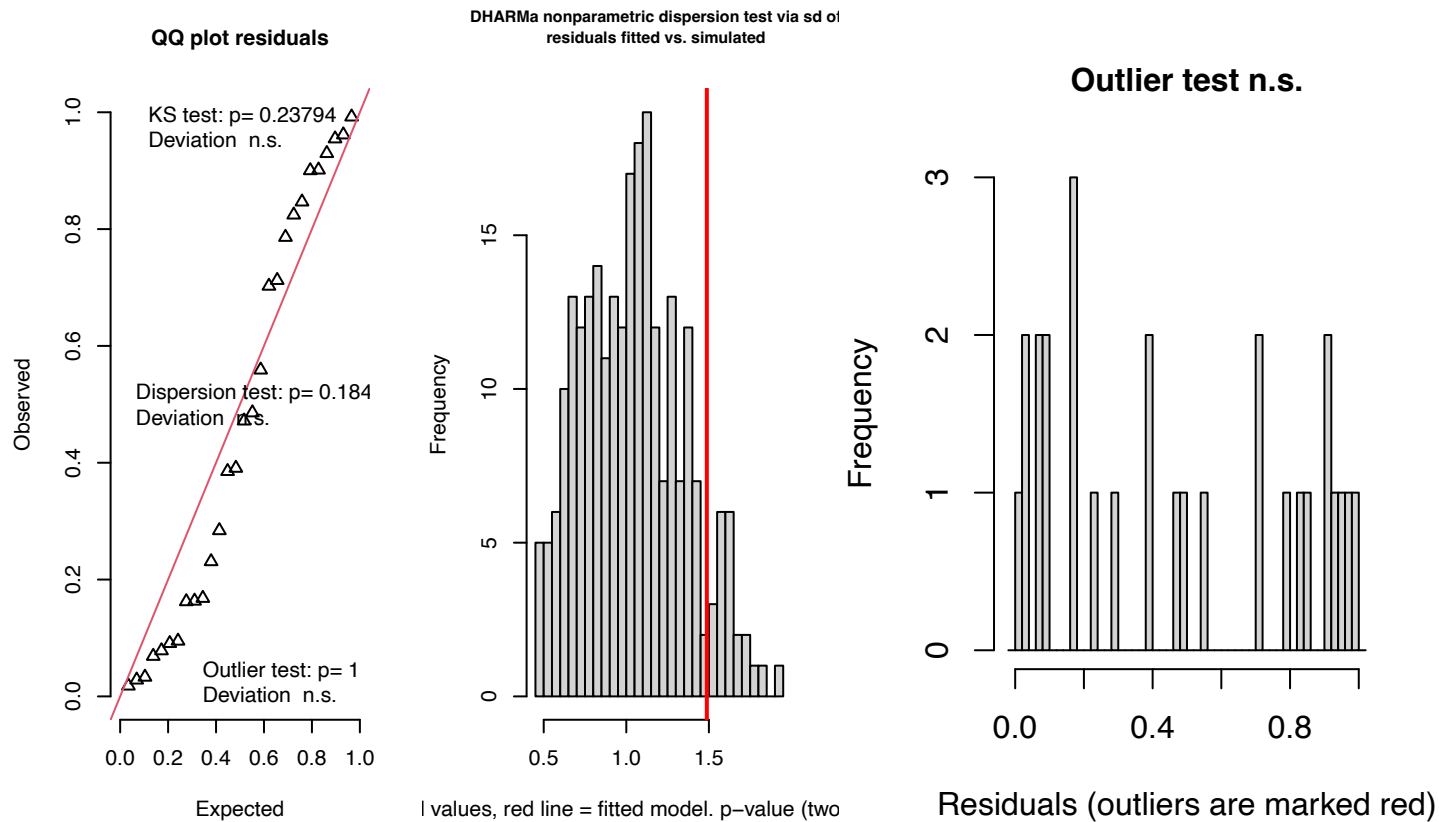
```
testZeroInflation(m30)
```

DHARMA zero-inflation test via comparison to expected zeros with simulation under H0 = fitted model



Simulated values, red line = fitted model. p-value (two.sided) = 0.048

```
##
## DHARMA zero-inflation test via comparison to expected zeros with
## simulation under H0 = fitted model
##
## data: simulationOutput
## ratioObsSim = 1.7675, p-value = 0.048
## alternative hypothesis: two.sided
testResiduals(m30)
```



```
## $uniformity
##
## Exact one-sample Kolmogorov-Smirnov test
##
## data: simulationOutput$scaledResiduals
## D = 0.18901, p-value = 0.2379
## alternative hypothesis: two-sided
##
##
## $dispersion
##
## DHARMA nonparametric dispersion test via sd of residuals fitted vs.
## simulated
##
## data: simulationOutput
## dispersion = 1.4335, p-value = 0.184
## alternative hypothesis: two.sided
##
##
## $outliers
##
## DHARMA bootstrapped outlier test
##
## data: simulationOutput
## outliers at both margin(s) = 0, observations = 28, p-value = 1
## alternative hypothesis: two.sided
## percent confidence interval:
## 0.00000 0.01875
```

```

## sample estimates:
## outlier frequency (expected: 0.00107142857142857 )
##
## $uniformity
##
## Exact one-sample Kolmogorov-Smirnov test
##
## data: simulationOutput$scaledResiduals
## D = 0.18901, p-value = 0.2379
## alternative hypothesis: two-sided
##
##
## $dispersion
##
## DHARMA nonparametric dispersion test via sd of residuals fitted vs.
## simulated
##
## data: simulationOutput
## dispersion = 1.4335, p-value = 0.184
## alternative hypothesis: two.sided
##
##
## $outliers
##
## DHARMA bootstrapped outlier test
##
## data: simulationOutput
## outliers at both margin(s) = 0, observations = 28, p-value = 1
## alternative hypothesis: two.sided
## percent confidence interval:
## 0.00000 0.01875
## sample estimates:
## outlier frequency (expected: 0.00107142857142857 )
##
##
## !!!! POTENTIAL ISSUES WITH ZERO INFLATION

# calculate the residuals for the model
m30 <- simulateResiduals(fittedModel = interaction_rej_molbig, plot = F)

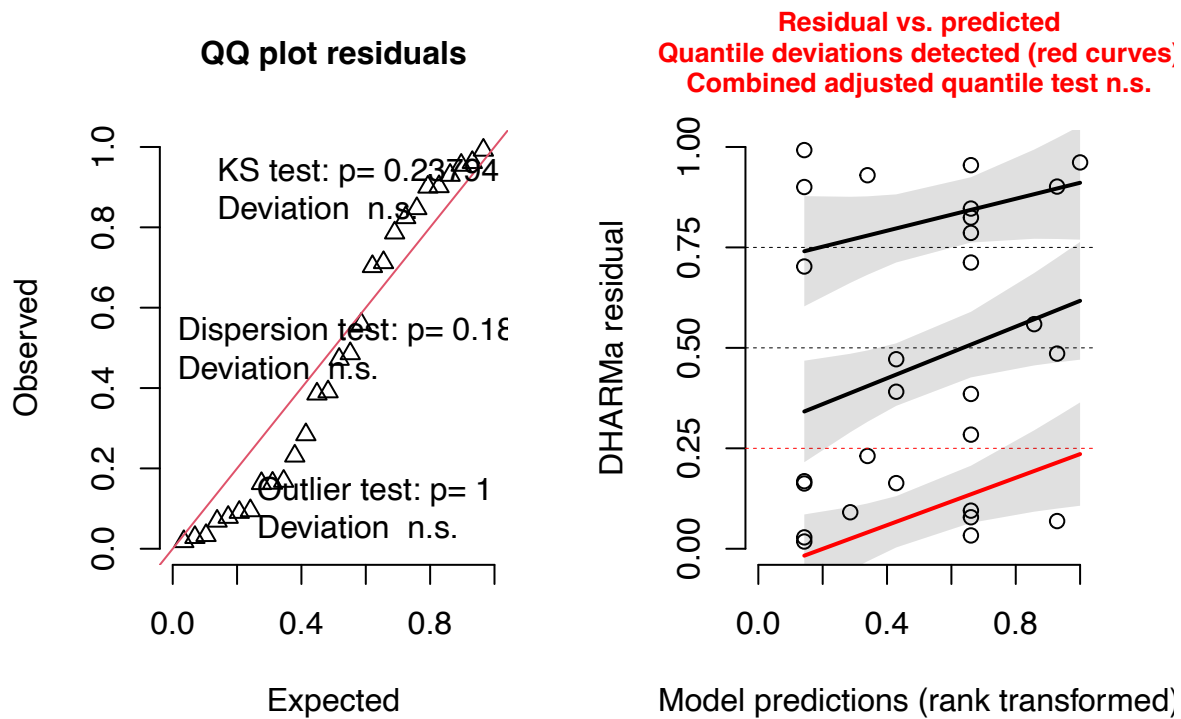
residuals(m30)

## [1] 0.16813357 0.28402805 0.38531230 0.16241866 0.23091928 0.09487309
## [7] 0.39080881 0.09080429 0.16336635 0.70251705 0.96153547 0.07838099
## [13] 0.06888305 0.02828293 0.90029568 0.55901121 0.90142674 0.99218892
## [19] 0.01798381 0.48573922 0.71238468 0.92968639 0.47174114 0.84671472
## [25] 0.95470641 0.82438473 0.03315419 0.78619368

#plot simulation output
plot(m30)

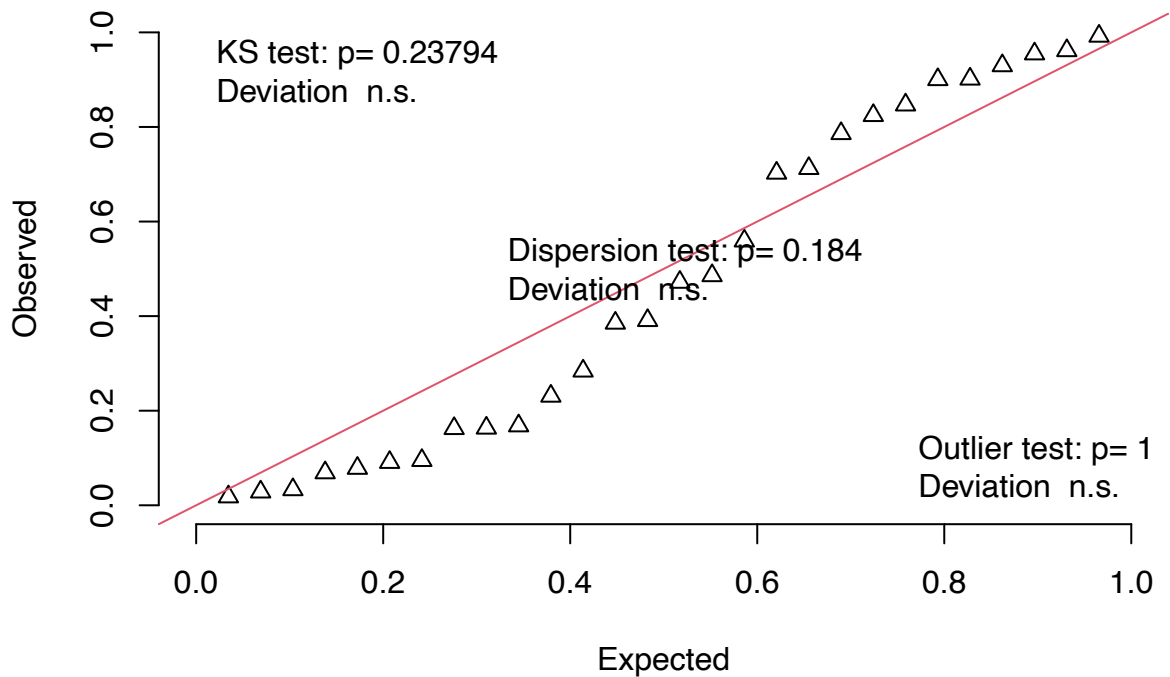
```

DHARMA residual



#tests if the overall distribution conforms to expectations.
`testUniformity(m30)`

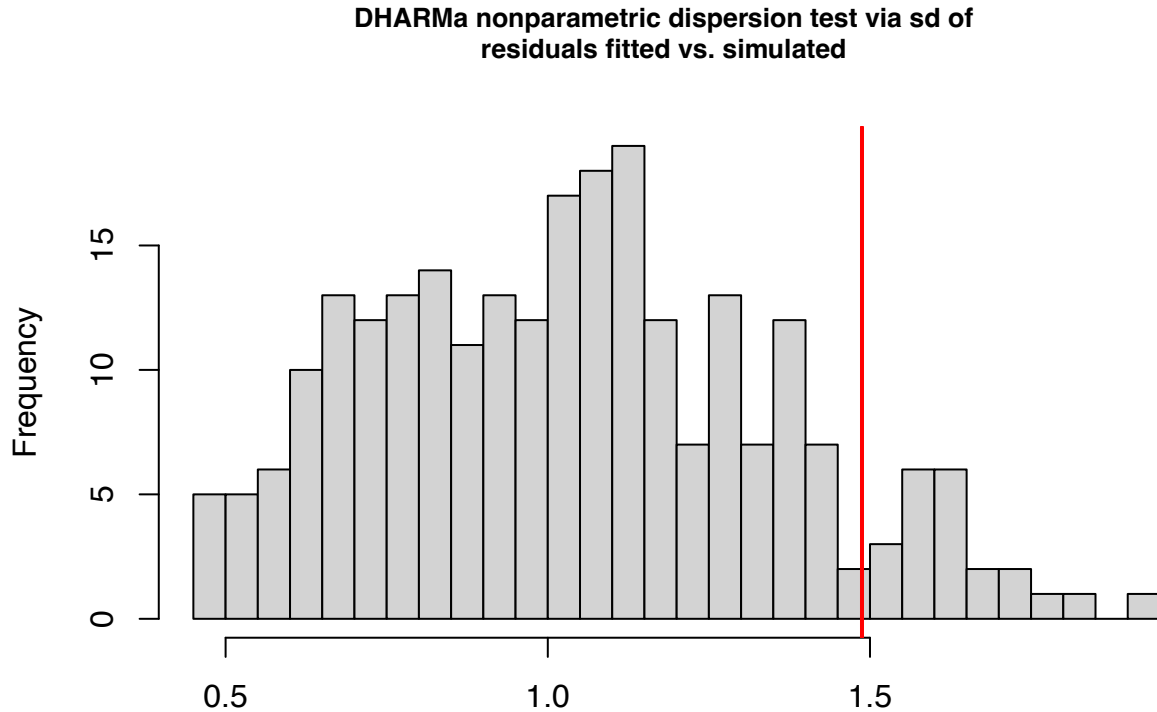
QQ plot residuals



 ## Exact one-sample Kolmogorov-Smirnov test

```
##
## data:  simulationOutput$scaledResiduals
## D = 0.18901, p-value = 0.2379
## alternative hypothesis: two-sided
```

```
testDispersion(m30)
```

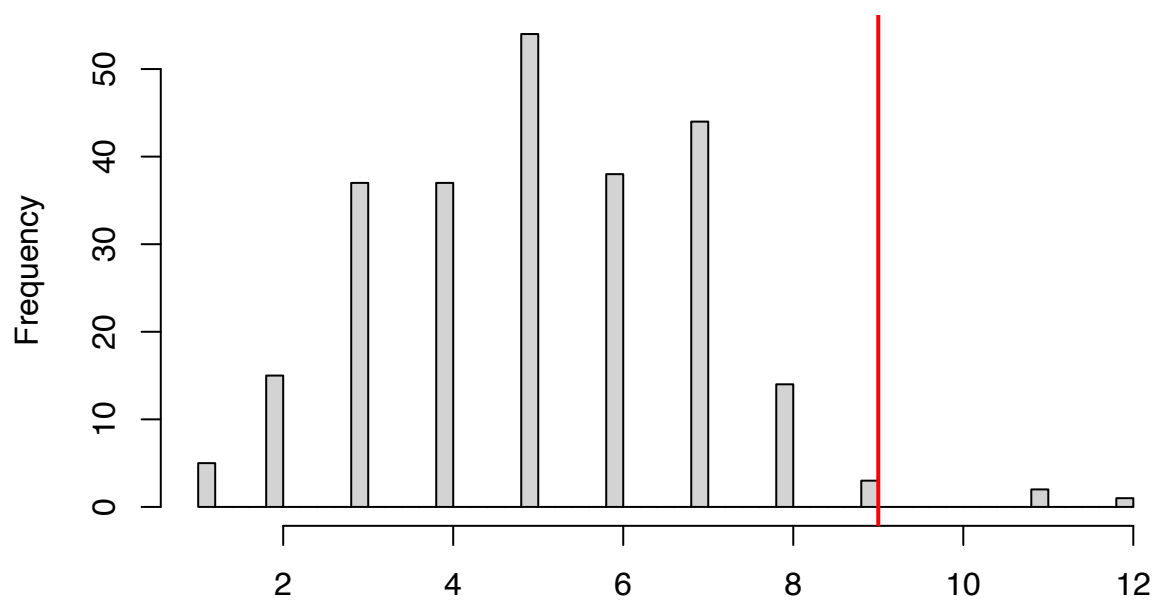


Simulated values, red line = fitted model. p-value (two.sided) = 0.184

```
##
## DHARMa nonparametric dispersion test via sd of residuals fitted vs.
## simulated
##
## data:  simulationOutput
## dispersion = 1.4335, p-value = 0.184
## alternative hypothesis: two.sided
```

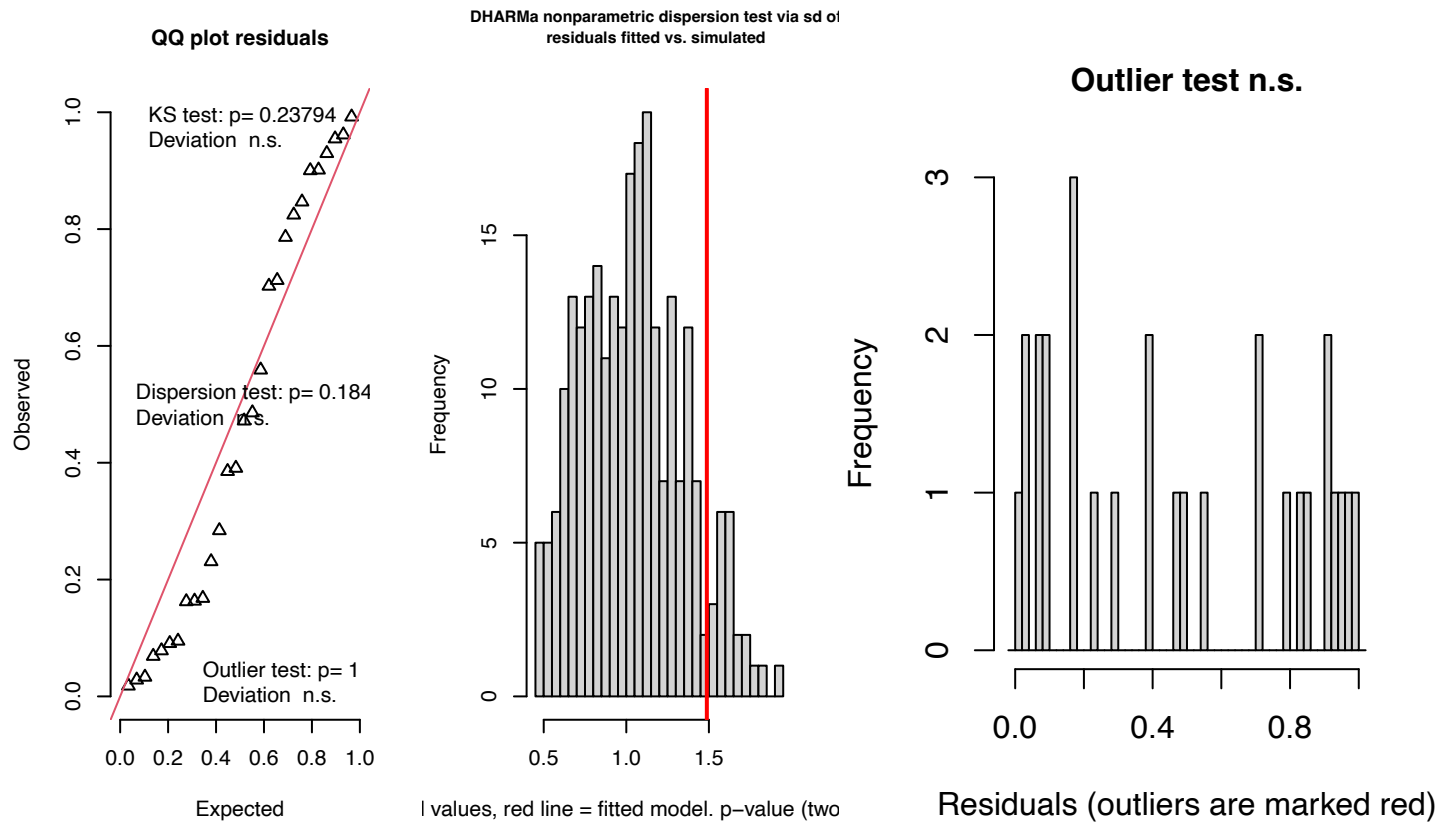
```
testZeroInflation(m30)
```

DHARMA zero-inflation test via comparison to expected zeros with simulation under H0 = fitted model



Simulated values, red line = fitted model. p-value (two.sided) = 0.048

```
##
## DHARMA zero-inflation test via comparison to expected zeros with
## simulation under H0 = fitted model
##
## data: simulationOutput
## ratioObsSim = 1.7675, p-value = 0.048
## alternative hypothesis: two.sided
testResiduals(m30)
```



```
## $uniformity
##
## Exact one-sample Kolmogorov-Smirnov test
##
## data: simulationOutput$scaledResiduals
## D = 0.18901, p-value = 0.2379
## alternative hypothesis: two-sided
##
##
## $dispersion
##
## DHARMA nonparametric dispersion test via sd of residuals fitted vs.
## simulated
##
## data: simulationOutput
## dispersion = 1.4335, p-value = 0.184
## alternative hypothesis: two.sided
##
##
## $outliers
##
## DHARMA bootstrapped outlier test
##
## data: simulationOutput
## outliers at both margin(s) = 0, observations = 28, p-value = 1
## alternative hypothesis: two.sided
## percent confidence interval:
## 0.00000000 0.03571429
```

```

## sample estimates:
## outlier frequency (expected: 0.00142857142857143 )
##
## 0

## $uniformity
##
## Exact one-sample Kolmogorov-Smirnov test
##
## data: simulationOutput$scaledResiduals
## D = 0.18901, p-value = 0.2379
## alternative hypothesis: two-sided
##
##
## $dispersion
##
## DHARMA nonparametric dispersion test via sd of residuals fitted vs.
## simulated
##
## data: simulationOutput
## dispersion = 1.4335, p-value = 0.184
## alternative hypothesis: two.sided
##
##
## $outliers
##
## DHARMA bootstrapped outlier test
##
## data: simulationOutput
## outliers at both margin(s) = 0, observations = 28, p-value = 1
## alternative hypothesis: two.sided
## percent confidence interval:
## 0.00000000 0.03571429
## sample estimates:
## outlier frequency (expected: 0.00142857142857143 )
##
## 0

# calculate the residuals for the model
m35 <- simulateResiduals(fittedModel = glm_int_molbig, plot = F)

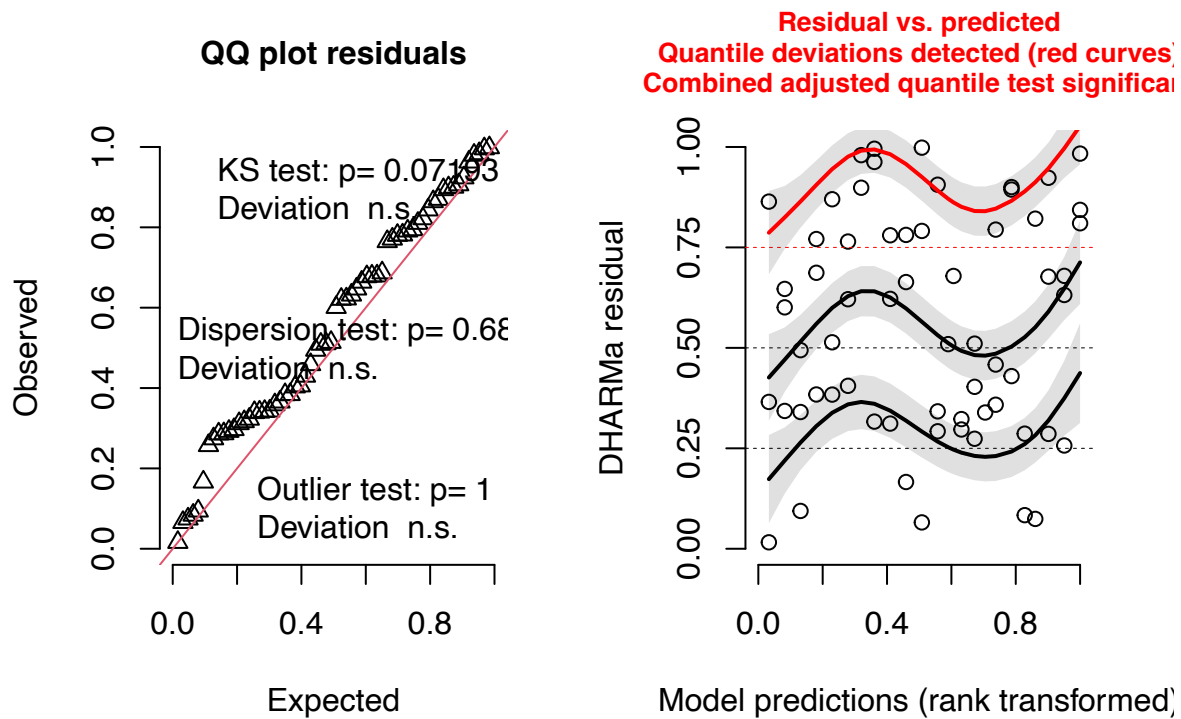
residuals(m35)

## [1] 0.77085650 0.25703781 0.67921933 0.50927810 0.90608934 0.29238867
## [7] 0.67728003 0.81008545 0.01603164 0.68688248 0.84348131 0.64666239
## [13] 0.92333984 0.86987254 0.79433496 0.31158318 0.38400010 0.06558162
## [19] 0.82157353 0.89397539 0.60080925 0.99829072 0.49427552 0.90013825
## [25] 0.40300872 0.38372877 0.98352099 0.79105178 0.86403006 0.09412743
## [31] 0.97969985 0.78024228 0.34300251 0.51052091 0.34028725 0.31646778
## [37] 0.27380127 0.62194290 0.76463253 0.36546265 0.96274790 0.89857954
## [43] 0.66374024 0.28591365 0.32272656 0.45845389 0.62105494 0.99560250
## [49] 0.34196969 0.78112316 0.63163049 0.40554907 0.16617980 0.51383855
## [55] 0.35850762 0.42973554 0.29680600 0.28646018 0.07427792 0.08368133
## [61] 0.33911773 0.67895374

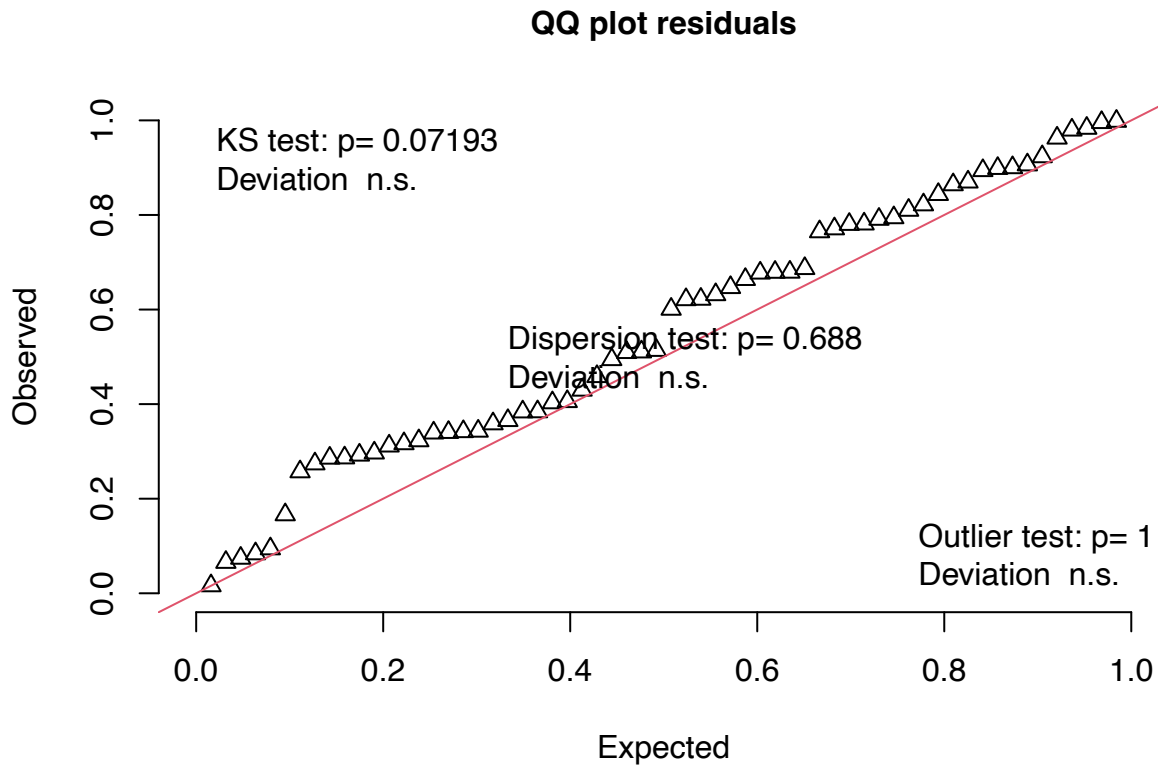
#plot simulation output
plot(m35)

```

DHARMA residual



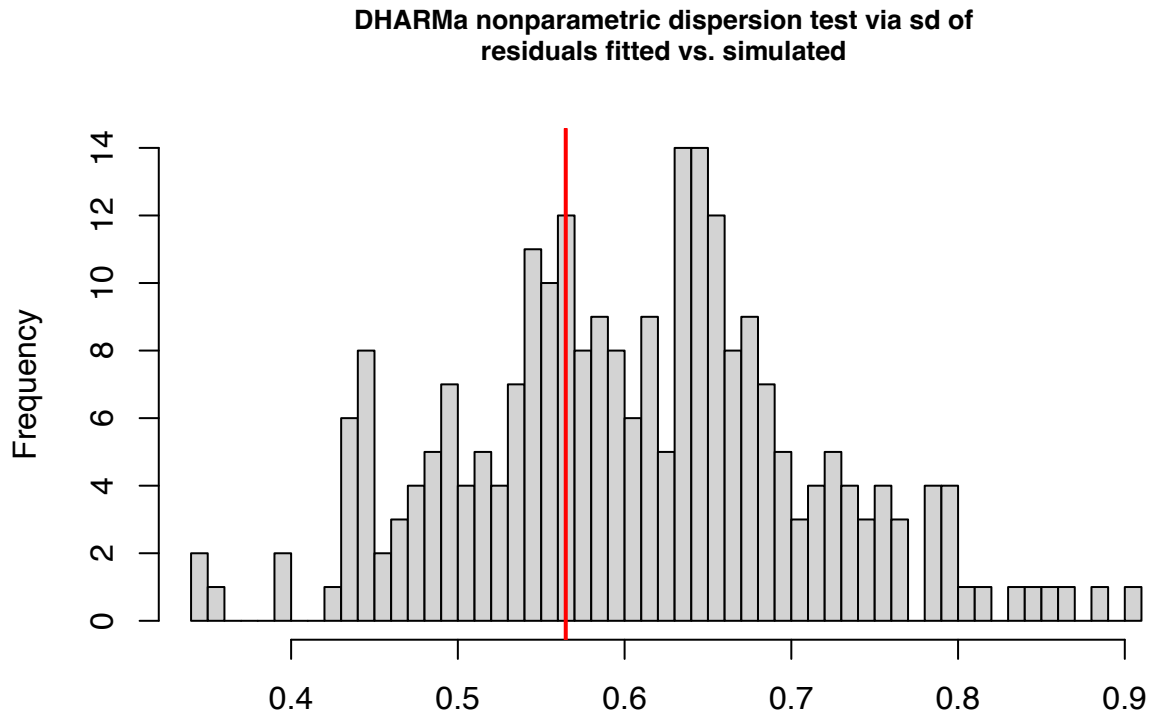
#tests if the overall distribution conforms to expectations.
`testUniformity(m35)`



 ## Exact one-sample Kolmogorov-Smirnov test

```
##
## data: simulationOutput$scaledResiduals
## D = 0.1609, p-value = 0.07193
## alternative hypothesis: two-sided
```

```
testDispersion(m35)
```

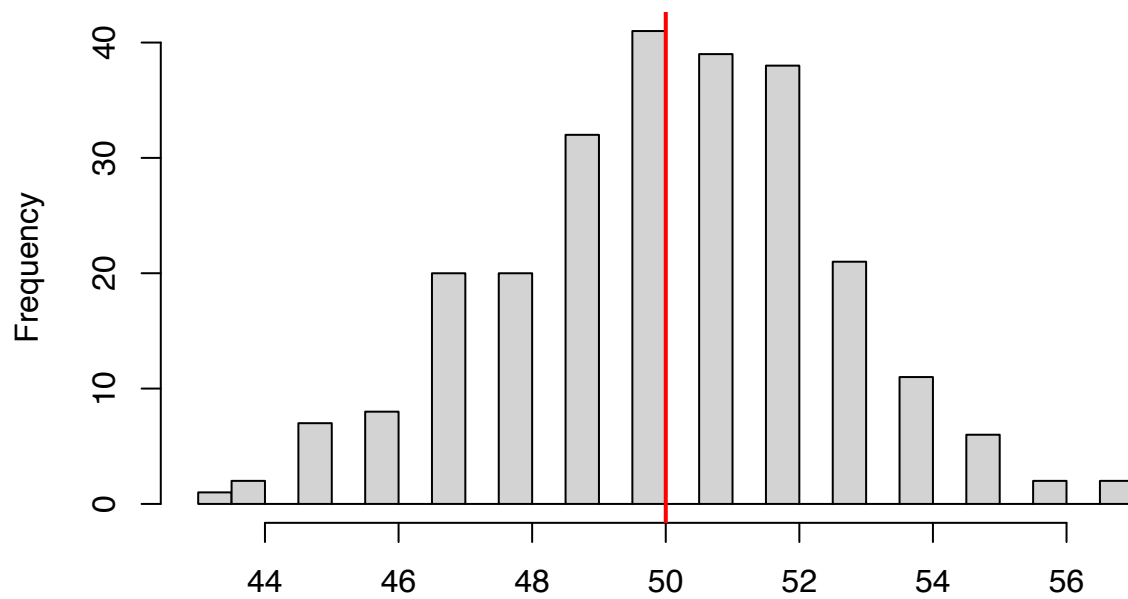


Simulated values, red line = fitted model. p-value (two.sided) = 0.688

```
##
## DHARMa nonparametric dispersion test via sd of residuals fitted vs.
## simulated
##
## data: simulationOutput
## dispersion = 0.93024, p-value = 0.688
## alternative hypothesis: two.sided
```

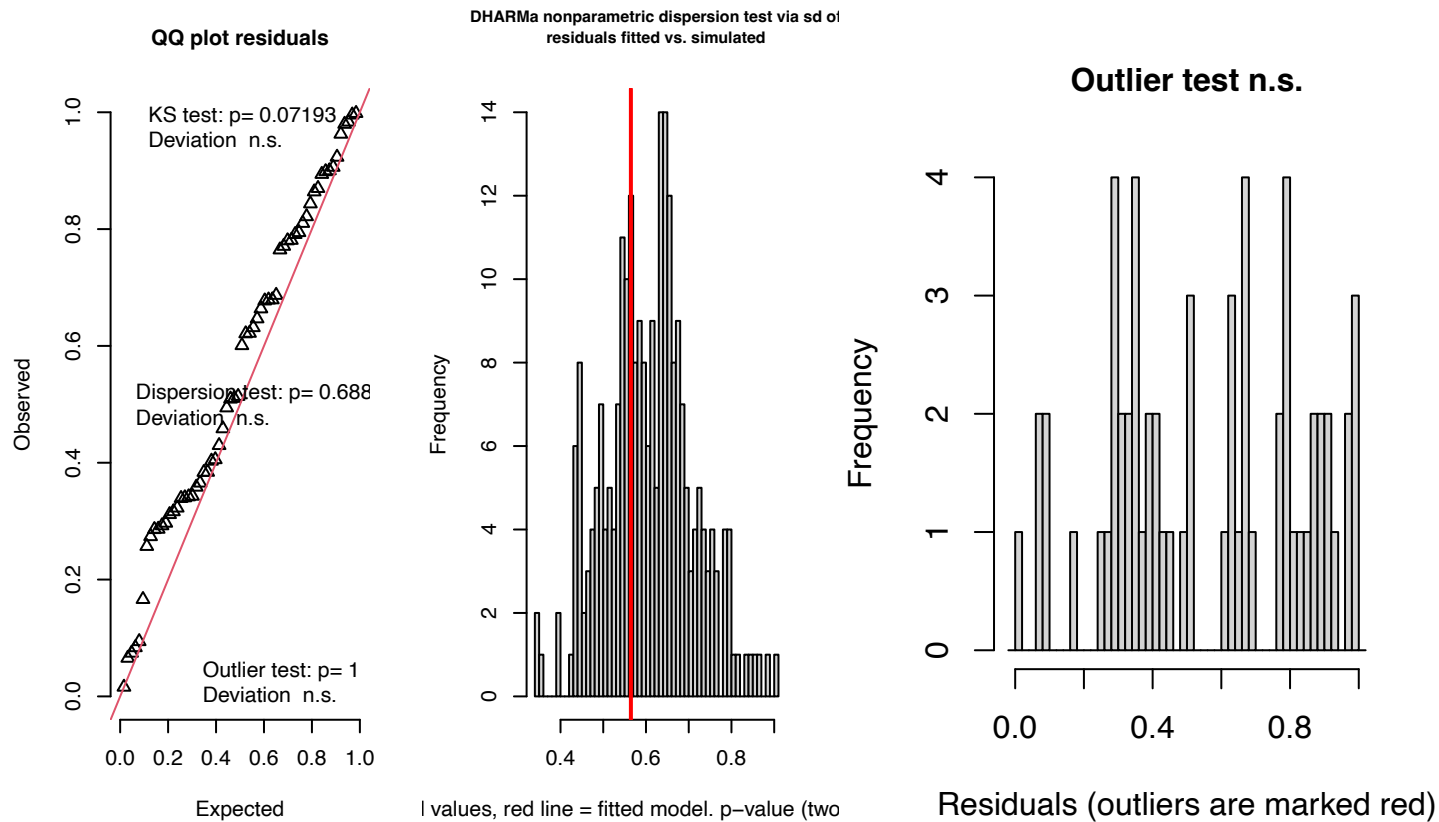
```
testZeroInflation(m35)
```

DHARMA zero-inflation test via comparison to expected zeros with simulation under H0 = fitted model



Simulated values, red line = fitted model. p-value (two.sided) = 1

```
##
## DHARMA zero-inflation test via comparison to expected zeros with
## simulation under H0 = fitted model
##
## data: simulationOutput
## ratioObsSim = 0.99522, p-value = 1
## alternative hypothesis: two.sided
testResiduals(m35)
```



```
## $uniformity
##
## Exact one-sample Kolmogorov-Smirnov test
##
## data: simulationOutput$scaledResiduals
## D = 0.1609, p-value = 0.07193
## alternative hypothesis: two-sided
##
##
## $dispersion
##
## DHARMA nonparametric dispersion test via sd of residuals fitted vs.
## simulated
##
## data: simulationOutput
## dispersion = 0.93024, p-value = 0.688
## alternative hypothesis: two.sided
##
##
## $outliers
##
## DHARMA bootstrapped outlier test
##
## data: simulationOutput
## outliers at both margin(s) = 0, observations = 62, p-value = 1
## alternative hypothesis: two.sided
## percent confidence interval:
## 0 0
```

```
## sample estimates:
## outlier frequency (expected: 0.00032258064516129 )
##
## $uniformity
##
## Exact one-sample Kolmogorov-Smirnov test
##
## data: simulationOutput$scaledResiduals
## D = 0.1609, p-value = 0.07193
## alternative hypothesis: two-sided
##
##
## $dispersion
##
## DHARMA nonparametric dispersion test via sd of residuals fitted vs.
## simulated
##
## data: simulationOutput
## dispersion = 0.93024, p-value = 0.688
## alternative hypothesis: two.sided
##
##
## $outliers
##
## DHARMA bootstrapped outlier test
##
## data: simulationOutput
## outliers at both margin(s) = 0, observations = 62, p-value = 1
## alternative hypothesis: two.sided
## percent confidence interval:
## 0 0
## sample estimates:
## outlier frequency (expected: 0.00032258064516129 )
##
## 0
```

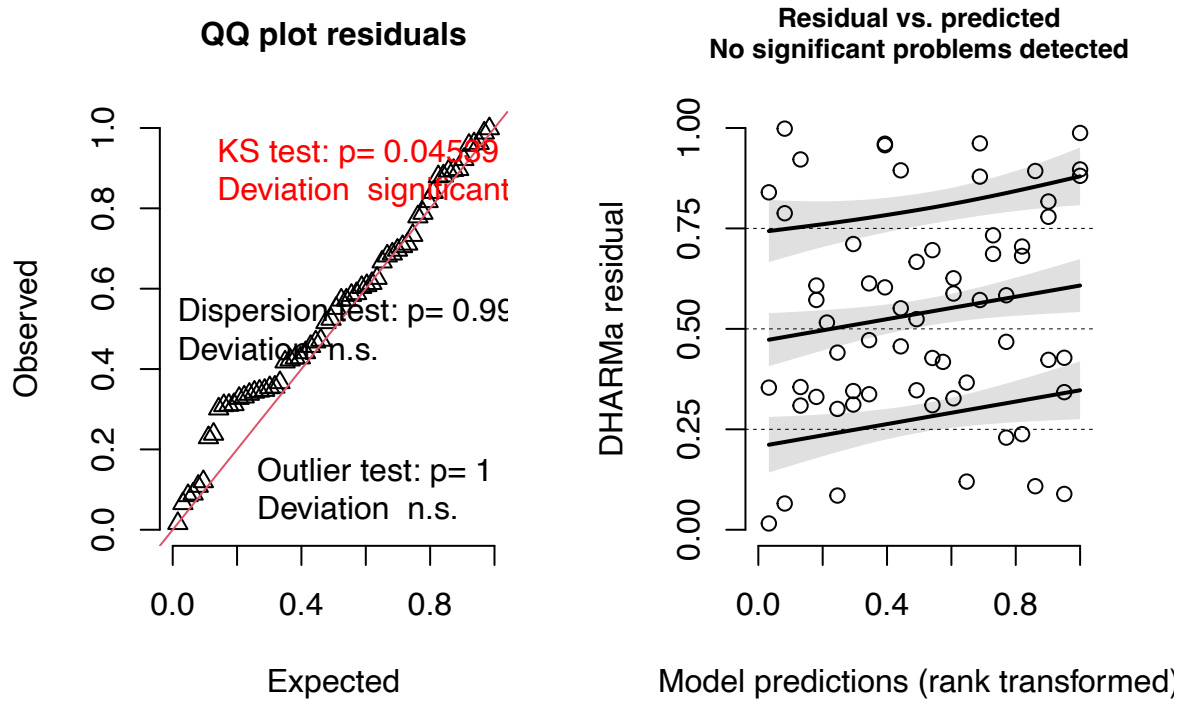
```
# calculate the residuals for the model
m36 <- simulateResiduals(fittedModel = glm_add_molbig, plot = F)
```

```
residuals(m36)
```

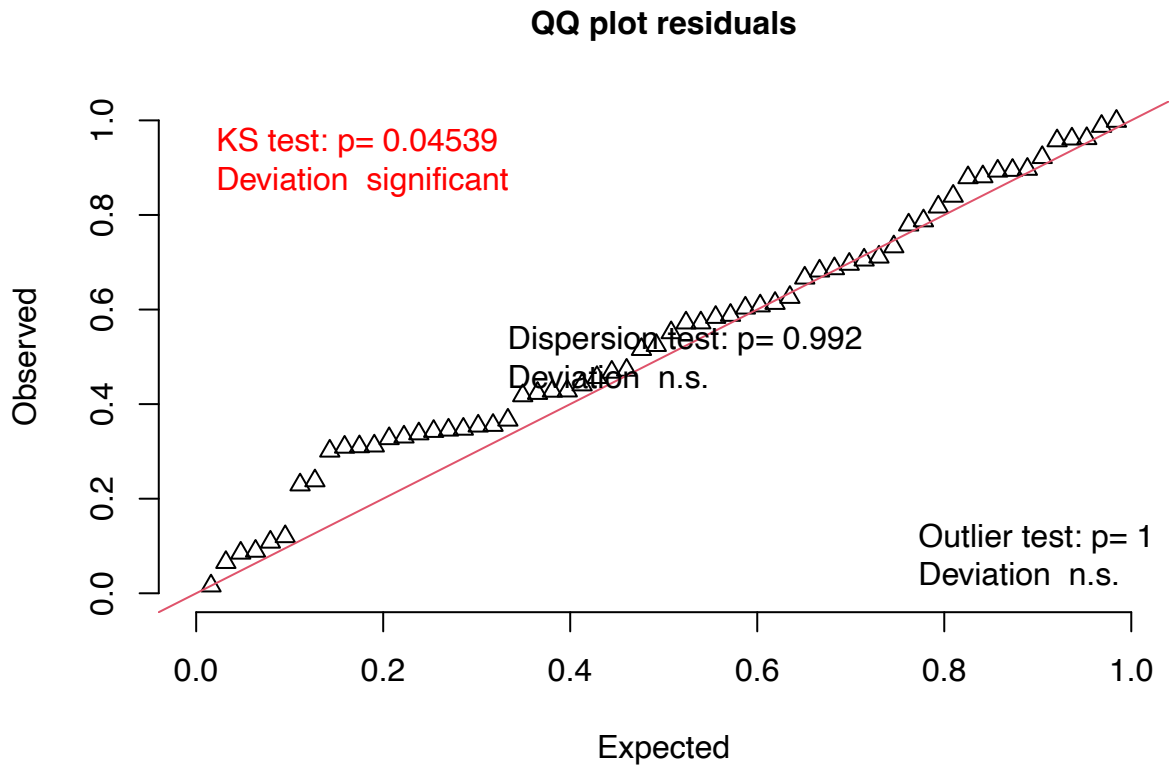
```
## [1] 0.62593548 0.42286865 0.81697790 0.51575198 0.92164453 0.30913340
## [7] 0.87898001 0.88093440 0.01571101 0.58797140 0.89674683 0.60786265
## [13] 0.96166992 0.69589803 0.89453075 0.22932522 0.32716809 0.06531717
## [19] 0.89294412 0.96081699 0.57197041 0.99829072 0.44089377 0.95702153
## [25] 0.47191695 0.31005285 0.98759339 0.78775573 0.83983721 0.08509120
## [31] 0.73281548 0.58362123 0.33065442 0.61316533 0.30081393 0.23798377
## [37] 0.33733526 0.46770106 0.66675957 0.35376785 0.68162552 0.68651477
## [43] 0.34248996 0.57182730 0.34524237 0.55117491 0.52417037 0.70488657
## [49] 0.35523575 0.42805549 0.77897829 0.34715001 0.08907237 0.42751368
## [55] 0.45648486 0.60328259 0.31164630 0.36635724 0.10832196 0.11990161
## [61] 0.41784148 0.71113164
```

```
#plot simulation output
plot(m36)
```

DHARMA residual



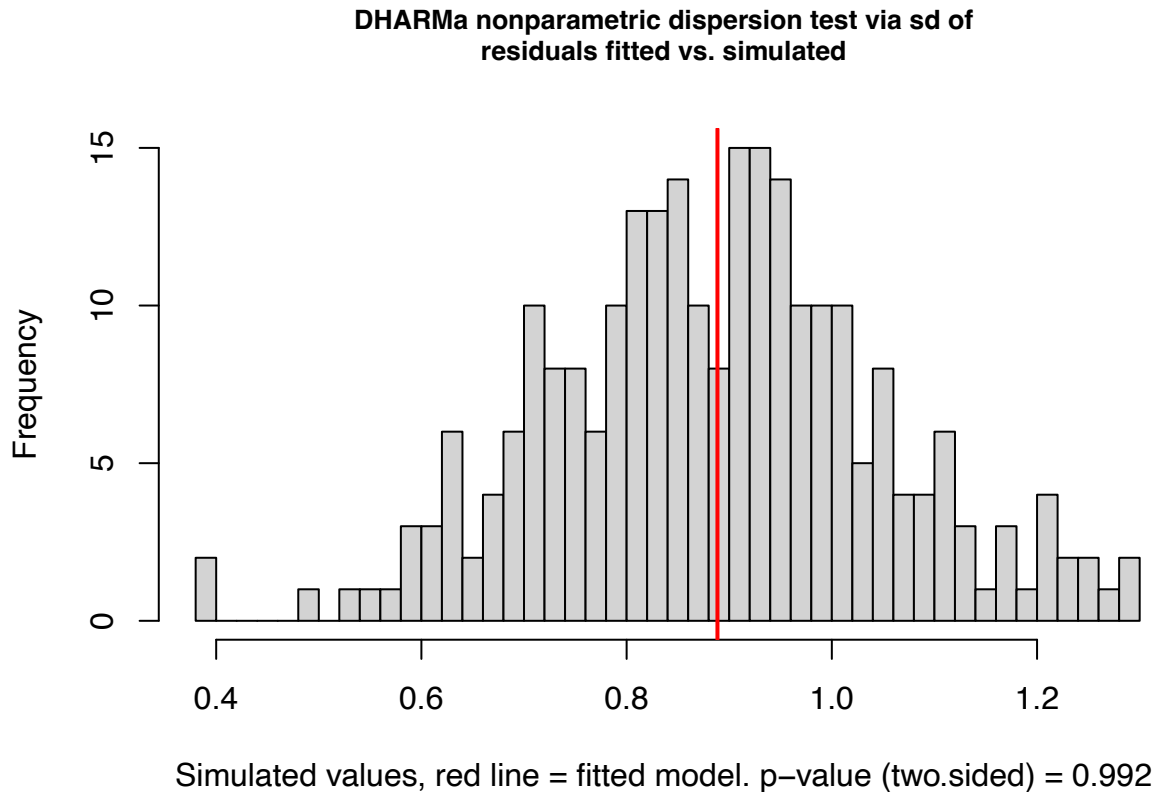
#tests if the overall distribution conforms to expectations.
`testUniformity(m36)`



 ## Exact one-sample Kolmogorov-Smirnov test

```
##
## data: simulationOutput$scaledResiduals
## D = 0.17178, p-value = 0.04539
## alternative hypothesis: two-sided
```

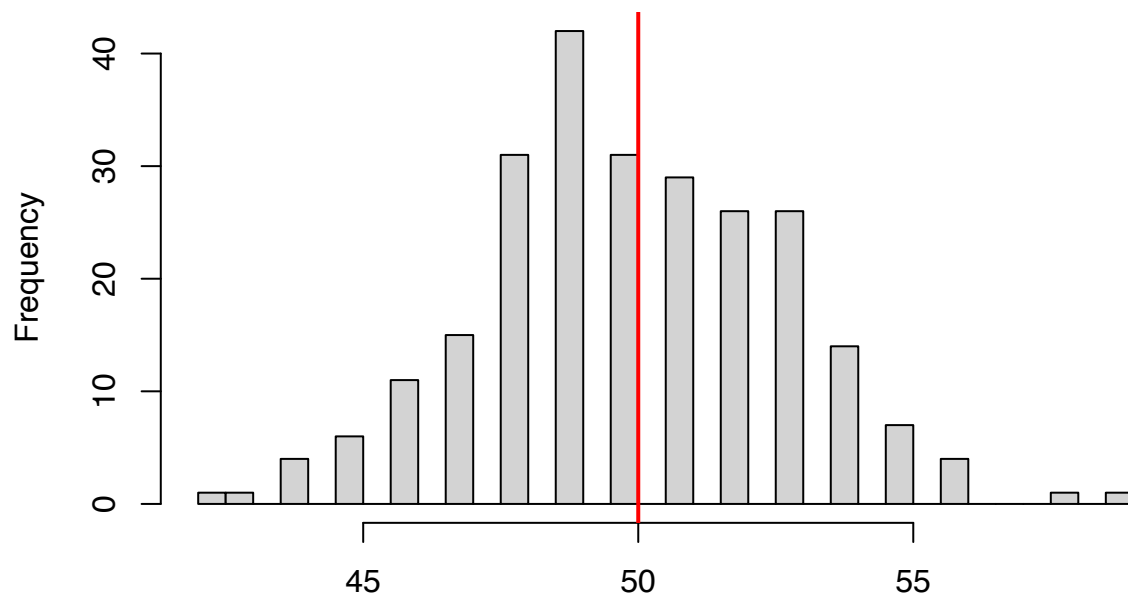
```
testDispersion(m36)
```



```
##
## DHARMa nonparametric dispersion test via sd of residuals fitted vs.
## simulated
##
## data: simulationOutput
## dispersion = 1.0036, p-value = 0.992
## alternative hypothesis: two.sided
```

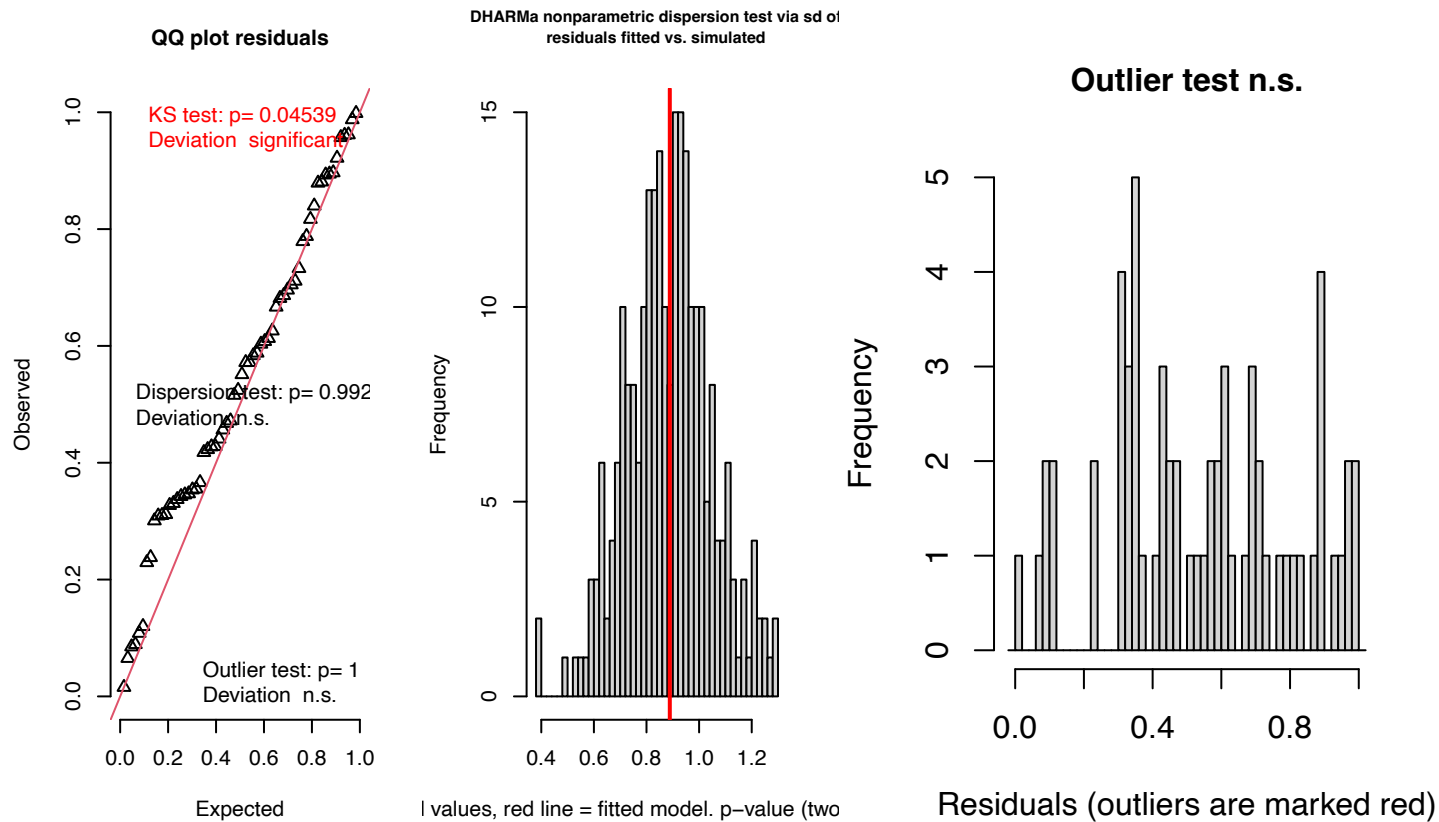
```
testZeroInflation(m36)
```

DHARMA zero-inflation test via comparison to expected zeros with simulation under H0 = fitted model



Simulated values, red line = fitted model. p-value (two.sided) = 1

```
##
## DHARMA zero-inflation test via comparison to expected zeros with
## simulation under H0 = fitted model
##
## data: simulationOutput
## ratioObsSim = 0.99769, p-value = 1
## alternative hypothesis: two.sided
testResiduals(m36)
```



```
## $uniformity
##
## Exact one-sample Kolmogorov-Smirnov test
##
## data: simulationOutput$scaledResiduals
## D = 0.17178, p-value = 0.04539
## alternative hypothesis: two-sided
##
##
## $dispersion
##
## DHARMA nonparametric dispersion test via sd of residuals fitted vs.
## simulated
##
## data: simulationOutput
## dispersion = 1.0036, p-value = 0.992
## alternative hypothesis: two.sided
##
##
## $outliers
##
## DHARMA bootstrapped outlier test
##
## data: simulationOutput
## outliers at both margin(s) = 0, observations = 62, p-value = 1
## alternative hypothesis: two.sided
## percent confidence interval:
## 0 0
```

```

## sample estimates:
## outlier frequency (expected: 0 )
##                                0

## $uniformity
##
## Exact one-sample Kolmogorov-Smirnov test
##
## data:  simulationOutput$scaledResiduals
## D = 0.17178, p-value = 0.04539
## alternative hypothesis: two-sided
##
##
## $dispersion
##
## DHARMA nonparametric dispersion test via sd of residuals fitted vs.
## simulated
##
## data:  simulationOutput
## dispersion = 1.0036, p-value = 0.992
## alternative hypothesis: two.sided
##
##
## $outliers
##
## DHARMA bootstrapped outlier test
##
## data:  simulationOutput
## outliers at both margin(s) = 0, observations = 62, p-value = 1
## alternative hypothesis: two.sided
## percent confidence interval:
## 0 0
## sample estimates:
## outlier frequency (expected: 0 )
##                                0

```